Specification

Renault / Nissan ODX Authoring Guidelines

Version	**1.0 dated 2017-12-29**
Status	draft
Author	Cont, Armin;Hummel, Markus;Hansen, Daniel
Copyright	2017 Vector Informatik GmbH

Released

Date	Name, Role

Inspected

Date	Name, Role

Modified

Version	Name	Chapter, short summary
0.1	Cont	Document created.
0.2	Cont	Update AGL to match latest ODX sample data.
0.3	Hummel, Cont	Extended AGL with all contents envisioned for the beta delivery.
0.4	Cont	Extended AGL chapter: ► Adapted short naming rules. ► Updated DTC modelling rules. ► Seed & key overriding. ► Usage of SPRMIB.
0.5	Cont	Added subsections of processes chapter.
0.6	Hansen, Cont	Extended AGL chapter: ► Adapted figures to reflect SHORT-NAMING rules (ongoing). ► Adapted figures to show constant values (ongoing). ► Added UDS $22 flavor supporting multiple DIDs in request. ► Updated UDS snapshot definition schemes. ► ODX-D support for 11 and 29 bit functional CAN identifiers. ► Merged Spec B and C diag layers. ► Updated rules / recos for usage of description elements. ► Mandated structure usage in flat services. ► Use of admin data. ► Modelling of UDS extended data records. ► Refined protocol selection impact on layer structure. ► LONG-NAME rules / DDT Data representation. ► Added descriptions of UDS $14. ► Revoked rules for overriding ISO 15765-2 comparams.

© 2017 Vector Informatik GmbH

0.7	Hansen	Added Service $34 (RequestDownload) Added Service $35 (RequestUpload) Added Service $36 (TransferData) Added Service $37 (RequestTransferExit) Added Service $23 (ReadMemoryByAddress) Added Service $3D (WriteMemoryByAddress) Added Service $28 (CommunicationControl) and added related TABLE Updated Service $3E (TesterPresent) Updated Service $85 (ControlDTCSetting) Updated graphical structure of Service $14 (ClearDiagnostic-Information) and $14 FFFFFF (ClearAllDiagnosticInformation) Updated available FUNCT-CLASSes Updated DIAG-LAYER(-CONTAINER) structure
1.0	Hummel, Hansen, Cont	Extended AGL chapter: ► Adapted figures to reflect introduced SHORT-NAMING rules. ► Adapted figures to show constant values. ► Recommended modelling of UDS snapshots. ► Spec UDS and Spec B/C support check facilities. ► Spec B/C supplier specific affectation IO ctrl parameters. ► Spec B/C “otherStatus” IO ctrl bytes. ► Spec B/C predefined special LIDs. ► Updated ODX-C rules. ► Updated ODX-V rules. ► Auto Identification. Put model level workflow descriptions into prose.

Content

1 Documents, Terms and Conventions 9

1.1 Related Documents 9

1.2 Terms, Acronyms and Abbreviations 9

1.3 Document Conventions 10

2 Authoring Guidelines 11

2.1 Preface 12

2.1.1 General 12
2.1.2 Definitions 12

2.2 General Requirements 13

2.2.1 XML-ID 13
2.2.2 OID 13
2.2.3 TID 13
2.2.4 ELEMENT-ID 14
2.2.5 COMPANY-DATA 15
2.2.6 ADMIN-DATA 15

2.3 ODX-D 17

2.3.1 DIAG-LAYERs 17
2.3.2 DIAG-LAYER-CONTAINERs 23
2.3.3 FUNCT-CLASS 25
2.3.4 TABLEs 26
2.3.5 DIAG-SERVICEs for UDS 44
2.3.6 DIAG-SERVICEs for B/C 80
2.3.7 GLOBAL-NEG-RESPONSE 113
2.3.8 DIAG-DATA-DICTIONARY-SPEC / PARAM 113
2.3.9 STATE-CHART 118

2.4 ODX-C 120

2.4.1 CAN 120
2.4.2 Ethernet 120

2.5 ODX-V 121

2.5.1 Elements Used 121
2.5.2 Top-Level Structure 121
2.5.3 Vehicle Models 121
2.5.4 Vehicle Information on Connectors 122
2.5.5 Vehicle Information on ECU Usage Scenarios 122

2.6 Annexes 124

2.6.1 Prefixes for XML-IDs of ODX Object Instances 124

© 2017 Vector Informatik GmbH

List of Figures

Figure 1: Hierarchy of DIAG-LAYERs 17
Figure 2 DIAG-LAYER-CONTAINERs 23
Figure 3 Specification of Diagnostic Sessions with TABLE Sessions 26
Figure 4 Specification of Static DIDs with TABLEs TA_S22 and TA_S2E 27
Figure 5 Example for I/O Control Enable Mask Record Definitions 30
Figure 6 Definition of Returnable, Resettable and Freezable UDS I/O Controls 31
Figure 7 Definition of Adjustable UDS IO Controls 32
Figure 8 DTC, Device and Failure Type Tables 34
Figure 9 Snapshot Tables 36
Figure 10 Extended Data Record TABLE 38
Figure 11 Tables for security access levels 41
Figure 12 Specification of ECU reset types with TABLE TA_S11 42
Figure 13 Structure of UDS DIAG-SERVICE DiagnosticSessionControl (0x10) 47
Figure 14 Structure of UDS DIAG-SERVICE ReadDataByIdentifier ($22) 48
Figure 15 Structure of DIAG-SERVICE ReadDataByIdentifier_MultiId ($22) 49
Figure 16 Structure of UDS DIAG-SERVICE SupportedReadableDIDs ($22) 49
Figure 17 Structure of UDS Service Read System Supplier Identifier (F18A) 50
Figure 18 Structure of UDS Service Read System Supplier ECU Software Number (F194) 51
Figure 19 Structure of UDS Service Read System Supplier ECU Version Number (F195) 52
Figure 20 Structure of UDS Service Read Diagnostic Version (F1A0) 52
Figure 21 Structure of UDS DIAG-SERVICE WriteDataByIdentifier ($2E) 53
Figure 22 Structure of UDS Service SupportedWriteableDIDs ($2E) 54
Figure 23 Structure of UDS DIAG-SERVICE IOControl_shortTermAdjustment ($2Fxxxx03) 55
Figure 24 Structure of UDS DIAG-SERVICE IOControl_returnControlToECU ($2Fxxxx00) 56
Figure 25 Structure of UDS DIAG-SERVICE IOControl_resetToDefault ($2Fxxxx01) 57
Figure 26 Structure of UDS DIAG-SERVICE IOControl_freezeCurrentState ($2Fxxxx02) 58
Figure 27 Structure of UDS Service SupportedControllableDIDs ($2F) 59
Figure 28 UDS ReadDtcInformation – ReportDtcByStatusMask ($1902) 61
Figure 29 UDS ReadDtcInformation – ReportDtcSnapshotRecordByDtcNumber ($1904) 63
Figure 30 UDS ReadDtcInformation – ReportDtcExtDataRecordByDtcNumber ($1906) 64
Figure 31 Structure of UDS DIAG-SERVICE ClearAllDiagnosticInformation ($14) 65
Figure 32 Structure of UDS DIAG-SERVICE Routine_Control_Start (0x3101) 66
Figure 33 Structure of UDS DIAG-SERVICE Routine_Control_Stop (0x3102) 67
Figure 34 Structure of UDS DIAG-SERVICE Routine_Control_Request_Results (0x3103) 68
Figure 35 Structure of UDS Service SupportedRoutineControllableDIDs ($31) 68
Figure 36 Structure of UDS DIAG-SERVICE SecurityAccess_requestSeed ($27) 69
Figure 37 Structure of UDS DIAG-SERVICE SecurityAccess_sendKey ($27) 70
Figure 38 Structure of UDS DIAG-SERVICE ECUReset (0x11) 71
Figure 39 UDS DIAG-SERVICE ControlDTCSetting ($85) 72

© 2017 Vector Informatik GmbH

Figure 40 UDS DIAG-SERVICE TesterPresent ($3E) ... 73
Figure 41 UDS DIAG-SERVICE RequestDownload ($34) ... 74
Figure 42 UDS DIAG-SERVICE RequestUpload ($35) ... 75
Figure 43 DIAG-SERVICE TransferData ($36) ... 76
Figure 44 UDS DIAG-SERVICE RequestTransferExit ($37) ... 76
Figure 45 UDS DIAG-SERVICE CommunicationControl ($28) ... 77
Figure 46 UDS DIAG-SERVICE ReadMemoryByAddress ($23) ... 78
Figure 47 UDS DIAG-SERVICE WriteMemoryByAddress ($3D) ... 79
Figure 48 Structure of B/C DIAG-SERVICE StartDiagnosticSession (0x10) ... 82
Figure 49 Structure of B/C DIAG-SERVICE ReadDataByLocalIdentifier ($21) ... 83
Figure 50 Structure of B/C DIAG-SERVICE SupportedReadableLIDs ($21) ... 83
Figure 51 Structure of B/C DIAG-SERVICE Read_Identification_80 ($2180) ... 84
Figure 52 Structure of B/C DIAG-SERVICE Read_Identification_83 ($2183) ... 85
Figure 53 Structure of B/C DIAG-SERVICE ReadDataByIdentifier ($22) ... 86
Figure 54 Structure of B/C DIAG-SERVICE ReadDataByIdentifier_MultiId ($22) ... 87
Figure 55 Structure of B/C DIAG-SERVICE SupportedReadableDIDs ($22) ... 87
Figure 56 Structure of B/C DIAG-SERVICE WriteDataByIdentifier ($2E) ... 88
Figure 57 Structure of B/C DIAG-SERVICE WriteDataByLocalIdentifier ($3B) ... 89
Figure 58 Structure of B/C DIAG-SERVICE SupportedWritebleLIDs ($3B) ... 90
Figure 59 Structure of B/C DIAG-SERVICE IOControlStartTemporary ($30) ... 91
Figure 60 Structure of B/C DIAG-SERVICE IOControlStartPermanent ($30) ... 92
Figure 61 Structure of B/C DIAG-SERVICE IOControlStatus ($30) ... 93
Figure 62 Structure of B/C DIAG-SERVICE IOControlStop ($30) ... 94
Figure 63 Structure of B/C DIAG-SERVICE SupportedIOControlableLIDs ($30) ... 95
Figure 64 B/C ReadDtcInformation – ReportDtcByStatusMask ($1902) ... 96
Figure 65 B/C ReadDtcInformation – ReportDtcSnapshotRecordByDtcNumber ($1904) ... 98
Figure 66 B/C ReadDtcInformation – ReportDtcExtDataRecordByDtcNumber ($1906) ... 99
Figure 67 Structure of B/C DIAG-SERVICE ClearAllDiagnosticInformation ($14) ... 101
Figure 68 Structure of B/C DIAG-SERVICE StartRoutineByLocalIdentifier (0x31) ... 101
Figure 69 Structure of B/C DIAG-SERVICE SupportedRoutineControlableLIDs ($31) ... 102
Figure 70 Structure of B/C DIAG-SERVICE StopRoutineByLocalIdentifier (0x32) ... 103
Figure 71 Structure of B/C DIAG-SERVICE SecurityAccess_requestSeed ($27) ... 104
Figure 72 Structure of B/C DIAG-SERVICE SecurityAccess_sendKey ($27) ... 105
Figure 73 Structure of B/C DIAG-SERVICE ECUReset (0x11) ... 106
Figure 74 B/C DIAG-SERVICE TesterPresent ($3E) ... 107
Figure 75 B/C DIAG-SERVICE RequestDownload ($34) ... 108
Figure 76 B/C DIAG-SERVICE RequestUpload ($35) ... 108
Figure 77 B/C DIAG-SERVICE TransferData ($36) ... 109
Figure 78 B/C DIAG-SERVICE RequestTransferExit ($37) ... 110
Figure 79 B/C DIAG-SERVICE CommunicationControl ($28) ... 111
Figure 80 B/C DIAG-SERVICE ReadMemoryByAddress ($23) ... 112
Figure 81 B/C DIAG-SERVICE WriteMemoryByAddress ($3D) ... 112

Figure 82 Structure of GLOBAL-NEG-RESPONSE 113

List of Tables

Table 1 Terms, Acronyms and Abbreviations 10
Table 2 Predefined ECU-SHARED-DATA Layers 18
Table 3 Predefined PROTOCOL Layers 19
Table 4 Predefined FUNCTIONAL-GROUP Layers 19
Table 5 Matching Parameters in ECU-VARIANT-PATTERNs of Renault ECUs 22
Table 6 Matching Parameters in ECU-VARIANT-PATTERNs of Nissan ECUs 22
Table 7 Predefined DIAG-LAYER-CONTAINERs available for reuse 24
Table 8 Renault / Nissan FUNCT-CLASSes 25
Table 9 Overview of UDS DIAG-SERVICEs and their FUNCT-CLASSes 46
Table 10 Overview of B/C DIAG-SERVICEs and their FUNCT-CLASSes 81
Table 11 Predefined Session States 119
Table 12 Predefined Security Access Levels 119
Table 13 Prefixes for XML-IDs of ODX Object Instances 125

© 2017 Vector Informatik GmbH

1 Documents, Terms and Conventions

1.1Related Documents

[1] DEA-SMD3-OutInj_Vector_ODX_OTX_Study_2015_v0 (original project RFQ from Renault)

[2] ISO 22901-1 Road vehicles - Open diagnostic data exchange (ODX) - Pt1: Data model specification

[3] Renault Standard 36-00-013/C – Diagnostic on CAN: Diagnostic Services Implementation (2010)

1.2Terms, Acronyms and Abbreviations

Term	Definition
AGL	Authoring Guideline.
BV	BASE-VARIANT (a major structural building block of ODX models).
Device	At Renault / Nissan a device represents a component inside an ECU, for which the ECU might store DTCs and/or snapshots.
DLC	DIAG-LAYER-CONTAINER
DOP	DATA-OBJECT-PROP.
DTC-DOP	ODX object representing diagnostic trouble codes.
ESD	ECU-SHARED-DATA (a major structural building block of ODX models).
EV	ECU-VARIANT (a major structural building block of ODX models).
FG, FGL	Functional Group Layer (a major structural building block of ODX models).
Function	Renault / Nissan term representing a common use case of ECUs (e.g. ABS, Airbag, etc.). Only exactly one ECU of a particular function is built into a vehicle. All ECUs belonging to a particular function share the same CAN-Identifiers and DoIP logical addresses. In ODX, they are defined inside of FUNCTIONAL-GROUPS contained in a function DIAG-LAYER-CONTAINER.
Function DLC	A DIAG-LAYER-CONTAINER containing FUNCTIONAL-GROUPs predefining the CAN-Identifiers and DoIP logical addresses of all ECUs belonging to the same function.
Global Snapshot	In most Renault / Nissan ECUs, there are snapshot DIDs which must be supported by each and every snapshot record. At Renault / Nissan, these snapshot DIDs are called “global snapshot DIDs”. The set of all global snapshot DIDs is called the “global snapshot” of an ECU.
Item List	Document internal to Nissan diagnostics department defining diagnostic contents of an ECU (e. g. DIDs).
PR, PRL	Protocol Layer (a major structural building block of ODX models).
RFQ package	Zip file containing interrelated diagnostic specifications of one version, together valid at one particular point of time, defining the presets valid at the beginning of

	an ECUs development and governing the development from thereon.
RFQ layer	ODX files which are delivered as part of RFQ packages, containing predefined ECU-SHARED-DATA and PROTOCOL layers to be reused for concrete ECU projects.
SDG	Special Data Group. An optional structural ODX component intended for extension of arbitrary ODX objects with OEM specific information not covered by the ODX standard.
SNREF	SHORT-NAME Reference.

Table 1 Terms, Acronyms and Abbreviations

1.3 Document Conventions

In this document, the following terminology applies:

- "Shall" expresses an obligatory/mandatory requirement.
- "Should" expresses a recommendation or an advice.
- "Must" expresses a legal or normative requirement.
- "Will" expresses a precautionary consideration or an additional/optional feature.
- "May" expresses a permitted practice/method, not to be considered as a requirement.

2

2 Authoring Guidelines

This chapter specifies the requirements for ODX descriptions used at Renault or Nissan. Its intention is to facilitate all ODX based diagnostic processes by defining rules and recommendations for authors of ODX models created to be used within Renault or Nissan.

© 2017 Vector Informatik GmbH

2.1 Preface

2.1.1 General

Authoring rules defined in ISO 22901-1 [2] remain unchanged.

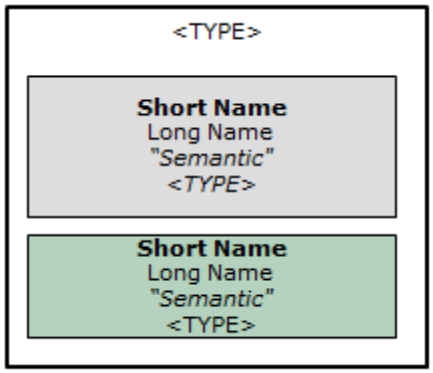
If this document does not address a specific optional element, the author is free to decide to use the element or not. If addressed, the author shall follow the specific rules defined in this document for the respective elements. Some optional elements are thereby deemed mandatory to use, others are forbidden to use.

The end of each chapter may contain a paragraph specifying the according implementation requirements. These requirements are defined as one of the following:

- [Rule] := Mandatory rule to be followed by ODX authors
- [Rule/xsd] := Mandatory rule which is covered by the Renault / Nissan ODX XML schema
- [Reco] := Recommendation

2.1.2 Definitions

This document makes widely use of figures to express structural DIAG-SERVICE definitions. The following rules apply to reading such a figure.



The figure contains the following information:

<TYPE> gives the Type of ODX-Element used (e.g. CODED-CONST, TABLE-KEY,..)

Short Name represents the illustrated ODX element's SHORT-NAME and. It is given in bold. Likewise, long name represents the element's LONG-NAME.

"Semantic" defines the value of the ODX SEMANTIC which is associated with the illustrated ODX element. It is given in italics between double quotes. The SEMANTIC value is intended to be used by client applications of the ODX data (e. g. testers, validation tools...) to discern or identify specific elements in Renault / Nissan ODX models.

indicates that the value of the element is defined at runtime

indicates that the value of the element is defined in the model as a constant (e.g. a service ID)

indicates that the element has primarily structural meaning

→ defines "structural" references (ODX-LINK)

-> defines "informational" references (SHORT-NAME reference)

2.2 General Requirements

2.2.1 XML-ID

XML-IDs support referencing elements within ODX documents. Additionally, the IDs of Renault / Nissan documents shall simplify the detection / identification of referenced elements. As a result, every XML ID should be composed according to the following specific rules.

First, we assume that *elem* is an ODX element which can carry an ID. We define the following:

PREFIX(*elem*) is the prefix of *elem* as defined in Annex A.

PARENT(*elem*) is the next outer parent ODX element, which carries an ID.

The XML ID of the ODX element *elem* is defined as following:

```
                 elem is instance
                 of ODX-CATEGORY       : No further building rule.

ID(elem):=       elem is a direct
                 child of an instance
                 of ODX-CATEGORY       : PREFIX(elem) + "." + elem.SHORT-NAME

                 any other instance    : ID(PARENT(elem)) + "." + PREFIX(elem) + "." + elem.SHORT-NAME
```

An XML-ID of a DATA-OBJECT-PROP “Speed” (=SHORT-NAME) inside the ECU-VARIANT “B_Sample” of a DIAG-LAYER-CONTAINER “ALL_VARIANTS” would look like:

ID(*elem*) = DLC.ALLVARIANTS.EV.B_Sample.DOP.Speed

Implementation requirements:

[Rule] In Renault / Nissan ODX models, hierarchical XML-IDs as specified above should be used.

[Rule] For predefined DLCs and contained layers distributed via Renault / Nissan RFQ packages, the XML-ID must encompass the version number of the DLC.

2.2.2 OID

In ODX, several objects might carry an Object-ID (OID). This OID allows identifying an object during its lifecycle.

For Renault / Nissan, there are no restrictions concerning the usage of OIDs.

2.2.3 TID

ODX uses Text Identifiers (TI) to support multi language aspects of diagnostic data. They are compiled into the runtime format.

For Renault / Nissan, there are no restrictions concerning the usage of TIs.

© 2017 Vector Informatik GmbH

2.2.4 ELEMENT-ID

2.2.4.1 SHORT-NAME

2.2.4.1.1 General SHORT-NAME Building Rules

ODX uses SHORT-NAMEs to qualify an element. Usually, diagnostic code generation from ODX uses SHORT-NAMEs to generate variable names amongst others. Therefore, it is not allowed to use SHORT-NAMEs with a digit as first character.

Additionally, ODX allows setting an element's SHORT-NAME to a value which is completely unrelated to the value of the element's LONG-NAME. Oftentimes, this is unfavorable with regards to automated processing of the ODX data.

Therefore, SHORT-NAMEs should be deduced from the corresponding LONG-NAMEs as follows:

- Romanize[1] LONG-NAME to restrict used character set of the SHORT-NAME to ASCII characters.
- Replace all characters different to [a-z A-Z 0-9 _] with the character '_'
- Remove sequences of character '_' by a single '_' character

As specified in [ISO 22901-1], the length of a SHORT-NAME is limited to 128 characters.

Implementation requirements:

[Rule] A SHORT-NAME must not have a digit as first character.

[Reco] The SHORT-NAME should be deduced from the LONG-NAME of an element.

2.2.4.1.2 Building Rules for Specific Elements

For certain elements, Renault / Nissan require that SHORT-NAMEs are constructed following a very specific scheme. This includes some elements of the following categories:

- TABLEs
- TABELE-ROWs
- STRUCTUREs
- PARAMs

Note: Not all elements of the above-mentioned categories require special SHORT-NAMEs. Whether a specific naming scheme has to be actually applied for a certain element, depends on where the according element is defined and in what context it is used (e. g. which DIAG-SERVICE references a certain TABLE, what TABLE contains a certain TABLE-ROW etc.).

The specific rules for those elements are defined in the according sections of chapters 2.3.4 and 2.3.5 and a summary of the specific rules is given in section 2.3.8.3. If no specific rule is explicitly defined for a certain element, the general rules and recommendations given in section 2.2.4.1.1 shall be applied.

2.2.4.2 LONG-NAME

LONG-NAMEs are optional in ODX. This is unfortunate, seen that – among other things – diagnostic testers normally use the LONG-NAME for the textual representation of diagnostic content.

Therefore, the LONG-NAME is mandatory at Renault / Nissan for every ODX element having an ELEMENT-ID handle. In addition, it must not be empty and comply with the LONG-NAMING rules detailed below:

Implementation requirements:

[1] Romanization should be done for the following character sets: Basic Latin, Latin 1 Supplement, Latin Extended-A, Latin Extended-B, Greek and Coptic, Cyrillic, Hiragana, Katakana. For details see http://www.unicode.com/chars.

[Rule] The LONG-NAME element is mandatory for ODX elements with an ELEMENT-ID.

[Rule] LONG-NAMEs shall not be empty.

[Rule] All uppercase representations of all LONG-NAMEs inside the same namespace shall be unique within that namespace.

[Rule] The uppercase representation of an ODX PARAM's LONG-NAME shall either be unique amongst all uppercase PARAM LONG-NAME representations of the whole ODX model or all PARAMs having the same uppercase LONG-NAME representation shall either all reference the same DOP or DOPs which only differ in their byte orders. Exception are STRUCTUREs which model alternate representation of the very same data structures used by flat services (see rule below).

[Rule] In case a flat service is modelled which is related to an identifier such as a DID or RID etc. and the according data structure is also modelled via the regular TABLE (e. g. an alternate representation of a data record is modelled with a flat service while the original representation available via TABLE TA_S22 is still maintained) then the LONG-NAMEs of the ODX STRUCTUREs representing the alternate models of the very same data structures shall be the same (while the SHORT-NAMEs will of course differ according to the short naming rules defined in this document).

[Reco] LONG-NAMEs of DOPs should reflect the conversions / data formats modelled by that DOP and only that information; specifically, LONG-NAMEs of DOPs should not contain information specific to the actual data stored in the ECU or transmitted between testers and the ECU on which the conversion is applied. An exception to this is the naming of TEXTTABLEs which generally do contain data specific information as a matter of fact.

[Reco] in contrast to this, LONG-NAMEs of PARAMs should reflect the actual data stored in the ECU or transmitted between testers and the ECU which is represented by the PARAMs; LONG-NAMEs of PARAMs should not contain information specific to the structure of or the interpretation of or the conversion applied to the data they represent.

[Rule] The length of a LONG-NAME is limited to 255 characters.

2.2.4.3 DESCRIPTION (DESC)

No restriction exists in Renault / Nissan ODX concerning the use of description elements. The XHTML subset supported by ODX inside of description elements may be used as the authors see fit. Authors however should be aware that descriptions at most elements will be lost when an ODX model is converted into DDT.

Implementation recommendations:

[Reco] For ODX models intended to be converted into DDT format, authors should avoid descriptions at elements other than TABLE-ROWs and PARAMs which directly correspond to DDT Data elements.

2.2.5 COMPANY-DATA

COMPANY-DATA shall not be used.

2.2.6 ADMIN-DATA

Implementation requirements:

[Rule] ADMIN-DATA shall be used and maintained at all DIAG-LAYER-CONTAINERs distributed with RFQ packages, at the Datas Format ESD, at ECU-VARIANT layers and in ODX-V files.

[Rule] ADMIN-DATA may not be used at elements other than DLCs distributed with RFQ packages, the Datas Format ESD, ECU-VARIANT layers and ODX-V files.

[Rule] At ECU-VARIANTs, only two revision states may be used: Officialised and Not-Officialised.

© 2017 Vector Informatik GmbH

2.3 ODX-D

2.3.1 DIAG-LAYERs

2.3.1.1 Layer Hierarchy

Figure 1 shows an exemplary setup for a UDS based ECU ("ABS") supporting CAN and Ethernet:

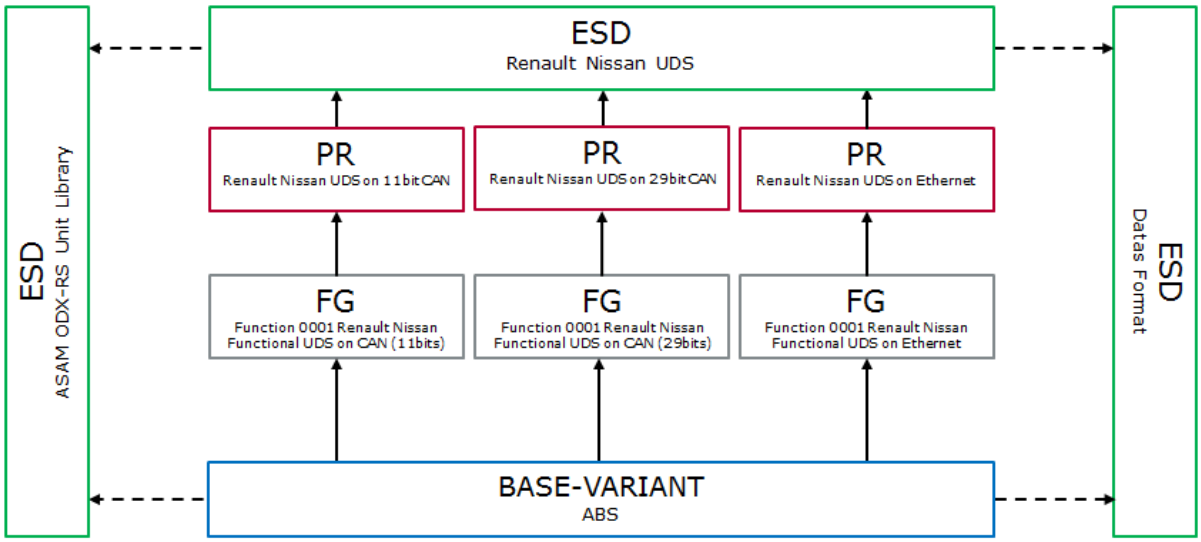


Figure 1: Hierarchy of DIAG-LAYERs

For other diagnostic specifications (such as Spec B, C etc.), instead of the "Renault Nissan Spec UDS" shared data layer and deriving protocol and functional group layers, according other layers from the RFQ packages are inherited by the BASE-VARIANT. Thereby, the basic structure remains the same: A top level ESD predefines the potential services and states and for each network potentially supported, there is a pair consisting of a protocol and functional group layer inheriting from that top level ESD.

Both the top-level ESDs and the deriving protocol and functional group layers are predefined by RFQ packages and intended to be reused as is in concrete ECU projects. One of the early results of those projects is the creation of a BASE-VARIANT for the ECU. The BASE-VARIANT inherits from all of the functional group layers associated with one of the networks actually supported by that ECU.

Both in the layers predefined in RFQ packages and in the ECU specific lower layers, authors may (and are encouraged to) make use of the predefined Datas Format ESD and the ASAM ODX-RS Unit Library.

2.3.1.2 ECU-SHARED-DATA

The table below defines the ECU-SHARED-DATAs and their content used in the context of Renault / Nissan ODX models.

LONG- / SHORT-NAME	Defined by	Diagnostic Content
Renault Nissan UDS / Renault_Nissan_UDS	RFQ Package	▶ Predefined UDS DIAG-SERVICEs. ▶ Session state model as STATE-CHART.
Renault Nissan BC / Renault_Nissan_BC	RFQ Package	▶ Predefined Spec B / Spec C DIAG-SERVICEs. ▶ Session state model as STATE-CHART.
Renault Nissan OBD / Renault_Nissan_OBD	RFQ Package	▶ Predefied ODX Mode DIAG-SERVICEs.

Datas Format / DatasFormat	Renault / Nissan Diagnosis Dpt.	► Predefined DOPs.
ODX-RS UNIT Library / ODX_RS_UNIT_LIB	ASAM	► Predefined physical units.

Table 2 Predefined ECU-SHARED-DATA Layers

Implementation requirements:

[Rule] ECU-SHARED-DATAs not listed in Table 2 must not be used in concrete ECU projects without prior consultation of Renault / Nissan diagnostic experts.

[Rule] All ECU-SHARED-DATAs predefined by a RFQ package and reused in one ODX model must originate from the same RFQ package.

[Rule] Likewise, ECU-SHARED-DATAs originating from one RFQ package shall not be reused by protocol layers originating from another RFQ package.

[Rule] ECU designers and suppliers shall not edit the contents of ECU-SHARED-DATAs (in case changes are needed, Renault / Nissan diagnostic experts can provide needed updates).

[Rule] The XML-ID of the layers listed in Table 2 must encompass the version number of the DLC in which they are contained (except for the Datas Format and ODX-RS Unit Lib ESDs).

[Reco] Whenever possible, DOPs shall be reused from the Datas Format ESD.

[Reco] Whenever possible, physical units shall be reused from the ODX-RS Unit Library.

2.3.1.3 PROTOCOL

The table below defines the PROTOCOLs together with the referenced COMPARAM-SPEC and the according PARENT-REF allowed in the context of Renault / Nissan ODX. They are predefined by a RFQ package in conjunction with the ECU-SHARED-DATAs they inherit from.

LONG- / SHORT-NAME	PARENT-REF	COMPARAM-SPEC-REF
Renault Nissan UDS on 11bit CAN / Renault_Nissan_UDS_on_11bit_CAN	Renault_Nissan_Spec_UDS	ISO_15765_3_on_ISO_15765_2
Renault Nissan UDS on 29bit CAN / Renault_Nissan_UDS_on_29bit_CAN	Renault_Nissan_Spec_UDS	ISO_15765_3_on_ISO_15765_2
Renault Nissan UDS on Ethernet / Renault_Nissan_UDS_on_Ethernet	Renault_Nissan_Spec_UDS	ISO_14229_5_on_ISO_13400_2
Renault Nissan BC on 11bit CAN / Renault_Nissan_BC_on_11bit_CAN	Renault_Nissan_Spec_BC	ISO_15765_3_on_ISO_15765_2
Renault Nissan BC on 29bit CAN / Renault_Nissan_BC_on_29bit_CAN	Renault_Nissan_Spec_BC	ISO_15765_3_on_ISO_15765_2
Renault Nissan BC on Ethernet / Renault_Nissan_BC_on_Ethernet	Renault_Nissan_Spec_BC	ISO_14229_5_on_ISO_13400_2
Renault Nissan OBD on 11bit CAN / Renault_Nissan_OBD_on_11bit_CAN	Renault_Nissan_OBD	ISO_15765_3_on_ISO_15765_2
Renault Nissan OBD on 29bit CAN / Renault_Nissan_OBD_on_29bit_CAN	Renault_Nissan_OBD	ISO_15765_3_on_ISO_15765_2

© 2017 Vector Informatik GmbH

Renault Nissan OBD on Ethernet / Renault_Nissan_OBD_on_Ethernet	Renault_Nissan_OBD	ISO_14229_5_on_ISO_13400_2

Table 3 Predefined PROTOCOL Layers

Implementation requirements:

[Rule] PROTOCOL layers other than the ones listed in Table 3 must not be used.

[Rule] All protocol layers used in the context of one ODX model shall originate from the same RFQ package.

[Rule] Likewise, protocol layers originating from one RFQ package shall not inherit from an ECU-SHARED-DATA layer originating from another RFQ package and they shall not be reused by functional group layers originating from another RFQ package.

[Rule] The XML-ID of the layers listed in Table 3 must encompass the version number of the DLC in which they are contained.

2.3.1.4 FUNCTIONAL-GROUP

The table below defines the predefined FUNCTIONAL-GROUP layers together with the PARENT-REF allowed in the context of Renault / Nissan ODX.

LONG- / SHORT-NAME	PARENT-REF
Function YYYY[2] Renault Nissan Functional UDS on CAN (11bit) / Function_YYYY_Renault_Nissan_Functional_UDS_on_CAN_11bit	Renault_Nissan_UDS_on_CAN
Function YYYY[2] Renault Nissan Functional UDS on CAN (29bit) / Function_YYYY_Renault_Nissan_Functional_UDS_on_CAN_29bit	Renault_Nissan_UDS_on_CAN
Function YYYY[2] Renault Nissan Functional UDS on Ethernet / Function_YYYY_Renault_Nissan_Functional_UDS_on_Ethernet	Renault_Nissan_UDS_on_Ethernet
Function YYYY[2] Renault Nissan Functional BC on 11bit CAN / Function_YYYY_Renault_Nissan_Functional_BC_on_11bit_CAN	Renault_Nissan_BC_on_CAN
Function YYYY[2] Renault Nissan Functional BC on 29bit CAN / Function_YYYY_Renault_Nissan_Functional_BC_on_29bit_CAN	Renault_Nissan_BC_on_CAN
Function YYYY[2] Renault Nissan Functional BC on Ethernet / Function_YYYY_Renault_Nissan_Functional_BC_on_Ethernet	Renault_Nissan_BC_on_Ethernet
Function YYYY[2] Renault Nissan Functional OBD on CAN (11bit) / Function_YYYY_Renault_Nissan_Functional_OBD_on_CAN_11bit	Renault_Nissan_OBD_on_CAN
Function YYYY[2] Renault Nissan Functional OBD on CAN (29bit) / Function_YYYY_Renault_Nissan_Functional_OBD_on_CAN_29bit	Renault_Nissan_OBD_on_CAN
Function YYYY[2] Renault Nissan Functional OBD on Ethernet / Function_YYYY_Renault_Nissan_Functional_OBD_on_Ethernet	Renault_Nissan_OBD_on_Ethernet

Table 4 Predefined FUNCTIONAL-GROUP Layers

[2] The YYYY in the DIAG-LAYER short and long names given in Table 4 shall be replaced by the respective Function number, specifying the use case of the described ECU (e.g. Airbag, ABS, etc.)

Implementation requirements:

[Rule] FUNCTIONAL-GROUP layers different to the ones listed in Table 4 must not be used.

[Rule] All functional group layers used in the context of one ODX model shall originate from the same RFQ package.

[Rule] Likewise, protocol layers originating from one RFQ package shall not inherit from a protocol layer originating from another RFQ package.

[Rule] The XML-ID of the layers listed in Table 4 must encompass the version number of the DLC in which they are contained.

2.3.1.5 BASE-VARIANT

BASE-VARIANTs store the ECU related common diagnostics of the normal function's application. Diagnostic content is both inherited from higher layers and defined locally.

Renault / Nissan ECU designers define the content of BASE-VARIANTs during initial ECU development. They inherit from one or more of the functional group layers defined in section 2.3.1.4. DIAG-SERVICEs are reused as predefined in the higher layers which originate from an RFQ package. BASE-VARIANTs in Renault / Nissan ODX usually do not contain local service definitions; exception to this is the optional use of so called flat services as described in section 2.3.5.2.

The typical content of a BASE-VARIANT encompasses:

- Definition of the actual session states and state transitions (cf. section 2.3.4.1)
- Definition of the actual security levels and level transitions (cf. section 2.3.4.8).
- Reuse of predefined DIDs and definition of ECU specific DIDs (cf. section 0).
- Definition of I/O controls (cf. section 2.3.4.4).
- Devices, DTCs, snapshot and extended data records (cf. section 2.3.4.5).
- Routines (cf. section 2.3.4.7).
- Flat services (cf. section 2.3.5.2).

Communication parameter related requirements are defined in section 2.4.

Implementation requirements:

[Rule] BASE-VARIANTs shall not contain local definitions of DIAG-SERVICEs, except for flat services as defined in section 2.3.5.2.

[Rule] They shall inherit from one or more of the functional group layers listed in section 2.3.1.4.[3]

[Rule] When a BASE-VARIANT inherits from the RFQ layer "Renault Nissan Spec UDS", then at least one of the two available UDS $22 DIAG-SERVICEs (ReadDataByIdentifier or ReadDataByIdentifier_MultiId) shall be marked as not inherited (depending on whether the described ECU supports reading out multiple DIDs at once or not).

[Rule] Each inherited or imported layer shall originate from the same RFQ package.

[Rule] If the described ECU provides identifier support check facilities and thus the according support check services are not explicitly marked as NOT-INHERITED, the ECU designers shall make sure that identifiers used for support check queries are not also described in the regular TABLEs of the according read, write, IO control and routine services (see section 2.3.4.13 for details on the different support check identifiers).

[Rule] For Spec B/C based ECUs, the related RFQ ESD predefines several identification services (cf. section 2.3.6.5). Only one of those services is applicable for one ECU at a time.

[3] Note that CAN based ECUs at Renault / Nissan often support communication both via 11 bit and 29 bit CAN. For such ECUs, it is perfectly fine to let the BASE-VARIANT inherit from both 11 and 29 bit CAN FUNCTIONAL-GROUP layers available from the RFQ packages (e. g. from both Renault_Nissan_Functional_UDS_on_CAN_11bit and Renault_Nissan_Functional_UDS_on_CAN_29bit).

Consequently, ECU designers shall explicitly mark all but one of those identification services as NOT-INHERITED.

[Reco] The Datas Format ESD may be imported; if possible, DOPs should be reused from there.

[Reco] The ODX-RS Unit Library may be imported; if possible, units should be reused from there.

2.3.1.6 ECU-VARIANT

ECU-VARIANTs represent distinct phases in the evolution of an ECU's diagnostics. Diagnostic content is mainly intended to be inherited from the BASE-VARIANT but content may also be defined locally.

DIAG-SERVICEs are reused as predefined in the higher layers which originate from an RFQ package. ECU-VARIANTs in Renault / Nissan ODX usually do not contain local service definitions; exception to this is the optional use of so called flat services as described in section 2.3.5.2.

The typical content of an ECU-VARIANT corresponds to the contents listed in section 2.3.1.5 for BASE-VARIANTs with the difference that contents should mainly be reused from the parent BASE-VARIANT and possibly not defined locally.

2.3.1.6.1 Evolution

During initial development, Renault / Nissan ECU designers define the diagnostic content of an ECU in the BASE-VARIANT. During this phase, a first ECU-VARIANT exists whose essential content is the auto idents of the described ECUs in the ECU-VARIANT-PATTERNs which correspond to versions of the ECU software which can all be diagnosed with the ECU-VARIANT at hand. During further evolution, additional auto idents should be added to that ECU-VARIANT as long as all implementation states represented by the auto idents can be correctly diagnosed with the ECU-VARIANT at hand.

Once an evolution of the diagnostic functionality of the described ECU necessitates the creation of a new ECU-VARIANT, diagnostic contents are maintained and described in that new ECU-VARIANT. This is essentially the case, when the mere addition of a new auto ident to the existing ECU-VARIANT (or a related change in its diagnostic contents) would no longer allow to correctly diagnose one of the implementation states of the ECU represented by one of the auto idents in the ECU-VARIANT (please refer to chapter 0 for details on the conditions which necessitate the creation of a new ECU-VARIANT).

2.3.1.6.2 Auto Identification

Renault / Nissan make use of several services which are used for auto identification:

- $2180 and $2183 for Spec B/C based ECUs.
- $22F1A0, $22F194, $22F195 and $22F18A for Spec UDS based ECUs.

In Renault / Nissan ODX models, these services are represented explicitly by distinct DIAG-SERVICEs which are predefined in the ESDs distributed with RFQ packages; see sections 2.3.6.5.3 and 2.3.6.5.4 for the Spec B/C DIAG-SERVICEs and the sections 2.3.5.5.3 to 0 for the Spec UDS DIAG-SERVICEs.

Note that Renault ECUs and Nissan ECUs use different identifiers and / or elements of the according data records for identification. The ECU-VARIANT-PATTERNs of ECU-VARIANTs shall reference the dedicated DIAG-SERVICEs and the respective PARAMs depending on whether the model describes a Renault only or a Nissan only or a Renault / Nissan ECU.

In case of a Renault only ECU, an ECU-VARIANT shall contain exactly one ECU-VARIANT-PATTERN for each implementation state of the ECU diagnosable by that ECU-VARIANT. That pattern in turn shall encompass four MATCHING-PARAMETERs as defined by Table 5.

Specification	Referenced DIAG-COMMs	Referenced PARAMs
Spec UDS	Read diagnosticVersion (F1A0) Read systemSupplierECUSoftwareNumber (F194)	diagnosticVersion systemSupplierECUSoftwareNumber

	Read systemSupplierECUVersionNumber (F195) Read systemSupplierIdentifier (F18A)	systemSupplierECUSoftwareVersionNumber systemSupplierIdentifier
Spec B/C	Read Identification (80) Read Identification (80) Read Identification (80) Read Identification (80)	DiagnosticIdentificationCode SupplierNumber.ITG SoftwareNumber EditionNumber

Table 5 Matching Parameters in ECU-VARIANT-PATTERNs of Renault ECUs

In case of a Nissan only ECU, an ECU-VARIANT shall contain exactly one ECU-VARIANT-PATTERN for each implementation state of the ECU diagnosable by that ECU-VARIANT. That pattern in turn shall encompass one MATCHING-PARAMETER as defined by Table 6.

Specification	Referenced DIAG-COMMs	Referenced PARAMs
Spec UDS	Read diagnosticVersion (F1A0)	diagnosticVersion
Spec B/C	Read Identification (83)	DiagnosticIdentificationCode

Table 6 Matching Parameters in ECU-VARIANT-PATTERNs of Nissan ECUs

If an ECU-VARIANT describes a joint Renault / Nissan ECU, the ECU-VARIANT shall contain exactly two ECU-VARIANT-PATTERNs for each implementation state of the ECU diagnosable by that ECU-VARIANT, one ECU-VARIANT-PATTERN as defined by Table 5 and one as defined by Table 6.

2.3.1.6.3 Implementation Requirements

[Rule] ECU-VARIANTs shall not contain local definitions of DIAG-SERVICEs, except for flat services as defined in section 2.3.5.2.

[Rule] Each inherited or imported layer shall originate from the same RFQ package.

[Rule] If the described ECU provides identifier support check facilities and thus the according support check services are not explicitly marked as NOT-INHERITED, the ECU designers shall make sure that identifiers used for support check queries are not also described in the regular TABLEs of the according read, write, IO control and routine services (see section 2.3.4.13 for details on the different support check identifiers).

[Rule] Renault / Nissan ODX models shall encompass as few ECU-VARIANTs as possible. If a new evolution state of an ECU needs to be described and the ECU-VARIANT at hand can be adapted accordingly in such a fashion that all previous evolution states can still be correctly diagnosed with it, ECU designers shall actually use that ECU-VARIANT for this purpose instead of creating a new ECU-VARIANT.

[Rule] ECU-VARIANTs describing a Renault only ECU shall contain exactly one ECU-VARIANT-PATTERN for each implementation state of the ECU diagnosable by that ECU-VARIANT as defined by Table 5.

[Rule] ECU-VARIANTs describing a Nissan only ECU shall contain exactly one ECU-VARIANT-PATTERN for each implementation state of the ECU diagnosable by that ECU-VARIANT as defined by Table 6.

[Rule] ECU-VARIANTs describing a joint Renault / Nissan ECU shall contain exactly two ECU-VARIANT-PATTERNs for each implementation state of the ECU diagnosable by that ECU-VARIANT, one as defined by Table 5 and one as defined by Table 6.

[Reco] The Datas Format ESD may be imported; if possible, DOPs should be reused from there.

[Reco] The ODX-RS Unit Library may be imported; if possible, units should be reused from there.

[Reco] An ECU-VARIANT should contain as little local diagnostic content descriptions as possible and inherit as much content as possible from the parent BASE-VARIANT.

2.3.2 DIAG-LAYER-CONTAINERs

ODX allows storing of one or more DIAG-LAYERs inside a DIAG-LAYER-CONTAINER. As every DIAG-LAYER-CONTAINER results in one ODX-D file, Renault / Nissan therefore introduces rules to manage DIAG-LAYER-CONTAINERs.

As indicated in Figure 2, Renault / Nissan ODX models store the DIAG-LAYERs which conceptually belong to a common use case (such as UDS diagnostics…) in two related types of DIAG-LAYER-CONTAINERs, resulting in two physical ODX files.

- One type of DIAG-LAYER-CONTAINER contains the ECU-SHARED-DATA corresponding to the Renault / Nissan diagnostic specifications (e.g. UDS, Spec B/C, etc.) as well as all available PROTOCOLs that inherit from this ECU-SHARED-DATA.
- The other type of DIAG-LAYER-CONTAINER contains one FUNCTIONAL-GROUP layer for each available PROTOCOL (currently three: 11bit CAN, 29bit CAN and Ethernet).

From the perspective of one ECU's BASE-VARIANT, at least one DIAG-LAYER-CONTAINER of the second type (referred to as "function DLC") is reused and thus indirectly also at least one DLC of the first type (referred to as "RFQ DLC").

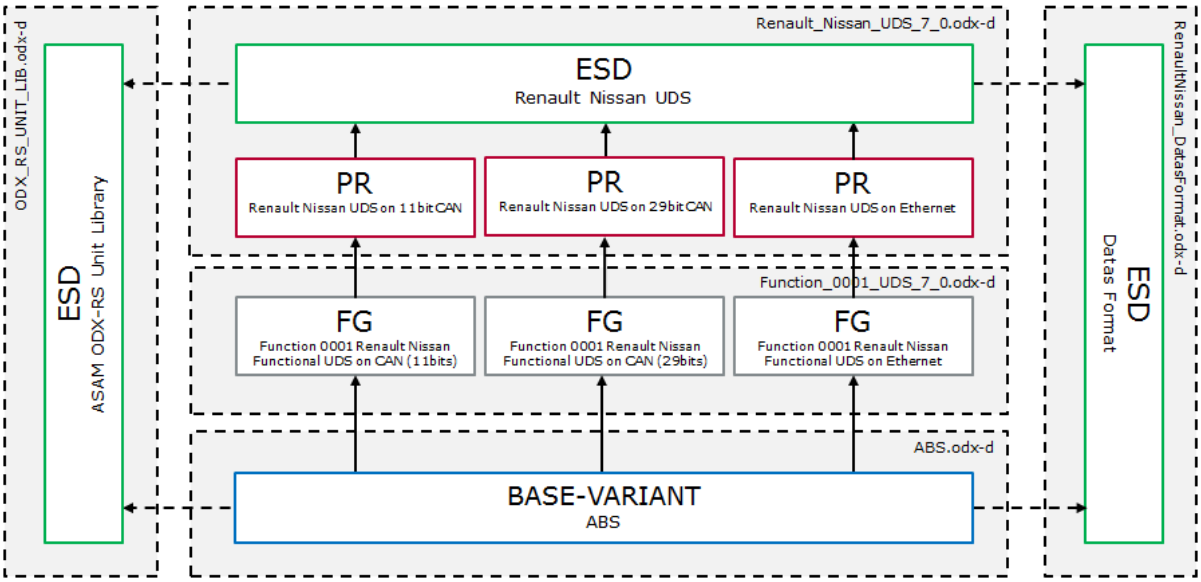


Figure 2 DIAG-LAYER-CONTAINERs

ODX descriptions for a concrete Renault / Nissan ECU encompass at least five DIAG-LAYER-CONTAINERs (and thus five ODX-D files):

- One DIAG-LAYER-CONTAINER (a RFQ DLC) containing the ECU-SHARED-DATA taken from an RFQ package corresponding to the Renault / Nissan diagnostic specification supported by the modelled ECU (e. g. Spec A, Spec B, Spec UDS…) and the available PROTOCOLs.
- One DIAG-LAYER-CONTAINER (a function DLC) containing the FUNCTIONAL-GROUPs that the modelled ECU can reuse; each FUNCTIONAL-GROUP layer references exactly one PROTOCOL.
- One DIAG-LAYER-CONTAINER created during ECU development encompassing base and ECU variant layers.
- The Datas Format ESD.
- The ASAM ODX-RS Unit Library.

Note: Besides ODX-D files, the containers also encompass files of other types (such as ODX-C, -V…).

For Renault / Nissan ODX models, the following DIAG-LAYER-CONTAINERs are available for reuse:

LONG- / SHORT-NAME[4]	Diag Spec	Contained DIAG-LAYERs
Renault Nissan UDS (*xx*) / Renault_Nissan_UDS_RFQ_*xx*	Spec UDS	Renault_Nissan_UDS Renault_Nissan_UDS_on_11bit_CAN Renault_Nissan_UDS_on_29bit_CAN Renault_Nissan_UDS_on_Ethernet
Renault Nissan BC (*xx*) / Renault_Nissan_ BC_*xx*	Spec B/C	Renault_Nissan_BC Renault_Nissan_BC_on_11bit_CAN Renault_Nissan_BC_on_29bit_CAN Renault_Nissan_BC_on_Ethernet
Renault Nissan OBD (*xx*) / Renault_Nissan_OBD_*xx*	OBD	Renault_Nissan_OBD Renault_Nissan_OBD_on_11bit_CAN Renault_Nissan_OBD_on_29bit_CAN Renault_Nissan_OBD_on_Ethernet
Function YYYY[5] ZZ[6] xx / Function_YYYY_ZZ_xx	all	Function_YYYY_Renault_Nissan_Functional_UDS_on_CAN_11bit Function_YYYY_Renault_Nissan_Functional_UDS_on_CAN_29bit Function_YYYY_Renault_Nissan_Functional_UDS_on_Ethernet Function_YYYY_Renault_Nissan_Functional_OBD_on_CAN_11bit Function_YYYY_Renault_Nissan_Functional_OBD_on_CAN_29bit Function_YYYY_Renault_Nissan_Functional_OBD_on_Ethernet Function_YYYY_Renault_Nissan_Functional_BC_on_CAN_11bit Function_YYYY_Renault_Nissan_Functional_BC_on_CAN_29bit Function_YYYY_Renault_Nissan_Functional_BC_on_Ethernet
Renault Nissan Datas Format / RenaultNissan_DatasFormat	all	DatasFormat
ODX-RS UNIT Library / ODX_RS_UNIT_LIB_DLC	all	ODX_RS_UNIT_LIB

Table 7 Predefined DIAG-LAYER-CONTAINERs available for reuse

Implementation requirements:

[Rule] The ODX model for one ECU must encompass at least five DIAG-LAYER-CONTAINERs: A RFQ DLC predefined by the RFQ package corresponding to the supported diagnostic specification, a function DLC (corresponding to the ECU function) containing all available FUNCT-GROUPs for the ECU, one DLC defined during ECU development containing all base and ECU variants, the DLCs containing the Datas Format ESD and the DLC containing the ASAM ODX-RS Unit Lib.

[Rule] The DIAG-LAYER-CONTAINERs which correspond to the Renault / Nissan diagnostic specifications supported by the ECU must have the respective names listed in Table 7.

[4] The *xx* in the DIAG-LAYER-CONTAINER short and long names given in Table 7 represent the version of the DLC.
[5] The YYYY in the DIAG-LAYER-CONTAINER short and long names given in Table 7 shall be replaced by the respective Function number, specifying the use case of the described ECU (e.g. Airbag, ABS, etc.)
[6] The ZZ in the DIAG-LAYER-CONTAINER short and long names given in Table 7 shall be replaced by the respective relevant diagnostic specification (e.g. UDS, OBD, Spec B/C, etc.)

[Rule] The file name of the odx-d file which contains a DIAG-LAYER-CONTAINER shall equal the contained DIAG-LAYER-CONTAINER's SHORT-NAME.

[Rule] The DIAG-LAYER-CONTAINER encompassing the base and ECU variants shall be named after the modelled ECU.

[Rule] The DIAG-LAYER-CONTAINER encompassing the FUNCTIONAL-GROUPs shall contain the number of the function the ECU is supposed to fulfill (e.g. Airbag, ABS, etc.), postfixed by the diagnostic specification (UDS, OBD, Spec B/C, etc) and the version of the DLC.

[Rule] The XML-ID of the DLCs listed in Table 7 must encompass their version number (exception: the DLCs containing the Datas Format ESD and the ODX-RS Unit Lib).

[Rule] It is not allowed to define DIAG-LAYER-CONTAINERs different to the ones above.

2.3.3 FUNCT-CLASS

For better structuring of DIAG-SERVICEs, the diagnostic use cases at Renault / Nissan are assigned to according FUNCT-CLASSes as given in Table 8:

Use case	FUNCT-CLASS SHORT-NAME
Diagnostic Session Control	Session
Static DIDs	DIDs
Static LIDs	LIDs
I/O Controls	I_O_Controls
Fault Memory	Fault_Memory
Routine Control	Routine_Control
ECU Reset	ECUReset
Tester Present	TesterPresent
Control DTC Setting	ControlDTCSetting
Security Access	SecurityAccess
Upload Download	UploadDownload
Memory	Memory
Support Check	SupportCheck

Table 8 Renault / Nissan FUNCT-CLASSes

Implementation requirements:

[Rule] FUNCT-CLASSes different to those listed in Table 8 shall not be used.

[Rule] FUNCT-CLASSes shall only be specified inside the predefined Renault / Nissan diagnostic specification ESDs listed in Table 2.

2.3.4 TABLEs

2.3.4.1 Diagnostic Sessions

Renault / Nissan ODX uses TABLE TA_S10 (Sessions) to specify the session types supported by an ECU. It is referenced dynamically by the DiagnosticSessionControl service ($10) predefined in the diagnostic specification ESDs which are part of Renault / Nissan RFQ packages.

The four session types defined by ISO UDS are thereby predefined by TABLE TA_S10 (Sessions) on ESD level. ECU designers are expected to override TABLE TA_S10 on variant level in order to specify different sets of session types for a concrete ECU.

The predefined session types can thereby be reused by using a TABLE-ROW-REF referencing the according table row from the ESD level. Other session types (e. g. supplier sessions) must be defined locally by introducing an according TABLE-ROW for that session in the overridden TABLE TA_S10 on variant level.

On ESD level, Renault / Nissan ODX also predefines a STATE-CHART defining all session types potentially supported by concrete ECUs together with all possible state transitions (cf. section 2.3.9).

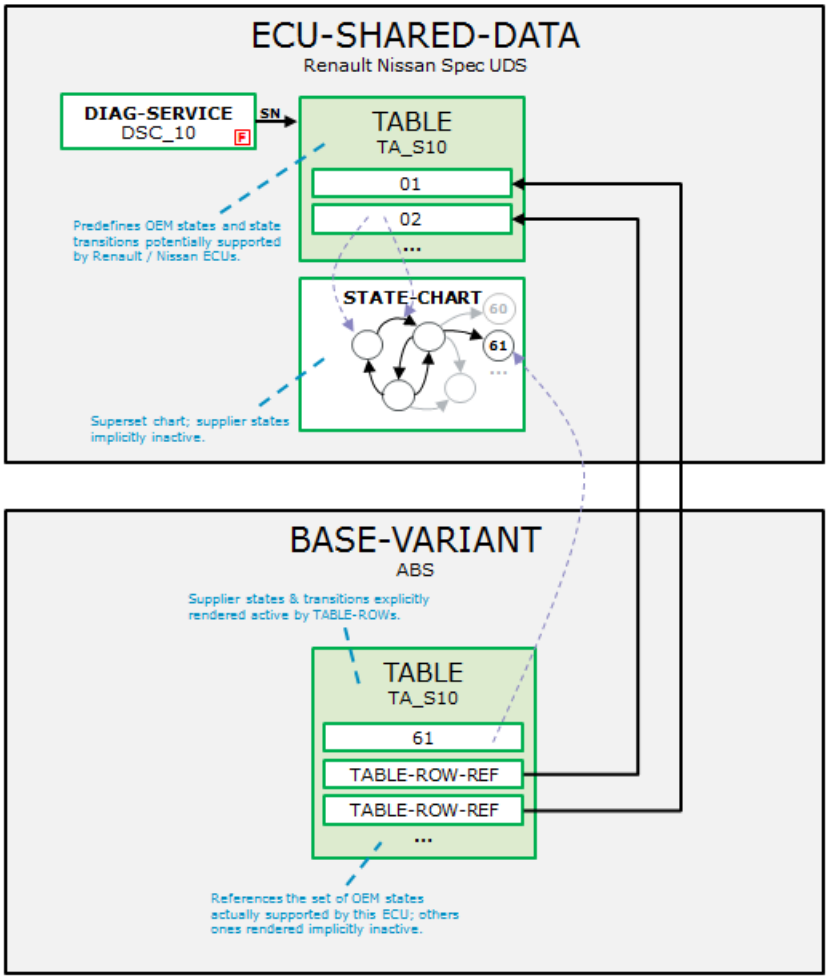


Figure 3 Specification of Diagnostic Sessions with TABLE Sessions

Implementation requirements:

[Rule] The diagnostic specification ESDs contained in Renault / Nissan RFQ packages shall encompass a TABLE with SHORT-NAME "TA_S10" ("Sessions") and one STATE-CHART of kind "SESSION" predefining all session types and state transitions potentially supported by a concrete ECU.

[Rule] That TABLE shall have the SHORT-NAME shall be "TA_S10" and LONG-NAME "Sessions".

[Rule] The DiagnosticSessionControl service ($10) shall reference that TABLE via SNREF.

[Rule] For ECUs which support different sets of session types and state transitions, TABLE Sessions shall be overridden on variant level.

[Rule] According state preconditions and state transitions of newly introduced TABLE-ROWs on variant level shall be reused from the STATE-CHART predefined on ESD level.

2.3.4.2 Static DIDs

As indicated in Figure 4, Renault / Nissan ODX uses two TABLEs to specify the static DIDs supported by an ECU: One TABLE for readable DIDs and one table for writable DIDs.

TABLE TA_S2E (DIDs Writable) is thereby dynamically referenced by the WriteDataByIdentifier service ($2E) and TABLE TA_S22 (DIDs Readable) by the ReadDataByIdentifier service ($22). On level of the diagnostic specification ESDs contained in Renault / Nissan RFQ packages, all DIDs predefined in the RFQ package are also specified in the two TABLES by according TABEL-ROWs.

Each TABLE-ROW references a STRUCTURE which describes the content of the DID's data record and specifies the diagnostic session preconditions by referencing the according states from the STATE-CHART also contained in the diagnostic specification ESD.

TABLE-ROWs of DIDs which must be implemented by concrete ECUs are thereby marked as mandatory. Likewise, DIDs whose data records must be implemented identically in all ECUs are marked as final at the according TABLE-ROW.

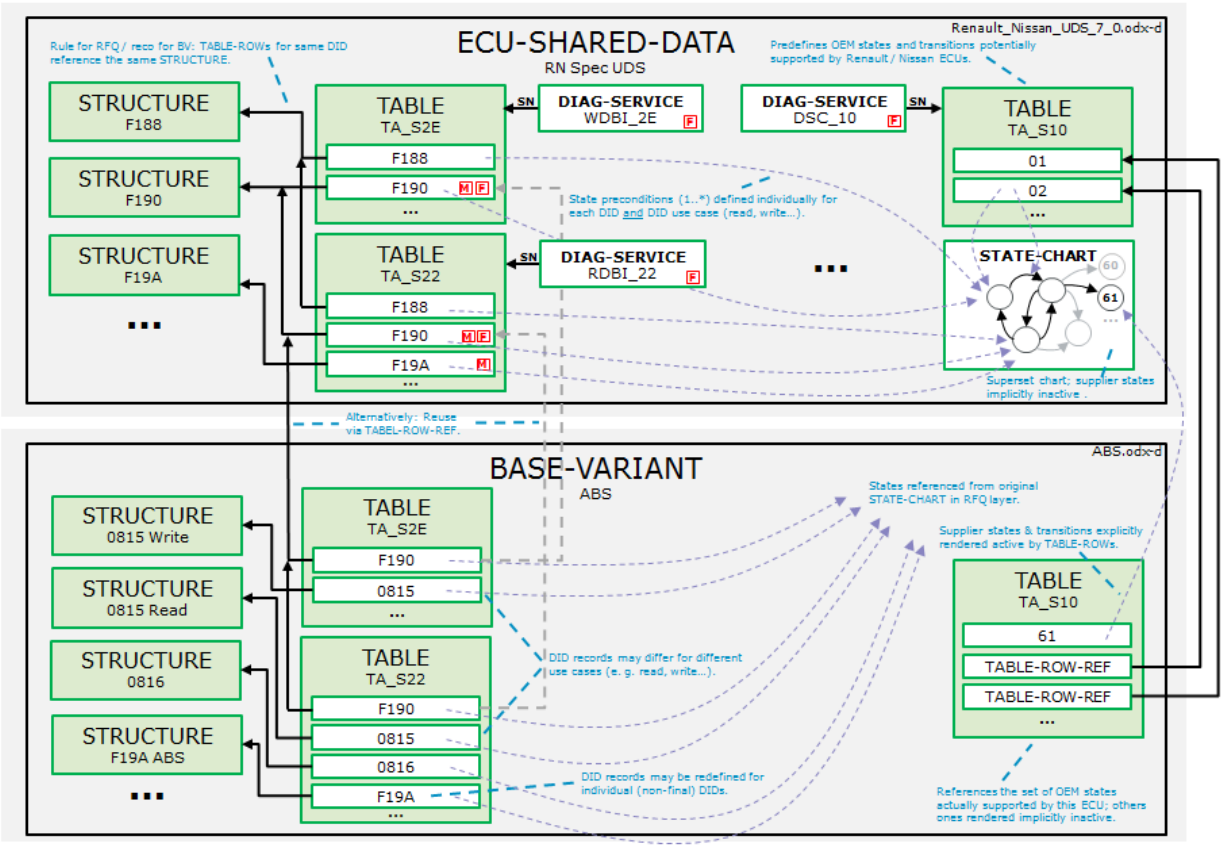


Figure 4 Specification of Static DIDs with TABLEs TA_S22 and TA_S2E

ECU Designers are expected to override the two TABLEs TA_S22 and TA_S2E at variant level. For DIDs predefined by the RFQ package, ECU designers have the choice to either reuse them by an according TABLE-ROW-REF referencing the original TABLE-ROW on ESD level or to introduce a new TABLE-ROW.

ECU specific DIDs or RFQ DIDs whose data record is different than specified at ESD level must be defined at BASE-VARIANT level by introducing a new TABLE-ROW in the overridden DID TABLEs. Diagnostic session preconditions shall thereby be specified by referencing the according states from the STATE-CHART predefined in the diagnostic specification ESD.

Implementation requirements:

[Rule] The diagnostic specification ESDs contained in Renault / Nissan RFQ packages shall encompass two TABLEs: "DIDs Readable" and "DIDs Writable".

[Rule] The TABLE for readable DIDs shall have the SHORT-NAME "TA_S22" and TABLE for writable DIDs "TA_S2E".

[Rule] The TABLE for readable DIDs shall have the LONG-NAME "DIDs Readable" and TABLE for writable DIDs "DIDs Writable".

[Rule] The SHORT-NAMEs of the TABLE-ROWs in TABLE "TA_S22" and in TABLE "TA_S2E" as well as the SHORT-NAMEs of the STRUCTUREs referenced by those TABLE-ROWs and the SHORT-NAMEs of the PARAMs in those STRUCTUREs shall follow the respective rules specified in section 2.3.8.3.

[Rule] The ReadDataByIdentifier ($22) and WriteDataByIdentifier ($2E) services shall reference those TABLEs via SNREF.

[Rule] The two DID TABLEs shall be overridden on variant level.

[Rule] They shall contain entries for all readable and writable DIDs implemented by the ECU.

[Rule] DIDs marked as mandatory on ESD level must have an according entry in the overridden TABLEs at variant level.

[Rule] Diagnostic session preconditions shall be specified by referencing the according states from the STATE-CHART predefined in the diagnostic specification ESD.

[Rule] The DID data records shall be specified by STRUCTUREs referenced from the according TABLE-ROWs in the DID TABLEs.

[Rule] The TABLEs TA_S22 and TA_S2E shall not contain entries for support check identifiers.

2.3.4.3 Static LIDs

Renault / Nissan ODX handles static LIDs in a similar manner to DIDs as given in the chapter before. It uses two TABLEs to specify the static LIDs supported by an ECU: One TABLE for readable LIDs and one table for writable LIDs.

TABLE TA_S3B (LIDs Writable) is thereby dynamically referenced by the WriteDataByLocalIdentifier service ($3B) and TABLE TA_S21 (LIDs Readable) by the ReadDataByLocalIdentifier service ($21). On level of the diagnostic specification ESDs contained in Renault / Nissan RFQ packages, all LIDs predefined in the RFQ package are also specified in the two TABLES by according TABEL-ROWs.

Each TABLE-ROW references a STRUCTURE which describes the content of the LID's data record and specifies the diagnostic session preconditions by referencing the according states from the STATE-CHART also contained in the diagnostic specification ESD.

TABLE-ROWs of LIDs which must be implemented by concrete ECUs are thereby marked as mandatory. Likewise, LIDs whose data records must be implemented identically in all ECUs are marked as final at the according TABLE-ROW.

ECU Designers are expected to override the two TABLEs TA_S21 and TA_S3B at variant level. For LIDs predefined by the RFQ package, ECU designers have the choice to either reuse them by an according TABLE-ROW-REF referencing the original TABLE-ROW on ESD level or to introduce a new TABLE-ROW.

ECU specific LIDs or RFQ LIDs whose data record is different than specified at ESD level must be defined at BASE-VARIANT level by introducing a new TABLE-ROW in the overridden LID TABLEs.

Diagnostic session preconditions shall thereby be specified by referencing the according states from the STATE-CHART predefined in the diagnostic specification ESD.

Implementation requirements:

[Rule] The diagnostic specification ESDs contained in Renault / Nissan RFQ packages shall encompass two TABLEs: "LIDs Readable" and "LIDs Writable".

[Rule] The TABLE for readable LIDs shall have the SHORT-NAME "TA_S21" and TABLE for writable LIDs "TA_S3B".

[Rule] The TABLE for readable LIDs shall have the LONG-NAME "LIDs Readable" and TABLE for writable LIDs "LIDs Writable".

[Rule] The SHORT-NAMEs of the TABLE-ROWs in TABLE "TA_S21" and in TABLE "TA_S3B" as well as the SHORT-NAMEs of the STRUCTUREs referenced by those TABLE-ROWs and the SHORT-NAMEs of the PARAMs in those STRUCTUREs shall follow the respective rules specified in section 2.3.8.3.

[Rule] The ReadDataByLocalIdentifier ($21) and WriteDataByLocalIdentifier ($3B) services shall reference those TABLEs via SNREF.

[Rule] The two LID TABLEs shall be overridden on variant level.

[Rule] They shall contain entries for all readable and writable LIDs implemented by the ECU.

[Rule] LIDs marked as mandatory on ESD level must have an according entry in the overridden TABLEs at variant level.

[Rule] Diagnostic session preconditions shall be specified by referencing the according states from the STATE-CHART predefined in the diagnostic specification ESD.

[Rule] The LID data records shall be specified by STRUCTUREs referenced from the according TABLE-ROWs in the LID TABLEs.

[Rule] The TABLEs TA_S21 and TA_S3B shall not contain entries for support check identifiers.

2.3.4.4 I/O Controls – UDS

The I/O Control related diagnostic services ($22, $2F) make use of several TABLEs related to:

- The data record's interpretation itself.
- The data record's "Control Enable Mask Record".

At Renault/Nissan, all I/O Control DIDs must support the diagnostic service $22 (ReadDataByIdentifier). As a result, the positive responses of all $2F services simply reference the TABLE TA_S22 by a SHORT-NAME-REF in order to enable the interpretation of the data record.

2.3.4.4.1 Control Enable Mask Record Definition

Beside the interpretation of the I/O Control DIDs, the author must also specify the "Control Enable Mask Records" of IO controllable identifiers using the TABLE with SHORT-NAME TA_S2F_CEMR ("DIDs I/O Control Enable Mask Record").

That TABLE shall contain a TABLE-ROW for each IO controllable DID, where the STRUCTURE referenced by the TABLE-ROW defines the actual control enable mask record of the respective DID (as e. g. the TABLE-ROW for the DIDs 0x0301 and 0x5009 in Figure 5).

For identifiers which only consist of a single parameter, the according TABLE-ROW in TABLE TA_S2F-CEMR shall not reference anything (as e. g. the TABLE-ROW for the DID 0x4B15 in Figure 5).

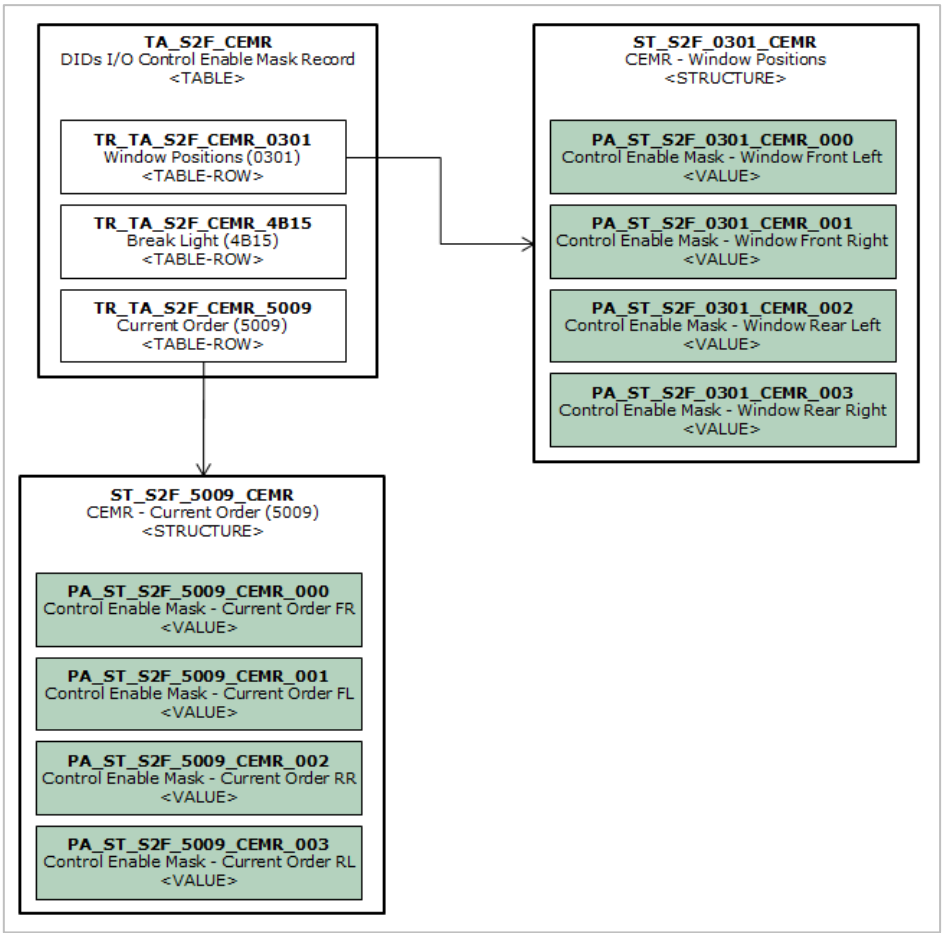


Figure 5 Example for I/O Control Enable Mask Record Definitions

Implementation requirements:

[Rule] The diagnostic specification ESDs contained in Renault / Nissan RFQ packages shall encompass a TABLE with SHORT-NAME "TA_S2F_CEMR". It may be overridden on variant level to specify the control enable mask records supported by the ECU.

[Rule] The SHORT-NAMEs of the TABLE-ROWs in TABLE "TA_S2F_CEMR" as well as the SHORT-NAMEs of the STRUCTUREs referenced by those TABLE-ROWs and the SHORT-NAMEs of the PARAMs in those STRUCTUREs shall follow the respective rules specified in section 2.3.8.3.

[Rule] In case of I/O Control DIDs having only one parameter, the according TABLE-ROW in TABLE "TA_S2F_CEMR" shall not define a reference to a STRUCTURE or a DATA-OBJECT-PROP.

[Rule] In case of I/O Control DIDs having more than one parameter, the according TABLE-ROW in TABLE "TA_S2F_CEMR" shall apply an ODX-LINK to a STRUCTURE describing the "Control Enable Mask Record".

[Rule] The author must define the session/security related state information at the TABLE-ROWs.

2.3.4.4.2 DID-Specific Definition of Supported IO Control Parameters

I/O Control DIDs at Renault/Nissan might support different InputOutputControlParameter (=sub-functions). Therefore, the author has to specify for each I/O Control which sub-function is supported by updating the TABLEs given in Figure 6 and Figure 7.

Renault / Nissan ODX models use four TABLEs to document which DIDs are usable with which of the four sub functions of the UDS IO control service $2F, that is: which DIDs are freezable, short-term adjustable, returnable and/or resettable. The control enable masks of the individual DIDs are defined by a separate, fifth TABLE, as described above in section 2.3.4.4.1.

As shown in Figure 6, the TABLEs for the freeze, reset and return sub functions merely consist of TABLE-ROW-REFs to that fifth, control enable mask TABLE:

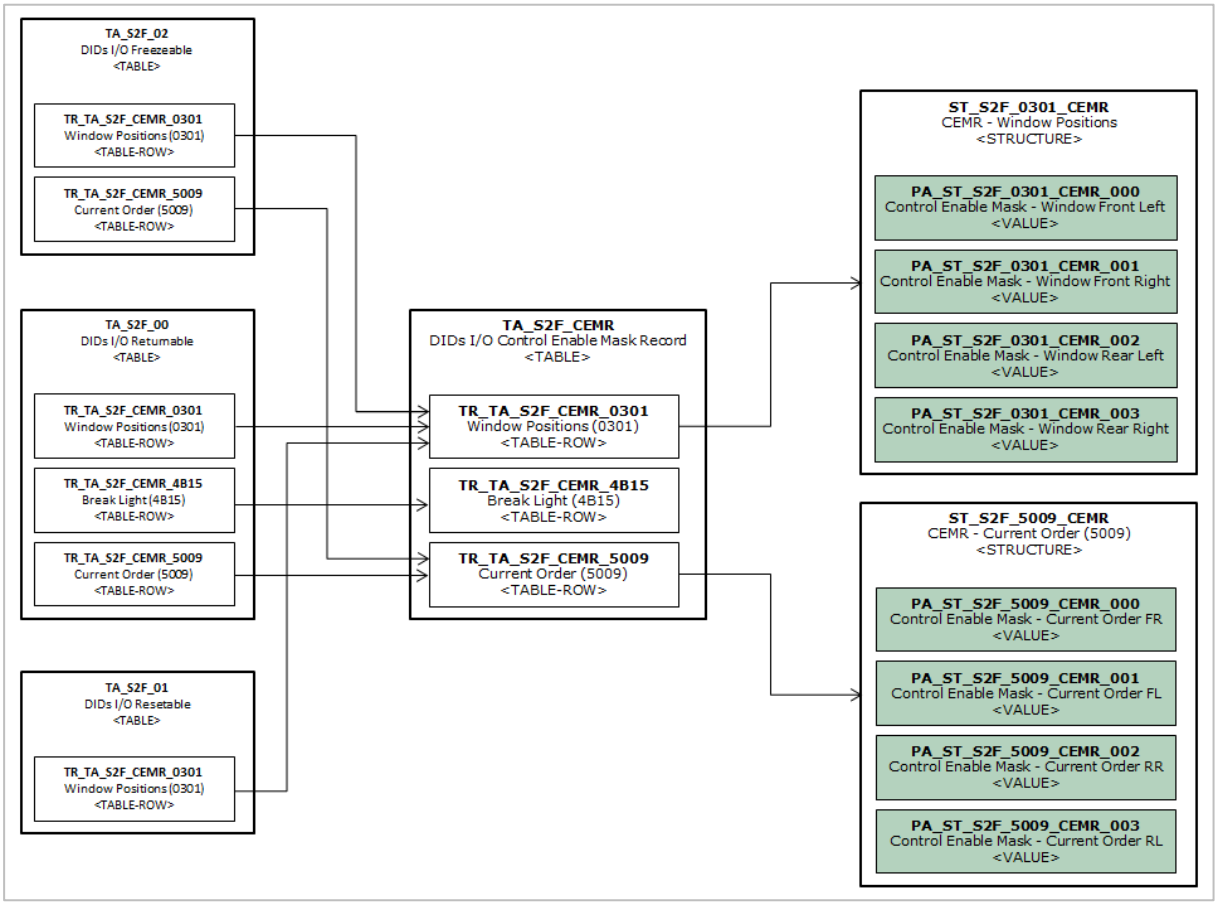


Figure 6 Definition of Returnable, Resettable and Freezable UDS I/O Controls

Implementation requirements:

[Rule] The diagnostic specification ESDs contained in Renault / Nissan RFQ packages shall encompass a TABLE with SHORT-NAME "TA_S2F_00". It is intended to be overridden on variant level to specify the returnable IO controls supported by the described ECU.

[Rule] The diagnostic specification ESDs contained in Renault / Nissan RFQ packages shall encompass a TABLE with SHORT-NAME "TA_S2F_01". It is intended to be overridden on variant level to specify the resettable IO controls supported by the described ECU.

[Rule] The diagnostic specification ESDs contained in Renault / Nissan RFQ packages shall encompass a TABLE with SHORT-NAME "TA_S2F_02". It is intended to be overridden on variant level to specify the freezable IO controls supported by the described ECU.

[Rule] The SHORT-NAMEs of TABLE-ROW-REFs contained in one of the TABLEs with SHORT-NAME "TA_S2F_00", "TA_S2F_01" or "TA_S2F_02" shall be "TR_TA_S2F_CEMR_{NNNN}", where {NNNN} shall be replaced with the four-digit hex code of the DID represented by the TABLE-ROW-REF.

[Rule] The TABLEs with SHORT-NAME "TA_S2F_00", "TA_S2F_01" or "TA_S2F_02" shall contain TABLE-ROW-REFs and no TABLE-ROWs. The TABLE-ROW-REFs shall target the according TABLE-ROWs in the TABLE with SHORT-NAME "TA_S2F_CEMR".

[Rule] The author must define the session/security related state information at each TABLE-ROW.

In contrast to this, the TABLE for the short-term adjustment sub function ("TA_S2F_03") consists of TABLE-ROWs, as shown in Figure 7. Each of those TABLE-ROWs references one STRUCTURE defining the I/O Control's data record and – if the according data record consists of more than one parameter – the STRUCTURE defining the I/O control's control enable mask record" (which shall be the same STRUCTURE also referenced by the afore-mentioned TABLE "TA_S2F_CEMR").

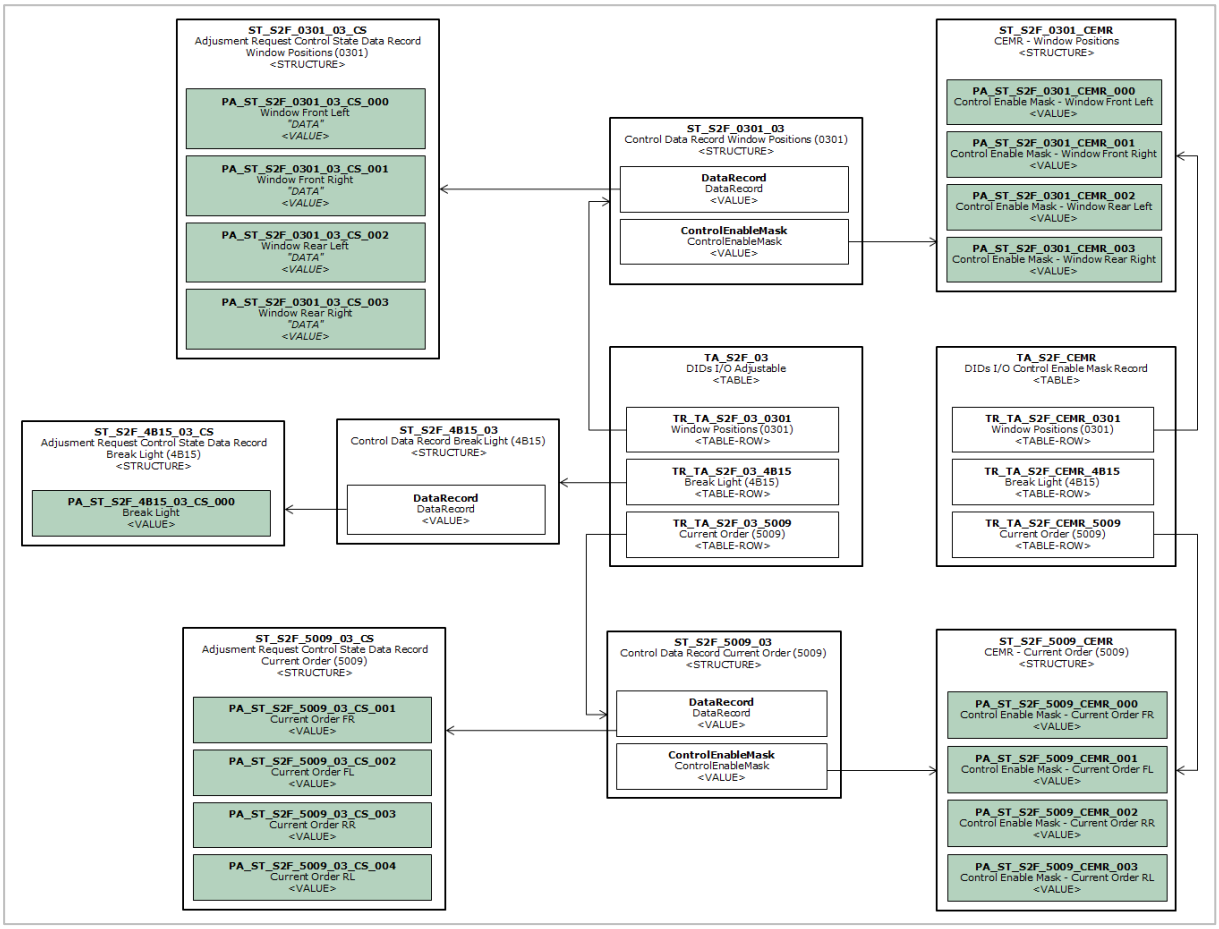


Figure 7 Definition of Adjustable UDS IO Controls

Implementation requirements:

[Rule] The diagnostic specification ESDs contained in Renault / Nissan RFQ packages shall encompass a TABLE with SHORT-NAME "TA_S2F_03". It is intended to be overridden on variant level to specify the short-term-adjustable IO controls supported by the described ECU.

[Rule] The author shall not use TABLE-ROW-REFs inside the TABLE "TA_S2F_03".

[Rule] The author must define the session/security related state information at each TABLE-ROW.

[Rule] Each TABLE-ROW of TABLE "TA_S2F_03" must reference a STUCTURE via an ODX-LINK.

[Rule] The SHORT-NAME of STRUCTUREs referenced by TABLE "TA_S2F_03" shall be "ST_S2F_{NNNN}_03", where {NNNN} shall be replaced by the four-digit hex code of the DID represented by the referencing TABLE-ROW.

[Rule] For controllable DIDs with only one parameter, the STRUCTURE referenced by the according TABLE-ROW of TABLE "TA_S2F_03" shall consist of only one PARAM referencing a STRUCTURE which represents the IO control's control state record.

[Rule] In case of a controllable DID with more than one parameter, the STRUCTURE referenced by the according TABLE-ROW of TABLE "TA_S2F_03" must contain a VALUE on BYTE-POSITION zero referencing a STRUCTURE which represents the IO control's control state record. In addition to this, it must contain a VALUE referencing the very same STRUCTURE which is also referenced by that DID's entry in TABLE "TA_S2F_CEMR" (that is: the STRUCTURE defining the control enable mask record for that DID).

[Rule] The SHORT-NAME of the STRUCTUREs representing control state records of IO controls shall be "ST_S2F_{NNNN}_03_CS", where {NNNN} shall be replaced by the four-digit hex code of the IO controllable DID whose control state record is represented by that STRUCTURE.

[Rule] The SHORT-NAMEs of the PARAMs in STRUCTUREs which represent control state records shall follow the respective rules specified in section 2.3.8.3.

2.3.4.5 I/O Controls – B/C

The I/O Control related diagnostic services related to B/C ($30, $21) make use of several TABLEs related to:

- Permanent and/or temporary control
- Stopping an I/O Control
- Requesting I/O status

At Renault/Nissan, it is not mandatory for I/O Controls to support the diagnostic service $21 (ReadDataByLocalIdentifier). But, the positive response for all I/O controls is structured in the same way for all I/O Controls in all sub-functions. As a result, the positive responses of all $30 services simply reference one TABLE TA_S30_PR by a SHORT-NAME-REF to enable the interpretation of the positive response.

Renault/Nissan uses the following set of independent TABLEs in scope of I/O Controls for UDS:

- IO Control – Temporary ("TA_S30_00")
  used for definition of request parameters on controlling temporary I/O
- IO Control – Permanent ("TA_S30_20")
  used for definition of request parameters on controlling permanent I/O
- IO Control - Status ("TA_S30_01")
  used for definition of request parameters on requesting the I/O's status
- IO Control - Stop ("TA_S30_11")
  used for definition of request parameters on stopping an I/O
- IO Control Other Control Status ("TA_S30_PR")
  used for additional response parameters in all I/O control "sub-functions", beside the ControlStatus ("Control in progress", "Control complete and OK", …)

Implementation requirements:

[Rule] The diagnostic specification ESDs for B/C contained in Renault / Nissan RFQ packages shall encompass five TABLEs with the following SHORT-NAMEs: "TA_S30_00", "TA_S30_20", "TA_S30_01", "TA_S30_11" and "TA_S30_PR". As no "generic" I/O exist at Renault/Nissan, these TABLEs contain placeholder data at the RFQ level. The user must override them on BASE-/ECU-VARIANT level to specify which control parameters are supported for the different controllable I/Os.

[Rule] The author must define the session/security related state information at each TABLE-ROW of the four request related TABLEs "TA_S30_00", "TA_S30_20", "TA_S30_01" and "TA_S30_11".

[Rule] The SHORT-NAMEs of the TABLE-ROWs in TABLEs "TA_S30_00", "TA_S30_20", "TA_S30_01", "TA_S30_11" and "TA_S30_PR" as well as the SHORT-NAMEs of the

STRUCTUREs referenced by those TABLE-ROWs and the SHORT-NAMEs of the PARAMs in those STRUCTUREs shall follow the respective rules specified in section 2.3.8.3.

2.3.4.6 Fault Memory Tables

The fault memory related services (such as $1902 and $1904 and the different OBD modes) make use of different TABLEs and TEXTTABLEs (and other ODX elements) in order to define an ECU's:

- Devices and failure types
- Valid combination of the latter two (representing the actual DTCs)
- OBD trouble codes
- Snapshot record numbers, snapshot records and the DIDs they are composed of
- Links between the snapshots and actual DTCs stored for them

They are defined at the RFQ layer (with placeholder data) and intended to be overridden at the variant layer. Figure 8 and Figure 9 provide a conceptual overview of these overridden constructs (with some exemplary sample data).

Note: Although not all of the ODX elements shown in these figures are ODX TABLEs (actually only the definitions of the DIDs making up individual snapshot records are based on ODX TABLEs), this section nonetheless explains all of the constructs to provide a full overview of the whole context.

2.3.4.6.1 DTC Definition

In order to define DTCs, Renault / Nissan ODX makes use of four different ODX elements:

- One TEXTTABLE "FailureTypes" yielding the failure types for which DTCs might be stored.
- One DTC-DOP "Devices" representing the devices inside the ECU for which DTCs might be stored.
- One DTC-DOP "ObdDtcRecord" yielding all supported OBD fault codes which might be stored.
- One DTC-DOP "DtcRecord" which defines the valid combinations of Devices, OBD Trouble Codes and Failure Types, yielding the actual three byte DTCs stored by the described ECU and reported by e. g. the UDS service $19.

Actually, at Renault / Nissan, the base DTC numbers of non-OBD DTCs always correspond to one of the device numbers defined by the DTC-DOP "Devices"; thus, a DTC is always implicitly related to certain device.

Likewise, the failure type bytes of the stored DTCs always correspond to one of the failure types defined by the TEXTTABLE "FailureTypes".

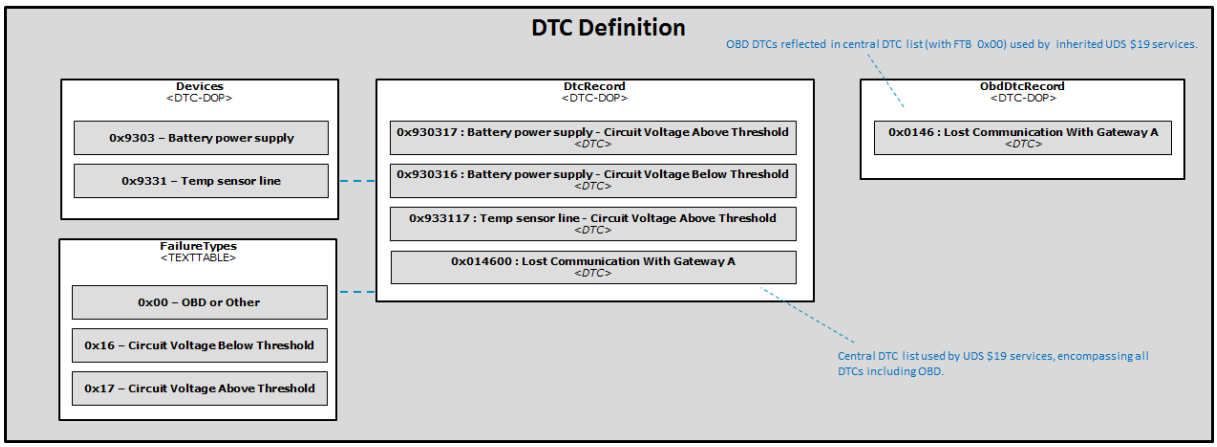


Figure 8 DTC, Device and Failure Type Tables

For OBD DTCs, the base DTC number corresponds to the according OBD trouble code in DTC-DOP "ObdDtcRecord" and the failure type byte is always 0x00.

The DTC-DOP "DtcRecord" is actually referenced by SHORT-NAME by the different sub functions of service $19. It encompasses all OBD and non-OBD DTCs which can be stored by the described ECU. The DTC-DOPs "Devices", "ObdDtcRecord" and "DtcRecord" are required to be overridden on BASE-VARIANT level to define the actual Devices, OBD and non-OBD DTCs of the respective ECU. The TEXTTABLE "FailureTypes" may be overridden on variant level to define the actual failure types supported by the respective ECU, but it is not required to do so.

Implementation requirements:

[Rule] On RFQ level, Renault / Nissan ODX models define the following four elements for the definition of DTCs: A TEXTTABLE with SHORT-NAME "FailureTypes", a DTC-DOP with SHORT-NAME "Devices", a DTC-DOP with SHORT-NAME "ObdDtcRecord" and a DTC-DOP with SHORT-NAME "DtcRecord".

[Rule] All three of the DTC-DOPs must be overridden on variant level, defining the actual DTCs implemented by the described ECU.

[Rule] The TEXTTABLE "FailureTypes" shall contain FailureTypes according to ISO15031-6 on the RFQ level. It may be overridden on variant level, but it is not required to do so.

[Rule] For each DTC defined in the DTC-DOP with SHORT-NAME "DtcRecord", there must be a corresponding DTC in the DTC-DOP with SHORT-NAME "Devices" whose TROUBLE-CODE value corresponds to the base DTC number (that is: the first two bytes of the TROUBLE-CODE) of the DTC in the DTC-DOP "DtcRecord".

[Rule] For each DTC defined in the DTC-DOP "DtcRecord", there must be a corresponding entry in the TEXTTABLE with SHORT-NAME "FailureTypes" whose value corresponds to the failure type byte (that is: the last byte of the TROUBLE-CODE) of the DTC in the DTC-DOP "DtcRecord".

[Rule] For each DTC defined in the DTC-DOP "DtcRecord" the TEXT element shall be composed of the TEXT element of the corresponding DTC in the DTC-DOP "Devices" and the text of the corresponding entry in the TEXTTABLE "FailureTypes", separated by the character sequence " – " (blank+minus+blank), except for DTCs whose failure type byte equals 0x00; for the latter ones, the TEXT element shall be a copy of the TEXT element of the corresponding DTC in the DTC-DOP "Devices".

[Reco] In case of a DTC whose FTB equals 0x00, one should avoid specifying the failure type as part of the TEXT of the corresponding DTC in the DTC-DOP "Devices", if there are other DTCs with the same Device number and a non-zero FTB; on the other side, if there are no additional DTCs with the same Device number, then Renault / Nissan does recommend to actually define the type of failure as part of the TEXT of the corresponding DTC in the DTC-DOP "Devices".

[Rule] For each DTC defined in the DTC-DOP "ObdDtcRecord", there shall be a corresponding DTC in the DTC-DOP "DtcRecord" which satisfies the following three conditions:

- The base DTC number (that is: the first two bytes of the TROUBLE-CODE) shall be the same as the TROUBLE-CODE of the DTC in DTC-DOP "ObdDtcRecord".
- The failure type byte (that is: the last byte of the TROUBLE-CODE) shall be 0x00.
- The TEXT element shall exactly correspond to the TEXT element of the corresponding DTC in DTC-DOP "ObdDtcRecord".

[Rule] DTC-DOP "DtcRecord" shall not contain any entry which is not either reflected in the DTC-DOP "Devices" or the DTC-DOP "ObdDtcRecord" as defined by the preceding rules (for process partner DTCs, according entries in DTC-DOP "Devices" shall be created).

[Rule] Authors shall use the DISPLAY-TROUBLE-CODE element of DTCs to express the DTC in SAE notation.

2.3.4.6.2 Snapshot Definition

In order to define snapshots, Renault / Nissan ODX makes at least use of three ODX elements:

- One ENV-DATA-DESC with SHORT-NAME "SnapshotRecordContents" used as enveloping parent element for the ENV-DATA elements representing the different kinds of snapshot records supported by the described ECU.
- One TEXTTABLE "DtcSnapshotRecordNumbersRQ" defining the different snapshot record numbers which may be specified in requests of service $1904.
- One TEXTTABLE "DtcSnapshotRecordNumbersPR" defining the different snapshot record numbers which may be returned by the ECU in the positive responses of service $1904.

Moreover, for each individual type of snapshot record supported by the described ECU at least one ENV-DATA element shall be defined on variant level representing the snapshot record. On RFQ level, one ENV-DATA element is defined representing an arbitrary snapshot record.

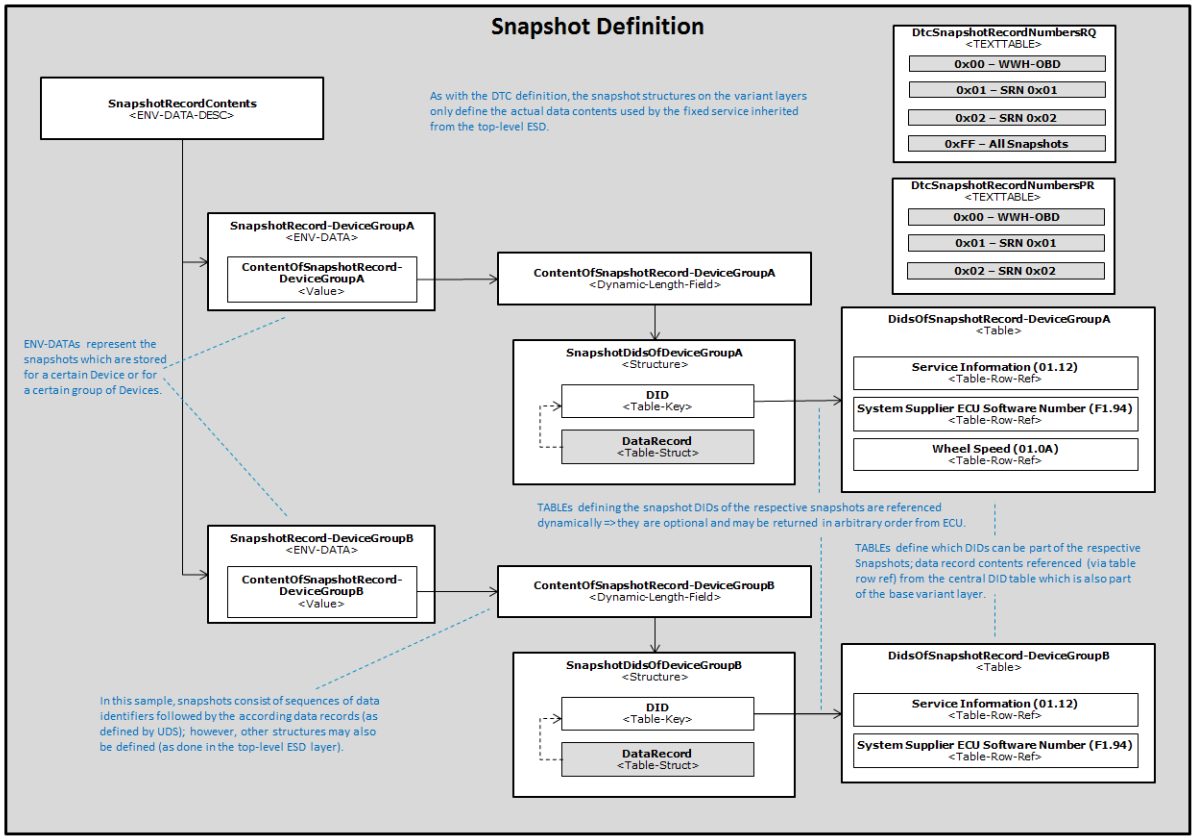


Figure 9 Snapshot Tables

In case of UDS based ECUs, it is **strongly** recommended to describe the contents of the individual snapshot records following the TABLE based approach as shown in Figure 9, where the individual DIDs which may be stored inside the respective snapshot records are defined by an according TABLE-ROW-REF referencing the according DID definition from TABLE "TA_S22" (cf. section 0).

Implementation requirements:

[Rule] On RFQ level, Renault / Nissan ODX models define the following four elements for the definition of snapshots: One TEXTTABLE with SHORT-NAME "DtcSnapshotRecord-NumbersRQ", one TEXTTABLE with SHORT-NAME "DtcSnapshotRecordNumbersPR", one ENV-DATA-DESC with SHORT-NAME "SnapshotRecordContents" and one ENV-DATA with SHORT-NAME "GenericEnvData".

[Rule] Except for the ENV-DATA "GenericEnvData", these elements shall be overridden on BASE-VARIANT level for defining the actual snapshot records stored by the described ECU.

[Rule] On BASE-VARIANT level the overridden TEXTTABLE "DtcSnapshotRecordNumbersRQ" shall contain entries for all possible snapshot record numbers which might be specified in requests of service $1904, including the number 0xFF used for requesting all snapshots and the number 0x00 for requesting only OBD relevant snapshots.

[Rule] Likewise, the overridden TEXTTABLE "DtcSnapshotRecordNumbersPR" shall contain entries for all possible snapshot record numbers which might be returned by the ECU in positive responses of service $1904.

[Rule] Except for the entry 0xFF in TEXTTABLE "DtcSnapshotRecordNumbersRQ", there must be a corresponding entry in TEXTTABLE "DtcSnapshotRecordNumbersPR" and all entries must match exactly the corresponding entries in TEXTTABLE "DtcSnapshotRecordNumbersPR".

[Rule] All ENV-DATAs referenced by the ENV-DATA-DESC "SnapshotRecordContents" overridden on BASE-VARIANT level shall be bound to one or more Devices or OBD Trouble Codes. That is: All DTC values specified for the ENV-DATA shall be two byte values for which an according entry exists either in the DTC-DOP "Devices" or in the DTC-DOP "ObdDtcRecord" or in both.

[Reco] For UDS based ECUs, snapshot records should be defined as shown in Figure 9, that is: The ENV-DATA content consists of DYNAMIC-LENGTH-FIELD which represents the sequence of data identifiers immediately followed by the respective data records. By defining the individual DIDs which might appear in that sequence by dynamically referencing an ODX TABLE as shown in that figure, the ECU is not restricted to returning the DIDs in a certain order in the response of $1904 and all described DIDs may be returned optionally.

2.3.4.6.3 Global Snapshot Record

In most ECUs, there are snapshot DIDs which must be supported by each and every snapshot record. At Renault / Nissan, these snapshot DIDs are called "global snapshot DIDs". The set of all global snapshot DIDs is called the "global snapshot" of an ECU. For such ECUs, the global snapshot shall be represented in the ODX data.

ECU designers must choose one of the following two approaches to represent the global snapshot:

- Add an ALL-VALUE ENV-DATA to the SnapshotRecordContents ENV-DATA-DESC.
- Use the optional element with SHORT-NAME "GlobalSnapshotRecord" to define the content of the global snapshot record.

The former solution shall be preferred, if possible: Prerequisite is that the described ECU actually returns global snapshot DIDs before any DTC-specific snapshot DIDs when snapshot records are queried (e. g. via UDS service $1904).

For other ECUs, at RFQ level (e. g. in layer "Renault Nissan Spec UDS") a STRUCTURE or TABLE with the SHORT-NAME "GlobalSnapshotRecord" may be present. This specifies all common data which must be part of all snapshot records defined for an ECU on variant level.

It is recommended that this is done by using a TABLE defining DIDs via TABLE-ROW-REFs referencing the according data records from TABLE "TA_S22" so that they are eligible for overriding on variant level as described in section 0.

Implementation requirements:

[Rule] If global snapshot DIDs are defined for a particular ECU, then the global snapshot shall be represented in the ODX data describing that ECU.

[Rule] If the ECU returns global snapshot DIDs before any DTC specific snapshot DIDs upon query of snapshot records, the approach of using an ALL-VALUE ENV-DATA shall be followed.

[Reco] If modelled with an ALL-VALUE ENV-DATA, the global snapshot record should be defined like the snapshot records shown in Figure 9; that is: The ALL-VALUE ENV-DATA content consists of a single value referencing an END-OF-PDU-FIELD which represents a sequence of data identifiers immediately followed by the respective data records. By defining the individual DIDs which might appear in that sequence by dynamically referencing an ODX TABLE as shown in that figure, the ECU is not restricted to returning the DIDs in a certain order in the response of $1904 and all DIDs may be returned optionally.

[Rule] If the approach using an ALL-VALUE ENV-DATA cannot be followed, then the global snapshot shall be represented by using an element with the SHORT-NAME "GlobalSnapshotRecord".

[Rule] The element "GlobalSnapshotRecord" may be overridden on variant level. In that case, the element on variant level must at least contain all the contents which the according element contains in the RFQ layer.

[Reco] If used, the element with SHORT-NAME "GlobalSnapshotRecord" should be a TABLE consisting of TABLE-ROW-REFs referencing DID definitions from the TABLE "TA-S22" (cf. section 0).

[Rule] All contents defined by the element "GlobalSnapshotRecord" must also be defined within all snapshot records definitions on variant level.

2.3.4.6.4 Extended Data Definition

At Renault/Nissan, all Extended Data Records are global (not DTC specific) and optional.

In order to define extended data records, Renault / Nissan ODX makes use of two elements:

- The TABLE with SHORT-NAME "ExtendedDataRecords", containing references to STRUCTUREs which define the actual contents of the extended data records.
- The TEXTTABLE with SHORT-NAME "ExtendedDataRecordsTableKeys" which is used as KEY-DOP for the TABLE mentioned above.

These two elements are predefined in the RFQ layer ESD. ECU designers are supposed to overwrite these elements on variant level to define the actual extended data records stored by the described ECU. All extended data records predefined on RFQ layer are optional.

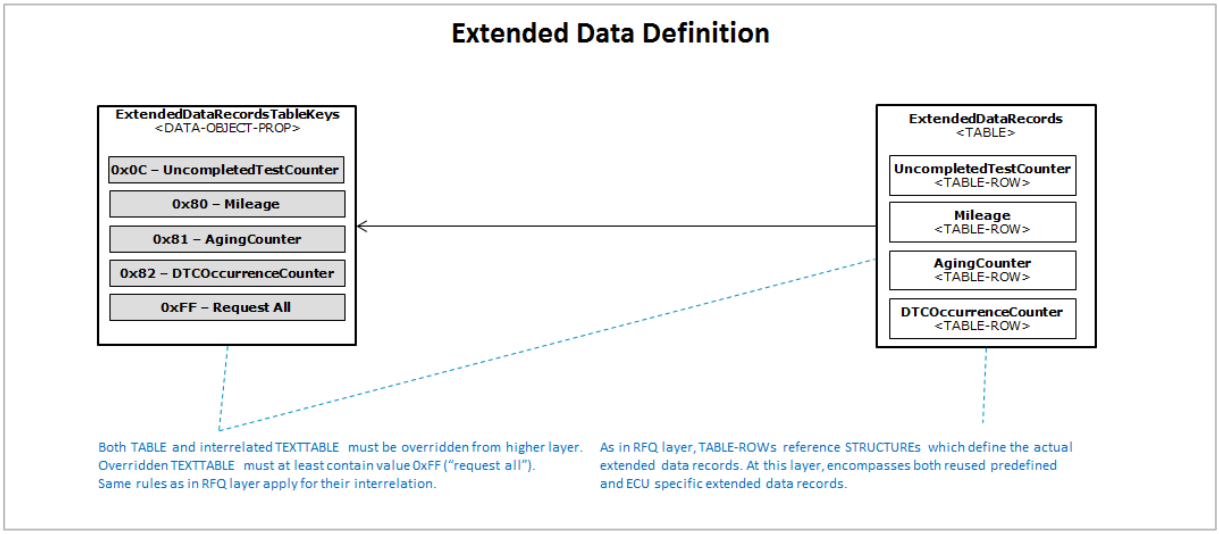


Figure 10 Extended Data Record TABLE

Implementation requirements:

[Rule] On RFQ level, Renault / Nissan ODX models define the following two elements for the definition of extended data records: One TABLE with SHORT-NAME "ExtendedData-Records" and one TEXTTABLE with SHORT-NAME "ExtendedDataRecordsTableKeys".

[Rule] Both of these elements shall be overridden on variant level for defining the actual extended data records stored by the described ECU.

[Rule] On variant level, the overridden TEXTTABLE "ExtendedDataRecordsTableKeys" shall contain entries for all possible extended data record numbers which might be contained in requests and positive responses of service $1906. It shall at least include the number 0xFF used for requesting all extended data records.

[Rule] Extended Data Records predefined in the RFQ level are optional, that is: it is not required to reuse them on variant level.

[Rule] For each entry in TEXTTABLE "ExtendedDataRecordsTableKeys", there must be a matching corresponding entry in TABLE "ExtendedDataRecords" and vice versa. Exception to this is the value 0xFF in the TEXTTABLE, which must not be contained in the TABLE.

[Rule] The TEXTTABLE "ExtendedDataRecordsTableKeys" shall be used as KEY-DOP of the TABLE "ExtendedDataRecords".

2.3.4.7 Routines – UDS

In Renault / Nissan ODX for UDS, six TABLEs are used to specify the optional control option records, info bytes and status records of the different routines implemented in the described ECUs:

- The TABLE with SHORT-NAME "TA_S31_01_RQ" to define the routine control option records used in the request of the "startRoutine" subfunction (0x01) of service $31.
- The TABLE with SHORT-NAME "TA_S31_01__PR" to define the routine info bytes and the routine status records of the positive response of the "startRoutine" subfunction (0x01).
- The TABLE with SHORT-NAME "TA_S31_SP_RQ " to define the routine control option records used in the request of the "stopRoutine" subfunction (0x02) of service $31.
- The TABLE with SHORT-NAME "TA_S31_SP_PR" to define the routine info bytes and the routine status records of the positive response of the "stopRoutine" subfunction (0x02).
- The TABLE with SHORT-NAME "TA_S31_RS_RQ" to define the routine control option records used in the request of the "requestRoutineResults" subfunction (0x03) of service $31.
- The TABLE with SHORT-NAME "TA_S31_RS_PR" to define the routine info bytes and the routine status records of the positive response of the "requestRoutineResults" subfunction (0x03).

These TABLEs are predefined on RFQ level (containing placeholder data) and must be overridden on BASE-VARIANT level in order to define the actual routine parameters of the described ECU.

Implementation requirements:

[Rule] The SHORT-NAME of the TABLE "RIDs Startable RQ" referenced by the REQUEST of the DIAG-SERVICE with SHORT-NAME "Routine_Control_Start" shall be "TA_S31_ST_RQ".

[Rule] The SHORT-NAME of the TABLE "RIDs Startable PR" which is referenced by the POS-RESPONSE of the DIAG-SERVICE with SHORT-NAME "Routine_Control_Start" shall be "TA_S31_ST_PR".

[Rule] The SHORT-NAME of the TABLE "RIDs Stoppable RQ" referenced by the REQUEST of the DIAG-SERVICE with SHORT-NAME "Routine_Control_Stop" shall be "TA_S31_SP_RQ".

[Rule] The SHORT-NAME of the TABLE "RIDs Stoppable PR" which is referenced by the POS-RESPONSE of the DIAG-SERVICE with SHORT-NAME "Routine_Control_Stop" shall be "TA_S31_SP_PR".

[Rule] The SHORT-NAME of the TABLE "RIDs Resultable RQ" which is referenced by the REQUEST of the DIAG-SERVICE with SHORT-NAME "Routine_Control_Request_Results" shall be "TA_S31_RS_RQ".

[Rule] The SHORT-NAME of the TABLE "RIDs Resultable PR" which is referenced by the POS-RESPONSE of the DIAG-SERVICE with SHORT-NAME "Routine_Control_Request_Results" shall be "TA_S31_RS_PR".

[Rule] If a routine supports subfunction "startRoutine" (0x01), then an according TABLE-ROW must exist both in TABLE "TA_S31_ST_RQ" and TABLE "TA_S31_ST_PR" for the routine, even if no parameter reference is given (indicating that the according optional records are not used for the routine at hand in that subfunction's request or response).

[Rule] If a routine supports subfunction "stopRoutine" (0x02), then an according TABLE-ROW must exist both in TABLE "TA_S31_SP_RQ" and TABLE "TA_S31_SP_PR" for the routine, even if no parameter reference is given (indicating that the according optional records are not used for the routine at hand in that subfunction's request or response).

[Rule] If a routine supports subfunction "requestRoutineResults" (0x03), then an according TABLE-ROW must exist both in TABLE "TA_S31_RS_RQ" and TABLE "TA_S31_RS_PR" for the routine, even if no parameter reference is given (indicating that the according optional records are not used for the routine at hand in that subfunction's request or response).

[Rule] The SHORT-NAMEs of the TABLE-ROWs in TABLEs "TA_S31_ST_RQ", "TA_S31_ST_PR", "TA_S31_SP_RQ", "TA_S31_SP_PR", "TA_S31_RS_RQ" and "TA_S31_RS_PR" as well as the SHORT-NAMEs of the STRUCTUREs referenced by those TABLE-ROWs and the SHORT-NAMEs of the PARAMs in those STRUCTUREs shall follow the respective rules specified in section 2.3.8.3.

2.3.4.8 Routines – B/C

In Renault / Nissan ODX for B/C, four TABLEs are used to specify the optional routine data records, routine optional response parameters and routine exit parameters of the different routines implemented in the described ECUs:

- The TABLE with SHORT-NAME "TA_S31_RQ" to define the routine data records used in the request of the "StartRoutineByLocalIdentifier" (service $31).
- The TABLE with SHORT-NAME "TA_S31_PR" to define the optional routine parameters (beside the routine entry status) used in the positive response of the "StartRoutineByLocalIdentifier" (service $31).
- The TABLE with SHORT-NAME "TA_S32_RQ" to define the routine exit parameters used in the request of the "StopRoutineByLocalIdentifier" (service $32).
- The TABLE with SHORT-NAME "TA_S32_PR" to define the optional routine exit parameters (beside the routine exit status) used in the positive response of the "StopRoutineByLocalIdentifier" (service $32).

These TABLEs are predefined on RFQ level (containing placeholder data) and must be overridden on BASE-VARIANT level to define the actual routine parameters of the described ECU.

Implementation requirements:

[Rule] The SHORT-NAME of TABLE "Routine – Start (Request)" referenced by the REQUEST of the DIAG-SERVICE with SHORT-NAME "StartRoutineByLocalIdentifier" shall be "TA_S31_RQ".

[Rule] The SHORT-NAME of TABLE "Routine – Start (Positive response)" referenced by the POS_RESPONSE of the DIAG-SERVICE with SHORT-NAME "StartRoutineByLocalIdentifier" shall be "TA_S31_PR".

[Rule] The SHORT-NAME of TABLE "Routine – Stop (Request)" referenced by the REQUEST of the DIAG-SERVICE with SHORT-NAME "StopRoutineByLocalIdentifier" shall be "TA_S32_RQ".

[Rule] The SHORT-NAME of TABLE "Routine – Stop (Positive response)" referenced by the REQUEST of the DIAG-SERVICE with SHORT-NAME "StopRoutineByLocalIdentifier" shall be "TA_S32_PR".

[Rule] If a routine identifier supports the service StartRoutineByLocalIdentifier, then an according TABLE-ROW must exist both in TABLE "TA_S31_RQ" and TABLE "TA_S31_PR" for the routine, even if no parameter reference is given (indicating that the according optional records are not used for the routine at hand in that subfunction's request or response).

[Rule] If a routine identifier supports the service StopRoutineByLocalIdentifier, then an according TABLE-ROW must exist both in TABLE "TA_S32_RQ" and TABLE "TA_S32_PR" for the routine, even if no parameter reference is given (indicating that the according optional records are not used for the routine at hand in that subfunction's request or response).

[Rule] The SHORT-NAMEs of the TABLE-ROWs in TABLEs "TA_S31_RQ", "TA_S31_PR", "TA_S32_RQ" and "TA_S32_PR" as well as the SHORT-NAMEs of the STRUCTUREs referenced by those TABLE-ROWs and the SHORT-NAMEs of the PARAMs in those STRUCTUREs shall follow the respective rules specified in section 2.3.8.3.

2.3.4.9 Security Access Levels

Renault / Nissan ODX uses the TABLEs with the SHORT_NAMEs "TA_S27_RS" ("SecurityAccess-requestSeed") and "TA-S27_SK" ("SecurityAccess-sendKey") to specify the security access levels supported by the described ECU. The TABLEs are referenced dynamically by the SecurityAccess services ($27) predefined in the diagnostic specification ESDs which are part of Renault / Nissan RFQ packages.

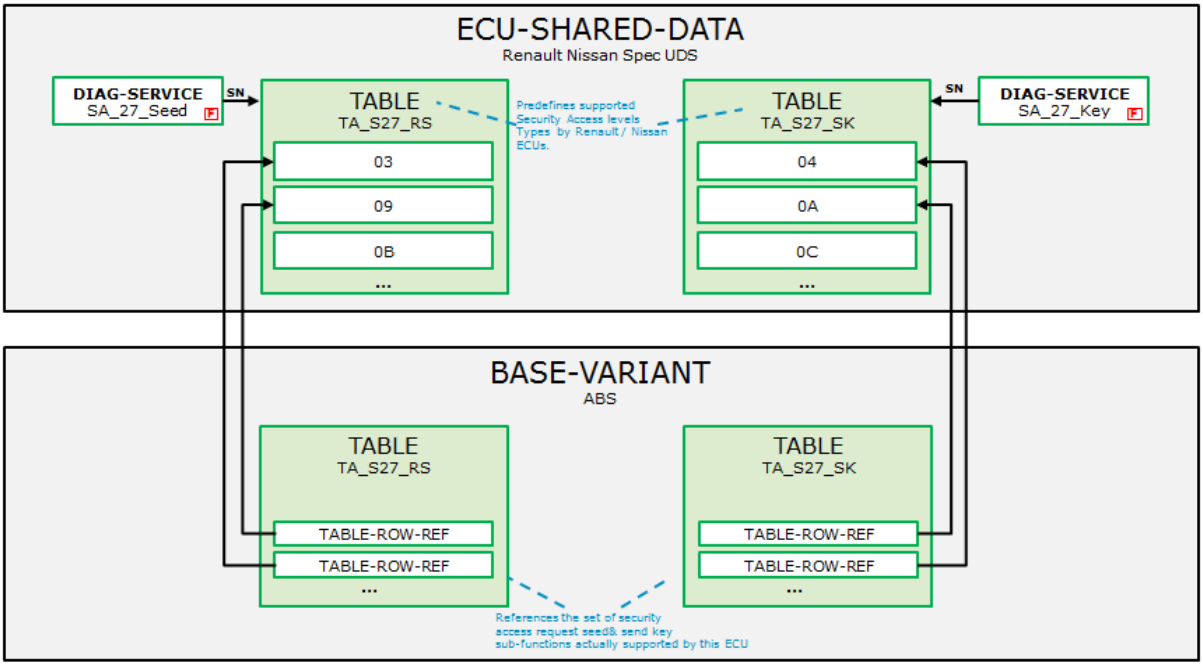


Figure 11 Tables for security access levels

The possible security models (encompassing the set of security access levels and the lengths of the seed and key parameters) are predefined per RFQ package. The services related to security access (request seed / send key) are predefined in the ESD and reference the TABLEs and the DOPs defining the seed and key lengths by SHORT-NAME-REF.

As a result, ECU designers are expected to override these TABLEs and DOPs at BASE-VARIANT level in order to specify different subsets of security access levels for a concrete ECU, but only as long as the overridden set is **in accordance with one of the security models predefined by Renault / Nissan**. Note that overriding on ECU-VARIANT level is not allowed in Renault / Nissan ODX.

The predefined security access related sub-functions can thereby be reused by simply creating a TABLE-ROW-REF referencing the according table row from the ESD level.

On ESD level, predefined STATE-CHARTs exist. Each TABLE-ROW of the TABLEs "TA_S27_RS" and "TA_S27_SK" must specify correct STATE-TRANSITION-/ and PRE-CONDITION-STATE-REFs targeting those STATE-CHARTs.

The seed and key lengths are predefined by the RFQ packages. They are independent from the security access levels inside the DIAG-SERVICEs (cf. section 2.3.5.11 for details on using and overriding seed and key definitions). Thus, the TABLE-ROWs of the TABLEs "TA_S27_RS" and "TA_S27_SK" must neither reference a STRUCTURE nor a DATA-OBJECT-PROP.

Implementation requirements:

[Rule] The diagnostic specification ESDs contained in Renault / Nissan RFQ packages shall encompass the TABLEs defining the supported SecurityAccess "request seed" and "send key" sub-functions.

[Rule] The SHORT-NAME of the TABLE which stores the request seed sub-functions shall be "TA_S27_RS".

[Rule] The TABLE storing the send key sub-functions shall have the SHORT-NAME "TA_S27_SK".

[Rule] TABLE-ROWs inside the two TABLEs "TA_S27_RS" and "TA_S27_SK" shall neither reference a STRUCTURE nor a DATA-OBJECT-PROP.

[Rule] The SecurityAccess services ($27) shall reference the TABLE "TA_S27_RS" or "TA_S27_SK" via SNREF.

[Rule] For ECUs which support different sets of security access levels, the two TABLEs "TA_S27_RS" and "TA_S27_SK" may be overridden at BASE-VARIANT level but must not be overridden at ECU-VARIANT level.

2.3.4.10 ECU-Reset Types

Renault / Nissan ODX uses a TABLE with SHORT-NAME TA_S11 ("ResetTypes") to specify the ECU reset types supported by an ECU. It is referenced dynamically by the ECUReset service ($11) predefined in the diagnostic specification ESDs which are part of Renault / Nissan RFQ packages.

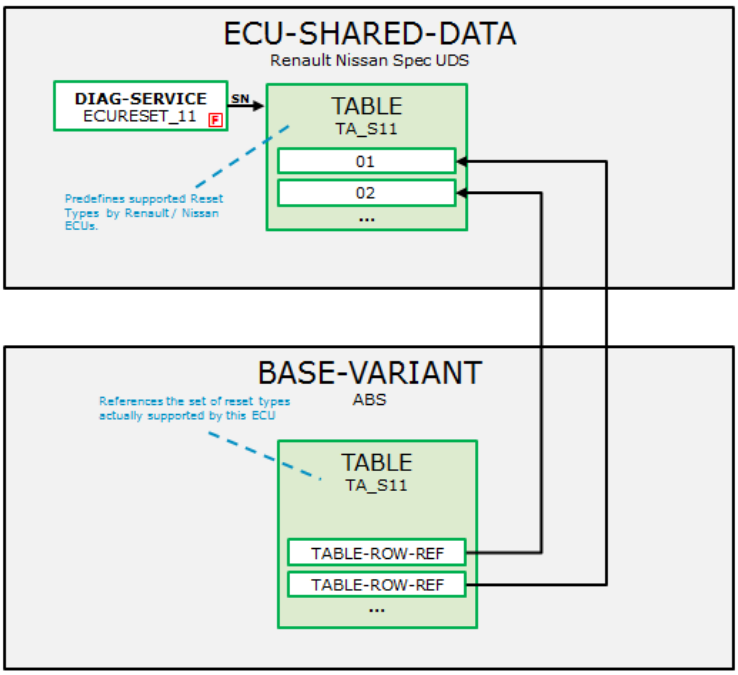


Figure 12 Specification of ECU reset types with TABLE TA_S11

Three ECU reset types defined by ISO UDS are thereby predefined by TABLE TA_S11 on ESD level. As indicated in Figure 12, ECU designers are expected to override TABLE TA_S11 on variant level in order to specify different sets of ECU reset types for a concrete ECU.

The predefined ECU reset types can thereby be reused by simply using a TABLE-ROW-REF referencing the according table row from the ESD level. Other ECU reset types are not allowed.

On ESD level, predefined STATE-CHARTs exist. Each TABLE-ROW of TABLE TA_S11 must specify correct STATE-TRANSITION-REFs into the default session and into the "locked state". The TABLE-ROWs must also specify the appropriate PRE-CONDITION-STATE-REFs.

As the reset services just request the reset type without any type depending parameters, the TABLE-ROWs of TABLE TA_S11 must neither reference a STRUCTURE nor a DATA-OBJECT-PROP.

Implementation requirements:

[Rule] The diagnostic specification ESDs contained in Renault / Nissan RFQ packages shall encompass a TABLE ResetTypes.

[Rule] That TABLE shall have the SHORT-NAME "TA_S11".

[Rule] TABLE-ROWs inside the TABLE "ResetTypes" shall neither reference a STRUCTURE nor a DATA-OBJECT-PROP.

[Rule] The EcuReset service ($11) shall reference that TABLE via SNREF.

[Rule] For ECUs which support different sets of reset types, the TABLE TA_S11 shall be overridden at variant level.

2.3.4.11 DTC Setting Types

Renault / Nissan ODX uses the TABLE with SHORT-NAME TA_S85 ("DTCSettingTypes") to specify the ControlDTCSetting types supported by an ECU. It is referenced dynamically by the ControlDTCSetting service ($85) predefined in the diagnostic specification ESDs which are part of Renault / Nissan RFQ packages.

Two setting types as defined by ISO UDS are thereby predefined in by TABLE TA_S85 on ESD level. The service to control the DTC setting which is predefined in the ESD references this TABLE by an ODX-LINK. As a result, ECU designers can't modify the TABLE TA_S85 on variant level.

On ESD level, predefined STATE-CHARTs exist. Each TABLE-ROW of TABLE TA_S85 shall specify the appropriate PRE-CONDITION-STATE-REFs targeting those STATE-CHARTs.

As the control DTC setting services just requests an according setting type (on/off) without any type depending parameters, the TABLE-ROWs of TABLE TA_S85 must neither reference a STRUCTURE nor a DATA-OBJECT-PROP.

Implementation requirements:

[Rule] The diagnostic specification ESDs contained in Renault / Nissan RFQ packages shall encompass a TABLE DTCSettingTypes.

[Rule] That TABLE shall have the SHORT-NAME "TA_S85".

[Rule] TABLE-ROWs inside the TABLE "TA_S85" shall neither reference a STRUCTURE nor a DATA-OBJECT-PROP.

[Rule] The ControlDTCSetting service ($85) shall reference that TABLE statically via ODX-LINK.

[Rule] ECU designers must not override TABLE "TA_S85" on variant layers.

2.3.4.12 Communication Control Types

Renault / Nissan ODX uses the TABLE with SHORT-NAME TA_S28 ("CommunicationControlTypes") to specify the CommunicationControl types supported by an ECU. It is referenced dynamically by the

CommunicationControl service ($28) predefined in the diagnostic specification ESDs which are part of Renault / Nissan RFQ packages.

The supported CommunicationControl types specify which sub-functions are available for the SERVICE "CommunicationControl" ($28). Since ECUs could support different sub-functions, ECU designers may modify the TABLE on variant level.

As the Communication Control services just request an according communication control type (sub-function identifier) without any type depending parameters, the TABLE-ROWs of TABLE CommunicationControlTypes must neither reference a STRUCTURE nor a DATA-OBJECT-PROP.

Implementation requirements:

[Rule] The diagnostic specification ESDs contained in Renault / Nissan RFQ packages shall encompass a TABLE CommunicationControlTypes.

[Rule] That TABLE shall have the SHORT-NAME "TA_S28".

[Rule] TABLE-ROWs inside the TABLE "TA_S28" shall neither reference a STRUCTURE nor a DATA-OBJECT-PROP.

[Rule] The CommunicationControl service ($28) shall reference that TABLE via SNREF.

[Rule] The TABLE may be overridden on variant level, but it is not required.

[Rule] The TABLE shall contain entries for all available sub-functions of the service "CommunicationControl" ($28).

[Rule] The TABLE shall use a 1-Byte HexDump DOP as its KEY-DOP.

2.3.4.13 Support Check TABLEs

The diagnostic specification ESDs distributed with Renault / Nissan RFQ packages predefine three TABLEs for the support check facilities:

- For Spec UDS, the according ESD defines the TABLE with SHORT-NAME "TA_SC_DID" ("DIDs Supportable") which is used by all support check services.
- For Spec B/C, the according ESD defines the two TABLEs with SHORT-NAMEs "TS_SC_DID" ("DIDs Supportable") and "TA_SC_LID" ("LIDs Supportable").

Implementation requirements:

[Rule] The SHORT-NAME of the TABLE "DIDs Supportable" referenced by the support check DIAG-SERVICEs for two byte identifiers shall be "TS_SC_DID".

[Rule] The SHORT-NAME of the TABLE "LIDs Supportable" referenced by the support check DIAG-SERVICEs for one byte identifiers shall be "TS_SC_LID".

[Rule] Both the request and the positive response of the support check DIAG-SERVICEs shall reference TABLE "TA_SC_DID" or "TA_SC_LID" statically.

[Rule] The TABLEs "TA_SC_DID" and "TA_SC_LID" must not be overridden on variant layers.

2.3.5 DIAG-SERVICEs for UDS

2.3.5.1 General DIAG-SERVICE Requirements

In ODX, DIAG-SERVICEs may reference multiple FUNCT-CLASS objects. For Renault / Nissan use cases this might confuse the user. Therefore, DIAG-SERVICEs shall only reference a single FUNCT-CLASS.

Renault / Nissan does not want to deviate from the ISO definition of negative responses. As a result, one GLOBAL-NEGATIVE-RESPONSE is predefined which handles all services according to ISO. ODX authors shall not define a NEG-RESPONSE on DIAG-SERVICE level.

In ODX it is possible to modify communication parameters at DIAG-COMM level. In Renault / Nissan ODX, this feature shall not be used in order to reduce possible issues in tester and/or future diagnostic code generation.

In order to be fully compliant to ISO UDS, it shall be possible to execute every UDS DIAG-SERVICE both physically and functionally.

To express the ability of suppressing the positive response via setting the SPRMIB, authors shall not use the POS-RESPONSE-SUPPRESSABLE element, except for those services where its usage is explicitly allowed by this document.

Implementation requirements:

[Rule] DIAG-SERVICEs shall have exactly one reference to a FUNCT-CLASS.

[Rule/xsd] DIAG-SERVICEs shall not reference a NEG-RESPONSE.

[Rule] ODX authors must not override communication parameters at DIAG-SERVICE level.

[Rule] ODX authors must not define ADDRESSING-MODE different to "FUNCTIONAL-OR-PHYSICAL" at DIAG-SERVICE level.

[Rule] Generally, ODX authors must not use the POS-RESPONSE-SUPPRESSABLE element at DIAG-SERVICEs (except for where its usage is explicitly allowed by these AGLs).

2.3.5.2 Flat Services

The guiding philosophy at Renault / Nissan is to predefine and fix all service structures in the RFQ packages. ECU designers and suppliers should not have to bother themselves with defining service structures but should rather concentrate on defining diagnostic contents.

However, the RFQ packages do not predefine all services defined by the RFQ specs in all possible flavors (e. g. more strict session & security preconditions as allowed by the standard and defined on RFQ layer) and for all conceivable use cases (e. g. writing of a data record which should consist of four individually editable parameters in engineering but only of one single parameter for writing in plant because the data record is read in its entirety from a barcode there).

For this, in Renault / Nissan ODX models, ECU designers may define DIAG-SERVICEs describing such a service on variant layer, given that it complies to the structuring rules defined below. Such a DIAG-SERVICE is called a "flat service".

Thereby, the following rules apply:

[Rule] Flat services may be defined on variant level only.

[Rule] Flat services shall have exactly one request and one positive response.

[Rule] Both the request and the positive response shall thereby consist of at most two PARAMs: The first one a CODED-CONST specifying the SID / SIDPR and the second a VALUE referencing a STRUCTURE representing the data transferred by the request / positive response.

[Rule] The SHORT-NAME of the STRUCTURE referenced by the flat service's request shall be "ST_FS{SID}_{#NNNN}_RQ", where {SID} shall be replaced by the referencing service's SID and {#NNNN} can be optional used and replaced to represent a sub function and / or identifier if several flat services for the same SID need to be defined.

[Rule] The SHORT-NAME of the STRUCTURE referenced by the flat service's positive response shall be "ST_FS{SID}_{#NNNN}_PR", where {SID} shall be replaced by the referencing service's SID and {#NNNN} can be optional used and replaced to represent a sub function and / or identifier if several flat services for the same SID need to be defined.

[Rule] The STRUCTUREs referenced by flat services shall consist of flat lists of CODED-CONSTs or VALUEs. The VALUEs may thereby only reference DATA-OBJECT-PROPs which are of type

IDENTICAL, LINEAR or TEXTTABLE; they must not reference complex elements such as STRUCTURES or TABLES.

[Rule] ECU designers shall not define flat services with a service ID not covered by the diagnostic specification the described ECU is based on.

[Rule] Flat services shall have the same functional class as the corresponding service predefined by the RFQ package.

[Rule] In case a flat service is modelled which is related to an identifier such as a DID or RID etc. and the according data structure is also modelled via the regular TABLE (e. g. an alternate representation of a data record is modelled with a flat service while the original representation available via TABLE TA_S22 is still maintained) then the LONG-NAMEs of the ODX STRUCTUREs representing the alternate models of the very same data structures shall be the same (while the SHORT-NAMEs will of course differ according to the short naming rules defined in this document).

2.3.5.3 DIAG-SERVICE Overview

The following table shows the UDS based diagnostic services and their FUNCT-CLASSes.

FUNCT-CLASS	UDS Service
Session	$10 (DiagnosticSessionControl)
DIDs	$22 (ReadDataByIdentifier)
	$2E (WriteDataByIdentifier)
I/O Controls	$2F (InputOutpuControlByIdentifier)
Fault Memory	$14 (ClearAllDiagnosticInformation)
	$19 (ReadDTCInformation)
Routine Control	$31 (RoutineControl)
ECUReset	$11 (ECUReset)
TesterPresent	$3E (TesterPresent)
ControlDTCSetting	$85 (ControlDTCSetting)
SecurityAccess	$27 (SecurityAccess)
Upload Download	$34 (RequestDownload)
	$35 (RequestUpload)
	$36 (TransferData)
	$37 (RequestTransferExit)
CommunicationControl	$28 (CommunicationControl)
Memory	$23 (ReadMemoryByAddress)
	$3D (WriteMemoryByAddress)

Table 9 Overview of UDS DIAG-SERVICEs and their FUNCT-CLASSes

2.3.5.4 DiagnosticSessionControl ($10)

Service $10 is predefined once in the Renault Nissan Spec UDS ESD available from RFQ packages. It is marked final – ECU designers and suppliers shall not override or redefine this service in lower layers.

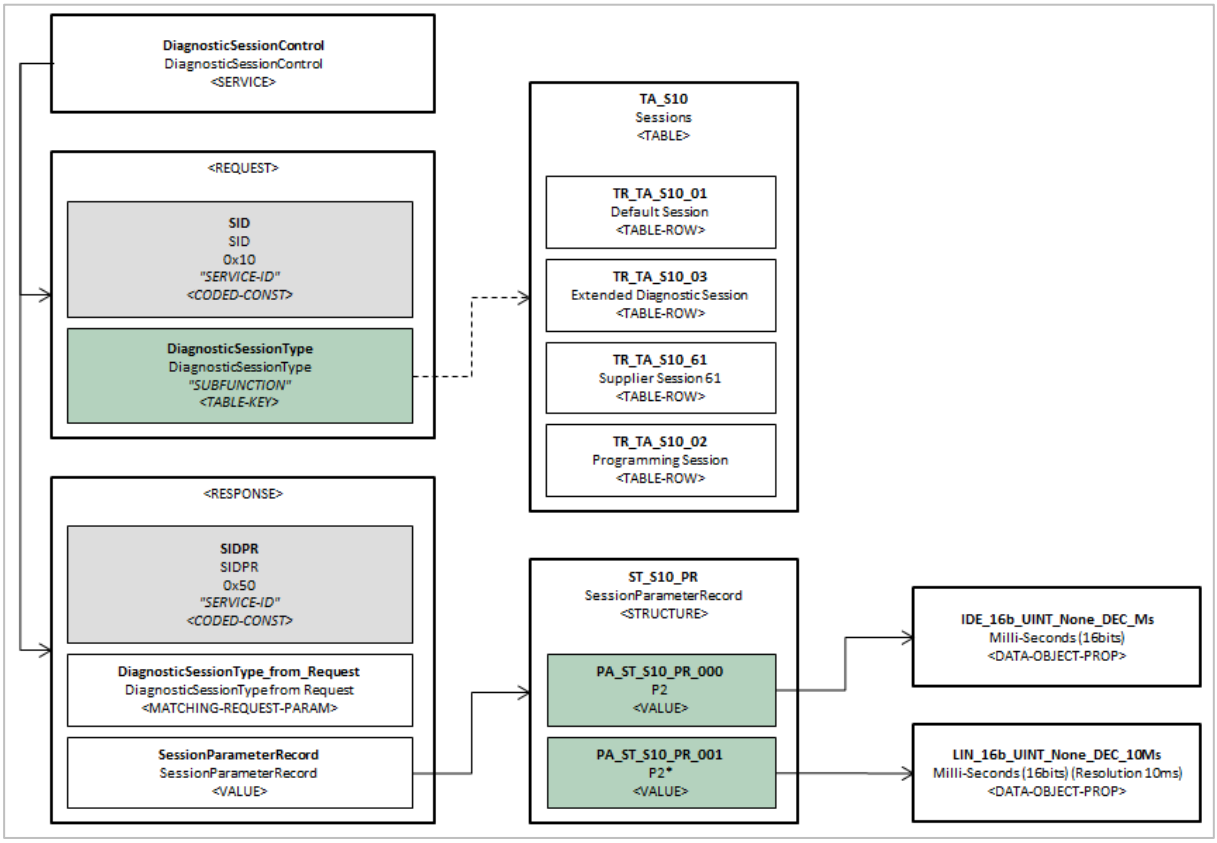


Figure 13 Structure of UDS DIAG-SERVICE DiagnosticSessionControl (0x10)

The service makes use of the predefined TABLE "TA_S10" ("Sessions") to specify the supported diagnostic session types. The following session types are thereby predefined:

- Default Session (0x01)
- Programming Session (0x02)
- Extended Diagnostic Session (0x03)
- Safety System Diagnostic Session (0x04)

ECU designers may override that TABLE to tailor these presets for ECUs which support a different set of session types. Please refer to section 2.3.4.1 for the exact tailoring rules.

Implementation requirements:

[Rule] The SHORT- and LONG-NAME of the DIAG-SERVICE shall be "DiagnosticSessionControl".

[Rule] The SEMANTIC attribute of the DIAG-SERVICE shall be set to "SESSION"

[Rule] The DIAG-SERVICE shall not use the DIAGNOSTIC-CLASS attribute.

[Rule] It shall be marked as IS-FINAL="true".

[Rule] The DIAG-SERVICE shall reference the FUNCT-CLASS "Session".

2.3.5.5 ReadDataByIdentifier ($22)

2.3.5.5.1 General Read Service

Service $22 is predefined in the Renault Nissan Spec UDS ESD available from RFQ packages. There are two definitions of $22, ReadDataByIdentifier and ReadDataByIdentifier_MultiId. The former is intended to be used when describing ECUs which only support reading out a single DID at a time, the latter when describing ECUs which support reading out multiple DIDs at once. Both are marked final – ECU designers and suppliers shall not override or redefine these services in lower layers.

Both services make use of the predefined TABLE "TA_S22" to specify both the data identifier(s) in request and response and the content of the associated data record(s). The TABLE in the Renault Nissan Spec UDS ESD encompasses the set of readable DIDs predefined by the RFQ package (of which Figure 14 merely shows a small subset).

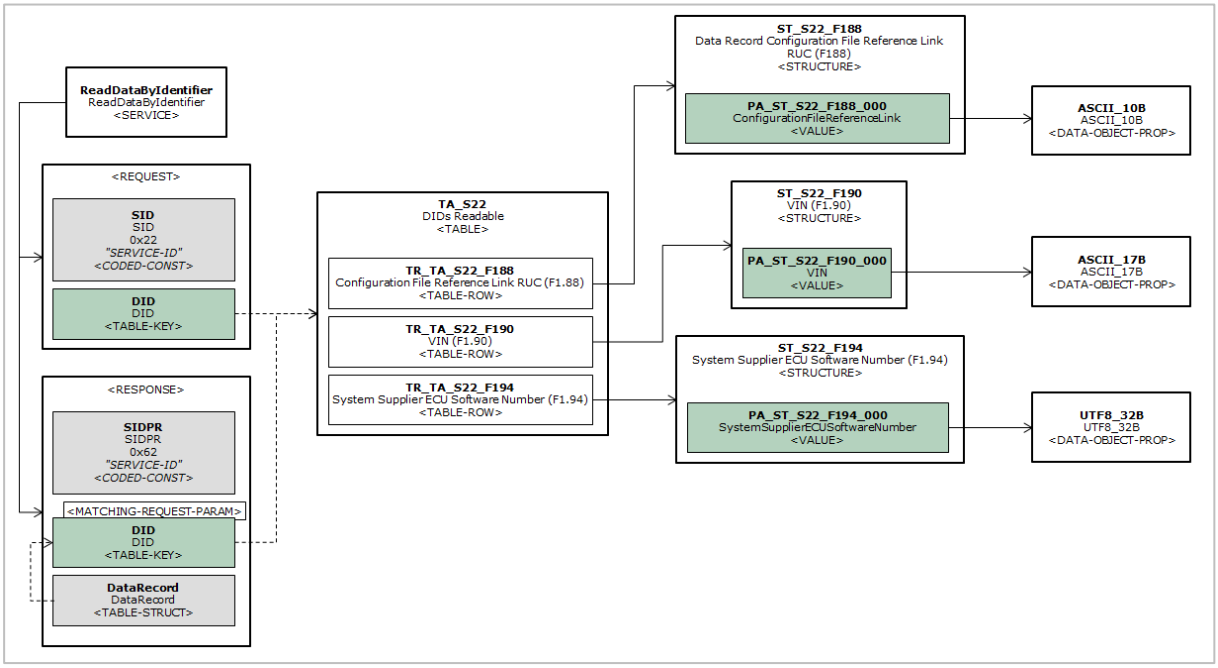


Figure 14 Structure of UDS DIAG-SERVICE ReadDataByIdentifier ($22)

ECU designers are allowed to override TABLE "TA_S22" to tailor this predefined set of readable DIDs for ECUs which support a different set of DIDs. Please refer to to section 0 for detailed tailoring rules (and a general overview of how DIDs can be defined and reused in lower layers).

Implementation requirements:

[Rule] The SHORT- and LONG_NAME of the two DIAG-SERVICEs representing UDS service $22 shall be "ReadDataByIdentifier" and "ReadDataByIdentifier_MultiId".

[Rule] The DIAG-SERVICEs shall be marked as IS-FINAL="true".

[Rule] The SEMANTIC attribute of the DIAG-SERVICEs shall be set to "STOREDDATAREAD".

[Rule] The DIAGNOSTIC-CLASS attribute shall not be used.

[Rule] The DIAG-SERVICEs shall reference the FUNCT-CLASS "DIDs".

[Rule] Both the request and the positive response of the DIAG-SERVICEs shall reference the TABLE "TA_S22" dynamically by SNREF.

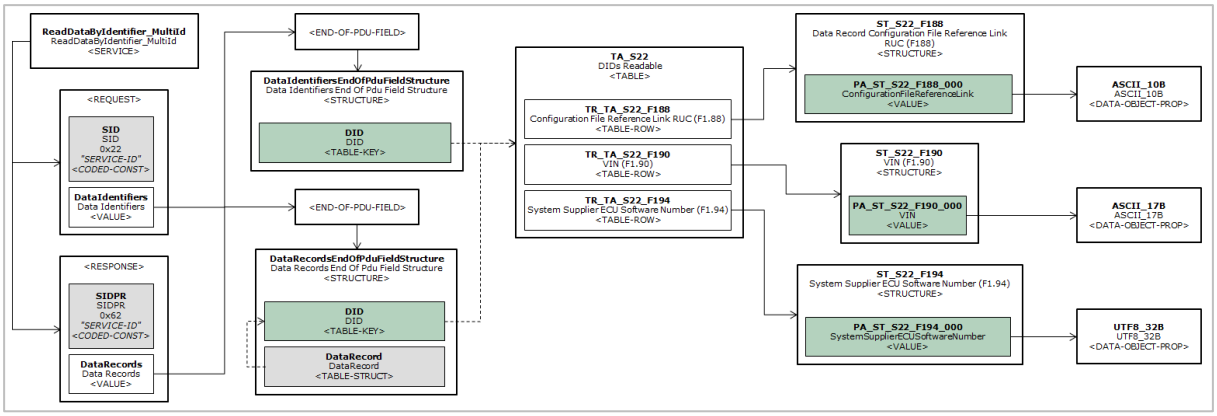


Figure 15 Structure of DIAG-SERVICE ReadDataByIdentifier_MultiId ($22)

2.3.5.5.2 Support Check

For checking of supported readable two-byte DIDs, one DIAG-SERVICE "SupportedReadableDIDs" (using service $22) is predefined in the Renault Nissan Spec UDS ESD available from RFQ packages.

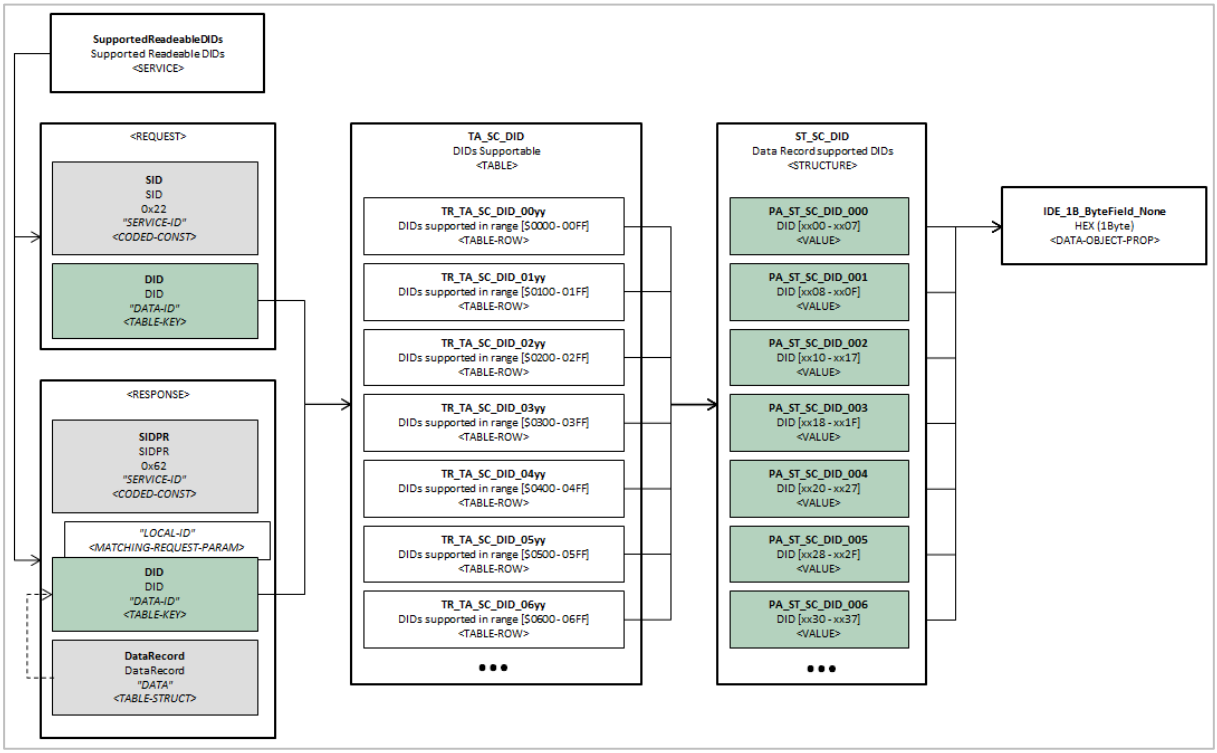


Figure 16 Structure of UDS DIAG-SERVICE SupportedReadableDIDs ($22)

Implementation requirements:

[Rule] The SHORT- and LONG-NAME of the DIAG-SERVICE shall be "SupportedReadableDIDs".

[Rule] The DIAG-SERVICEs shall be marked as IS-FINAL="true".

[Rule] The SEMANTIC attribute of the DIAG-SERVICEs shall be set to "STOREDDATA".

[Rule] The DIAGNOSTIC-CLASS attribute shall not be used.

[Rule] The DIAG-SERVICEs shall reference the FUNCT-CLASS "Support Check".

[Rule] Both the request and the positive response of the DIAG-SERVICEs shall reference the TABLE "TA_SC_DID" statically. The TABLE must not be overridden on variant layers.

[Rule] The service shall be marked as NOT-INHERITED on variant layers if the support check is not implemented in the described ECU.

2.3.5.5.3 Identification - Read System Supplier Identifier (F18A)

The Renault / Nissan ESDs distributed with RFQ packages predefine four UDS RDBI services for identification purposes. The DIAG-SERVICE "Read_systemSupplierIdentifier_F18A" is one of them.

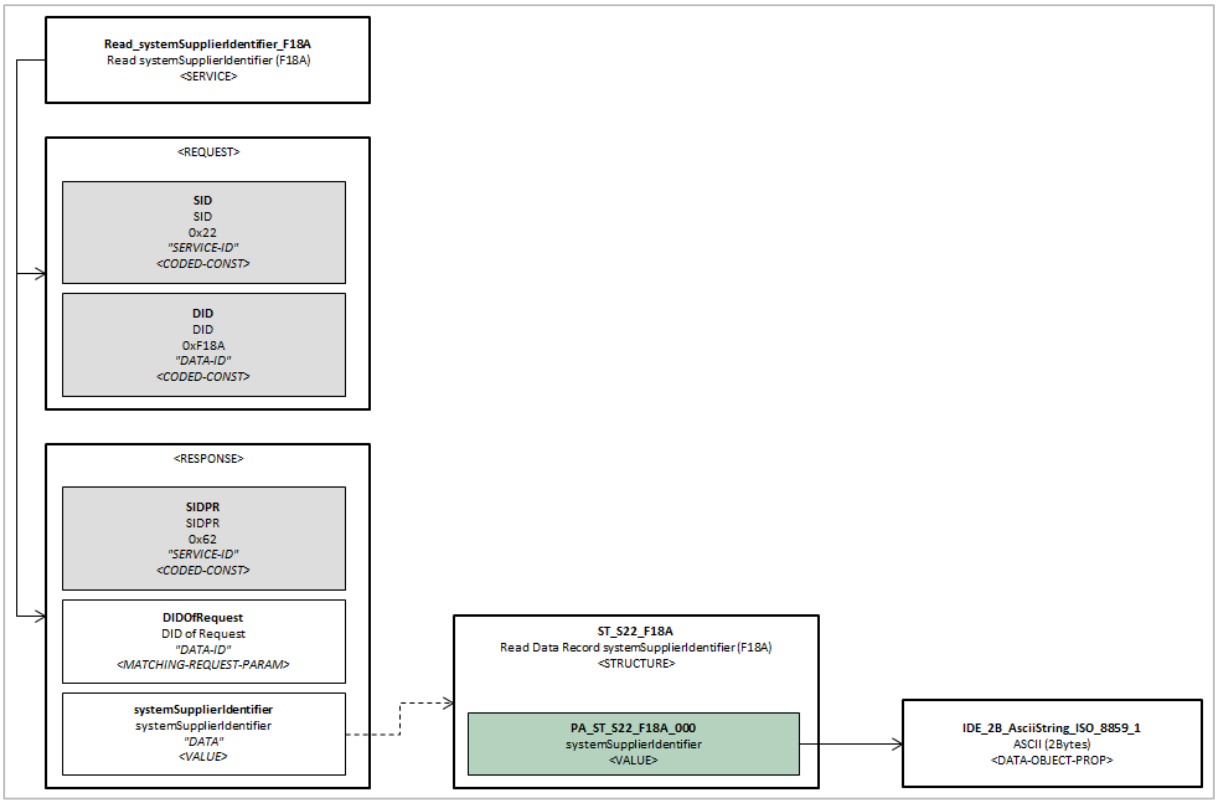


Figure 17 Structure of UDS Service Read System Supplier Identifier (F18A)

Implementation requirements:

[Rule] The SHORT-NAME of the UDS service $22 intended to read DID 0xF18A for identification purposes shall be "Read_systemSupplierIdentifier_F18A".

[Rule] The LONG-NAME of the UDS service $22 intended to read DID 0xF18A for identification purposes shall be "Read systemSupplierIdentifier (F18A)".

[Rule] The DIAG-SERVICE shall be marked as IS-FINAL="true".

[Rule] The SEMANTIC attribute of the DIAG-SERVICEs shall be set to "IDENTIFICATION".

[Rule] The DIAGNOSTIC-CLASS attribute shall be "VARIANTIDENTIFICATION".

[Rule] The DIAG-SERVICEs shall reference the FUNCT-CLASS "Identification".

[Rule] The positive response of the DIAG-SERVICEs shall reference STRUCTURE "ST_S22_F18A" by SNREF.

2.3.5.5.4 Identification - Read System Supplier ECU Software Number (F194)

The Renault / Nissan ESDs distributed with RFQ packages predefine four UDS RDBI services for identification purposes. The DIAG-SERVICE "Read_systemSupplierECUSoftwareNumber_F194" is one of them.

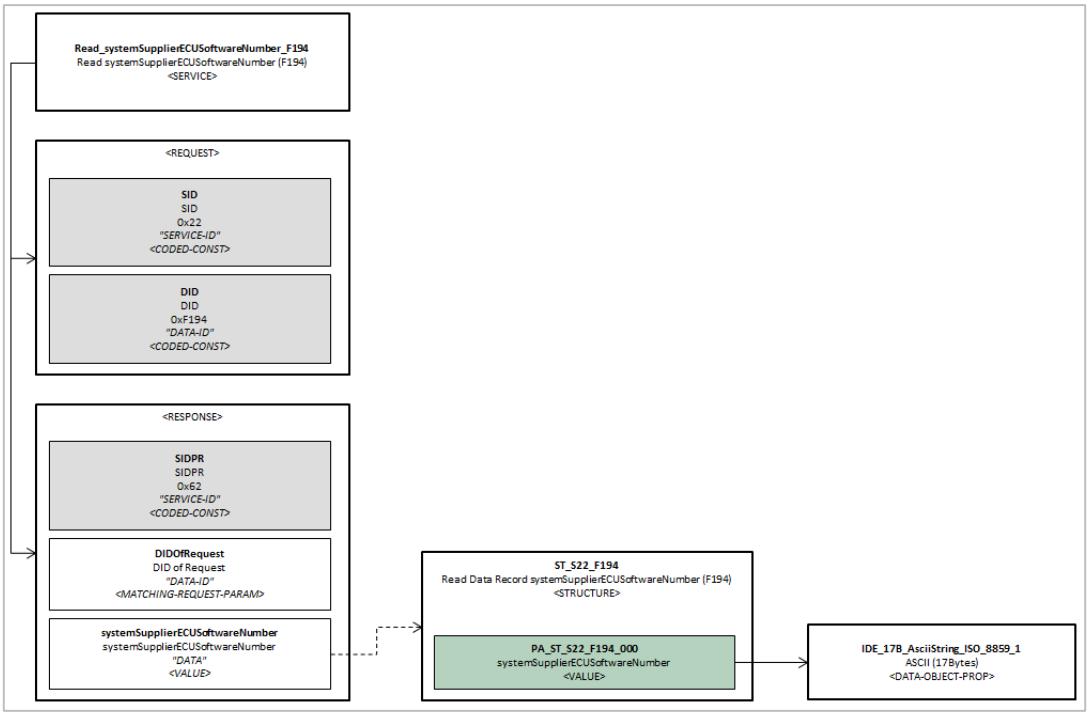


Figure 18 Structure of UDS Service Read System Supplier ECU Software Number (F194)

Implementation requirements:

[Rule] The SHORT-NAME of the UDS service $22 intended to read DID 0xF194 for identification purposes shall be "Read_systemSupplierECUSoftwareNumber_F194".

[Rule] The LONG-NAME of the UDS service $22 intended to read DID 0xF194 for identification purposes shall be "Read systemSupplierECUSoftwareNumber (F194)".

[Rule] The DIAG-SERVICE shall be marked as IS-FINAL="true".

[Rule] The SEMANTIC attribute of the DIAG-SERVICEs shall be set to "IDENTIFICATION".

[Rule] The DIAGNOSTIC-CLASS attribute shall be "VARIANTIDENTIFICATION".

[Rule] The DIAG-SERVICEs shall reference the FUNCT-CLASS "Identification".

[Rule] The positive response of the DIAG-SERVICEs shall reference STRUCTURE "ST_S22_F194" by SNREF.

2.3.5.5.5 Identification - Read System Supplier ECU Version Number (F195)

The Renault / Nissan ESDs distributed with RFQ packages predefine four UDS RDBI services for identification purposes. The DIAG-SERVICE "Read_systemSupplierECUVersionNumber_F195" is one of them.

Implementation requirements:

[Rule] The SHORT-NAME of the UDS service $22 intended to read DID 0xF195 for identification purposes shall be "Read_systemSupplierECUVersionNumber_F195".

[Rule] The LONG-NAME of the UDS service $22 intended to read DID 0xF195 for identification purposes shall be "Read systemSupplierECUVersionNumber (F195)".

[Rule] The DIAG-SERVICE shall be marked as IS-FINAL="true".

[Rule] The SEMANTIC attribute of the DIAG-SERVICEs shall be set to "IDENTIFICATION".

[Rule] The DIAGNOSTIC-CLASS attribute shall be "VARIANTIDENTIFICATION".

[Rule] The DIAG-SERVICEs shall reference the FUNCT-CLASS "Identification".

[Rule] The positive response of the DIAG-SERVICEs shall reference STRUCTURE "ST_S22_F195" by SNREF.

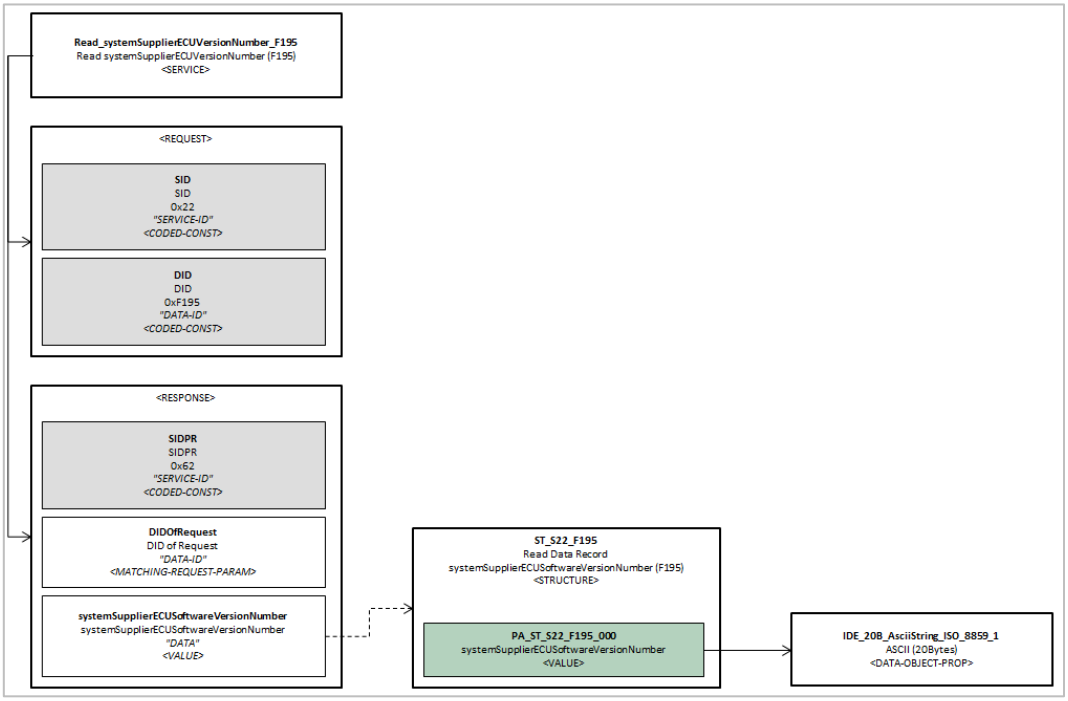


Figure 19 Structure of UDS Service Read System Supplier ECU Version Number (F195)

2.3.5.5.6 Identification - Read Diagnostic Version (F1A0)

The Renault / Nissan ESDs distributed with RFQ packages predefine four UDS RDBI services for identification purposes. The DIAG-SERVICE "Read_diagnosticVersion_F1A0" is one of them.

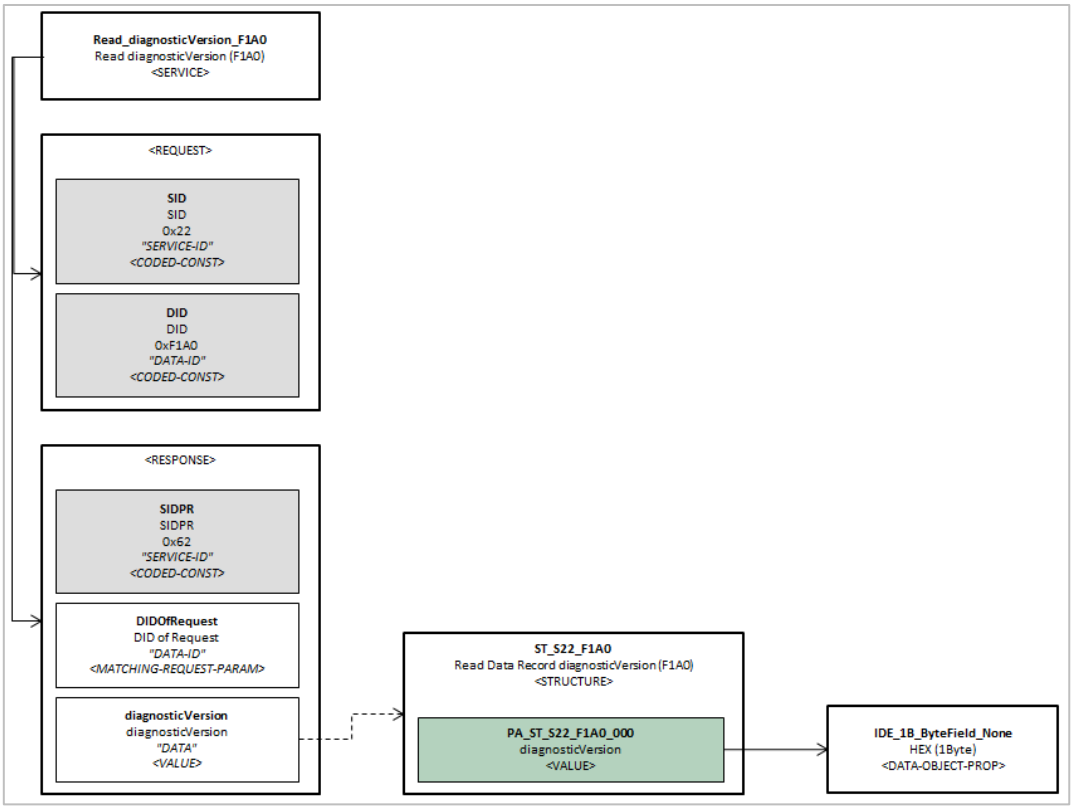


Figure 20 Structure of UDS Service Read Diagnostic Version (F1A0)

Implementation requirements:

[Rule] The SHORT-NAME of the UDS service $22 intended to read DID 0xF1A0 for identification purposes shall be “Read_diagnosticVersion_F1A0”.

[Rule] The LONG-NAME of the UDS service $22 intended to read DID 0xF1A0 for identification purposes shall be “Read diagnosticVersion (F1A0)”.

[Rule] The DIAG-SERVICE shall be marked as IS-FINAL=”true”.

[Rule] The SEMANTIC attribute of the DIAG-SERVICEs shall be set to “IDENTIFICATION”.

[Rule] The DIAGNOSTIC-CLASS attribute shall be “VARIANTIDENTIFICATION”.

[Rule] The DIAG-SERVICEs shall reference the FUNCT-CLASS “Identification”.

[Rule] The positive response of the DIAG-SERVICEs shall reference STRUCTURE “ST_S22_F1A0” by SNREF.

2.3.5.6 WriteDataByIdentifier ($2E)

2.3.5.6.1 General Write Service

Service $2E is predefined once in the Renault Nissan Spec UDS ESD available from RFQ packages. It is marked final – ECU designers and suppliers shall not override or redefine this service in lower layers.

The service makes use of the predefined TABLE “DIDs_Writeable” to specify both the data identifier in request and response and the content of the associated data record. The TABLE in the Renault Nissan Spec UDS ESD encompasses the set of writable DIDs predefined by the RFQ package (of which Figure 21 merely shows a small subset).

ECU designers are allowed to override TABLE “DIDs_Writeable” to tailor this predefined set of writable DIDs for ECUs which support a different set of DIDs. Please refer to to section 0 for detailed tailoring rules (and a general overview of how DIDs can be defined and reused in lower layers).

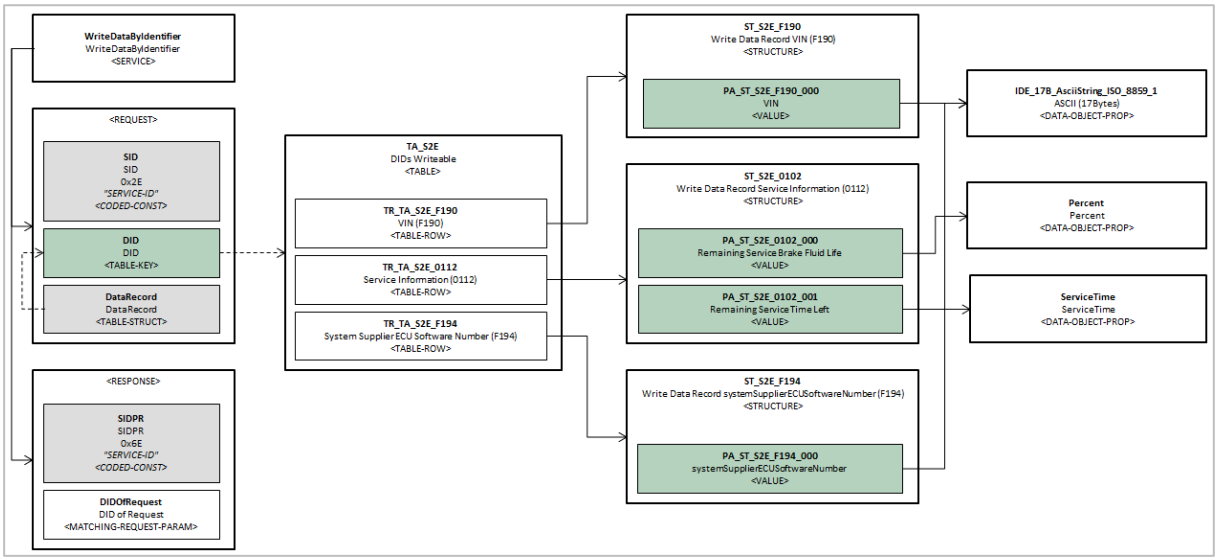


Figure 21 Structure of UDS DIAG-SERVICE WriteDataByIdentifier ($2E)

Implementation requirements:

[Rule] The SHORT- and LONG_NAME of the DIAG-SERVICE shall be “WriteDataByIdentifier”.

[Rule] The DIAG-SERVICE shall be marked as IS-FINAL=”true”.

[Rule] The SEMANTIC attribute of the DIAG-SERVICE shall be set to “STOREDDATAWRITE”.

[Rule] The DIAGNOSTIC-CLASS attribute shall not be used.

[Rule] The DIAG-SERVICE shall reference the FUNCT-CLASS "DIDs".

[Rule] The request of the DIAG-SERVICE shall reference TABLE "TA_S2E" dynamically by SNREF.

2.3.5.6.2 Support Check

For checking of supported writeable two-byte DIDs, one DIAG-SERVICE "SupportedWriteableDIDs" (using service $2E) is predefined in the Renault Nissan Spec UDS ESD available from RFQ packages.

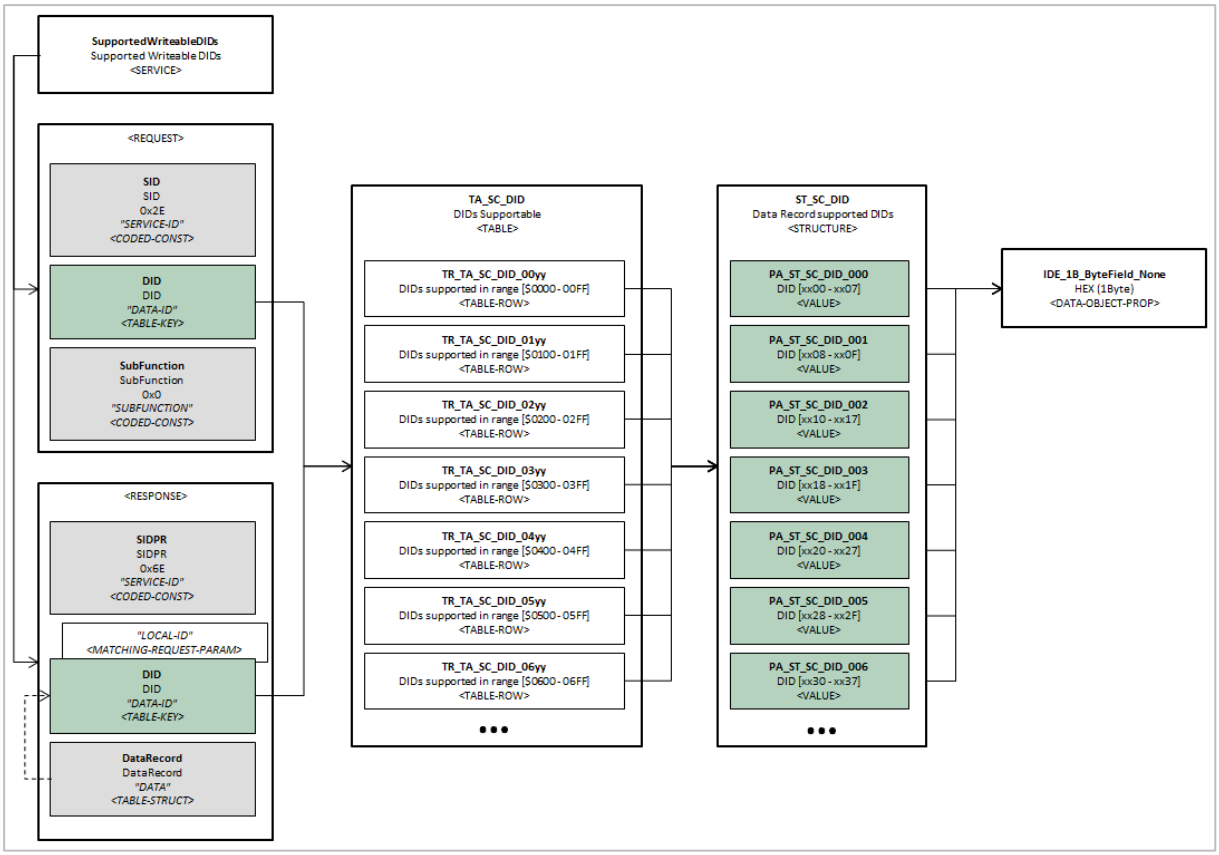


Figure 22 Structure of UDS Service SupportedWriteableDIDs ($2E)

Implementation requirements:

[Rule] The SHORT- and LONG-NAME of the DIAG-SERVICE shall be "SupportedWriteableDIDs".

[Rule] The DIAG-SERVICEs shall be marked as IS-FINAL="true".

[Rule] The SEMANTIC attribute of the DIAG-SERVICEs shall be set to "STOREDDATA".

[Rule] The DIAGNOSTIC-CLASS attribute shall not be used.

[Rule] The DIAG-SERVICEs shall reference the FUNCT-CLASS "Support Check".

[Rule] Both the request and the positive response of the DIAG-SERVICEs shall reference the TABLE "TA_SC_DID" statically. The TABLE must not be overridden on variant layers.

[Rule] The service shall be marked as NOT-INHERITED on variant layers if the support check is not implemented in the described ECU.

2.3.5.7 InputOutputControlByIdentifier ($2F)

As I/O Controls might support different InputOutputControlParameters at Renault and Nissan, it is required to define each InputOutputControlParameter with its own DIAG-SERVICE.

2.3.5.7.1 IOControl - shortTermAdjustment

Service $2F using the InputOutputControlParameter 0x03 ("shortTermAdjustment") is predefined once in the Renault Nissan Spec UDS ESD available from RFQ packages. It is marked final – ECU designers and suppliers shall not override or redefine this service in lower layers.

The service makes use of the predefined TABLEs "TA_S2F_03" and "TA_S2F_PR" to specify both the data identifier's data record in the response and the content of the data record together with the "Control Enable Mask Record" in the request. The TABLEs in the Renault Nissan Spec UDS ESD encompasses the set of adjustable I/O DIDs predefined by the RFQ package.

ECU designers may override both TABLEs to tailor this predefined set of I/O Controls for ECUs which support a different set of DIDs. Please refer to section 2.3.4.4 for detailed tailoring rules (and a general overview of how I/O Controls can be defined).

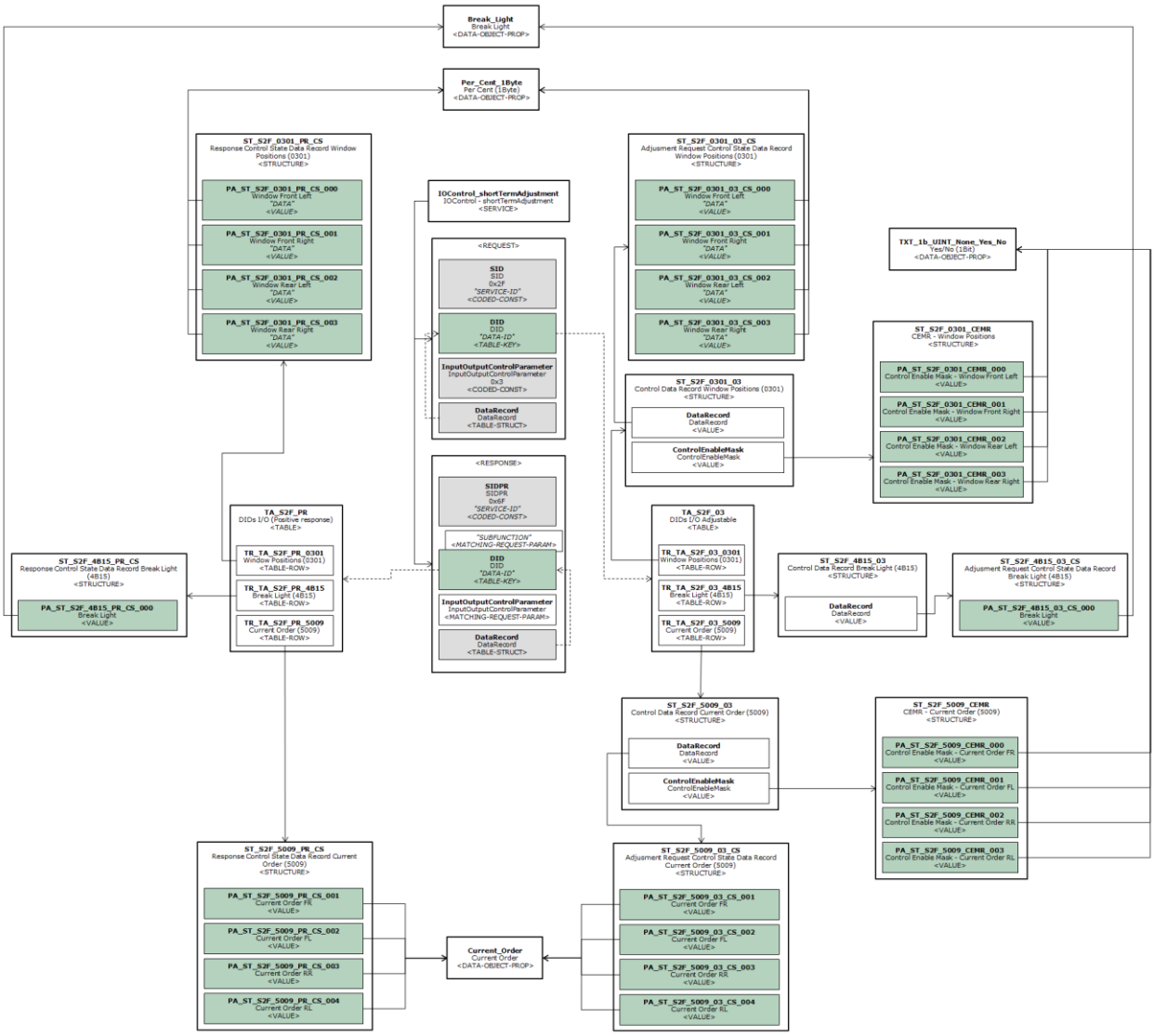


Figure 23 Structure of UDS DIAG-SERVICE IOControl_shortTermAdjustment ($2Fxxxx03)

Implementation requirements:

[Rule] The SHORT-NAME of the DIAG-SERVICE shall be "IOControl_shortTermAdjustment".

[Rule] The LONG-NAME of the DIAG-SERVICE shall be "IOControl - shortTermAdjustment".

[Rule] The DIAG-SERVICE shall be marked as IS-FINAL="true".

[Rule] The SEMANTIC attribute of the DIAG-SERVICE shall be set to “CONTROL”.

[Rule] The DIAGNOSTIC-CLASS attribute shall not be used.

[Rule] The DIAG-SERVICE shall reference the FUNCT-CLASS “I_O_Controls”.

[Rule] The request of the DIAG-SERVICE shall reference the TABLE “TA_S2F_03” dynamically by SNREF.

[Rule] The response of the DIAG-SERVICE shall reference the TABLE “TA_S2F_PR” dynamically by SNREF.

2.3.5.7.2 IOControl - returnControlToECU

Service $2F using the InputOutputControlParameter 0x00 (“returnControlToECU”) is predefined once in the Renault Nissan Spec UDS ESD available from RFQ packages. It is marked final – ECU designers and suppliers shall not override or redefine this service in lower layers.

The service makes use of the predefined TABLEs “TA_S2F_00” and “TA_S2F_PR” to specify both the data identifier’s data record in the response and the content of the “Control Enable Mask Record” in the request. The TABLE in the Renault Nissan Spec UDS ESD encompasses the set of returnable I/O DIDs predefined by the RFQ package.

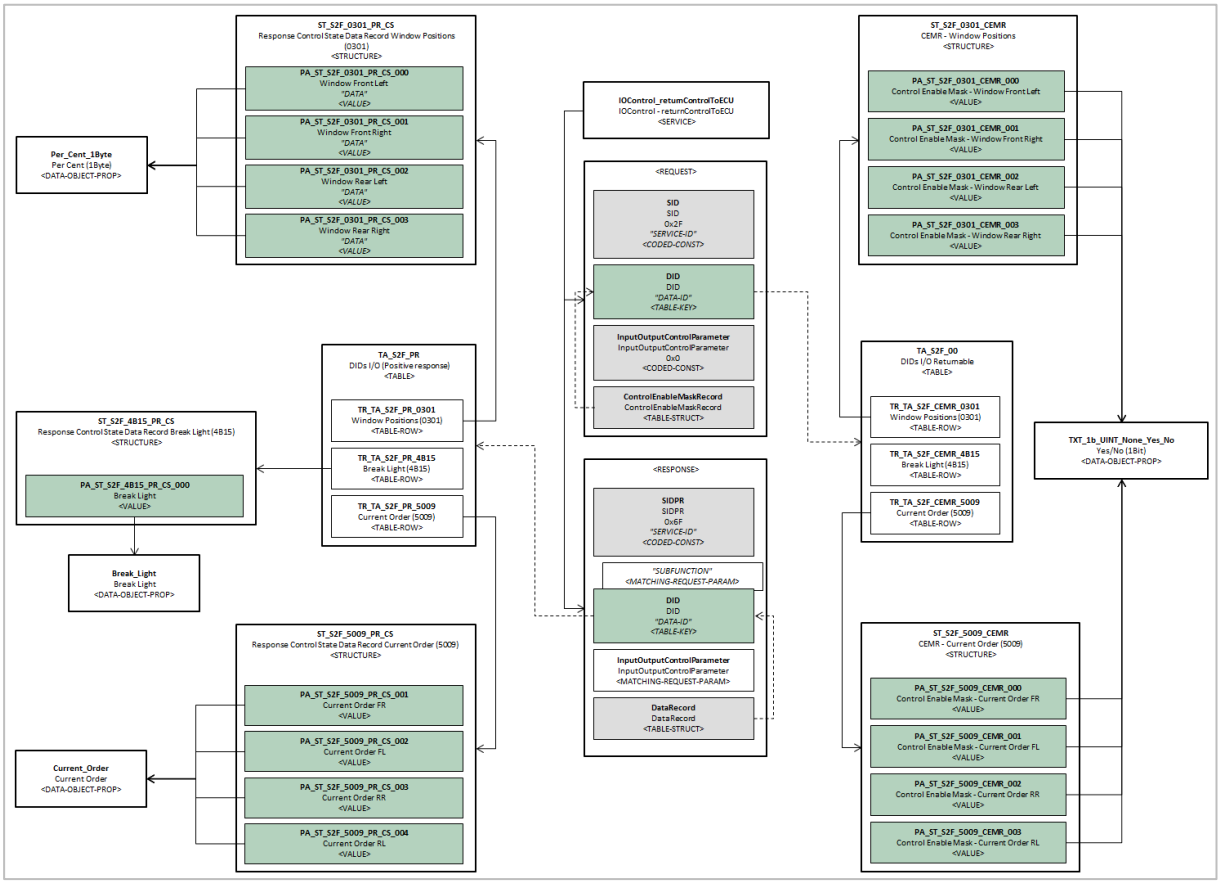


Figure 24 Structure of UDS DIAG-SERVICE IOControl_returnControlToECU ($2Fxxxx00)

ECU designers may override both TABLEs to tailor this predefined set of I/O Controls for ECUs which support a different set of controllable DIDs. Please refer to section 2.3.4.4 for detailed tailoring rules (and a general overview of how I/O Controls can be defined).

Implementation requirements:

[Rule] The SHORT-NAME of the DIAG-SERVICE shall be “IOControl_returnControlToECU”.

[Rule] The LONG-NAME of the DIAG-SERVICE shall be “IOControl - returnControlToECU”.

[Rule] The DIAG-SERVICE shall be marked as IS-FINAL="true".

[Rule] The SEMANTIC attribute of the DIAG-SERVICE shall be set to "CONTROL".

[Rule] The DIAGNOSTIC-CLASS attribute shall not be used.

[Rule] The DIAG-SERVICE shall reference the FUNCT-CLASS "I_O_Controls".

[Rule] The request of the DIAG-SERVICE shall reference the TABLE "TA_S2F_00" dynamically by SNREF.

[Rule] The response of the DIAG-SERVICE shall reference the TABLE "TA_S2F_PR" dynamically by SNREF.

2.3.5.7.3 IOControl - resetToDefault

Service $2F using the InputOutputControlParameter 0x01 ("resetToDefault") is predefined once in the Renault Nissan Spec UDS ESD available from RFQ packages. It is marked final – ECU designers and suppliers shall not override or redefine this service in lower layers.

The service makes use of the predefined TABLEs "TA_S2F_00" and "TA_S2F_PR" to specify both the data identifier's data record in the response and the content of the "Control Enable Mask Record" in the request. The TABLE in the Renault Nissan Spec UDS ESD encompasses the set of resettable I/O DIDs predefined by the RFQ package.

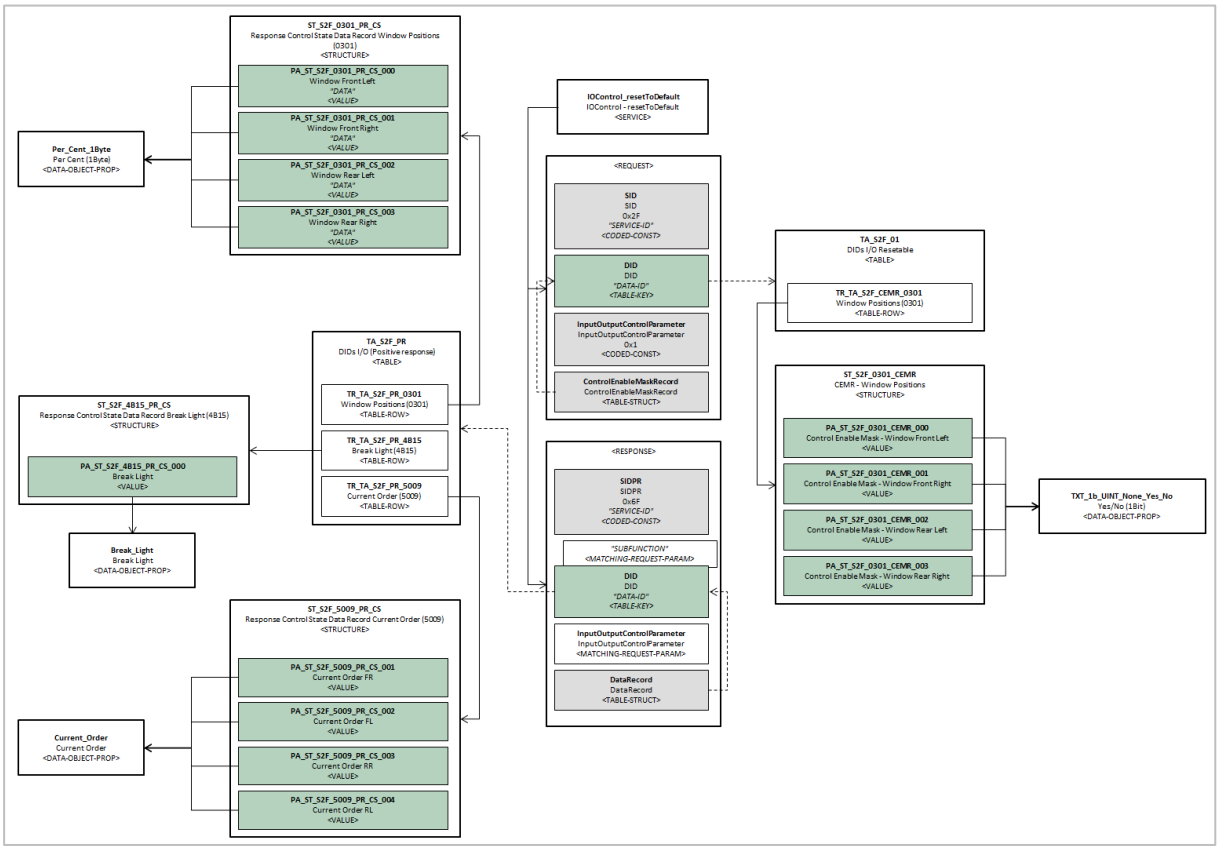


Figure 25 Structure of UDS DIAG-SERVICE IOControl_resetToDefault ($2Fxxxx01)

ECU designers are allowed to override both TABLEs to tailor this predefined set of I/O Controls for ECUs which support a different set of controllable DIDs. Please refer to section 2.3.4.4 for detailed tailoring rules (and a general overview of how I/O Controls can be defined).

Implementation requirements:

[Rule] The SHORT-NAME of the DIAG-SERVICE shall be "IOControl_resetToDefault".

[Rule] The LONG-NAME of the DIAG-SERVICE shall be “IOControl - resetToDefault”.

[Rule] The DIAG-SERVICE shall be marked as IS-FINAL=”true”.

[Rule] The SEMANTIC attribute of the DIAG-SERVICE shall be set to “CONTROL”.

[Rule] The DIAGNOSTIC-CLASS attribute shall not be used.

[Rule] The DIAG-SERVICE shall reference the FUNCT-CLASS “I_O_Controls”.

[Rule] The request of the DIAG-SERVICE shall reference the TABLE “TA_S2F_01” dynamically by SNREF.

[Rule] The response of the DIAG-SERVICE shall reference the TABLE “TA_S2F_PR” dynamically by SNREF.

2.3.5.7.4 IOControl - freezeCurrentState

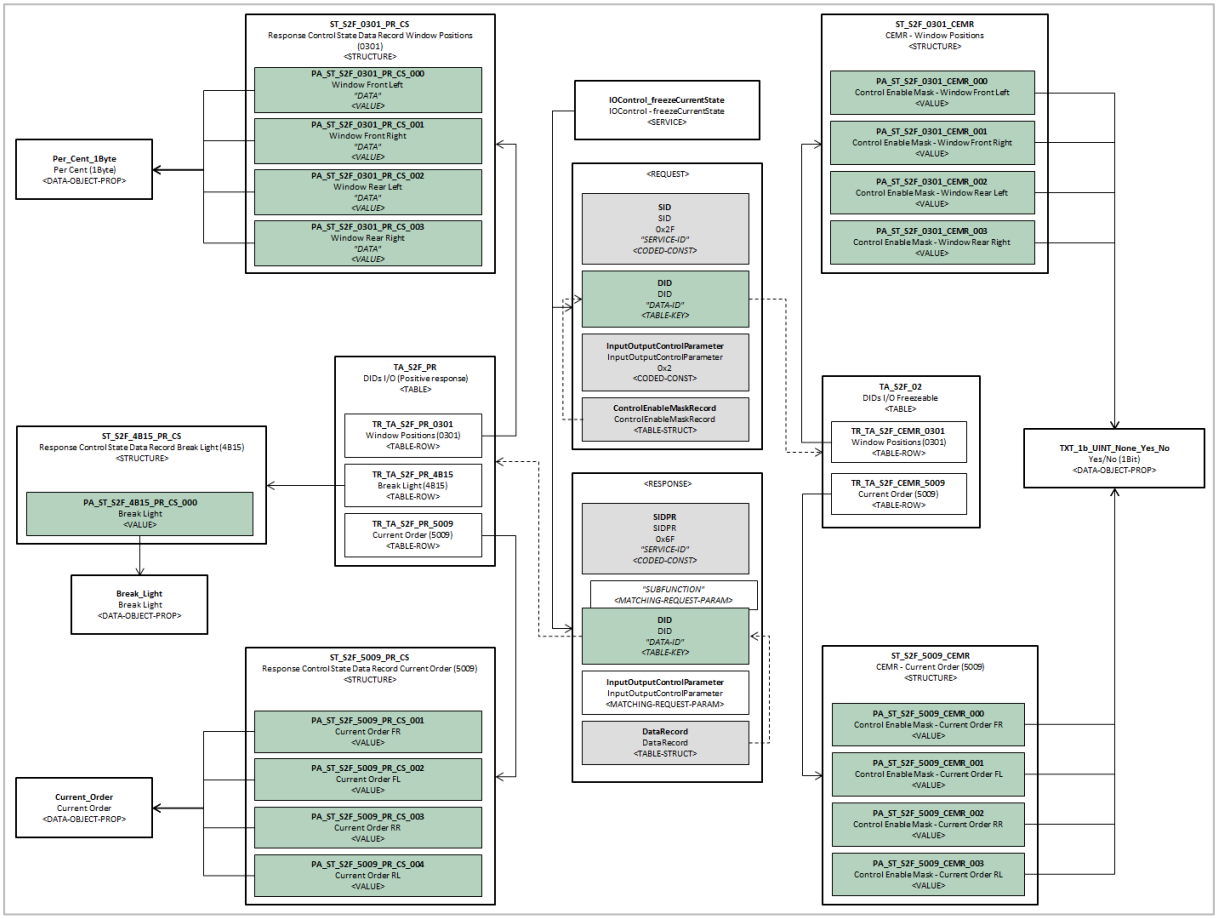


Figure 26 Structure of UDS DIAG-SERVICE IOControl_freezeCurrentState ($2Fxxxx02)

Service $2F using the InputOutputControlParameter 0x02 (“freezeCurrentState”) is predefined once in the Renault Nissan Spec UDS ESD available from RFQ packages. It is marked final – ECU designers and suppliers shall not override or redefine this service in lower layers.

The service makes use of the predefined TABLEs “TA_S2F_00” and “TA_S2F_PR” to specify both the data identifier’s data record the response and the content of the “Control Enable Mask Record” in the request. The TABLEs in the Renault Nissan Spec UDS ESD encompasses the set of freezable I/O DIDs predefined by the RFQ package.

ECU designers may override both TABLEs to tailor this predefined set of I/O Controls for ECUs which support a different set of controllable DIDs. Please refer to section 2.3.4.4 for detailed tailoring rules (and a general overview of how I/O Controls can be defined).

Implementation requirements:

[Rule] The SHORT-NAME of the DIAG-SERVICE shall be “IOControl_freezeCurrentState”.

[Rule] The LONG-NAME of the DIAG-SERVICE shall be “IOControl - freezeCurrentState”.

[Rule] The DIAG-SERVICE shall be marked as IS-FINAL=”true”.

[Rule] The SEMANTIC attribute of the DIAG-SERVICE shall be set to “CONTROL”.

[Rule] The DIAGNOSTIC-CLASS attribute shall not be used.

[Rule] The DIAG-SERVICE shall reference the FUNCT-CLASS “I_O_Controls”.

[Rule] The request of the DIAG-SERVICE shall reference the TABLE “TA_S2F_02” dynamically by SNREF.

[Rule] The response of the DIAG-SERVICE shall reference the TABLE “TA_S2F_PR” dynamically by SNREF.

2.3.5.7.5 Support Check

For checking of supported controllable two-byte DIDs, one DIAG-SERVICE “SupportedControllable-DIDs” (using service $2F) is predefined in the Renault Nissan Spec UDS ESD available from RFQ packages.

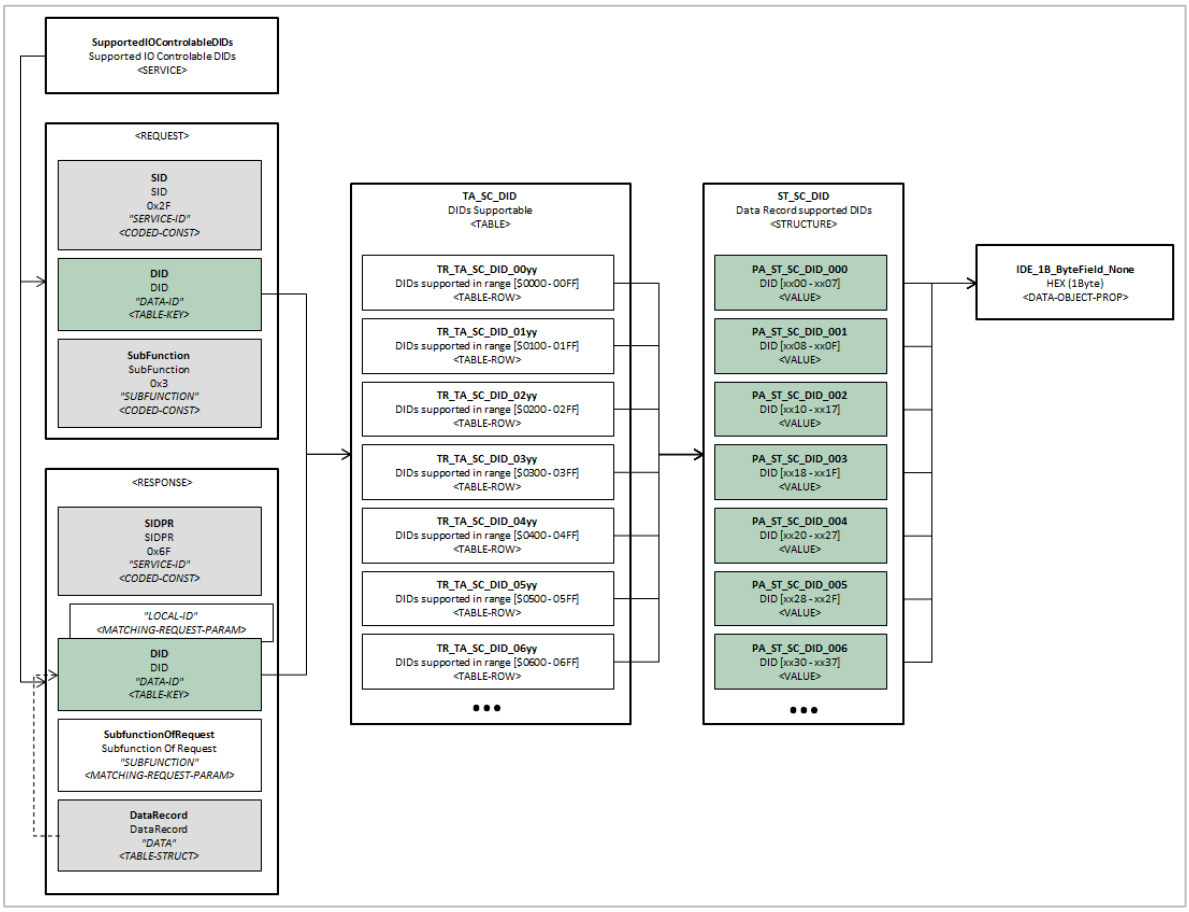


Figure 27 Structure of UDS Service SupportedControllableDIDs ($2F)

Implementation requirements:

[Rule] SHORT- and LONG-NAME of the DIAG-SERVICE shall be “SupportedControllableDIDs”.

[Rule] The DIAG-SERVICEs shall be marked as IS-FINAL=”true”.

[Rule] The SEMANTIC attribute of the DIAG-SERVICEs shall be set to “STOREDDATA”.

[Rule] The DIAGNOSTIC-CLASS attribute shall not be used.

[Rule] The DIAG-SERVICEs shall reference the FUNCT-CLASS "Support Check".

[Rule] Both the request and the positive response of the DIAG-SERVICEs shall reference the TABLE "TA_SC_DID" statically. The TABLE must not be overridden on variant layers.

[Rule] The service shall be marked as NOT-INHERITED on variant layers if the support check is not implemented in the described ECU.

2.3.5.8 ReadDtcInformation ($19)

The different sub functions of service $19 reference different TABLEs, TEXTTABLEs and other ODX elements for the definition of the associated fault data (such as DTCs, snapshots etc.) and the different interrelations between that data. Please cf. also section 2.3.4.6 for an overview focusing on those referenced data structures.

2.3.5.8.1 ReportDtcByStatusMask (subfunction 0x02)

Service $1902 ("ReadDtcInformation – ReportDtcByStatusMask") is predefined once in the Renault Nissan Spec UDS ESD available from RFQ packages. It is marked final – ECU designers and suppliers shall not override or redefine this service in lower layers.

As shown in Figure 28, the definition of service $1902 makes use of the following ODX elements defining associated fault memory data:

- The STRUCTURE with SHORT-NAME "DtcStatusMask" which defines the individual DTC status bits used both in the request and the positive response.
- The STRUCTURE with SHORT-NAME "DtcStatusAvailabilityMask" which defines the DTC status bits returned in the positive response reflecting which status bits are supported by the ECU in the first place.
- The DTC-DOP with SHORT-NAME "DtcRecord" specifying the possible DTCs stored by in the ECU's fault memory.

All of these elements are predefined on RFQ level in the ESD "Renault Nissan Spec UDS". The latter two however are referenced via SHORT-NAME and are intended to be overridden on BASE-VARIANT level.

For the STRUCTURE "DtcStatusAvailabilityMask" this may be done optionally. The DTC-DOP with SHORT-NAME "DtcRecord" however must be overridden at BASE-VARIANT level. On RFQ level, it only contains a placeholder for the actual DTC data. Please refer to section 2.3.4.6 for a description of the detailed rules applicable for overriding this DTC-DOP and for defining DTCs of an ECU.

Implementation requirements:

[Rule] The SHORT-NAME of the DIAG-SERVICE shall be "ReadDtcInformation_ReportDtc-ByStatusMask".

[Rule] The LONG-NAME of the DIAG-SERVICE shall be "ReadDtcInformation-ReportDtcByStatus-Mask".

[Rule] The DIAG-SERVICE shall be marked as IS-FINAL="true".

[Rule] The SEMANTIC attribute of the DIAG-SERVICE shall be set to "FAULTREAD".

[Rule] The DIAGNOSTIC-CLASS attribute shall not be used.

[Rule] The DIAG-SERVICE shall reference the FUNCT-CLASS "Fault_Memory".

[Rule] The positive response of the DIAG-SERVICE shall reference the STRUCTURE with SHORT-NAME "DtcStatusAvailabilityMask" dynamically by SNREF.

[Rule] The positive response of the DIAG-SERVICE shall reference the DTC-DOP "DtcRecord" dynamically by SNREF.

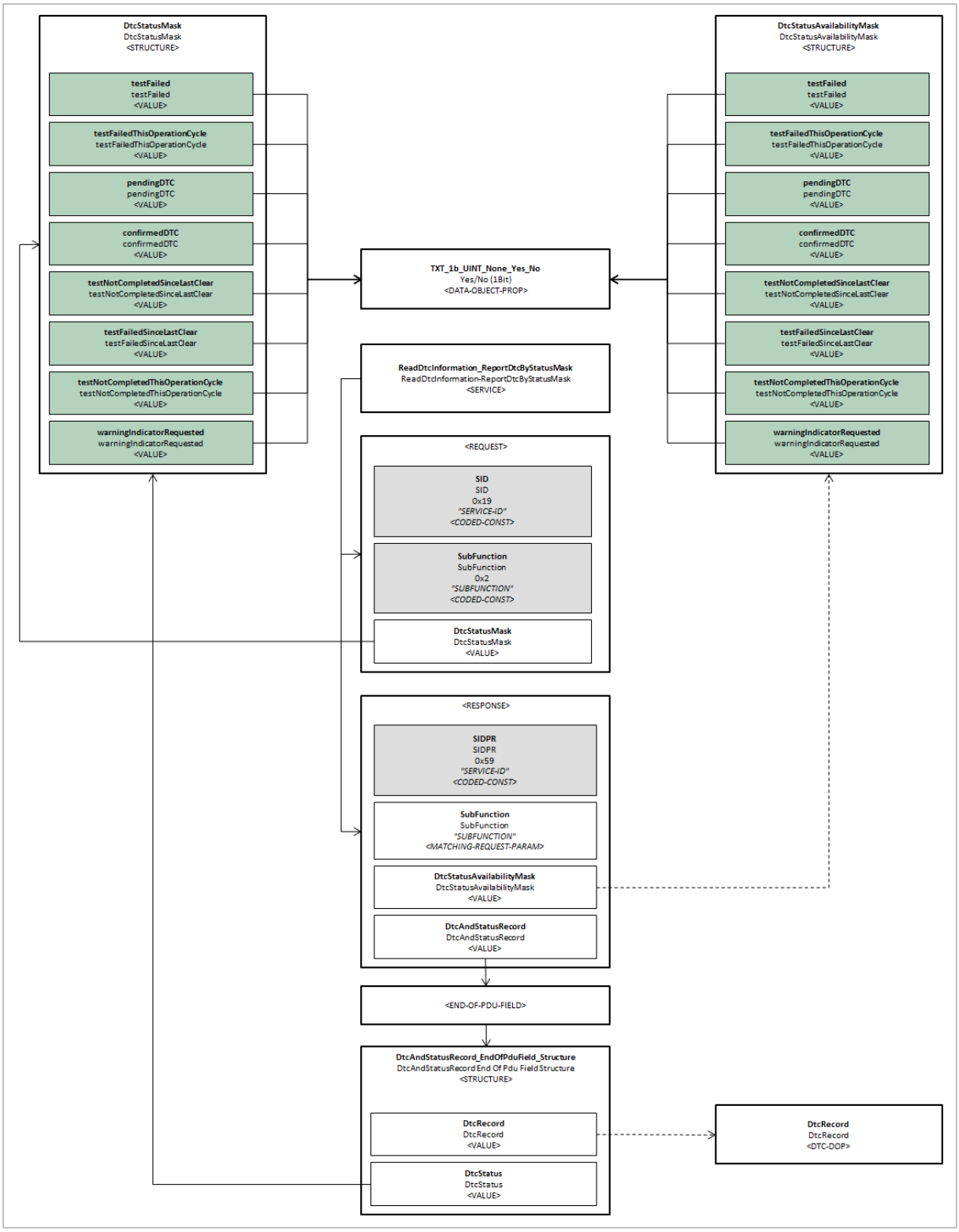


Figure 28 UDS ReadDtcInformation – ReportDtcByStatusMask ($1902)

2.3.5.8.2 ReportDtcSnapshotRecordByDtcNumber (subfunction 0x04)

Service $1904 ("ReadDtcInformation – ReportDtcSnapshotRecordByDtcNumber") is predefined once in the Renault Nissan Spec UDS ESD available from RFQ packages. It is marked final – ECU designers and suppliers shall not override or redefine this service in lower layers.

As shown in Figure 29, the definition of service $1904 makes use of the following ODX elements defining associated fault memory data:

- The STRUCTURE with SHORT-NAME "DtcStatusMask" which defines the individual DTC status bits used in the positive response.
- The DTC-DOP with SHORT-NAME "DtcRecord" specifying the possible DTCs stored by in the ECU's fault memory which can be queried in the request of $1904.
- The DTC-DOP with SHORT-NAME "Devices" specifying the different devices implemented in the ECU for which DTCs might be stored (and their possible base DTC values).
- The TEXTTABLE with SHORT-NAME "FailureTypes" specifying the possible fault types for which DTCs might be stored (and their possible failure type bytes).
- The TEXTTABLE with SHORT-NAME "DtcSnapshotRecordNumbersRQ" defining the possible snapshot record number values which may be specified in the request of service $1904.
- The TEXTTABLE with SHORT-NAME "DtcSnapshotRecordNumbersPR" defining the possible snapshot record number values returned in the positive response of service $1904.
- The ENV-DATA-DESC with SHORT-NAME "SnapshotRecordContents" defining the different snapshot records which might be stored by the described ECU.

All of these elements are predefined at RFQ level in the ESD "Renault Nissan Spec UDS". Only the first one is not referenced via SHORT-NAME. All other elements must be overridden on variant level to define the devices, failure types, DTCs and snapshot records and snapshot record numbers actually supported by the described ECUs in the context of service $1904 (for on RFQ level, these elements contain placeholder data only).

Exempt from this is the TEXTTABLE with SHORT-NAME "FailureTypes": It may be overridden on variant level, but it is not required to do so, for this TEXTTABLE contains valid FailureTypes according to ISO15031-6 on the RFQ level already. Please refer to section 2.3.4.6 for a description of the detailed rules applicable for overriding these elements and for defining the associated fault memory data.

Implementation requirements:

[Rule] The SHORT-NAME of the DIAG-SERVICE shall be "ReadDtcInformation_ReportDtcSnapshotRecordByDtcNumber".

[Rule] The LONG-NAME of the DIAG-SERVICE shall be "ReadDtcInformation-ReportDtcSnapshotRecordByDtcNumber".

[Rule] The DIAG-SERVICE shall be marked as IS-FINAL="true".

[Rule] The SEMANTIC attribute of the DIAG-SERVICE shall be set to "FAULTREAD".

[Rule] The DIAGNOSTIC-CLASS attribute shall not be used.

[Rule] The DIAG-SERVICE shall reference the FUNCT-CLASS "Fault_Memory".

[Rule] The request of the DIAG-SERVICE shall reference the DTC-DOP with SHORT-NAME "DtcRecord" dynamically by SNREF.

[Rule] The request of the DIAG-SERVICE shall reference the TEXTTABLE with SHORT-NAME "DtcSnapshotRecordNumbersRQ" dynamically by SNREF.

[Rule] The positive response of the DIAG-SERVICE shall reference the TEXTTABLE with SHORT-NAME "DtcSnapshotRecordNumbersPR" dynamically by SNREF.

[Rule] The positive response of the DIAG-SERVICE shall reference the DTC-DOP with SHORT-NAME "Devices" dynamically by SNREF.

[Rule] The positive response of the DIAG-SERVICE shall reference the TEXTTABLE with SHORT-NAME "FailureTypes" dynamically by SNREF.

[Rule] The positive response of the DIAG-SERVICE shall reference the ENV-DATA-DESC with SHORT-NAME "SnapshotRecordContents" dynamically by SNREF.

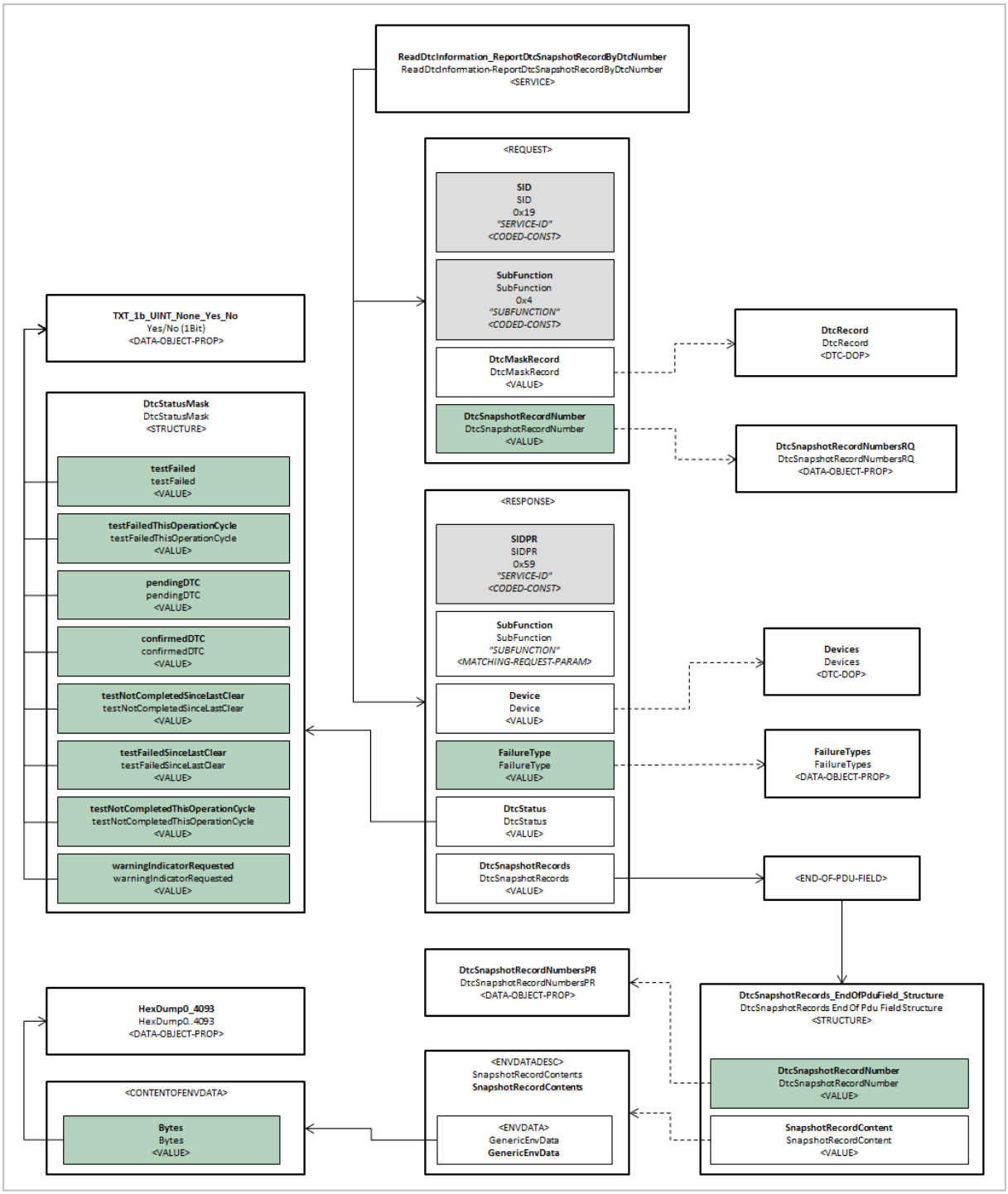


Figure 29 UDS ReadDtcInformation – ReportDtcSnapshotRecordByDtcNumber ($1904)

2.3.5.8.3 ReportDtcExtDataRecordByDtcNumber (subfunction 0x06)

Service $1906 ("ReadDtcInformation – ReportDtcExtDataRecordByDtcNumber") is predefined once in the Renault Nissan Spec UDS ESD available from RFQ packages. It is marked final – ECU designers and suppliers shall not override or redefine this service in lower layers.

As shown in Figure 30, the definition of service $1906 makes use of the following ODX elements defining associated fault memory data:

- The STRUCTURE with SHORT-NAME "DtcStatusMask" which defines the individual DTC status bits used in the positive response.

- The DTC-DOP with SHORT-NAME “DtcRecord” specifying the possible DTCs stored by in the ECU’s fault memory which can be queried in the request of $1906.
- The DTC-DOP with SHORT-NAME “Devices” specifying the different devices implemented in the ECU for which DTCs might be stored (and their possible base DTC values).
- The TEXTTABLE with SHORT-NAME “FailureTypes” specifying the possible fault types for which DTCs might be stored (and their possible failure type bytes).
- The DOP with SHORT-NAME “ExtendedDataRecordsTableKeys” defining the possible extended data record number values which may be specified in the request of service $1906.
- The TABLE with SHORT-NAME “ExtendedDataRecords” defining the possible extended data record number values returned in the positive response of service $1906.

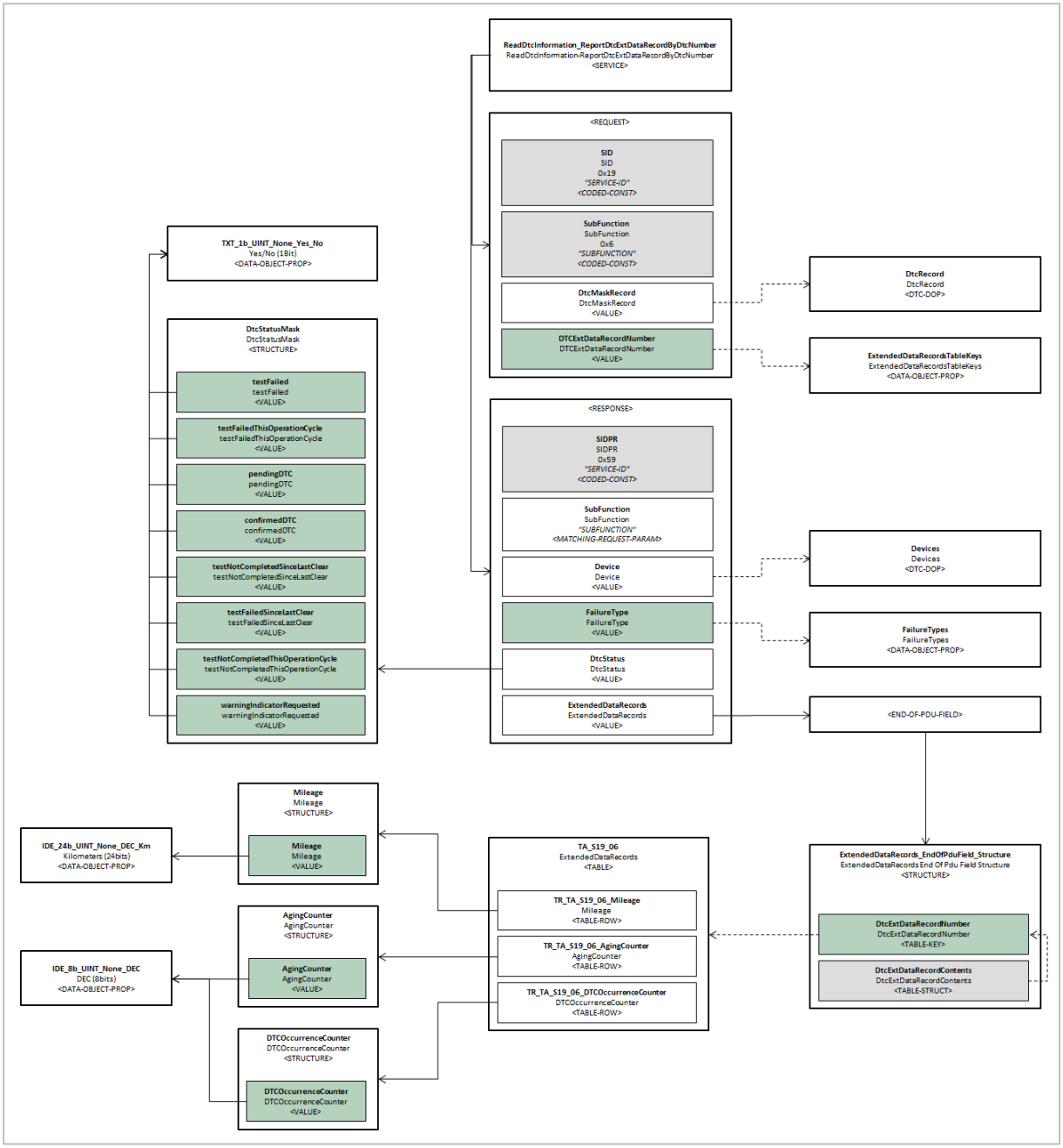


Figure 30 UDS ReadDtcInformation – ReportDtcExtDataRecordByDtcNumber ($1906)

All of these elements are predefined at RFQ level in the ESD “Renault Nissan Spec UDS”. Only the first one is not referenced via SHORT-NAME. All other elements must be overridden on variant level to define the devices, failure types, DTCs and extended data records and extended data record numbers

actually supported by the described ECUs in the context of service $1906 (for on RFQ level, these elements contain placeholder data only). Exempt from this is the TEXTTABLE with SHORT-NAME "FailureTypes": It may be overridden on variant level, but it is not required to do so, since this TEXTTABLE contains valid FailureTypes according to ISO15031-6 on the RFQ level already.

Please refer to section 2.3.4.6 for a description of the detailed rules applicable for overriding these elements and for defining the associated fault memory data.

Implementation requirements:

[Rule] The SHORT-NAME of the DIAG-SERVICE shall be "ReadDtcInformation_ReportDtcExtDataRecordByDtcNumber".

[Rule] The LONG-NAME of the DIAG-SERVICE shall be "ReadDtcInformation-ReportDtcExtDataRecordByDtcNumber".

[Rule] The DIAG-SERVICE shall be marked as IS-FINAL="true".

[Rule] The SEMANTIC attribute of the DIAG-SERVICE shall be set to "FAULTREAD".

[Rule] The DIAGNOSTIC-CLASS attribute shall not be used.

[Rule] The DIAG-SERVICE shall reference the FUNCT-CLASS "Fault_Memory".

[Rule] The request of the DIAG-SERVICE shall reference the DTC-DOP with SHORT-NAME "DtcRecord" dynamically by SNREF.

[Rule] The request of the DIAG-SERVICE shall reference the DOP with SHORT-NAME "ExtendedDataRecordsTableKeys" dynamically by SNREF.

[Rule] The positive response of the DIAG-SERVICE shall reference the DTC-DOP with SHORT-NAME "Devices" dynamically by SNREF.

[Rule] The positive response of the DIAG-SERVICE shall reference the TEXTTABLE with SHORT-NAME "FailureTypes" dynamically by SNREF.

[Rule] The positive response of the DIAG-SERVICE shall reference the TABLE with SHORT-NAME "ExtendedDataRecords" dynamically by SNREF.

2.3.5.9 ClearAllDiagnosticInformation ($14FFFFFF)

Service $14FFFFFF ("ClearAllDiagnosticInformation") is predefined once in the Renault Nissan Spec UDS ESD available from RFQ packages. It is marked final – ECU designers and suppliers shall not override or redefine this service in lower layers.

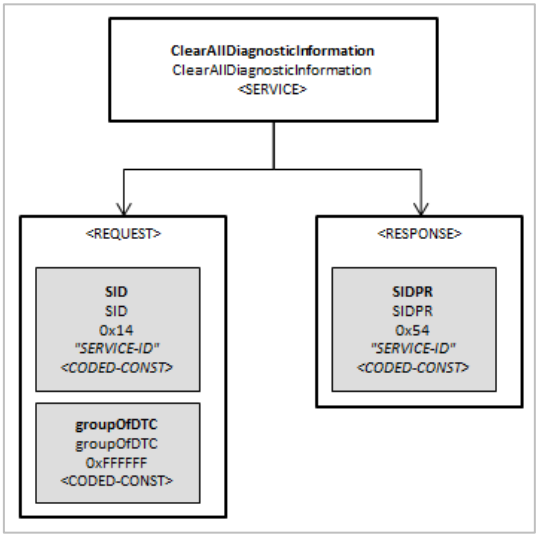


Figure 31 Structure of UDS DIAG-SERVICE ClearAllDiagnosticInformation ($14)

Implementation requirements:

[Rule] The SHORT-NAME of the DIAG-SERVICE shall be “ClearAllDiagnosticInformation”.

[Rule] The LONG-NAME of the DIAG-SERVICE shall be “ClearAllDiagnosticInformation”.

[Rule] The DIAG-SERVICE shall be marked as IS-FINAL=”true”.

[Rule] The SEMANTIC attribute of the DIAG-SERVICE shall be set to “FAULTCLEAR”.

[Rule] The DIAGNOSTIC-CLASS attribute shall not be used.

[Rule] The DIAG-SERVICE shall reference the FUNCT-CLASS “Fault_Memory”.

2.3.5.10 RoutineControl ($31)

The RFQ level ESD “Renault Nissan Spec UDS” predefines a total of three different routine control DIAG-SERVICEs, one for each of the subfunctions defined by ISO UDS. Each of these services makes use of two TABLEs intended to be overridden on BASE-VARIANT level in order to define the optional routine control records, info bytes and status records (cf. section 2.3.4.7 for details on these TABLEs).

2.3.5.10.1 StartRoutine (subfunction 0x01)

Service $3101 (“RoutineControl – StartRoutine”) is predefined once in the Renault Nissan Spec UDS ESD available from RFQ packages. It is marked final – ECU designers and suppliers shall not override or redefine this service in lower layers.

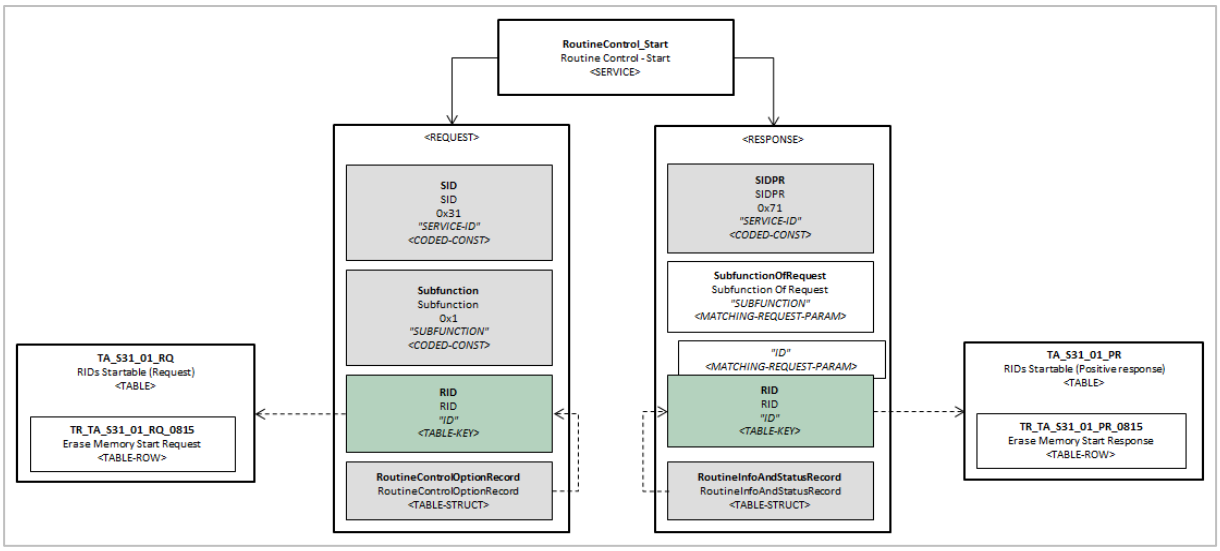


Figure 32 Structure of UDS DIAG-SERVICE Routine_Control_Start (0x3101)

Implementation requirements:

[Rule] The SHORT-NAME of the DIAG-SERVICE shall be “Routine_Control_Start”.

[Rule] The LONG-NAME of the DIAG-SERVICE shall be “Routine Control Start”.

[Rule] The DIAG-SERVICE shall be marked as IS-FINAL=”true”.

[Rule] The SEMANTIC attribute of the DIAG-SERVICE shall be set to “ROUTINE”.

[Rule] The DIAGNOSTIC-CLASS attribute shall not be used.

[Rule] The DIAG-SERVICE shall reference the FUNCT-CLASS “Routine_Control”.

[Rule] The request of the DIAG-SERVICE shall reference the TABLE with SHORT-NAME “TA_S31_01_RQ” dynamically by SNREF.

[Rule] The positive response of the DIAG-SERVICE shall reference the TABLE with SHORT-NAME "TA_S31_01_PR" dynamically by SNREF.

2.3.5.10.2 StopRoutine (subfunction 0x02)

Service $3102 ("RoutineControl – StopRoutine") is predefined once in the Renault Nissan Spec UDS ESD available from RFQ packages. It is marked final – ECU designers and suppliers shall not override or redefine this service in lower layers.

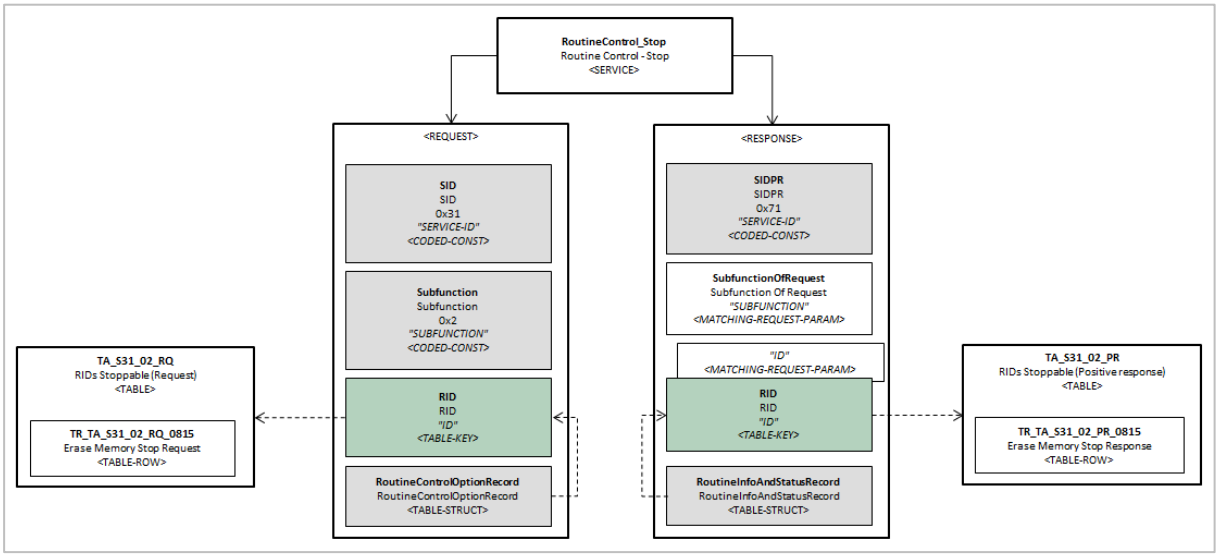


Figure 33 Structure of UDS DIAG-SERVICE Routine_Control_Stop (0x3102)

Implementation requirements:

[Rule] The SHORT-NAME of the DIAG-SERVICE shall be "Routine_Control_Stop".

[Rule] The LONG-NAME of the DIAG-SERVICE shall be "Routine Control Stop".

[Rule] The DIAG-SERVICE shall be marked as IS-FINAL="true".

[Rule] The SEMANTIC attribute of the DIAG-SERVICE shall be set to "ROUTINE".

[Rule] The DIAGNOSTIC-CLASS attribute shall not be used.

[Rule] The DIAG-SERVICE shall reference the FUNCT-CLASS "Routine_Control".

[Rule] The request of the DIAG-SERVICE shall reference the TABLE with SHORT-NAME "TA_S31_02_RQ" dynamically by SNREF.

[Rule] The positive response of the DIAG-SERVICE shall reference the TABLE with SHORT-NAME "TA_S31_02_PR" dynamically by SNREF.

2.3.5.10.3 RequestRoutineResults (subfunction 0x03)

Service $3103 ("RoutineControl – RequestRoutineResults") is predefined once in the Renault Nissan Spec UDS ESD available from RFQ packages. It is marked final – ECU designers and suppliers shall not override or redefine this service in lower layers.

Implementation requirements:

[Rule] The SHORT-NAME of the DIAG-SERVICE shall be "Routine_Control_Request_Results".

[Rule] The LONG-NAME of the DIAG-SERVICE shall be "Routine Control Request Results".

[Rule] The DIAG-SERVICE shall be marked as IS-FINAL="true".

[Rule] The SEMANTIC attribute of the DIAG-SERVICE shall be set to "ROUTINE".

[Rule] The DIAGNOSTIC-CLASS attribute shall not be used.

[Rule] The DIAG-SERVICE shall reference the FUNCT-CLASS “Routine_Control”.

[Rule] The request of the DIAG-SERVICE shall reference the TABLE with SHORT-NAME “TA_S31_03_RQ” dynamically by SNREF.

[Rule] The positive response of the DIAG-SERVICE shall reference the TABLE with SHORT-NAME “TA_S31_03_PR” dynamically by SNREF.

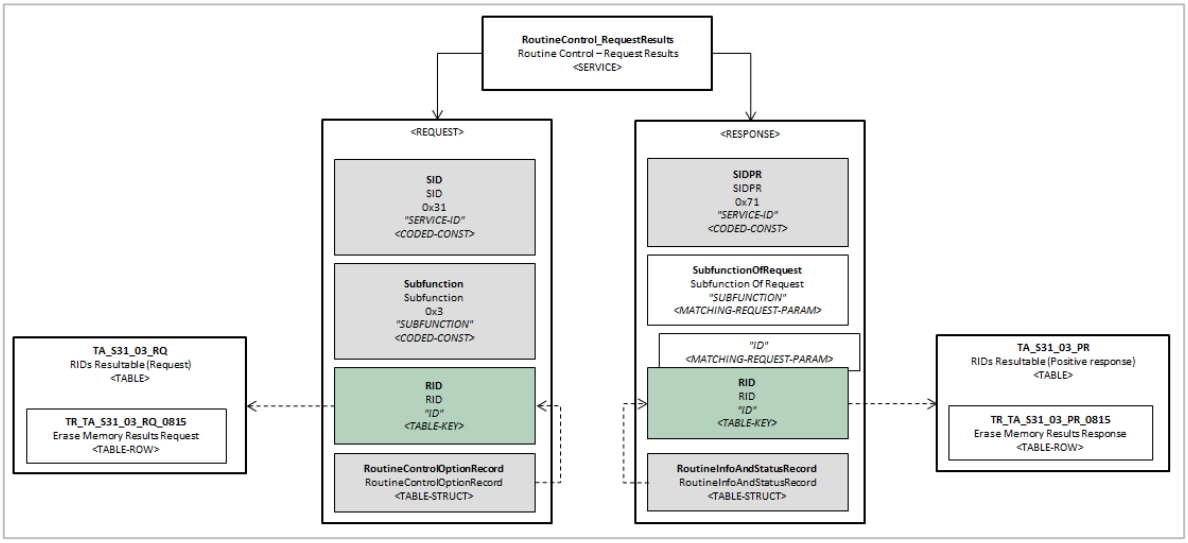


Figure 34 Structure of UDS DIAG-SERVICE Routine_Control_Request_Results (0x3103)

2.3.5.10.4 Support Check

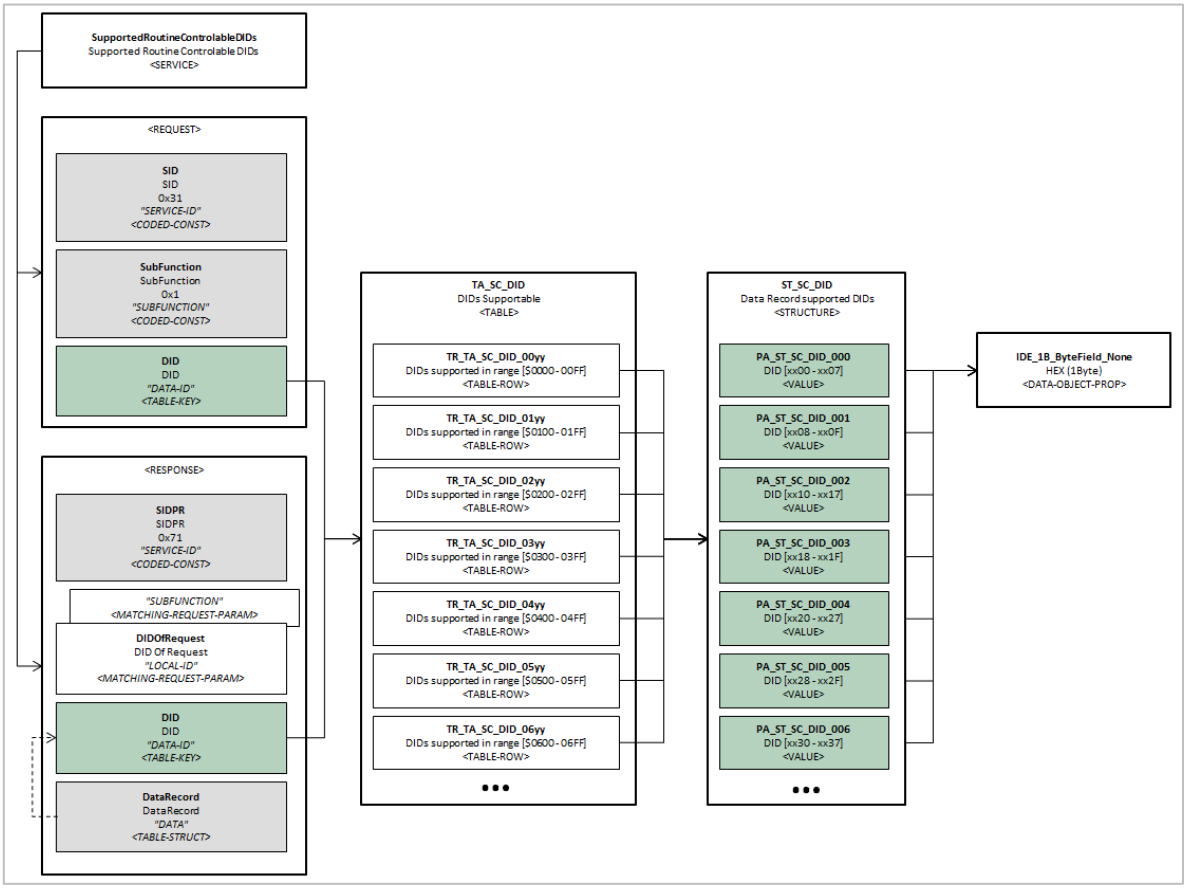


Figure 35 Structure of UDS Service SupportedRoutineControllableDIDs ($31)

For checking of supported RIDs, one DIAG-SERVICE "SupportedRoutineControlableDIDs" [sic!] (using service $31) is predefined in the Renault Nissan Spec UDS ESD available from RFQ packages.

Implementation requirements:

[Rule] SHORT- and LONG-NAME of the DIAG-SERVICE shall be "SupportedRoutineControlable-DIDs" [sic!].

[Rule] The DIAG-SERVICEs shall be marked as IS-FINAL="true".

[Rule] The SEMANTIC attribute of the DIAG-SERVICEs shall be set to "STOREDDATA".

[Rule] The DIAGNOSTIC-CLASS attribute shall not be used.

[Rule] The DIAG-SERVICEs shall reference the FUNCT-CLASS "Support Check".

[Rule] Both the request and the positive response of the DIAG-SERVICEs shall reference the TABLE "TA_SC_DID" statically. The TABLE must not be overridden on variant layers.

[Rule] The service shall be marked as NOT-INHERITED on variant layers if the support check is not implemented in the described ECU.

2.3.5.11 SecurityAccess ($27)

In UDS, requesting a seed and sending a key make use of services with slightly differing service specifications. Therefore, the two use cases are supported by two different DIAG-SERVICEs.

2.3.5.11.1 Security Access – Request Seed

Service $27 to request a security access seed (using odd numbers as sub-function) is predefined once in the Renault Nissan ESD available from RFQ packages. It is marked final – ECU designers and suppliers shall not override or redefine this service in lower layers.

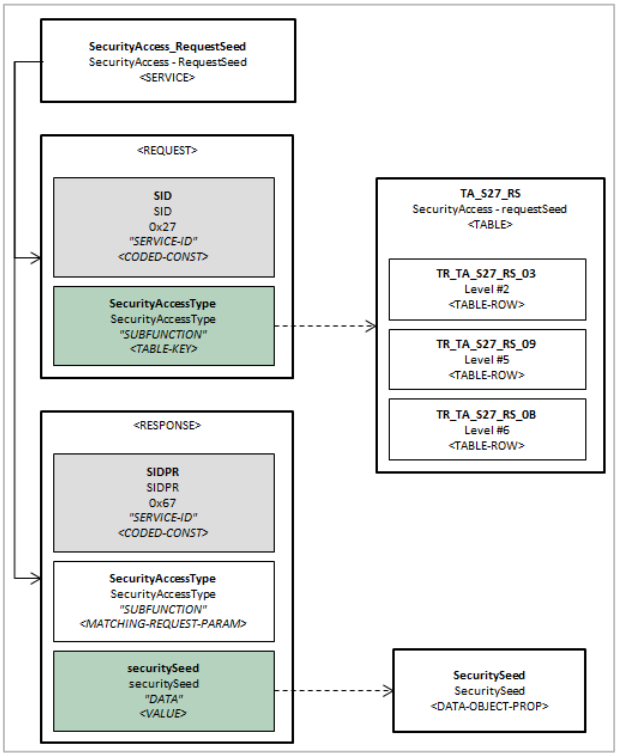


Figure 36 Structure of UDS DIAG-SERVICE SecurityAccess_requestSeed ($27)

The service makes use of the predefined TABLE "TA_S27_RS" to specify security access related sub-functions for the request. The TABLE in the Renault Nissan ESD encompasses the set of security access levels predefined by the RFQ package.

ECU designers may override the TABLE "TA_S27_RS" to tailor this predefined set of security levels for ECUs which support a different set of levels but only as long as the overridden set is **in accordance with one of the security models predefined by Renault / Nissan**. Please refer to the section 2.3.4.9 for detailed tailoring rules.

The positive response of the DIAG-SERVICE defines a VALUE "securitySeed" which references the predefined DATA-OBJECT-PROP "SecuritySeed" for all security access request seed sub-functions. It is not possible to define different seed lengths for different request seed sub functions, but ECU designers may override the DATA-OBJECT-PROP "SecuritySeed" to specify a different seed length for all sub functions. The used seed length must thereby be **in accordance with one of the security models predefined by Renault / Nissan**.

Implementation requirements:

[Rule] The SHORT-NAME of the DIAG-SERVICE shall be "SecurityAccess_requestSeed".

[Rule] The LONG-NAME of the DIAG-SERVICE shall be "SecurityAccess - requestSeed".

[Rule] The DIAG-SERVICE shall be marked as IS-FINAL="true".

[Rule] The SEMANTIC attribute of the DIAG-SERVICE shall be set to "SECURITY".

[Rule] The DIAGNOSTIC-CLASS attribute shall not be used.

[Rule] The DIAG-SERVICE shall reference the FUNCT-CLASS "SecurityAccess".

[Rule] The request of the DIAG-SERVICE shall reference the TABLE "TA_S27_RS" dynamically by SNREF.

[Rule] The securitySeed param of the DIAG-SERVICE's POS-RESPONSE shall reference the DATA-OBJECT-PROP "SecuritySeed" by SNREF.

2.3.5.11.2 Security Access – Send Key

Service $27 to send a security access key (using even numbers as sub-function) is predefined once in the Renault Nissan ESD available from RFQ packages. It is marked final – ECU designers and suppliers shall not override or redefine this service in lower layers.

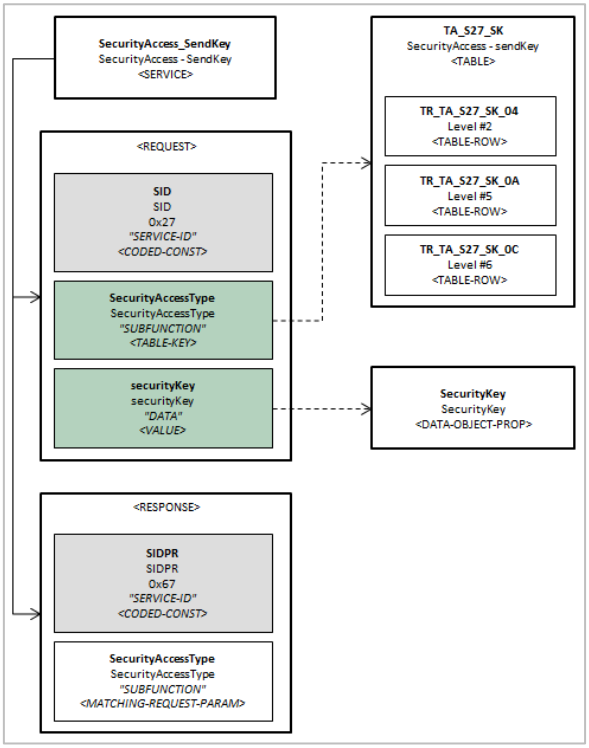


Figure 37 Structure of UDS DIAG-SERVICE SecurityAccess_sendKey ($27)

The service makes use of the predefined TABLE "TA_S27_SK" to specify security access related sub-function for the request. The TABLE in the Renault Nissan ESD encompasses the set of security access levels predefined by the RFQ package.

ECU designers may override the TABLE "TA_S27_SK " to tailor this predefined set of security levels for ECUs which support a different set of levels but only as long as the overridden set is **in accordance with one of the security models predefined by Renault / Nissan**. Please refer to the section 2.3.4.9 for detailed tailoring rules.

The request of the DIAG-SERVICE defines a VALUE "securityKey" which references the predefined DATA-OBJECT-PROP "SecurityKey" for all security access send key sub-functions. It is not possible to define different key lengths for different request seed sub-functions, but ECU designers may override the DATA-OBJECT-PROP "SecurityKey" to specify a different seed length for all sub functions if the used key length is **in accordance with one of the security models predefined by Renault / Nissan**.

Implementation requirements:

[Rule] The SHORT-NAME of the DIAG-SERVICE shall be "SecurityAccess_sendKey".

[Rule] The LONG-NAME of the DIAG-SERVICE shall be "SecurityAccess - sendKey".

[Rule] The DIAG-SERVICE shall be marked as IS-FINAL="true".

[Rule] The SEMANTIC attribute of the DIAG-SERVICE shall be set to "SECURITY".

[Rule] The DIAGNOSTIC-CLASS attribute shall not be used.

[Rule] The DIAG-SERVICE shall reference the FUNCT-CLASS "SecurityAccess".

[Rule] The request of the DIAG-SERVICE shall reference the TABLE "TA_S27_SK" dynamically by SNREF.

[Rule] The securityKey param of the DIAG-SERVICE's POS-RESPONSE shall reference the DATA-OBJECT-PROP "SecurityKey" by SNREF.

2.3.5.12 EcuReset ($11)

Service $11 is predefined once in the Renault Nissan Spec UDS ESD available from RFQ packages. It is marked final – ECU designers and suppliers shall not override or redefine this service in lower layers.

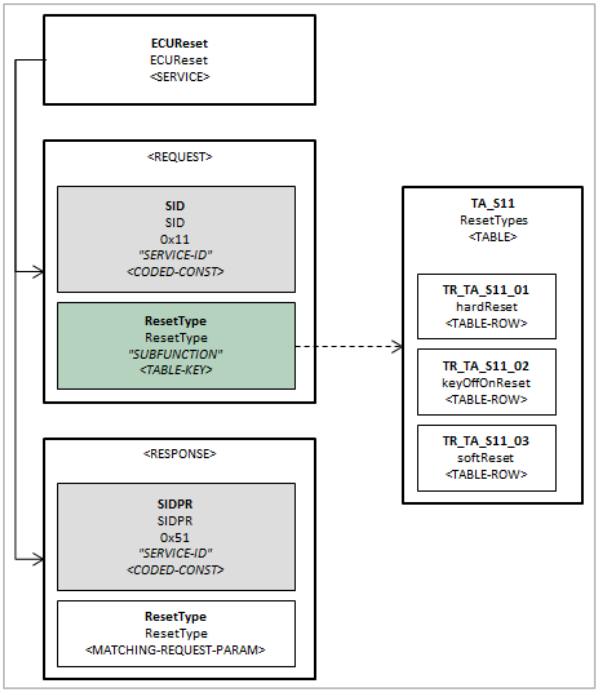


Figure 38 Structure of UDS DIAG-SERVICE ECUReset (0x11)

The service makes use of the predefined TABLE "TA_S11" to specify the supported reset types. The following reset types are thereby predefined:

- Hard Reset (0x01)
- Key Off/On Reset (0x02)
- Soft Reset (0x03)

ECU designers may override that TABLE to tailor these presets for ECUs which support a different set of reset types. Please refer to section 2.3.4.10 for the exact tailoring rules.

Implementation requirements:

[Rule] The SHORT- and LONG-NAME of the DIAG-SERVICE shall be "ECUReset".

[Rule] The SEMANTIC attribute of the DIAG-SERVICE shall be set to "FUNCTION"

[Rule] The DIAG-SERVICE shall not use the DIAGNOSTIC-CLASS attribute.

[Rule] It shall be marked as IS-FINAL="true".

[Rule] The DIAG-SERVICE shall reference the FUNCT-CLASS "ECUReset".

[Rule] The request of the DIAG-SERVICE shall reference the TABLE "TA_S11" dynamically by SNREF.

2.3.5.13 ControlDTCSetting ($85)

Service $85 ("ControlDTCSetting") is predefined once in the Renault Nissan Spec UDS ESD available from RFQ packages. It is marked final – ECU designers and suppliers shall not override or redefine this service in lower layers.

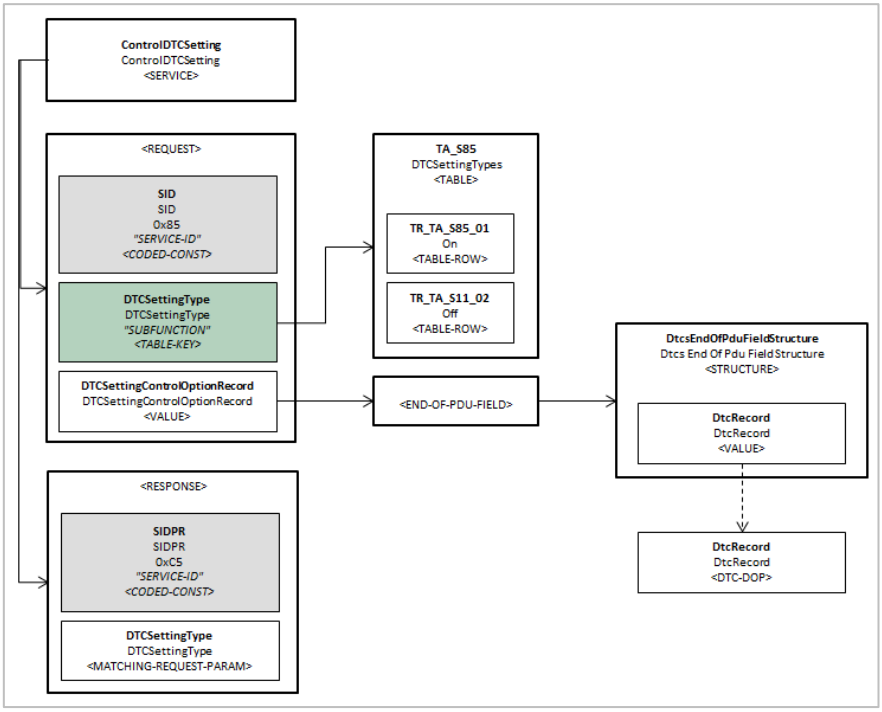


Figure 39 UDS DIAG-SERVICE ControlDTCSetting ($85)

The service makes use of the predefined TABLE "TA_S85". The following DTC setting types are thereby defined:

- On (0x01)
- Off (0x02)

ECU designers must not override that TABLE.

As shown in Figure 39, the definition of service $85 makes use of the following ODX elements:

- The Parameter with SHORT-NAME "DTCSettingControlOptionRecord" contains an END-OF-PDU-FIELD, which allows to specify one or more DTC-RECORDs, to which the DIAG-SERVICE shall be applied.
- For this, the END-OF-PDU-FIELD reuses the central DTC-DOP which defines the DTCs supported by the described ECU (cf. section 2.3.4.6.1).

Implementation requirements:

[Rule] The SHORT- and LONG-NAME of the DIAG-SERVICE shall be "ControlDTCSetting".

[Rule] The SEMANTIC attribute of the DIAG-SERVICE shall be set to "FUNCTION"

[Rule] The DIAG-SERVICE shall not use the DIAGNOSTIC-CLASS attribute.

[Rule] It shall be marked as IS-FINAL="true".

[Rule] The DIAG-SERVICE shall reference the FUNCT-CLASS "ControlDTCSetting".

[Rule] The DIAG-SERVICE shall reference TABLE "TA_S85" statically via ODX-LINK.

[Rule] ECU designers must not override TABLE "TA_S85" on variant layers.

2.3.5.14 TesterPresent ($3E)

Service $3E ("TesterPresent") is predefined once in the Renault Nissan Spec UDS ESD available from RFQ packages. It is marked final – ECU designers and suppliers shall not override or redefine this service in lower layers.

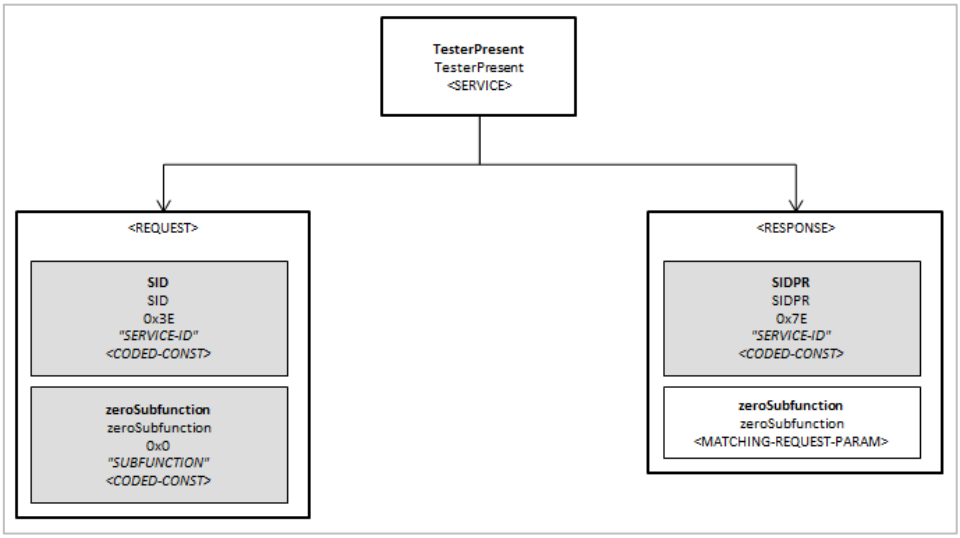


Figure 40 UDS DIAG-SERVICE TesterPresent ($3E)

Implementation requirements:

[Rule] The DIAG-SERVICE shall have the SHORT-/LONG-NAME "TesterPresent".

[Rule] The SEMANTIC attribute of the DIAG-SERVICE shall be set to "TESTERPRESENT".

[Rule] The DIAG-SERVICE shall not use the DIAGNOSTIC-CLASS attribute.

[Rule] The DIAG-SERVICE shall reference the FUNCT-CLASS "TesterPresent"

[Rule] The DIAG-SERVICE must use POS-RESPONSE-SUBPRESSABLE to express the SPRMIB.

[Rule] It shall be marked as IS-FINAL="true".

[Rule] It shall be marked as IS-MANDATORY="true".

2.3.5.15 RequestDownload ($34)

Service $34 ("Request Download") is predefined once in the Renault Nissan Spec UDS ESD available from RFQ packages. It is marked final – ECU designers and suppliers shall not override or redefine this service in lower layers.

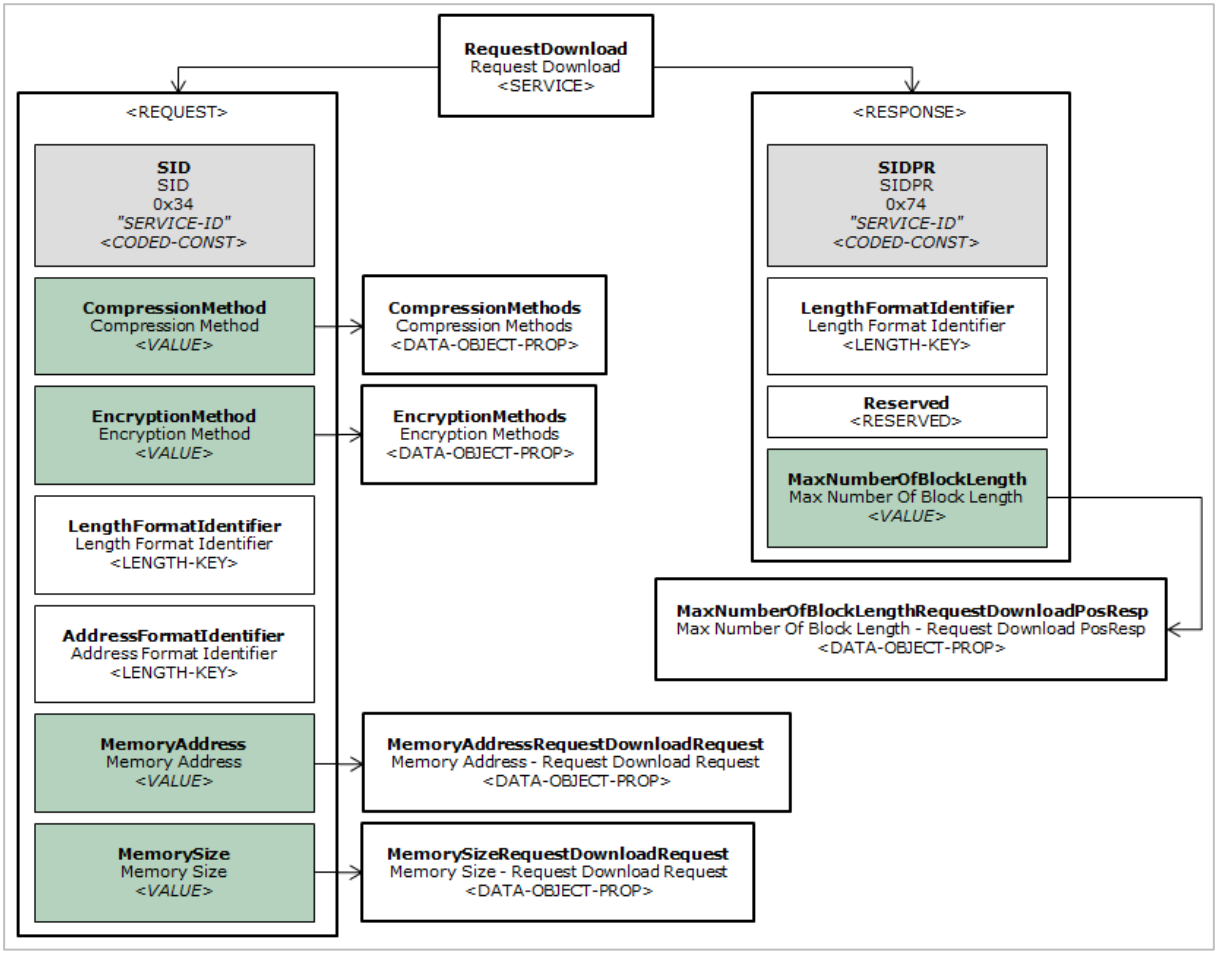


Figure 41 UDS DIAG-SERVICE RequestDownload ($34)

As shown in Figure 41, the definition of service $34 makes use of the following ODX elements defining the formatting of the data:

- The LENGTH-KEY with SHORT-NAME "LengthFormatIdentifier" specifying the length (in number of bytes) of the respective parameter (MemorySize for Request, MaxNumberOfBlockLength for Response).
- The LENGTH-KEY with SHORT-NAME "AddressFormatIdentifier" specifying the length (in number of bytes) of the parameter MemoryAddress in Request. In the response, this parameter is restricted to a reserved.

Implementation requirements:

[Rule] The DIAG-SERVICE shall have the SHORT-NAME "RequestDownload".

[Rule] The DIAG-SERVICE shall have the LONG-NAME "Request Download".

[Rule] The DIAG-SERVICE shall be marked as IS-FINAL="true".

[Rule] The SEMANTIC attribute of the DIAG-SERVICE shall be set to "DOWNLOAD".

[Rule] The DIAGNOSTIC-CLASS attribute shall not be used.

[Rule] The DIAG-SERVICE shall reference the FUNCT-CLASS "Upload Download".

2.3.5.16 RequestUpload ($35)

Service $35 ("Request Upload") is predefined once in the Renault Nissan Spec UDS ESD available from RFQ packages. It is marked final – ECU designers and suppliers shall not override or redefine this service in lower layers.

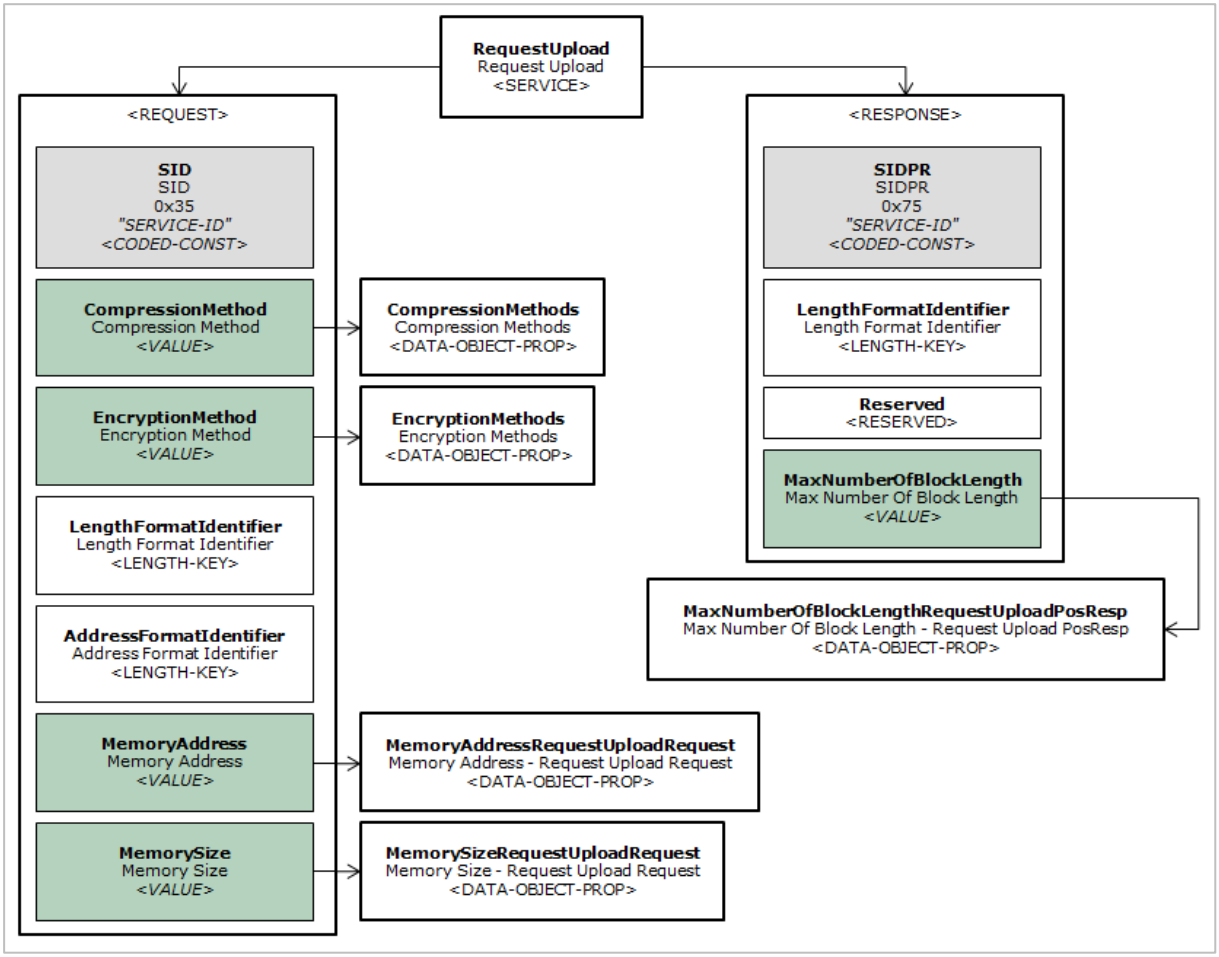


Figure 42 UDS DIAG-SERVICE RequestUpload ($35)

As shown in Figure 42, the definition of service $35 makes use of the following ODX elements defining the formatting of the data:

- The LENGTH-KEY with SHORT-NAME "LengthFormatIdentifier" specifying the length (in number of bytes) of the respective parameter (MemorySize for Request, MaxNumberOfBlockLength for Response).
- The LENGTH-KEY with SHORT-NAME "AddressFormatIdentifier" specifying the length (in number of bytes) of the parameter MemoryAddress in Request. In the response, this parameter is restricted to a reserved.

Implementation requirements:

[Rule] The DIAG-SERVICE shall have the SHORT-NAME "RequestUpload".

[Rule] The DIAG-SERVICE shall have the LONG-NAME "Request Upload".

[Rule] The DIAG-SERVICE shall be marked as IS-FINAL="true".

[Rule] The SEMANTIC attribute of the DIAG-SERVICE shall be set to "UPLOAD".

[Rule] The DIAGNOSTIC-CLASS attribute shall not be used.

[Rule] The DIAG-SERVICE shall reference the FUNCT-CLASS "Upload Download".

2.3.5.17 TransferData ($36)

Service $36 ("Transfer Data") is predefined once in the Renault Nissan Spec UDS ESD available from RFQ packages. It is marked final – ECU designers and suppliers shall not override or redefine this service in lower layers.

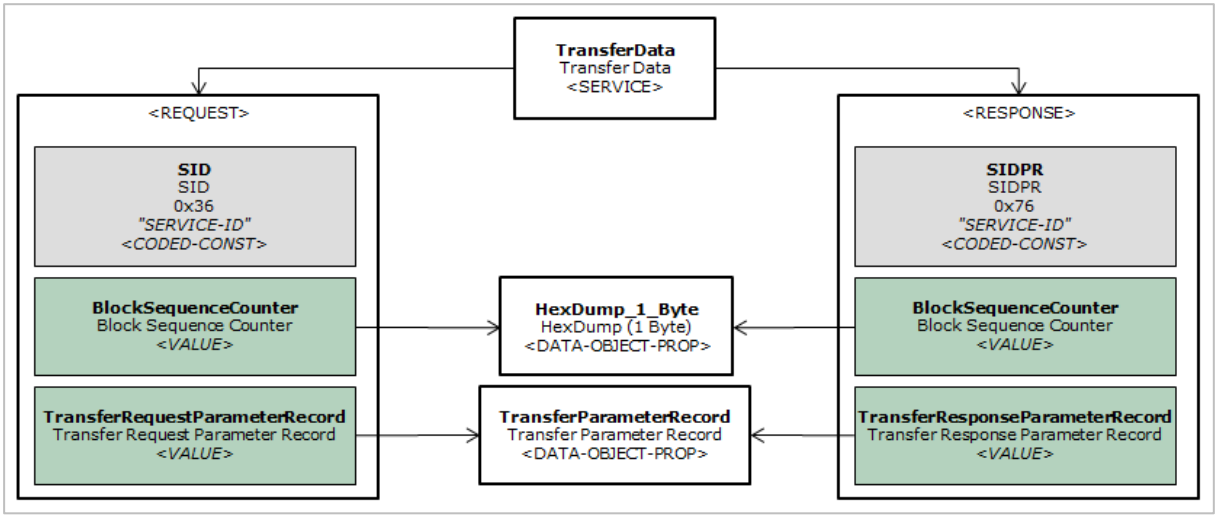


Figure 43 DIAG-SERVICE TransferData ($36)

The ODX elements used by service $36 are shown in Figure 43.

Implementation requirements:

[Rule] The DIAG-SERVICE shall have the SHORT-NAME "TransferData".

[Rule] The DIAG-SERVICE shall have the LONG-NAME "Transfer Data".

[Rule] The DIAG-SERVICE shall be marked as IS-FINAL="true".

[Rule] The SEMANTIC attribute of the DIAG-SERVICE shall not be used.

[Rule] The DIAGNOSTIC-CLASS attribute shall not be used.

[Rule] The DIAG-SERVICE shall reference the FUNCT-CLASS "Upload Download".

2.3.5.18 RequestTransferExit ($37)

Service $37 ("Request Transfer Exit") is predefined once in the Renault Nissan Spec UDS ESD available from RFQ packages. It is marked final – ECU designers and suppliers shall not override or redefine this service in lower layers.

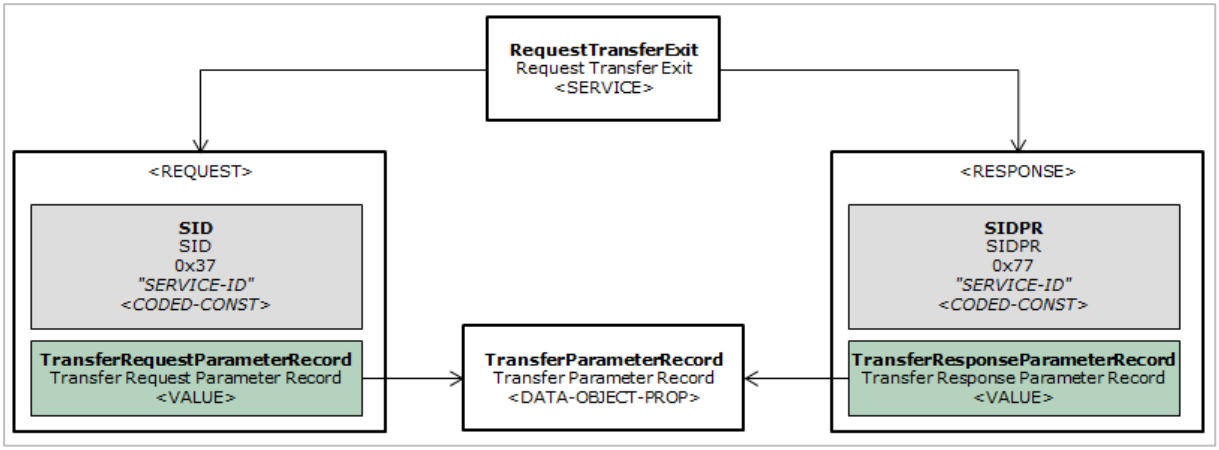


Figure 44 UDS DIAG-SERVICE RequestTransferExit ($37)

The ODX elements used by service $37 are shown in Figure 44.

Implementation requirements:

[Rule] The DIAG-SERVICE shall have the SHORT-NAME “RequestTransferExit”.

[Rule] The DIAG-SERVICE shall have the LONG-NAME “Request Transfer Exit”.

[Rule] The DIAG-SERVICE shall be marked as IS-FINAL=”true”.

[Rule] The SEMANTIC attribute of the DIAG-SERVICE shall not be used.

[Rule] The DIAGNOSTIC-CLASS attribute shall not be used.

[Rule] The DIAG-SERVICE shall reference the FUNCT-CLASS “Upload Download”.

2.3.5.19 CommunicationControl ($28)

Service $28 (“CommunicationControl”) is predefined once in the Renault Nissan Spec UDS ESD available from RFQ packages. It is marked final – ECU designers and suppliers shall not override or redefine this service in lower layers.

The service makes use of the predefined TABLE “TA_S28” to specify the supported sub-functions. The following sub-functions are thereby predefined:

- enableRxAndTx (0x00)
- enableRxAndDisableTx (0x01)
- disableRxAndEnableTx (0x02)
- disableRxAndTx(0x03)

ECU designers may override TABLE “TA_S28” on variant layers to tailor these presets for ECUs which support a different set of sub-functions.

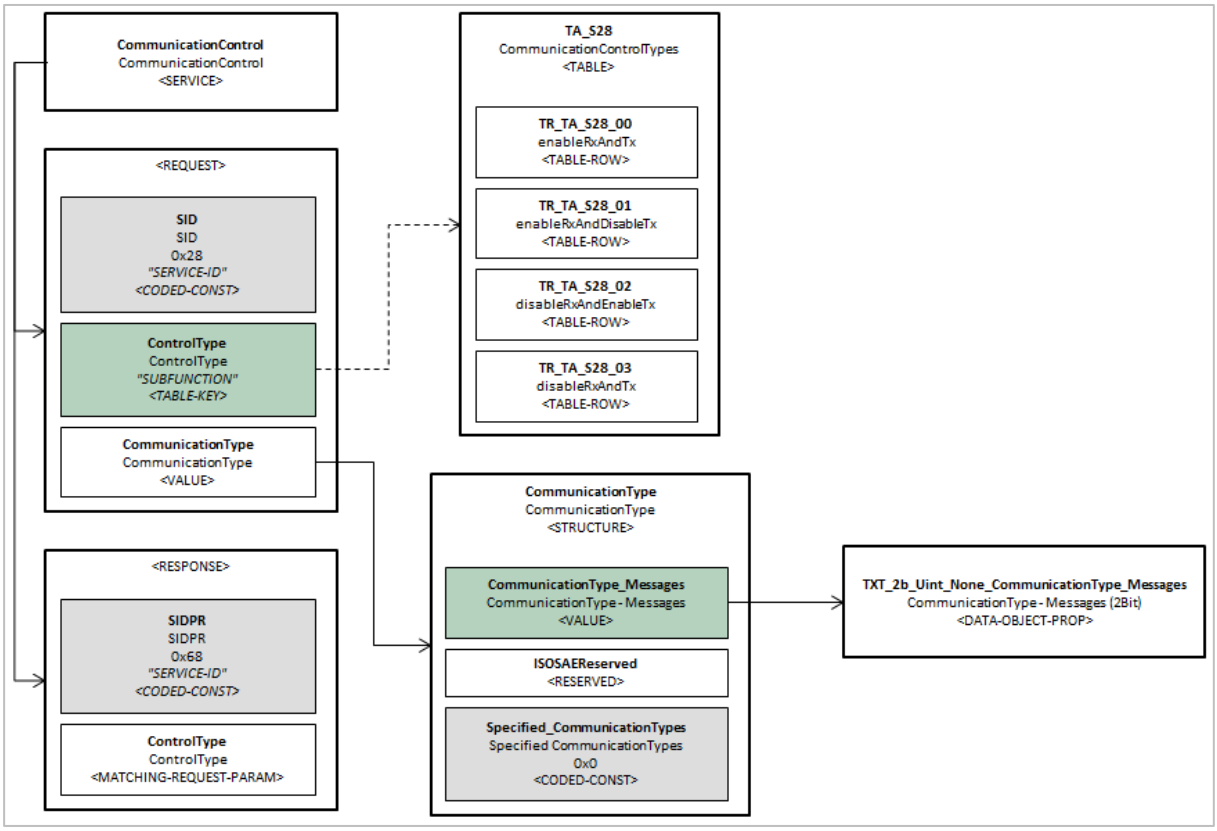


Figure 45 UDS DIAG-SERVICE CommunicationControl ($28)

The ODX elements used by service $28 are shown in Figure 45.

Implementation requirements:

[Rule] The DIAG-SERVICE shall have the SHORT-/LONG-NAME "CommunicationControl".

[Rule] The DIAG-SERVICE shall be marked as IS-FINAL="true".

[Rule] The SEMANTIC attribute of the DIAG-SERVICE shall be set to "COMMUNICATION".

[Rule] The DIAGNOSTIC-CLASS attribute shall not be used.

[Rule] The DIAG-SERVICE shall reference the FUNCT-CLASS "CommunicationControl".

[Rule] The request of the DIAG-SERVICE shall reference the TABLE "TA_S28" dynamically by SNREF.

2.3.5.20 ReadMemoryByAddress ($23)

Service $23 ("ReadMemoryByAddress") is predefined once in the Renault Nissan Spec UDS ESD available from RFQ packages. It is marked final – ECU designers and suppliers shall not override or redefine this service in lower layers.

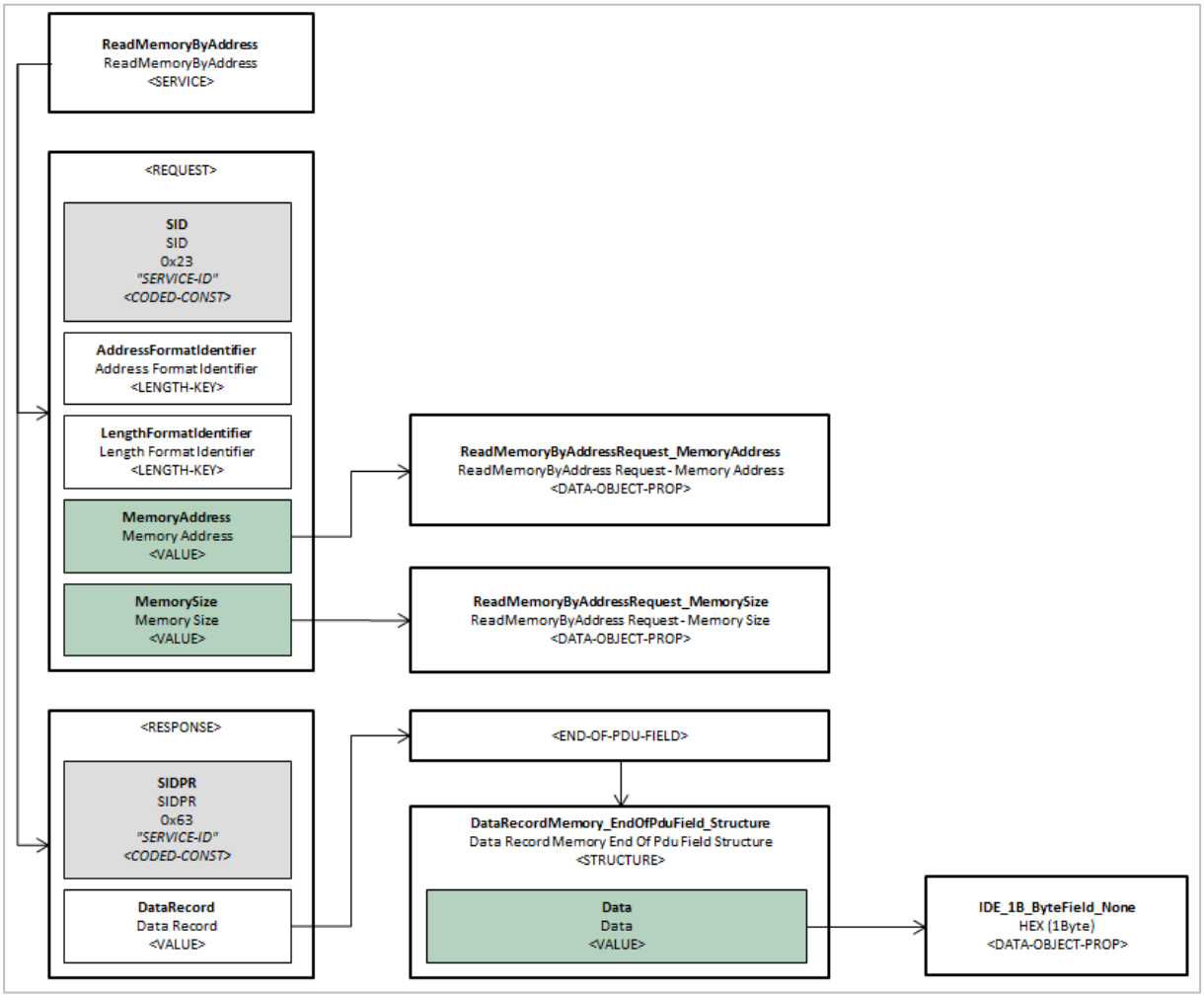


Figure 46 UDS DIAG-SERVICE ReadMemoryByAddress ($23)

The ODX elements used by service $23 are shown in Figure 46.

Implementation requirements:

[Rule] The DIAG-SERVICE shall have the SHORT-/LONG-NAME "ReadMemoryByAddress".

[Rule] The DIAG-SERVICE shall be marked as IS-FINAL="true".

[Rule] The SEMANTIC attribute of the DIAG-SERVICE shall be set to “MEMORYREAD”.

[Rule] The DIAGNOSTIC-CLASS attribute shall not be used.

[Rule] The DIAG-SERVICE shall reference the FUNCT-CLASS “Memory”.

2.3.5.21 WriteMemoryByAddress ($3D)

Service $3D (“WriteMemoryByAddress”) is predefined once in the Renault Nissan Spec UDS ESD available from RFQ packages. It is marked final – ECU designers and suppliers shall not override or redefine this service in lower layers.

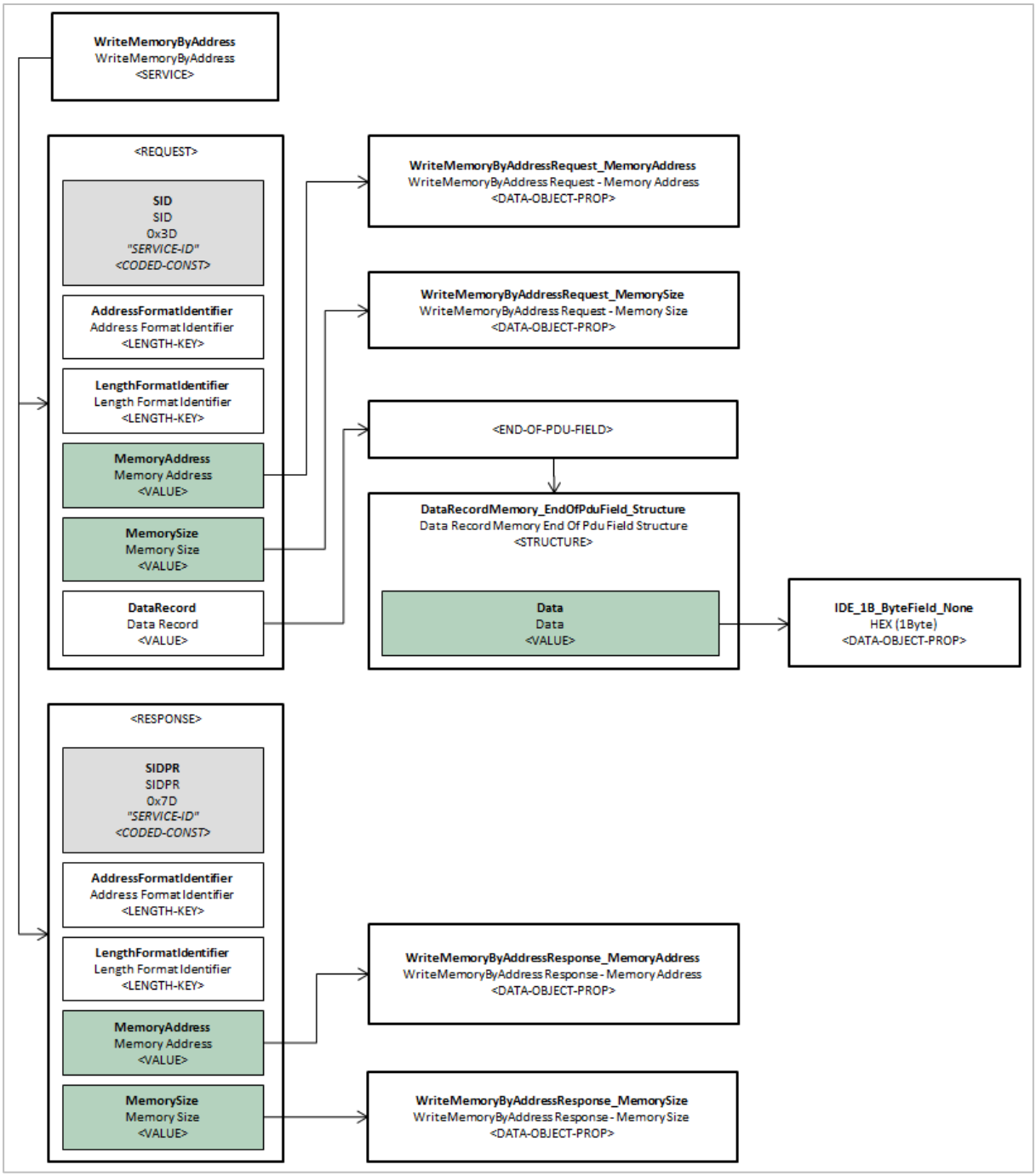


Figure 47 UDS DIAG-SERVICE WriteMemoryByAddress ($3D)

The ODX elements used by service $3D are shown in Figure 47.

Implementation requirements:

[Rule] The DIAG-SERVICE shall have the SHORT-/LONG-NAME “WriteMemoryByAddress”.

[Rule] The DIAG-SERVICE shall be marked as IS-FINAL=”true”.

[Rule] The SEMANTIC attribute of the DIAG-SERVICE shall be set to “MEMORYWRITE”.

[Rule] The DIAGNOSTIC-CLASS attribute shall not be used.

[Rule] The DIAG-SERVICE shall reference the FUNCT-CLASS “Memory”.

2.3.6 DIAG-SERVICEs for B/C

2.3.6.1 General DIAG-SERVICE Requirements

In ODX, DIAG-SERVICEs may reference multiple FUNCT-CLASS objects. For Renault / Nissan use cases this might confuse the user. Therefore, DIAG-SERVICEs shall only reference a single FUNCT-CLASS.

Renault / Nissan does not want to deviate from the ISO definition of negative responses. As a result, one GLOBAL-NEGATIVE-RESPONSE is predefined which handles all services according to ISO. ODX authors shall not define a NEG-RESPONSE on DIAG-SERVICE level.

In ODX it is possible to modify communication parameters at DIAG-COMM level. In Renault / Nissan ODX, this feature shall not be used to reduce possible issues in tester and/or future diagnostic code generation.

It shall be possible to execute every B/C DIAG-SERVICE only physically.

To express the ability of suppressing the positive response via setting the SPRMIB, authors shall not use the POS-RESPONSE-SUPPRESSABLE element, except for those services where its usage is explicitly allowed by this document.

Implementation requirements:

[Rule] DIAG-SERVICEs shall have exactly one reference to a FUNCT-CLASS.

[Rule/xsd] DIAG-SERVICEs shall not reference a NEG-RESPONSE.

[Rule] ODX authors must not override communication parameters at DIAG-SERVICE level.

[Rule] ODX authors must not define ADDRESSING-MODE different to “PHYSICAL” at DIAG-SERVICE level.

[Rule] Generally, ODX authors must not use the POS-RESPONSE-SUPPRESSABLE element at DIAG-SERVICEs (except for where its usage is explicitly allowed by these AGLs).

2.3.6.2 Flat Services

The rules defined in chapter 2.3.5.2 for Spec UDS ODX models apply also for Spec B/C ODX models.

2.3.6.3 DIAG-SERVICE Overview

The following table shows the UDS based use cases together with their diagnostic services and possible FUNCT-CLASS.

FUNCT-CLASS	UDS Service
Session	$10 (StartDiagnosticSession)
LIDs	$21 (ReadDataByLocalIdentifier)
	$3B (WriteDatByLocalIdentifier)

DIDs	$22 (ReadDataByIdentifier)
	$2E (WriteDataByIdentifier)
I/O Controls	$30 (InputOutpuControlByLocalIdentifier)
Fault Memory	$14 (ClearDiagnosticInformation)
	$14 (ClearAllDiagnosticInformation)
	$19 (ReadDTCInformation)
Routine Control	$31 (StartRoutineByLocalIdentifier)
	$32 (StopRoutineByLocalIdentifier)
ECUReset	$11 (ECUReset)
TesterPresent	$3E (TesterPresent)
SecurityAccess	$27 (SecurityAccess)
Upload Download	$34 (RequestDownload)
	$35 (RequestUpload)
	$36 (TransferData)
	$37 (RequestTransferExit)
CommunicationControl	$28 (CommunicationControl)
Memory	$23 (ReadMemoryByAddress)
	$3D (WriteMemoryByAddress)

Table 10 Overview of B/C DIAG-SERVICEs and their FUNCT-CLASSes

2.3.6.4 StartDiagnosticSession ($10)

Service $10 is predefined once in the Renault Nissan Spec B/C ESD available from RFQ packages. It is marked final – ECU designers and suppliers shall not override or redefine this service in lower layers.

The service makes use of the predefined TABLE “Sessions” to specify the supported diagnostic session types. The following session types are thereby predefined:

- DefaultSession (0x81)
- ProgrammingSession (0x85)
- DevelopmentMode (0x86)
- ExtendedDiagnosticSession (0xC0)

ECU designers can override that TABLE to tailor these presets for ECUs which support a different set of session types. Please refer to section 2.3.4.1 for the exact tailoring rules.

Implementation requirements:

[Rule] The SHORT- and LONG-NAME of the DIAG-SERVICE shall be “StartDiagnosticSession”.

[Rule] The SEMANTIC attribute of the DIAG-SERVICE shall be set to “SESSION”

[Rule] The DIAG-SERVICE shall not use the DIAGNOSTIC-CLASS attribute.

[Rule] It shall be marked as IS-FINAL="true".

[Rule] The DIAG-SERVICE shall reference the FUNCT-CLASS "Session".

[Rule] The request of the DIAG-SERVICE shall reference the TABLE "TA_S10" dynamically by SNREF.

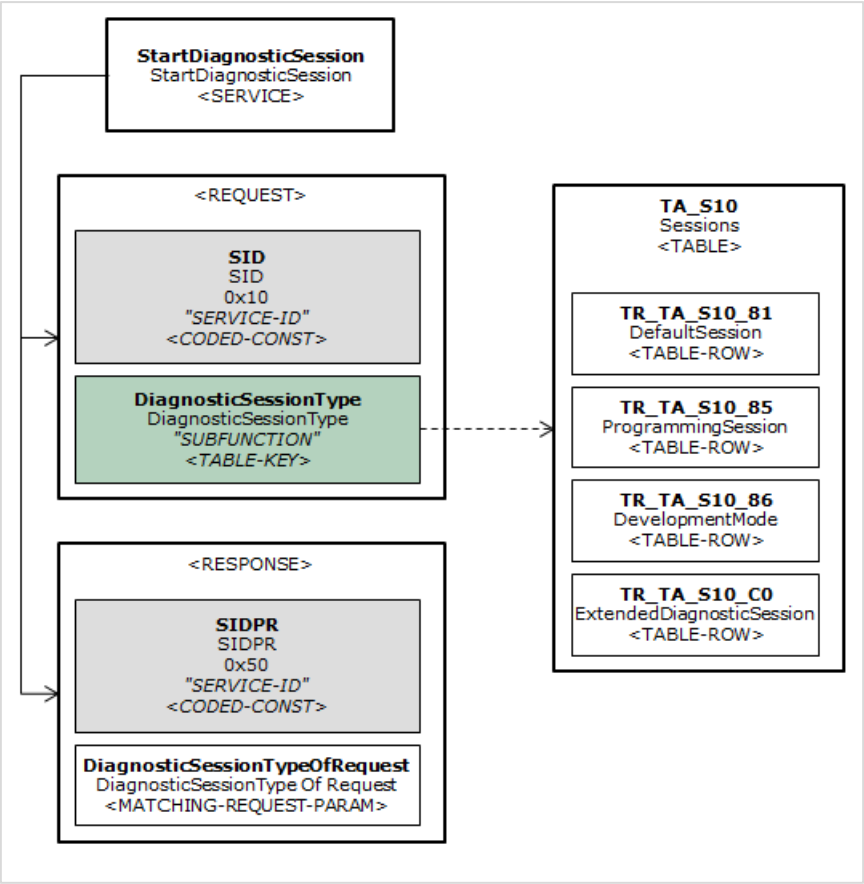


Figure 48 Structure of B/C DIAG-SERVICE StartDiagnosticSession (0x10)

2.3.6.5 ReadDataByLocalIdentifier ($21)

2.3.6.5.1 General Read Service

Service $21 is predefined in the Renault Nissan Spec B/C ESD available from RFQ packages. The service is marked final – ECU designers and suppliers shall not override or redefine this service in lower layers.

The service makes use of the predefined TABLE "TA_S21" to specify both the local identifier(s) in request and response and the content of the associated data record(s). The TABLE in the Renault Nissan Spec B/C ESD encompasses the set of readable LIDs predefined by the RFQ package (of which Figure 49 merely shows a small subset).

ECU designers can override TABLE "TA_S21" to tailor this predefined set of readable LIDs for ECUs which support a different set of LIDs. Please refer to to section 2.3.4.3 for detailed tailoring rules (and a general overview of how LIDs can be defined and reused in lower layers).

Implementation requirements:

[Rule] The SHORT- and LONG-NAME of the DIAG-SERVICE representing B/C service $21 shall be "ReadDataByLocalIdentifier".

[Rule] The DIAG-SERVICEs shall be marked as IS-FINAL="true".

[Rule] The SEMANTIC attribute of the DIAG-SERVICEs shall be set to "STOREDDATAREAD".

[Rule] The DIAGNOSTIC-CLASS attribute shall not be used.

[Rule] The DIAG-SERVICEs shall reference the FUNCT-CLASS “LIDs”.

[Rule] Both the request and the positive response of the DIAG-SERVICEs shall reference the TABLE “TA_S21” dynamically by SNREF.

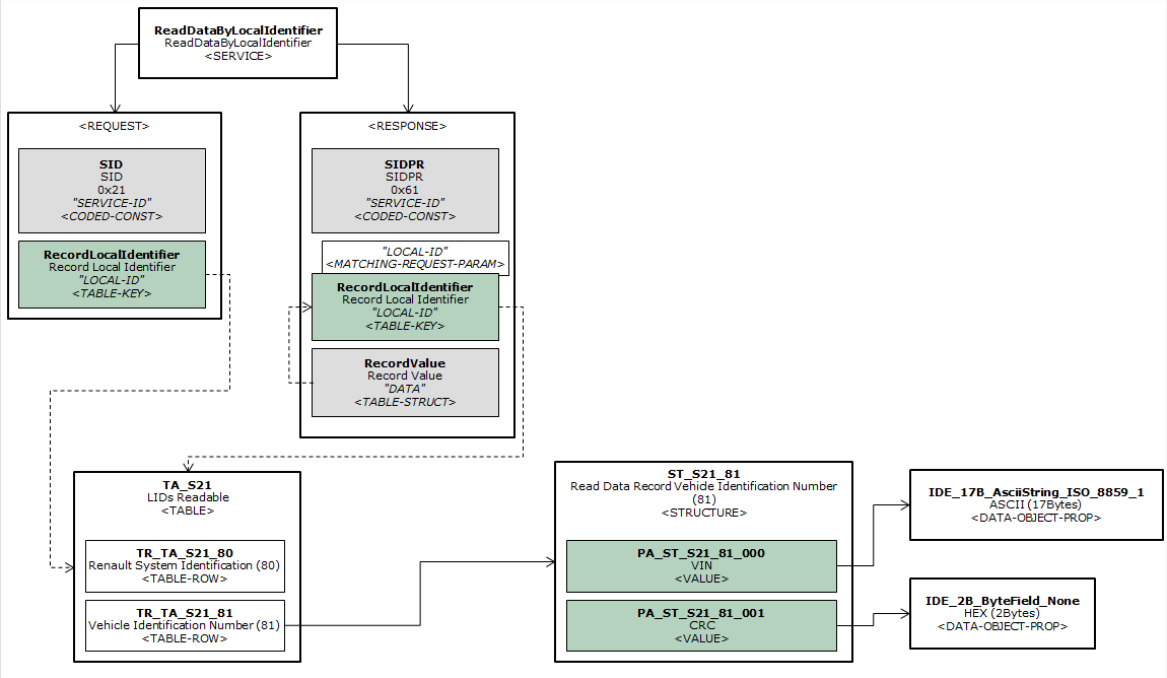


Figure 49 Structure of B/C DIAG-SERVICE ReadDataByLocalIdentifier ($21)

2.3.6.5.2 Support Check

For checking of supported readable one-byte LIDs, one DIAG-SERVICE “SupportedReadableLIDs” (using service $21) is predefined in the Renault Nissan Spec B/C ESD available from RFQ packages.

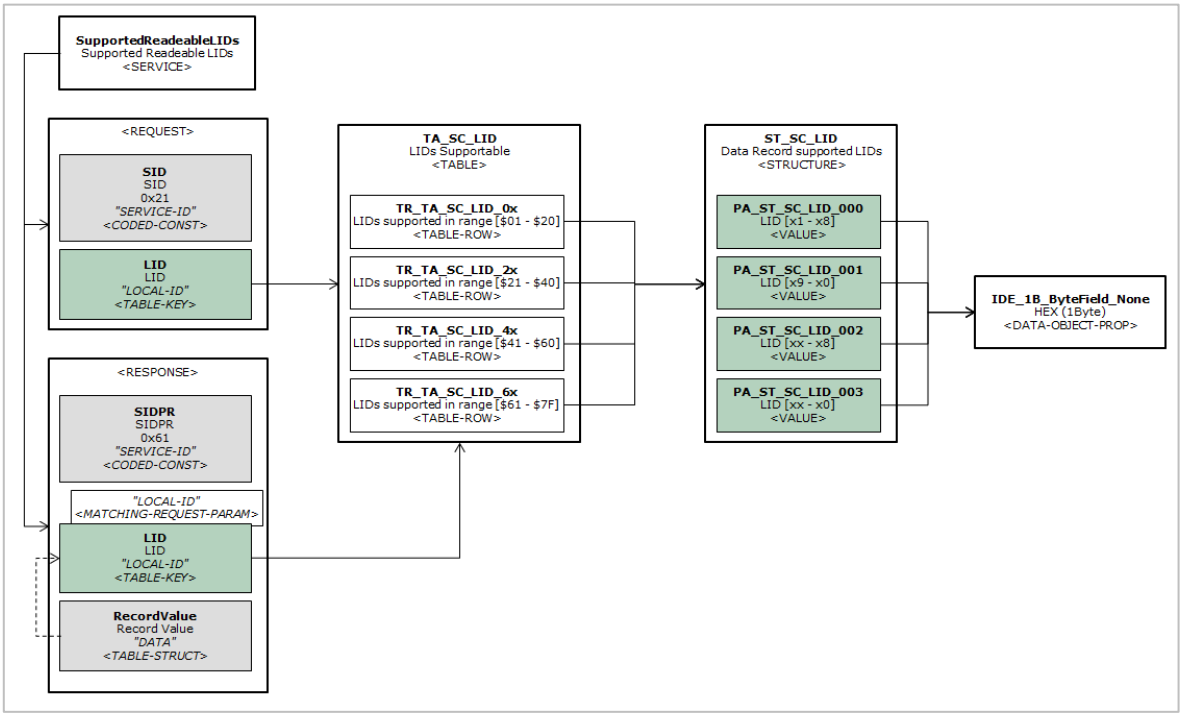


Figure 50 Structure of B/C DIAG-SERVICE SupportedReadableLIDs ($21)

Implementation requirements:

[Rule] The SHORT- and LONG-NAME of the DIAG-SERVICE shall be “SupportedReadableLIDs”.

[Rule] The DIAG-SERVICEs shall be marked as IS-FINAL=”true”.

[Rule] The SEMANTIC attribute of the DIAG-SERVICEs shall be set to “STOREDDATAREAD”.

[Rule] The DIAGNOSTIC-CLASS attribute shall not be used.

[Rule] The DIAG-SERVICEs shall reference the FUNCT-CLASS “Support Check”.

[Rule] Both the request and the positive response of the DIAG-SERVICEs shall reference the TABLE “TA_SC_LID” statically. The TABLE shall not be overridden on variant layers.

[Rule] The service shall be marked as NOT-INHERITED on variant layers if the support check is not implemented in the described ECU.

2.3.6.5.3 Identification – Read Identification 80

For reading of Renault related ECU identification data, Renault uses the predefined LID $80. This LID provides access to information like “spare part number”, “diagnostic identification code”, “supplier number” and many more. Please refer to [3] for details to the service specification.

The ODX data uses this service $2180 also for identifying the ECU variant. For details of ECU variant identification, please refer to chapter 2.3.1.6.

The service $2180 is predefined once in the Renault Nissan Spec B/C ESD available from RFQ packages. It is marked final – ECU designers and suppliers shall not override or redefine this service in lower layers.

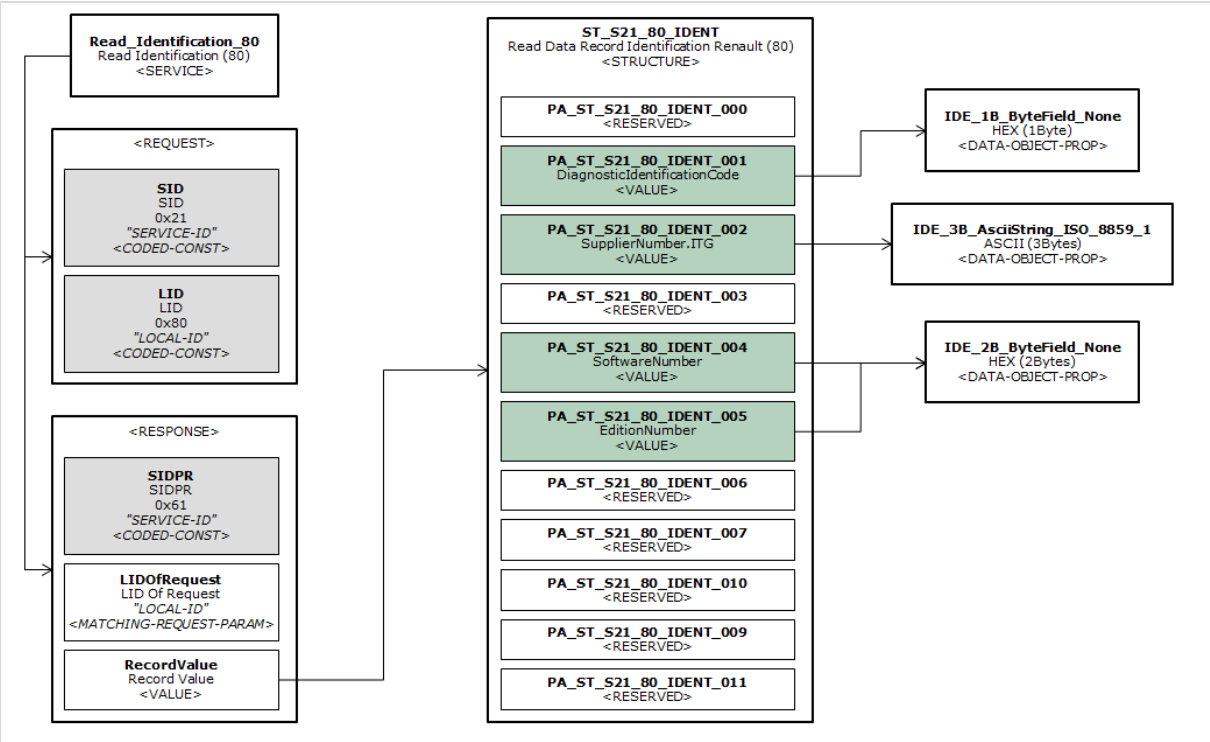


Figure 51 Structure of B/C DIAG-SERVICE Read_Identification_80 ($2180)

Implementation requirements:

[Rule] The SHORT-NAME of the DIAG-SERVICE shall be “Read_Identification_80”.

[Rule] The LONG-NAME of the DIAG-SERVICE shall be “Read Identification (80)”.

[Rule] The DIAG-SERVICEs shall be marked as IS-FINAL=”true”.

[Rule] The SEMANTIC attribute of the DIAG-SERVICEs shall be set to “IDENTIFICATION”.

[Rule] The DIAGNOSTIC-CLASS attribute shall be set to “VARIANTIDENTIFICATION”.

[Rule] The DIAG-SERVICEs shall reference the FUNCT-CLASS “Identification”.

[Rule] The positive response of the DIAG-SERVICE shall reference the STRUCTURE “ST_21_80_IDENT” (used for interpretation) by ODX-LINK.

[Rule] The author of ODX data shall not mark this service as “not inherited” on ECU-VARIANT-LAYER level.

2.3.6.5.4 Identification – Read Identification 83

For reading of Nissan related ECU identification data, Nissan uses the predefined LID $83. This LID provides access to information like “spare part number”, “diagnostic identification code”, “supplier number” and many more. Please refer to [3] for details to the service specification.

The ODX data uses this service $2183 also for identifying the ECU variant. For details of ECU variant identification, please refer to chapter 2.3.1.6.

The service $2183 is predefined once in the Renault Nissan Spec B/C ESD available from RFQ packages. It is marked final – ECU designers and suppliers shall not override or redefine this service in lower layers.

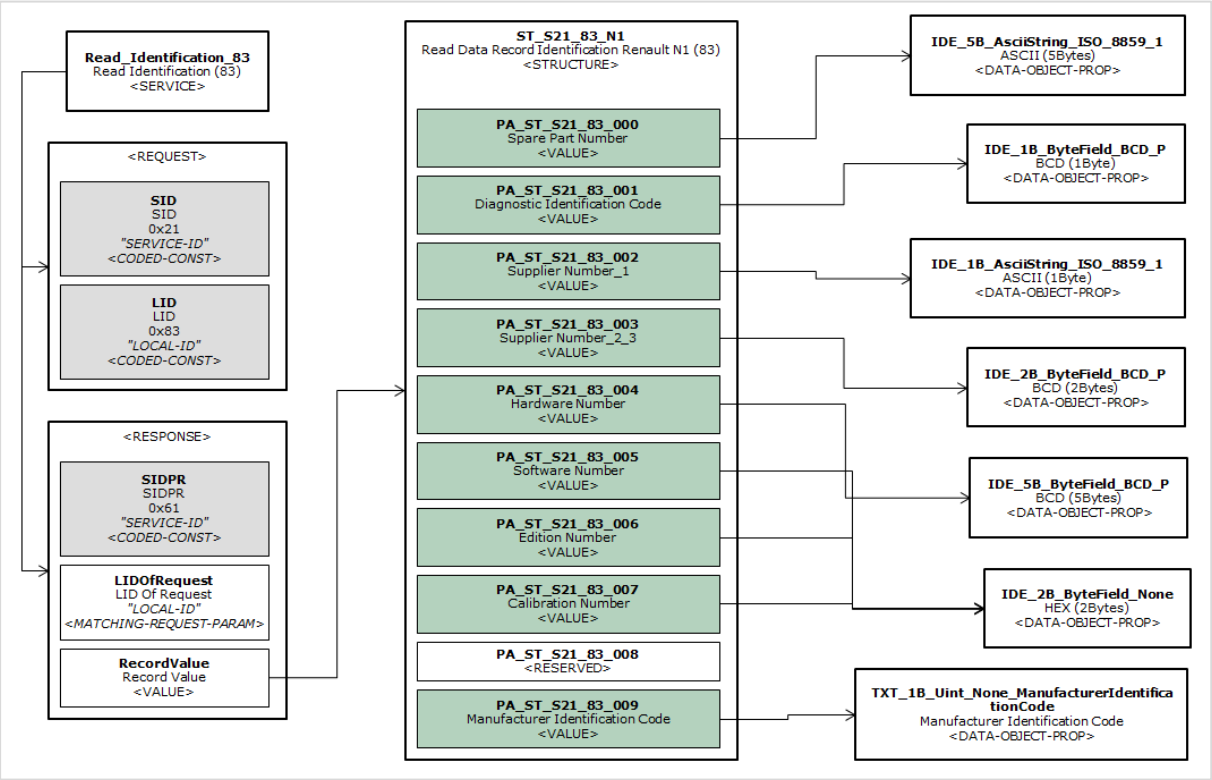


Figure 52 Structure of B/C DIAG-SERVICE Read_Identification_83 ($2183)

Implementation requirements:

[Rule] The SHORT-NAME of the DIAG-SERVICE shall be “Read_Identification_83”.

[Rule] The LONG-NAME of the DIAG-SERVICE shall be “Read Identification (83)”.

[Rule] The DIAG-SERVICEs shall be marked as IS-FINAL=”true”.

[Rule] The SEMANTIC attribute of the DIAG-SERVICEs shall be set to “IDENTIFICATION”.

[Rule] The DIAGNOSTIC-CLASS attribute shall be set to “VARIANTIDENTIFICATION”.

[Rule] The DIAG-SERVICEs shall reference the FUNCT-CLASS "Identification".

[Rule] The positive response of the DIAG-SERVICE shall reference the STRUCTURE "ST_21_83_N1" (used for interpretation) by ODX-LINK.

[Rule] The author of ODX data shall not mark this service as "not inherited" on ECU-VARIANT-LAYER level.

2.3.6.6 ReadDataByIdentifier ($22)

2.3.6.6.1 General Read Service

Service $22 is predefined in the Renault Nissan Spec B/C ESD available from RFQ packages. There are two definitions of $22, ReadDataByIdentifier and ReadDataByIdentifier_MultiId. The former is intended to be used when describing ECUs which only support reading out a single DID at a time, the latter when describing ECUs which support reading out multiple DIDs at once. Both are marked final – ECU designers and suppliers shall not override or redefine these services in lower layers.

Both services make use of the predefined TABLE "TA_S22" to specify both the data identifier(s) in request and response and the content of the associated data record(s). The TABLE in the Renault Nissan Spec B/C ESD encompasses the set of readable DIDs predefined by the RFQ package (of which Figure 53 merely shows a small subset).

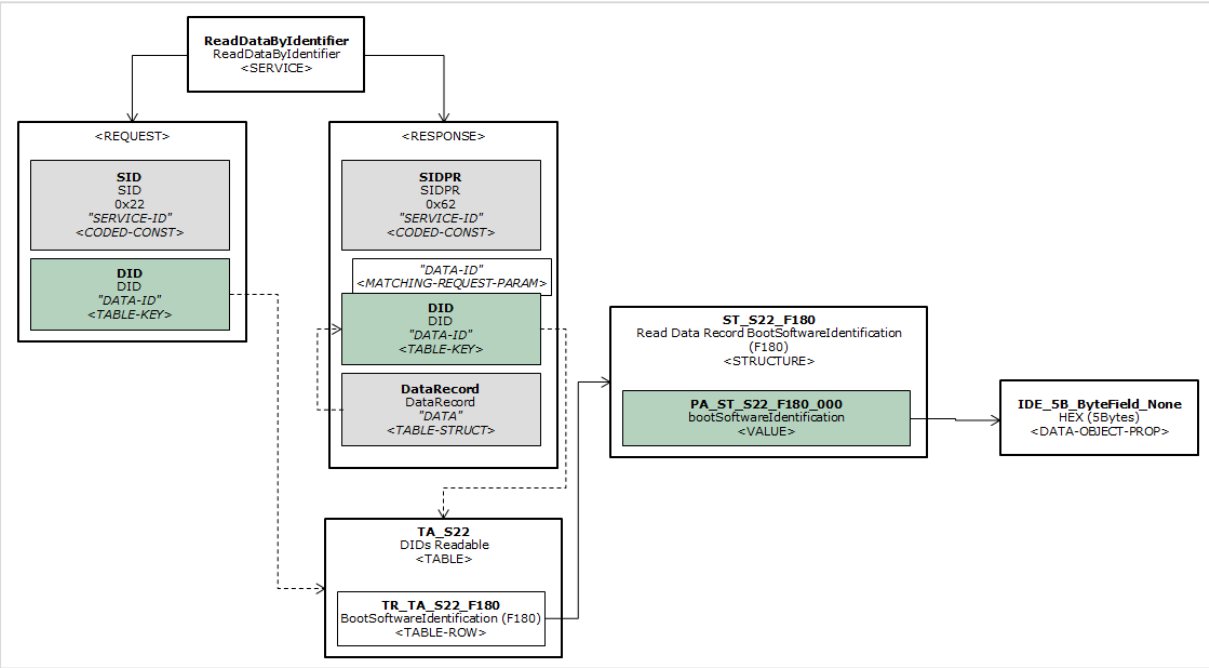


Figure 53 Structure of B/C DIAG-SERVICE ReadDataByIdentifier ($22)

ECU designers can override TABLE "TA_S22" to tailor this predefined set of readable DIDs for ECUs which support a different set of DIDs. Please refer to to section 0 for detailed tailoring rules (and a general overview of how DIDs can be defined and reused in lower layers).

Implementation requirements:

[Rule] The SHORT- and LONG_NAME of the two DIAG-SERVICEs representing B/C service $22 shall be "ReadDataByIdentifier" and "ReadDataByIdentifier_MultiId".

[Rule] The DIAG-SERVICEs shall be marked as IS-FINAL="true".

[Rule] The SEMANTIC attribute of the DIAG-SERVICEs shall be set to "STOREDDATAREAD".

[Rule] The DIAGNOSTIC-CLASS attribute shall not be used.

[Rule] The DIAG-SERVICEs shall reference the FUNCT-CLASS "DIDs".

[Rule] Both the request and the positive response of the DIAG-SERVICEs shall reference the TABLE "TA_S22" dynamically by SNREF.

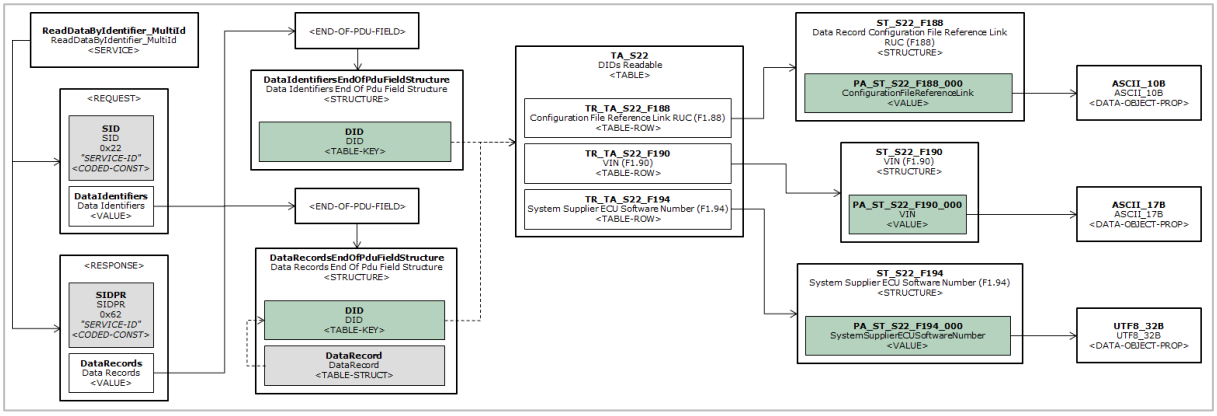


Figure 54 Structure of B/C DIAG-SERVICE ReadDataByIdentifier_MultiId ($22)

2.3.6.6.2 Support Check

For checking of supported readable two-byte DIDs, one DIAG-SERVICE "SupportedReadableDIDs" (using service $22) is predefined in the Renault Nissan Spec B/C ESD available from RFQ packages.

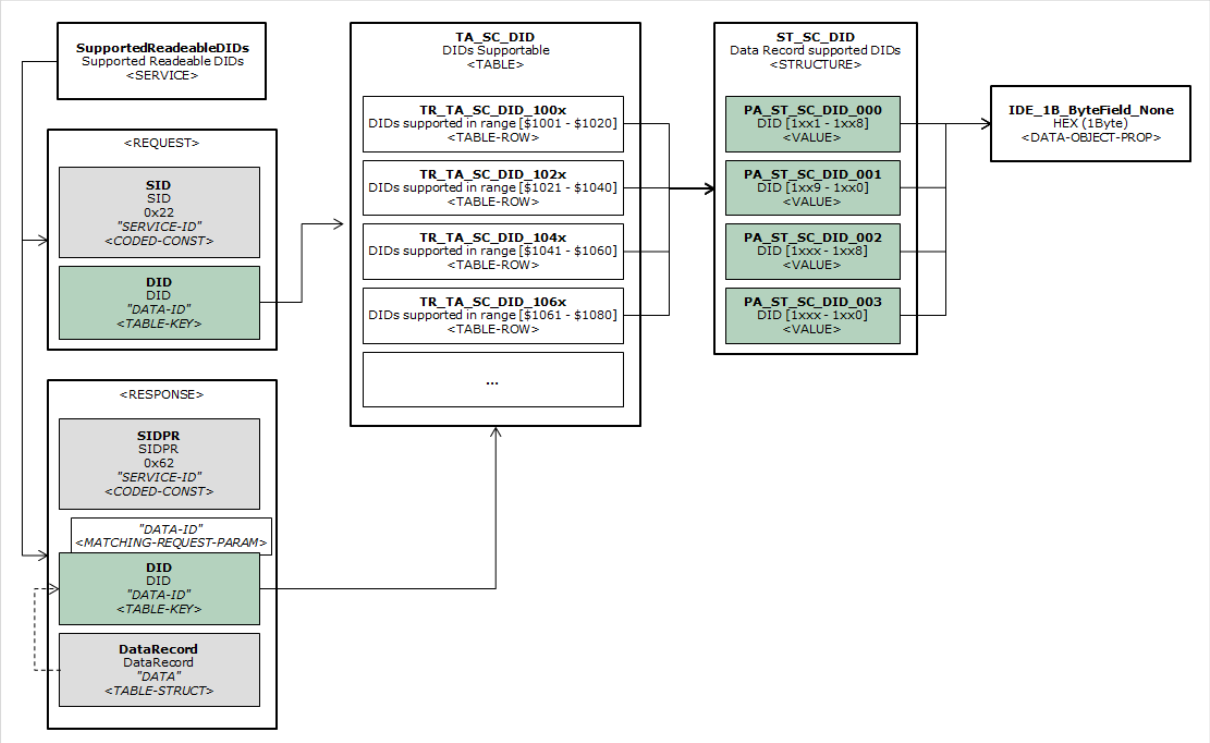


Figure 55 Structure of B/C DIAG-SERVICE SupportedReadableDIDs ($22)

Implementation requirements:

[Rule] The SHORT- and LONG-NAME of the DIAG-SERVICE shall be "SupportedReadableDIDs".

[Rule] The DIAG-SERVICEs shall be marked as IS-FINAL="true".

[Rule] The SEMANTIC attribute of the DIAG-SERVICEs shall be set to "STOREDDATAREAD".

[Rule] The DIAGNOSTIC-CLASS attribute shall not be used.

[Rule] The DIAG-SERVICEs shall reference the FUNCT-CLASS "Support Check".

[Rule] Both the request and the positive response of the DIAG-SERVICEs shall reference the TABLE "TA_SC_DID" statically. The TABLE shall not be overridden on variant layers.

[Rule] The service shall be marked as NOT-INHERITED on variant layers if the support check is not implemented in the described ECU.

2.3.6.7 WriteDataByIdentifier ($2E)

Service $2E is predefined once in the Renault Nissan Spec B/C ESD available from RFQ packages. It is marked final – ECU designers and suppliers shall not override or redefine this service in lower layers.

The service makes use of the predefined TABLE "TA_S2E" to specify both the data identifier in request and response and the content of the associated data record. The TABLE in the Renault Nissan Spec UDS ESD encompasses the set of writable DIDs predefined by the RFQ package (of which Figure 21 merely shows a small subset).

ECU designers can override TABLE "TA_S2E" to tailor this predefined set of writable DIDs for ECUs which support a different set of DIDs. Please refer to to section 0 for detailed tailoring rules (and a general overview of how DIDs can be defined and reused in lower layers).

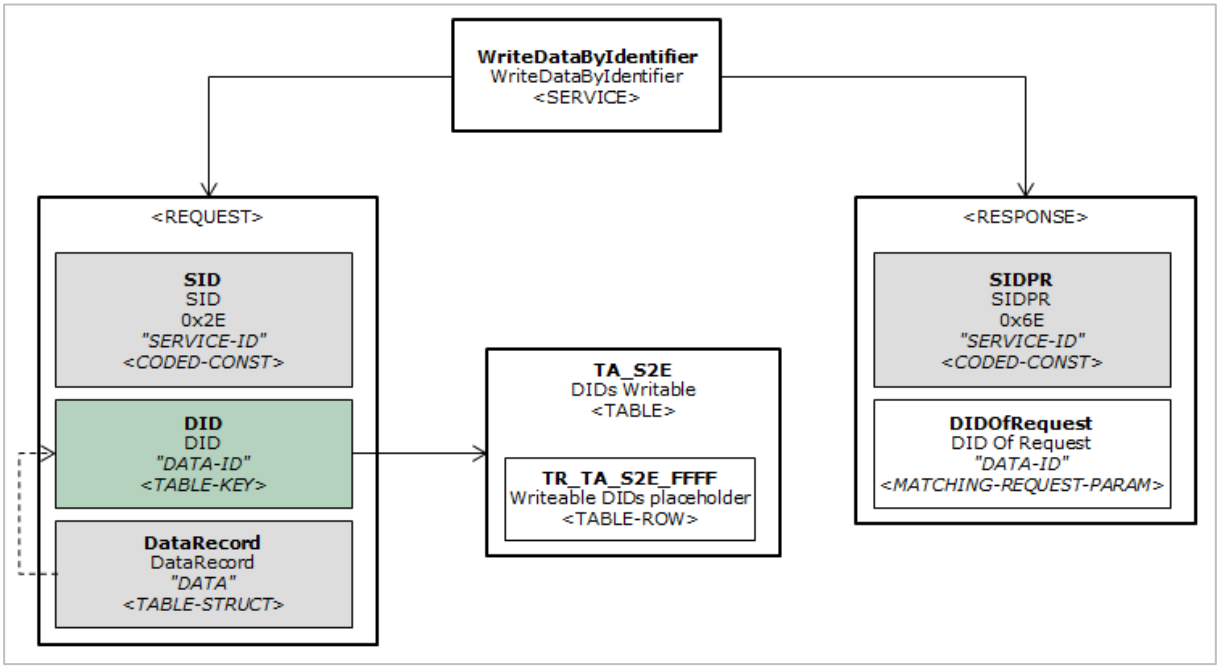


Figure 56 Structure of B/C DIAG-SERVICE WriteDataByIdentifier ($2E)

Implementation requirements:

[Rule] The SHORT- and LONG_NAME of the DIAG-SERVICE shall be "WriteDataByIdentifier".

[Rule] The DIAG-SERVICE shall be marked as IS-FINAL="true".

[Rule] The SEMANTIC attribute of the DIAG-SERVICE shall be set to "STOREDDATAWRITE".

[Rule] The DIAGNOSTIC-CLASS attribute shall not be used.

[Rule] The DIAG-SERVICE shall reference the FUNCT-CLASS "DIDs".

[Rule] The request of the DIAG-SERVICE shall reference the TABLE "TA_S2E" dynamically by SNREF.

2.3.6.8 WriteDataByLocalIdentifier ($3B)

2.3.6.8.1 General Write Service

Service $3B is predefined once in the Renault Nissan Spec B/C ESD available from RFQ packages. It is marked final – ECU designers and suppliers shall not override or redefine this service in lower layers.

The service makes use of the predefined TABLE "TA_S3B" to specify both the local identifier in request and response and the content of the associated data record. The TABLE in the Renault Nissan Spec B/C ESD encompasses the set of writable LIDs predefined by the RFQ package (of which Figure 57 merely shows a small subset).

ECU designers can override TABLE "TA_S3B" to tailor this predefined set of writable LIDs for ECUs which support a different set of LIDs. Please refer to to section 2.3.4.3 for detailed tailoring rules (and a general overview of how LIDs can be defined and reused in lower layers).

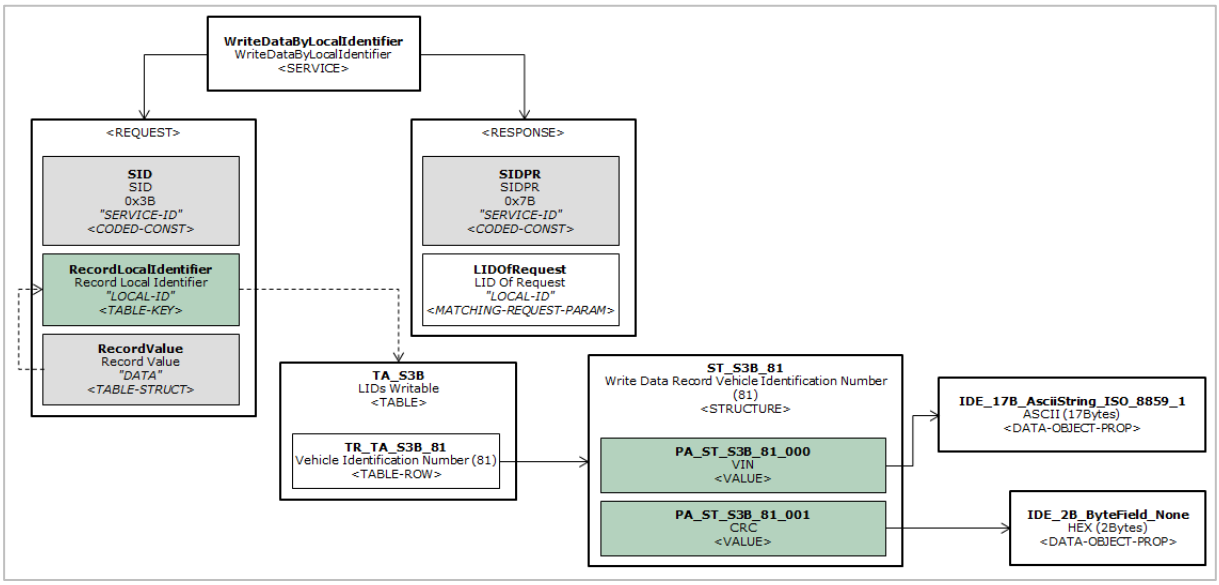


Figure 57 Structure of B/C DIAG-SERVICE WriteDataByLocalIdentifier ($3B)

Implementation requirements:

[Rule] The SHORT- and LONG-NAME of the DIAG-SERVICE shall be "WriteDataByLocalIdentifier".

[Rule] The DIAG-SERVICE shall be marked as IS-FINAL="true".

[Rule] The SEMANTIC attribute of the DIAG-SERVICE shall be set to "STOREDDATAWRITE".

[Rule] The DIAGNOSTIC-CLASS attribute shall not be used.

[Rule] The DIAG-SERVICE shall reference the FUNCT-CLASS "LIDs".

[Rule] The request of the DIAG-SERVICE shall reference the TABLE "TA_S3B" dynamically by SNREF.

2.3.6.8.2 Support Check

For checking of supported writeable one-byte LIDs, one DIAG-SERVICE "SupportedWriteableLIDs" (using service $3B) is predefined in the Renault Nissan Spec B/C ESD available from RFQ packages.

Implementation requirements:

[Rule] The SHORT- and LONG-NAME of the DIAG-SERVICE shall be "SupportedWriteableLIDs".

[Rule] The DIAG-SERVICEs shall be marked as IS-FINAL="true".

[Rule] The SEMANTIC attribute of the DIAG-SERVICEs shall be set to “STOREDDATAWRITE”.

[Rule] The DIAGNOSTIC-CLASS attribute shall not be used.

[Rule] The DIAG-SERVICEs shall reference the FUNCT-CLASS “Support Check”.

[Rule] Both the request and the positive response of the DIAG-SERVICEs shall reference the TABLE “TA_SC_LID” statically. The TABLE shall not be overridden on variant layers.

[Rule] The service shall be marked as NOT-INHERITED on variant layers if the support check is not implemented in the described ECU.

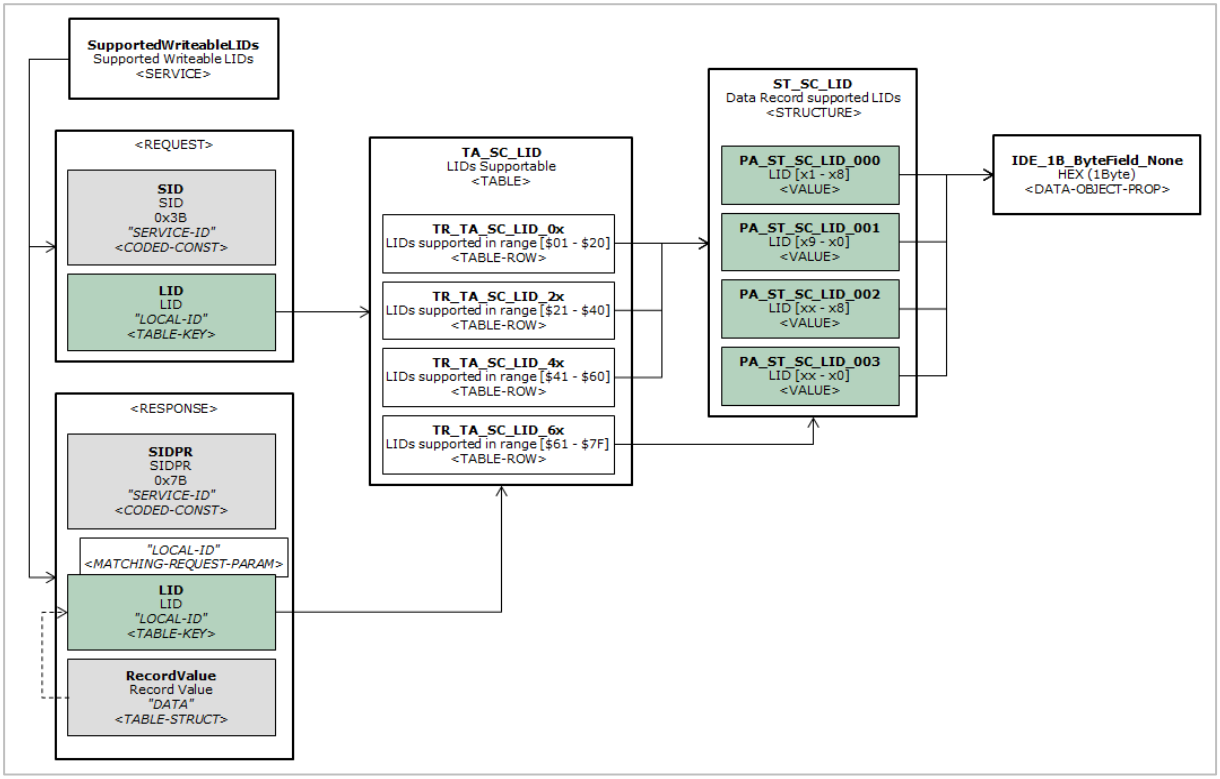


Figure 58 Structure of B/C DIAG-SERVICE SupportedWritebleLIDs ($3B)

2.3.6.9 InputOutputControlByLocalIdentifier ($30)

As Spec B/C I/O Controls might support different ControlOptions at Renault and Nissan, it is required to define each ControlOption with its own DIAG-SERVICE.

The following ControlOptions exits:

- Start Temporary Control ($00)
- Start Permanent Control ($20)
- Status Request ($01)
- Stop Control ($11)

The supplier specific affectation IO control parameters $FA to $FE defined by Renault / Nissan Spec B/C may be used but are not supported by the associated IO control TABLE structures predefined in the Spec B/C ESD distributed with RFQ packages. For ECUs which implement such IO control parameters, according flat services shall be defined on variant layer.

In contrast to this, the “otherStatus” bytes optionally available with IO control services as defined by Spec B/C are supported via the TABLE structures defined by the Spec B/C ESD. However, Renault / Nissan assumes that for ECUs which actually making use of these “otherStatus” bytes, the structure of these bytes does not depend on the IO control status parameter returned by the positive response of

the IO control service. For ECUs where such a dependency exists, flat services shall be defined on variant layer to model this fact.

Finally, if a flat service is used to model the support for a particular LID, then that very same LID shall not also be defined in the IO control TABLE structures predefined by the Spec B/C ESD.

2.3.6.9.1 StartTemporaryControl ($00)

Service $30 in B/C using the ControlOption 0x00 ("StartTemporaryControl") is predefined once in the Renault Nissan Spec B/C ESD available from RFQ packages. It is marked final – ECU designers and suppliers shall not override or redefine this service in lower layers.

The service makes use of the predefined TABLEs TA_S30_00 (IO Control – Temporary) and "TA_S30_PR" to specify both, the I/O's data record in the response and the content of the I/O in the request.

ECU designers can override both TABLEs: "TA_S30_00" (IO Control – Temporary) and "TA_S30_RP" to tailor the set of I/O Controls. Please refer to section 2.3.4.5 for detailed tailoring rules (and a general overview of how I/O Controls can be defined).

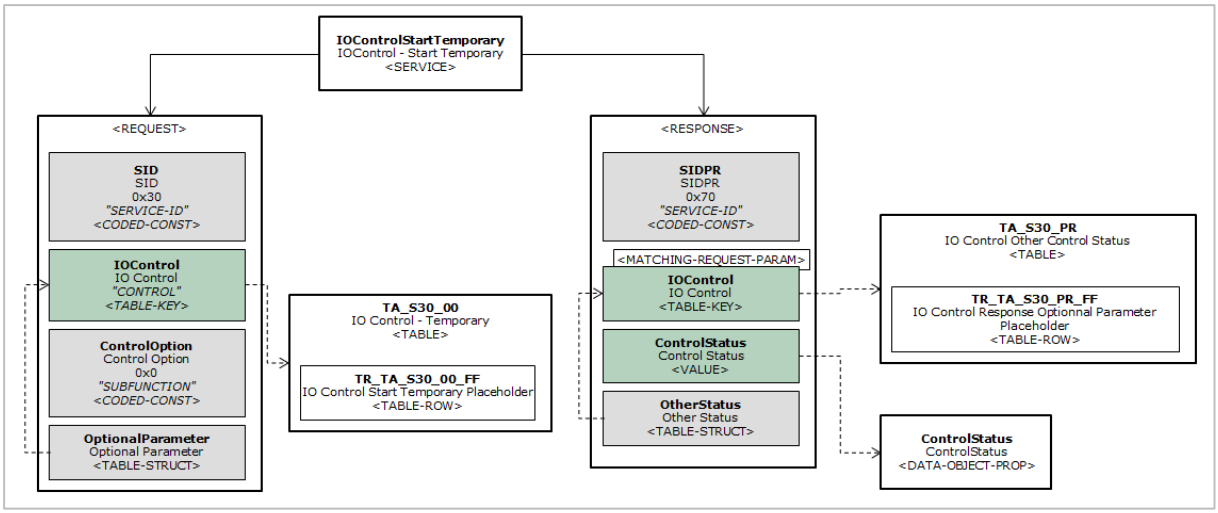


Figure 59 Structure of B/C DIAG-SERVICE IOControlStartTemporary ($30)

Implementation requirements:

[Rule] The SHORT-NAME of the DIAG-SERVICE shall be "IOControlStartTemporary".

[Rule] The LONG-NAME of the DIAG-SERVICE shall be "IOControl - Start Temporary".

[Rule] The DIAG-SERVICE shall be marked as IS-FINAL="true".

[Rule] The SEMANTIC attribute of the DIAG-SERVICE shall be set to "CONTROL".

[Rule] The DIAGNOSTIC-CLASS attribute shall not be used.

[Rule] The DIAG-SERVICE shall reference the FUNCT-CLASS "I_O_Controls".

[Rule] The request of the DIAG-SERVICE shall reference the TABLE "TA_S30_00" dynamically by SNREF.

[Rule] The response of the DIAG-SERVICE shall reference the TABLE "TA_S30_PR" dynamically by SNREF.

[Rule] The response of the DIAG-SERVICE shall reference the DATA-OBJECT-PROP "ControlStatus" by SNREF.

2.3.6.9.2 StartPermanentControl ($20)

Service $30 in B/C using the ControlOption 0x20 ("StartPermanentControl") is predefined once in the Renault Nissan Spec B/C ESD available from RFQ packages. It is marked final – ECU designers and suppliers shall not override or redefine this service in lower layers.

The service makes use of the predefined TABLEs TA_S30_20 (IO Control – Permanent) and "TA_S30_PR" to specify both, the I/O's data record in the response and the content of the I/O in the request.

ECU designers can override both TABLEs: "TA_S30_20" (IO Control – Temporary) and "TA_S30_PR" to tailor the set of I/O Controls. Please refer to section 2.3.4.5 for detailed tailoring rules (and a general overview of how I/O Controls can be defined).

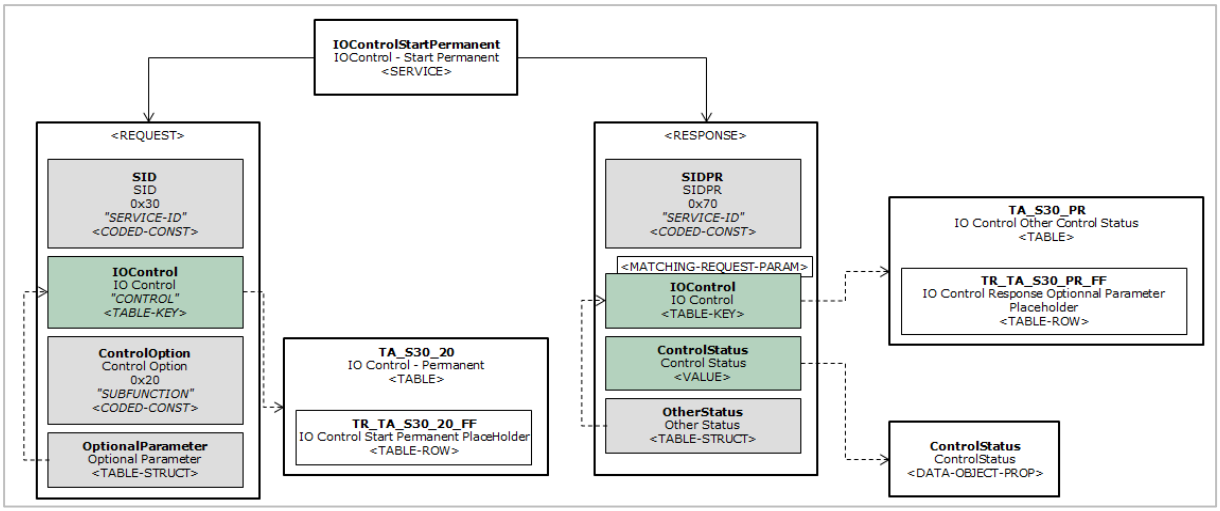


Figure 60 Structure of B/C DIAG-SERVICE IOControlStartPermanent ($30)

Implementation requirements:

[Rule] The SHORT-NAME of the DIAG-SERVICE shall be "IOControlStartPermanent".

[Rule] The LONG-NAME of the DIAG-SERVICE shall be "IOControl - Start Permanent".

[Rule] The DIAG-SERVICE shall be marked as IS-FINAL="true".

[Rule] The SEMANTIC attribute of the DIAG-SERVICE shall be set to "CONTROL".

[Rule] The DIAGNOSTIC-CLASS attribute shall not be used.

[Rule] The DIAG-SERVICE shall reference the FUNCT-CLASS "I_O_Controls".

[Rule] The request of the DIAG-SERVICE shall reference the TABLE "TA_S30_20" dynamically by SNREF.

[Rule] The response of the DIAG-SERVICE shall reference the TABLE "TA_S30_PR" dynamically by SNREF.

[Rule] The response of the DIAG-SERVICE shall reference the DATA-OBJECT-PROP "ControlStatus" by SNREF.

2.3.6.9.3 StatusRequest ($01)

Service $30 in B/C using the ControlOption 0x01 ("StatusRequest") is predefined once in the Renault Nissan Spec B/C ESD available from RFQ packages. It is marked final – ECU designers and suppliers shall not override or redefine this service in lower layers.

The service makes use of the predefined TABLEs TA_S30_01 (IO Control – Status) and "TA_S30_PR" to specify both, the I/O's data record in the response and the content of the I/O in the request.

ECU designers can override both TABLEs: "TA_S30_01" (IO Control – Status) and "TA_S30_PR" to tailor the set of I/O Controls. Please refer to section 2.3.4.5 for detailed tailoring rules (and a general overview of how I/O Controls can be defined).

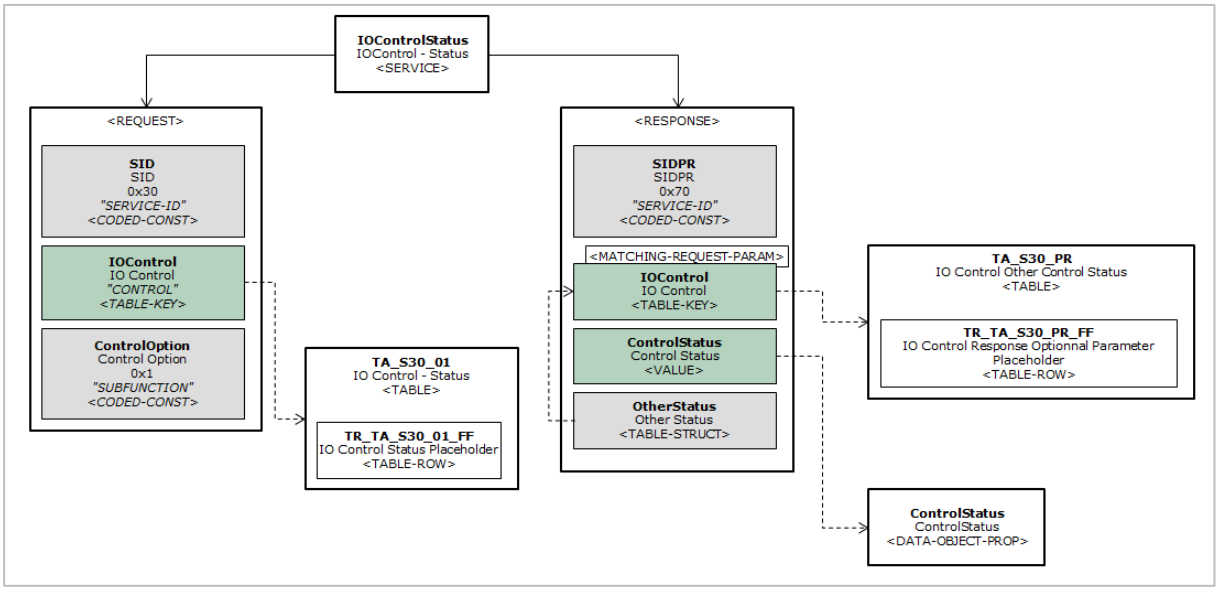


Figure 61 Structure of B/C DIAG-SERVICE IOControlStatus ($30)

Implementation requirements:

[Rule] The SHORT-NAME of the DIAG-SERVICE shall be "IOControlStatus".

[Rule] The LONG-NAME of the DIAG-SERVICE shall be "IOControl - Status".

[Rule] The DIAG-SERVICE shall be marked as IS-FINAL="true".

[Rule] The SEMANTIC attribute of the DIAG-SERVICE shall be set to "CONTROL".

[Rule] The DIAGNOSTIC-CLASS attribute shall not be used.

[Rule] The DIAG-SERVICE shall reference the FUNCT-CLASS "I_O_Controls".

[Rule] The request of the DIAG-SERVICE shall reference the TABLE "TA_S30_01" dynamically by SNREF.

[Rule] The response of the DIAG-SERVICE shall reference the TABLE "TA_S30_PR" dynamically by SNREF.

[Rule] The response of the DIAG-SERVICE shall reference the DATA-OBJECT-PROP "ControlStatus" by SNREF.

2.3.6.9.4 StopControl ($11)

Service $30 in B/C using the ControlOption 0x11 ("StopControl") is predefined once in the Renault Nissan Spec B/C ESD available from RFQ packages. It is marked final – ECU designers and suppliers shall not override or redefine this service in lower layers.

The service makes use of the predefined TABLEs TA_S30_11 (IO Control – Status) and "TA_S30_PR" to specify both, the I/O's data record in the response and the content of the I/O in the request.

ECU designers can override both TABLEs: "TA_S30_11" (IO Control – Status) and "TA_S30_PR" to tailor the set of I/O Controls. Please refer to section 2.3.4.5 for detailed tailoring rules (and a general overview of how I/O Controls can be defined).

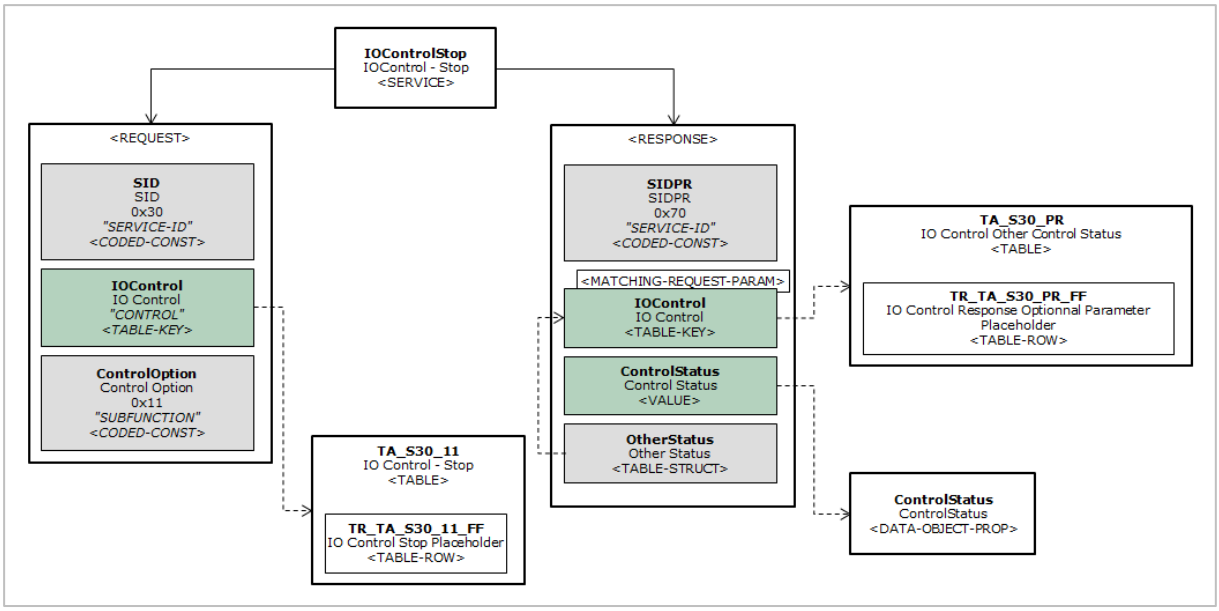


Figure 62 Structure of B/C DIAG-SERVICE IOControlStop ($30)

Implementation requirements:

[Rule] The SHORT-NAME of the DIAG-SERVICE shall be "IOControlStop".

[Rule] The LONG-NAME of the DIAG-SERVICE shall be "IOControl - Stop".

[Rule] The DIAG-SERVICE shall be marked as IS-FINAL="true".

[Rule] The SEMANTIC attribute of the DIAG-SERVICE shall be set to "CONTROL".

[Rule] The DIAGNOSTIC-CLASS attribute shall not be used.

[Rule] The DIAG-SERVICE shall reference the FUNCT-CLASS "I_O_Controls".

[Rule] The request of the DIAG-SERVICE shall reference the TABLE "TA_S30_11" dynamically by SNREF.

[Rule] The response of the DIAG-SERVICE shall reference the TABLE "TA_S30_PR" dynamically by SNREF.

[Rule] The response of the DIAG-SERVICE shall reference the DATA-OBJECT-PROP "ControlStatus" by SNREF.

2.3.6.9.5 Support Check

For checking of supported controllable one-byte I/O Control LIDs, one DIAG-SERVICE "SupportedIOControlableLIDs" (using service $30) is predefined in the Renault Nissan Spec B/C ESD available from RFQ packages.

Implementation requirements:

[Rule] The SHORT- and LONG-NAME of the DIAG-SERVICE shall be "SupportedIOControlable-LIDs".

[Rule] The DIAG-SERVICEs shall be marked as IS-FINAL="true".

[Rule] The SEMANTIC attribute of the DIAG-SERVICEs shall be set to "CONTROL".

[Rule] The DIAGNOSTIC-CLASS attribute shall not be used.

[Rule] The DIAG-SERVICEs shall reference the FUNCT-CLASS "Support Check".

[Rule] Both the request and the positive response of the DIAG-SERVICEs shall reference the TABLE "TA_SC_LID" statically. The TABLE shall not be overridden on variant layers.

[Rule] The service shall be marked as NOT-INHERITED on variant layers if the support check is not implemented in the described ECU.

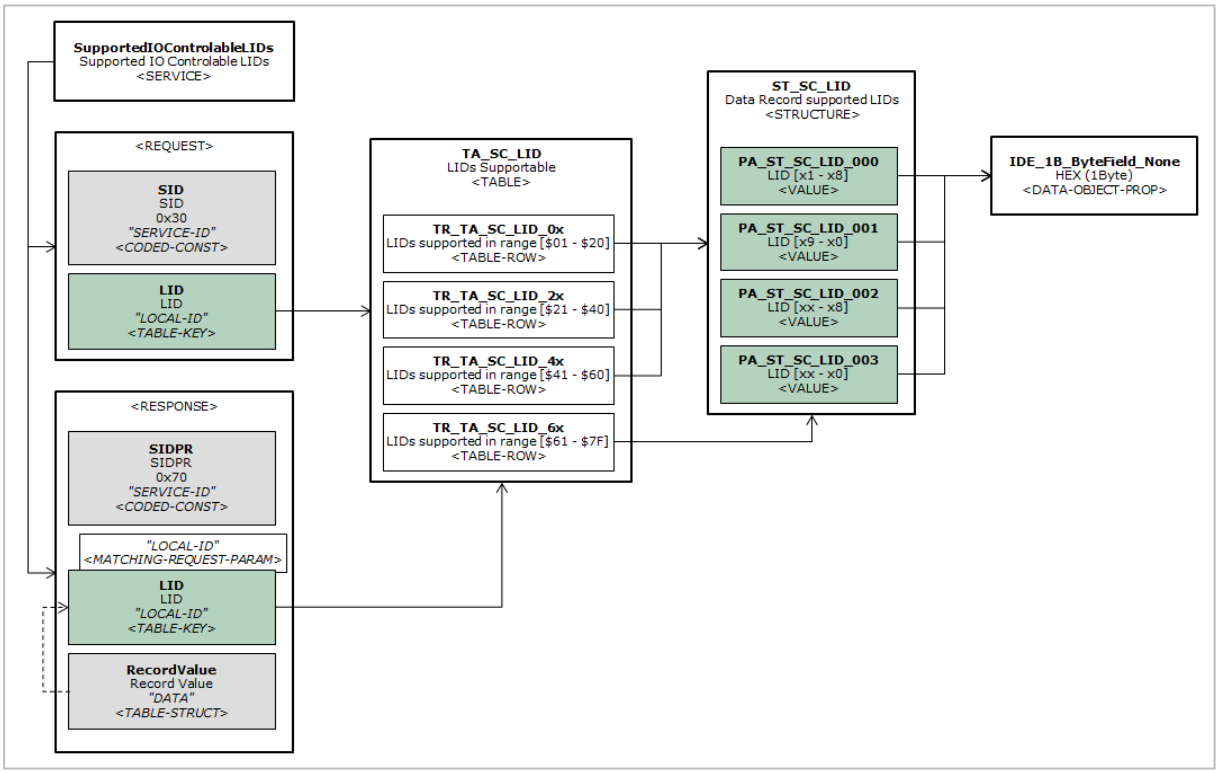


Figure 63 Structure of B/C DIAG-SERVICE SupportedIOControlableLIDs ($30)

2.3.6.10 ReadDtcInformation ($19)

The different sub functions of service $19 reference different TABLEs, TEXTTABLEs and other ODX elements for the definition of the associated fault data (such as DTCs, snapshots etc.) and the different interrelations between that data. Please cf. also section 2.3.4.6 for an overview focusing on those referenced data structures.

2.3.6.10.1 ReportDtcByStatusMask (subfunction 0x02)

Service $1902 ("ReadDtcInformation – ReportDtcByStatusMask") is predefined once in the Renault Nissan Spec B/C ESD available from RFQ packages. It is marked final – ECU designers and suppliers shall not override or redefine this service in lower layers.

As shown in Figure 64, the definition of service $1902 makes use of the following ODX elements defining associated fault memory data:

- The STRUCTURE with SHORT-NAME "DtcStatusMask" which defines the individual DTC status bits used both in the request and the positive response.
- The STRUCTURE with SHORT-NAME "DtcStatusAvailabilityMask" which defines the DTC status bits returned in the positive response reflecting which status bits are supported by the ECU in the first place.
- The DTC-DOP with SHORT-NAME "DtcRecord" specifying the possible DTCs stored by in the ECU's fault memory.

These elements are predefined at RFQ level in the ESD "Renault Nissan Spec B/C". The latter two however are referenced via SHORT-NAME and are intended to be overridden on BASE-VARIANT level.

For the STRUCTURE "DtcStatusAvailabilityMask" this may be done optionally. The DTC-DOP with SHORT-NAME "DtcRecord" however must be overridden at BASE-VARIANT level. On RFQ level, it only

contains a placeholder for the actual DTC data. Please refer to section 2.3.4.6 for a description of the detailed rules applicable for overriding this DTC-DOP and for defining DTCs of an ECU.

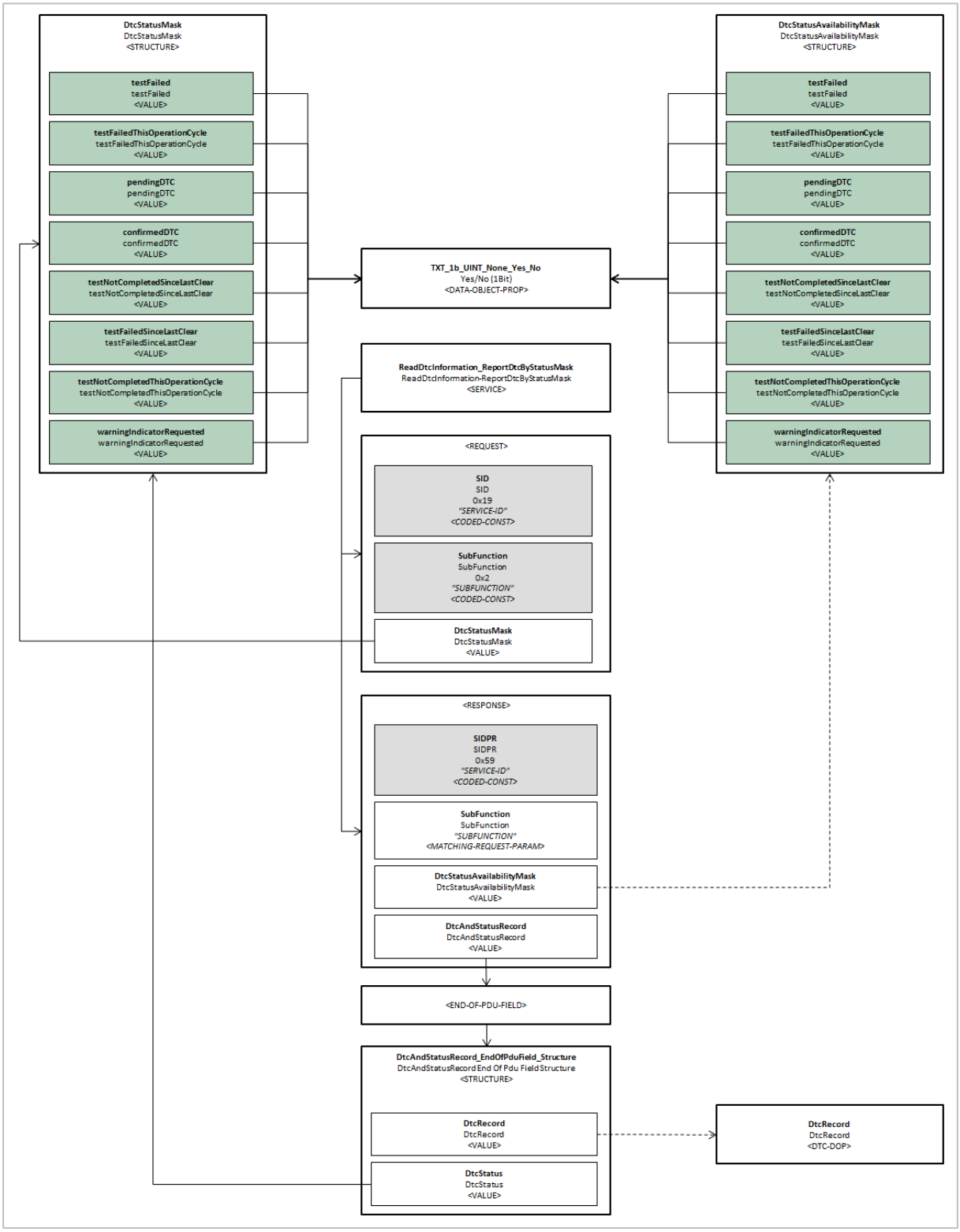


Figure 64 B/C ReadDtcInformation – ReportDtcByStatusMask ($1902)

Implementation requirements:

[Rule] The SHORT-NAME of the DIAG-SERVICE shall be "ReadDtcInformation_ReportDtc-ByStatusMask".

[Rule] The LONG-NAME of the DIAG-SERVICE shall be "ReadDtcInformation-ReportDtcByStatus-Mask".

[Rule] The DIAG-SERVICE shall be marked as IS-FINAL="true".

[Rule] The SEMANTIC attribute of the DIAG-SERVICE shall be set to "FAULTREAD".

[Rule] The DIAGNOSTIC-CLASS attribute shall not be used.

[Rule] The DIAG-SERVICE shall reference the FUNCT-CLASS "Fault_Memory".

[Rule] The positive response of the DIAG-SERVICE shall reference the STRUCTURE with SHORT-NAME "DtcStatusAvailabilityMask" dynamically by SNREF.

[Rule] The positive response of the DIAG-SERVICE shall reference the DTC-DOP "DtcRecord" dynamically by SNREF.

2.3.6.10.2 ReportDtcSnapshotRecordByDtcNumber (subfunction 0x04)

Service $1904 ("ReadDtcInformation – ReportDtcSnapshotRecordByDtcNumber") is predefined once in the Renault Nissan Spec B/C ESD available from RFQ packages. It is marked final – ECU designers and suppliers shall not override or redefine this service in lower layers.

As shown in Figure 65, the definition of service $1904 makes use of the following ODX elements defining associated fault memory data:

- The STRUCTURE with SHORT-NAME "DtcStatusMask" which defines the individual DTC status bits used in the positive response.
- The DTC-DOP with SHORT-NAME "DtcRecord" specifying the possible DTCs stored by in the ECU's fault memory which can be queried in the request of $1904.
- The DTC-DOP with SHORT-NAME "Devices" specifying the different devices implemented in the ECU for which DTCs might be stored (and their possible base DTC values).
- The TEXTTABLE with SHORT-NAME "FailureTypes" specifying the possible fault types for which DTCs might be stored (and their possible failure type bytes).
- The TEXTTABLE with SHORT-NAME "DtcSnapshotRecordNumbersRQ" defining the possible snapshot record number values which may be specified in the request of service $1904.
- The TEXTTABLE with SHORT-NAME "DtcSnapshotRecordNumbersPR" defining the possible snapshot record number values returned in the positive response of service $1904.
- The ENV-DATA-DESC with SHORT-NAME "SnapshotRecordContents" defining the different snapshot records which might be stored by the described ECU.

These elements are predefined at RFQ level in the ESD "Renault Nissan Spec B/C". Only the first one is not referenced via SHORT-NAME. All other elements must be overridden on variant level to define the devices, failure types, DTCs and snapshot records and snapshot record numbers actually supported by the described ECUs in the context of service $1904 (for on RFQ level, these elements contain placeholder data only). Exempt from this is the TEXTTABLE with SHORT-NAME "FailureTypes": It may be overridden on variant level, but it is not required to do so, since this TEXTTABLE contains valid FailureTypes according to ISO15031-6 on the RFQ level already. Please refer to section 2.3.4.6 for a description of the detailed rules applicable for overriding these elements and for defining the associated fault memory data.

Implementation requirements:

[Rule] The SHORT-NAME of the DIAG-SERVICE shall be "ReadDtcInformation_ReportDtcSnapshotRecordByDtcNumber".

[Rule] The LONG-NAME of the DIAG-SERVICE shall be "ReadDtcInformation-ReportDtcSnapshotRecordByDtcNumber".

[Rule] The DIAG-SERVICE shall be marked as IS-FINAL="true".

[Rule] The SEMANTIC attribute of the DIAG-SERVICE shall be set to "FAULTREAD".

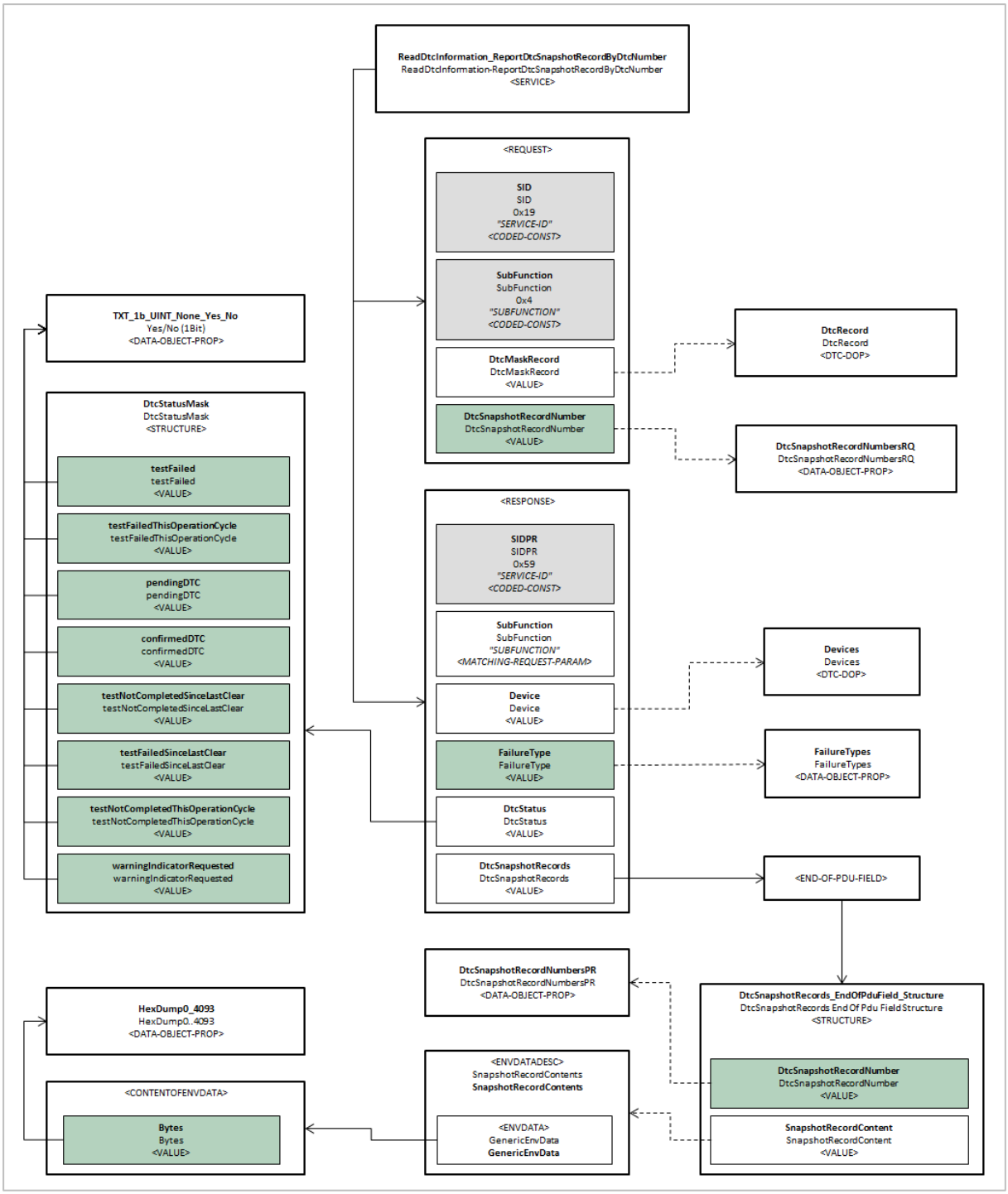


Figure 65 B/C ReadDtcInformation – ReportDtcSnapshotRecordByDtcNumber ($1904)

[Rule] The DIAGNOSTIC-CLASS attribute shall not be used.

[Rule] The DIAG-SERVICE shall reference the FUNCT-CLASS “Fault_Memory”.

[Rule] The request of the DIAG-SERVICE shall reference the DTC-DOP with SHORT-NAME “DtcRecord” dynamically by SNREF.

[Rule] The request of the DIAG-SERVICE shall reference the TEXTTABLE with SHORT-NAME “DtcSnapshotRecordNumbersRQ” dynamically by SNREF.

[Rule] The positive response of the DIAG-SERVICE shall reference the TEXTTABLE with SHORT-NAME “DtcSnapshotRecordNumbersPR” dynamically by SNREF.

[Rule] The positive response of the DIAG-SERVICE shall reference the DTC-DOP with SHORT-NAME “Devices” dynamically by SNREF.

[Rule] The positive response of the DIAG-SERVICE shall reference the TEXTTABLE with SHORT-NAME “FailureTypes” dynamically by SNREF.

[Rule] The positive response of the DIAG-SERVICE shall reference the ENV-DATA-DESC with SHORT-NAME “SnapshotRecordContents” dynamically by SNREF.

2.3.6.10.3 ReportDtcExtDataRecordByDtcNumber (subfunction 0x06)

Service $1906 (“ReadDtcInformation – ReportDtcExtDataRecordByDtcNumber”) is predefined once in the Renault Nissan Spec B/C ESD available from RFQ packages. It is marked final – ECU designers and suppliers shall not override or redefine this service in lower layers.

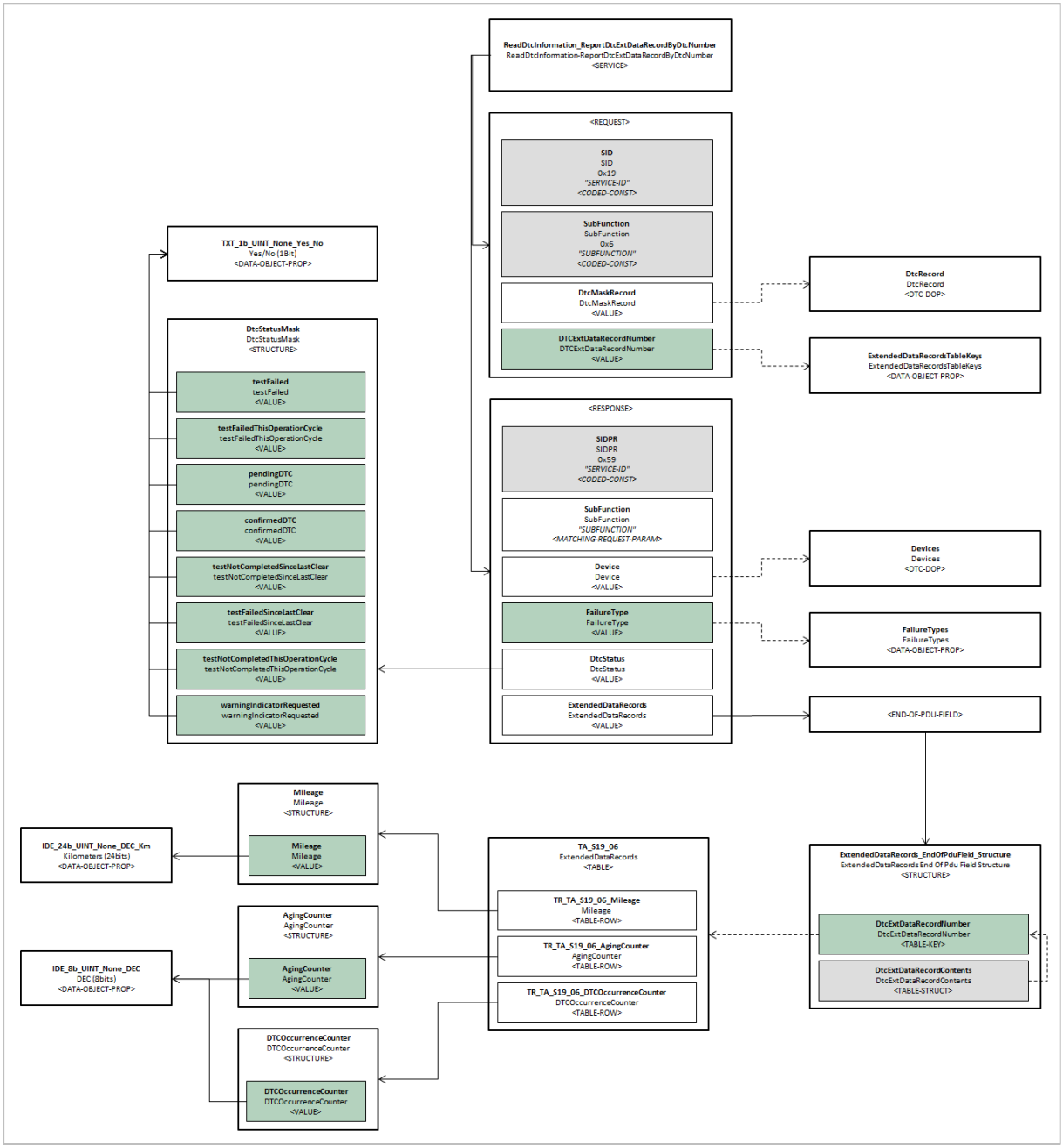


Figure 66 B/C ReadDtcInformation – ReportDtcExtDataRecordByDtcNumber ($1906)

As shown in Figure 66, the definition of service $1906 makes use of the following ODX elements defining associated fault memory data:

- The STRUCTURE with SHORT-NAME "DtcStatusMask" which defines the individual DTC status bits used in the positive response.
- The DTC-DOP with SHORT-NAME "DtcRecord" specifying the possible DTCs stored by in the ECU's fault memory which can be queried in the request of $1906.
- The DTC-DOP with SHORT-NAME "Devices" specifying the different devices implemented in the ECU for which DTCs might be stored (and their possible base DTC values).
- The TEXTTABLE with SHORT-NAME "FailureTypes" specifying the possible fault types for which DTCs might be stored (and their possible failure type bytes).
- The DOP with SHORT-NAME "ExtendedDataRecordsTableKeys" defining the possible extended data record number values which may be specified in the request of service $1906.
- The TABLE with SHORT-NAME "ExtendedDataRecords" defining the possible extended data record number values returned in the positive response of service $1906.

These elements are predefined at RFQ level in the ESD "Renault Nissan Spec B/C". Only the first one is not referenced via SHORT-NAME. All other elements must be overridden on variant level to define the devices, failure types, DTCs and extended data records and extended data record numbers supported by the described ECUs in the context of service $1906 (for on RFQ level, these elements contain placeholder data only). Exempt from this is the TEXTTABLE with SHORT-NAME "FailureTypes": It may be overridden on variant level, but it is not required to do so, since this TEXTTABLE contains valid FailureTypes according to ISO15031-6 on the RFQ level already.

Please refer to section 2.3.4.6 for a description of the detailed rules applicable for overriding these elements and for defining the associated fault memory data.

Implementation requirements:

[Rule] The SHORT-NAME of the DIAG-SERVICE shall be "ReadDtcInformation_ReportDtcExtDataRecordByDtcNumber".

[Rule] The LONG-NAME of the DIAG-SERVICE shall be "ReadDtcInformation-ReportDtcExtDataRecordByDtcNumber".

[Rule] The DIAG-SERVICE shall be marked as IS-FINAL="true".

[Rule] The SEMANTIC attribute of the DIAG-SERVICE shall be set to "FAULTREAD".

[Rule] The DIAGNOSTIC-CLASS attribute shall not be used.

[Rule] The DIAG-SERVICE shall reference the FUNCT-CLASS "Fault_Memory".

[Rule] The request of the DIAG-SERVICE shall reference the DTC-DOP with SHORT-NAME "DtcRecord" dynamically by SNREF.

[Rule] The request of the DIAG-SERVICE shall reference the DOP with SHORT-NAME "ExtendedDataRecordsTableKeys" dynamically by SNREF.

[Rule] The positive response of the DIAG-SERVICE shall reference the DTC-DOP with SHORT-NAME "Devices" dynamically by SNREF.

[Rule] The positive response of the DIAG-SERVICE shall reference the TEXTTABLE with SHORT-NAME "FailureTypes" dynamically by SNREF.

[Rule] The positive response of the DIAG-SERVICE shall reference the TABLE with SHORT-NAME "ExtendedDataRecords" dynamically by SNREF.

2.3.6.11 ClearAllDiagnosticInformation ($14FFFFFF)

Service $14FFFFFF ("ClearAllDiagnosticInformation") is predefined once in the Renault Nissan Spec B/C ESD available from RFQ packages. It is marked final – ECU designers and suppliers shall not override or redefine this service in lower layers.

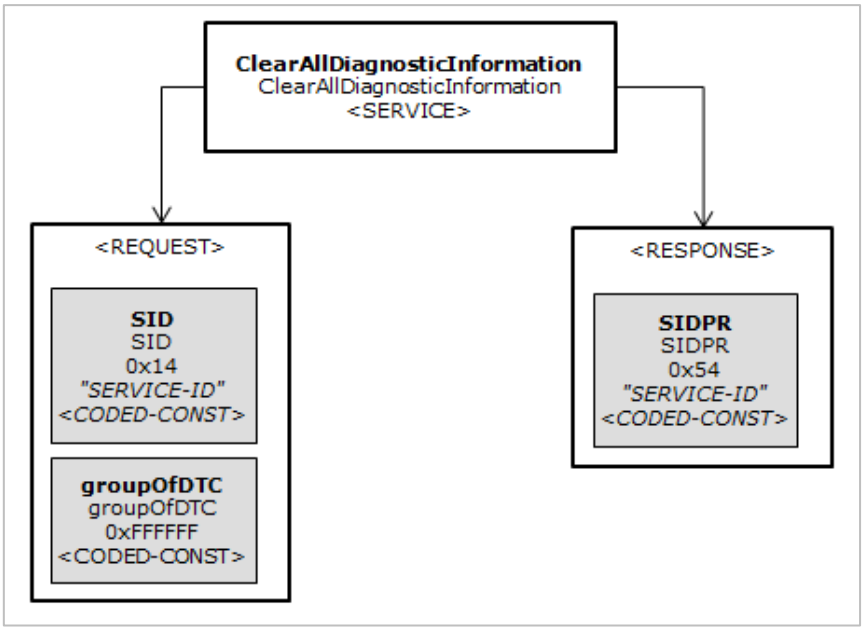


Figure 67 Structure of B/C DIAG-SERVICE ClearAllDiagnosticInformation ($14)

Implementation requirements:

[Rule] The SHORT-NAME of the DIAG-SERVICE shall be "ClearAllDiagnosticInformation".

[Rule] The LONG-NAME of the DIAG-SERVICE shall be "ClearAllDiagnosticInformation".

[Rule] The DIAG-SERVICE shall be marked as IS-FINAL="true".

[Rule] The SEMANTIC attribute of the DIAG-SERVICE shall be set to "FAULTCLEAR".

[Rule] The DIAGNOSTIC-CLASS attribute shall not be used.

[Rule] The DIAG-SERVICE shall reference the FUNCT-CLASS "Fault_Memory".

2.3.6.12 StartRoutineByLocalIdentifier ($31)

2.3.6.12.1 General Start Routine Service

Service $31 ("StartRoutineByLocalIdentifier") is predefined once in the Renault Nissan Spec B/C ESD available from RFQ packages. It is marked final – ECU designers and suppliers shall not override or redefine this service in lower layers.

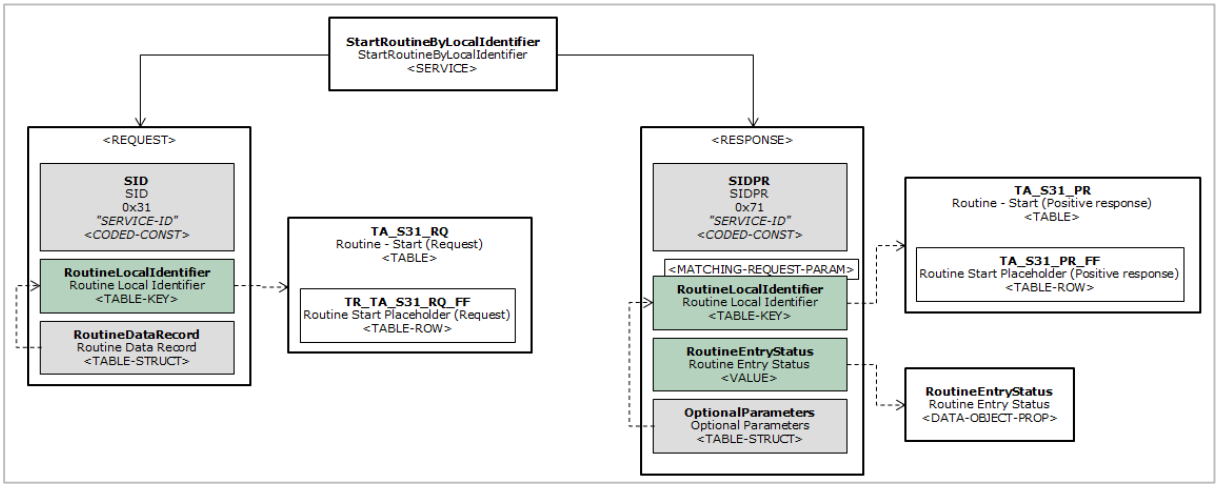


Figure 68 Structure of B/C DIAG-SERVICE StartRoutineByLocalIdentifier (0x31)

Implementation requirements:

[Rule] The SHORT- and LONG-NAME of the DIAG-SERVICE shall be "StartRoutineByLocalIdentifier".

[Rule] The DIAG-SERVICE shall be marked as IS-FINAL="true".

[Rule] The SEMANTIC attribute of the DIAG-SERVICE shall be set to "ROUTINE".

[Rule] The DIAGNOSTIC-CLASS attribute shall not be used.

[Rule] The DIAG-SERVICE shall reference the FUNCT-CLASS "Routine_Control".

[Rule] The request of the DIAG-SERVICE shall reference the TABLE with SHORT-NAME "TA_S31_RQ" dynamically by SNREF.

[Rule] The positive response of the DIAG-SERVICE shall reference the TABLE with SHORT-NAME "TA_S31_PR" dynamically by SNREF.

2.3.6.12.2 Support Check

For checking of supported one-byte routine controllable LIDs, one DIAG-SERVICE "SupportedRoutine-ControlableLIDs" (using service $31) is predefined in the Renault Nissan Spec B/C ESD available from RFQ packages.

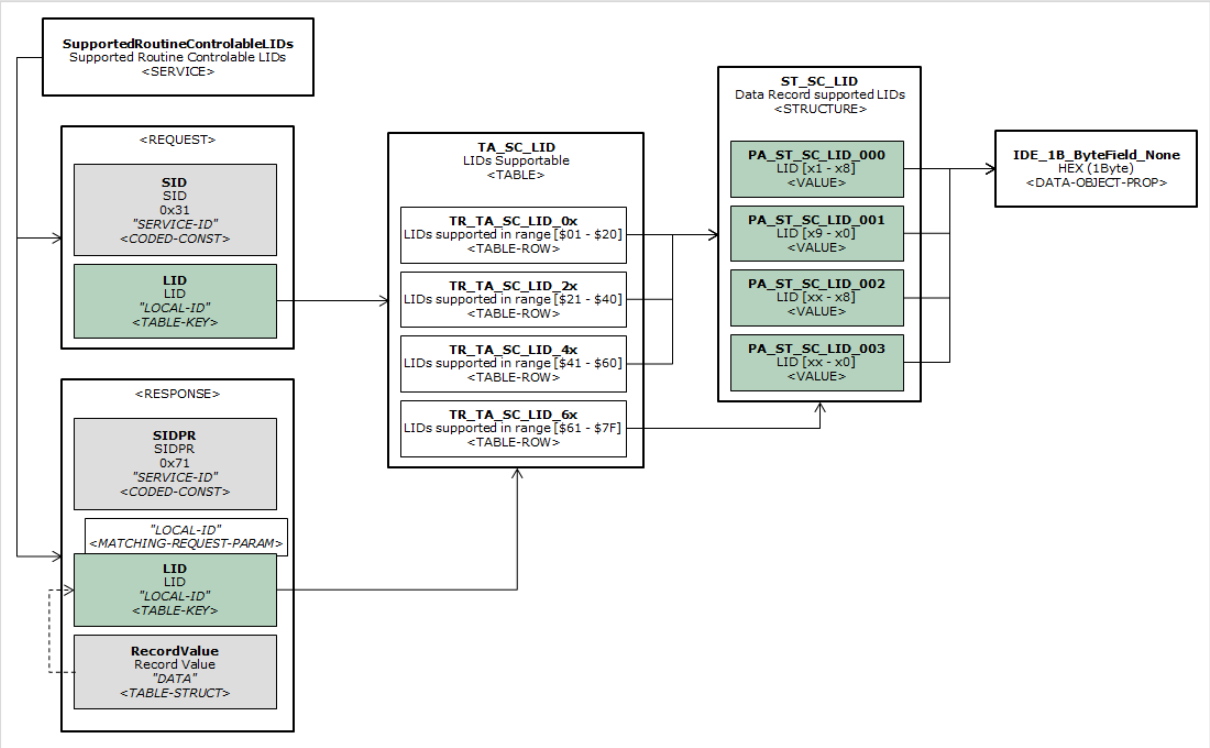


Figure 69 Structure of B/C DIAG-SERVICE SupportedRoutineControlableLIDs ($31)

Implementation requirements:

[Rule] The SHORT- and LONG-NAME of the DIAG-SERVICE shall be "SupportedRoutine-ControlableLIDs".

[Rule] The DIAG-SERVICEs shall be marked as IS-FINAL="true".

[Rule] The SEMANTIC attribute of the DIAG-SERVICEs shall be set to "ROUTINE".

[Rule] The DIAGNOSTIC-CLASS attribute shall not be used.

[Rule] The DIAG-SERVICEs shall reference the FUNCT-CLASS “Support Check”.

[Rule] Both the request and the positive response of the DIAG-SERVICEs shall reference the TABLE “TA_SC_LID” statically. The TABLE shall not be overridden on variant layers.

[Rule] The service shall be marked as NOT-INHERITED on variant layers if the support check is not implemented in the described ECU.

2.3.6.13 StopRoutineByLocalIdentifier ($32)

Service $32 (“StopRoutineByLocalIdentifier”) is predefined once in the Renault Nissan Spec B/C ESD available from RFQ packages. It is marked final – ECU designers and suppliers shall not override or redefine this service in lower layers.

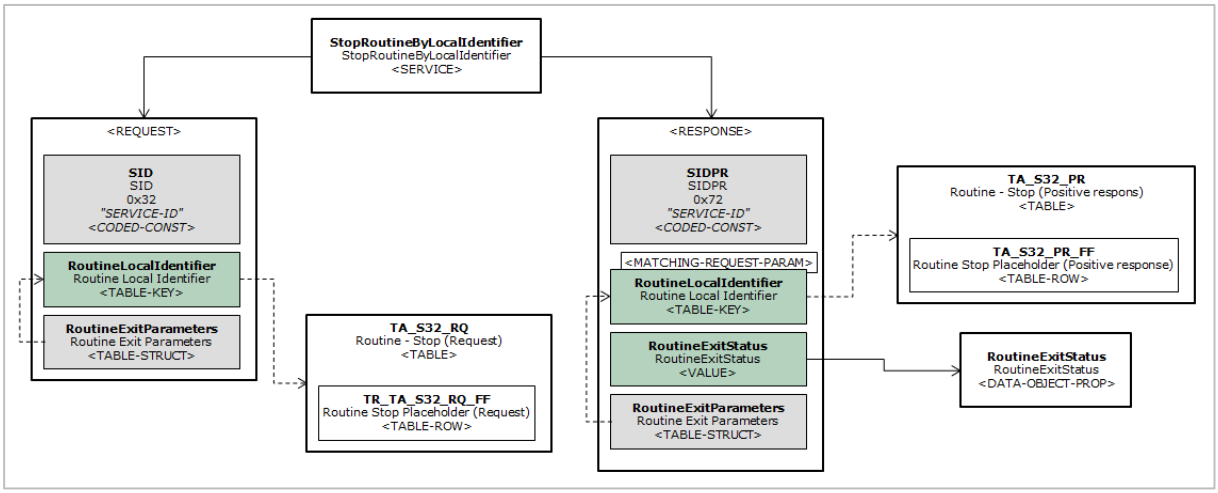


Figure 70 Structure of B/C DIAG-SERVICE StopRoutineByLocalIdentifier (0x32)

Implementation requirements:

[Rule] The SHORT-NAME of the DIAG-SERVICE shall be “StopRoutineByLocalIdentifier”.

[Rule] The LONG-NAME of the DIAG-SERVICE shall be “StopRoutineByLocalIdentifier”.

[Rule] The DIAG-SERVICE shall be marked as IS-FINAL=”true”.

[Rule] The SEMANTIC attribute of the DIAG-SERVICE shall be set to “ROUTINE”.

[Rule] The DIAGNOSTIC-CLASS attribute shall not be used.

[Rule] The DIAG-SERVICE shall reference the FUNCT-CLASS “Routine_Control”.

[Rule] The request of the DIAG-SERVICE shall reference the TABLE with SHORT-NAME “TA_S32_RQ” dynamically by SNREF.

[Rule] The positive response of the DIAG-SERVICE shall reference the TABLE with SHORT-NAME “TA_S32_PR” dynamically by SNREF.

2.3.6.14 SecurityAccess ($27)

In UDS as in B/C, requesting a seed and sending a key make use of services with slightly differing service specifications. Therefore, the two use cases are supported by two different DIAG-SERVICEs.

2.3.6.14.1 SecurityAccess – request seed

Service $27 to request a security access seed (using odd numbers as sub-function) is predefined once in the Renault Nissan Spec B/C ESD available from RFQ packages. It is marked final – ECU designers and suppliers shall not override or redefine this service in lower layers.

The service makes use of the predefined TABLE "TA_S27_RS" to specify security access related sub-function for the request. The TABLE in the Renault Nissan Spec B/C ESD encompasses the set of security access levels predefined by the RFQ package.

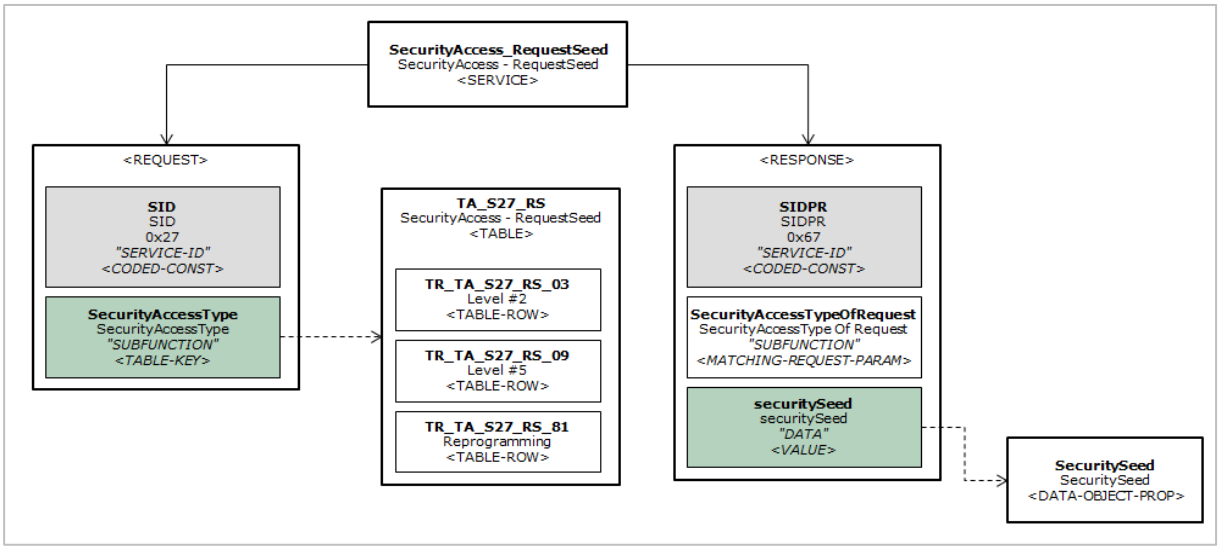


Figure 71 Structure of B/C DIAG-SERVICE SecurityAccess_requestSeed ($27)

ECU designers may override the TABLE "TA_S27_RS" to tailor this predefined set of security levels for ECUs which support a different set of levels but only as long as the overridden set is **in accordance with one of the security models predefined by Renault / Nissan**. Please refer to the section 2.3.4.9 for detailed tailoring rules.

The positive response of the DIAG-SERVICE defines a VALUE "securitySeed" which references the predefined DATA-OBJECT-PROP "SecuritySeed" for all security access request seed sub-functions. It is not possible to define different seed lengths for different request seed sub functions, but ECU designers may override the DATA-OBJECT-PROP "SecuritySeed" to specify a different seed length for all sub functions. The used seed length must thereby be **in accordance with one of the security models predefined by Renault / Nissan**.

Implementation requirements:

[Rule] The SHORT-NAME of the DIAG-SERVICE shall be "SecurityAccess_requestSeed".

[Rule] The LONG-NAME of the DIAG-SERVICE shall be "SecurityAccess - requestSeed".

[Rule] The DIAG-SERVICE shall be marked as IS-FINAL="true".

[Rule] The SEMANTIC attribute of the DIAG-SERVICE shall be set to "SECURITY".

[Rule] The DIAGNOSTIC-CLASS attribute shall not be used.

[Rule] The DIAG-SERVICE shall reference the FUNCT-CLASS "SecurityAccess".

[Rule] The request of the DIAG-SERVICE shall reference the TABLE "TA_S27_RS" dynamically by SNREF.

[Rule] The securitySeed param of the DIAG-SERVICE's POS-RESPONSE shall reference the DATA-OBJECT-PROP "SecuritySeed" by SNREF.

2.3.6.14.2 SecurityAccess – send key

Service $27 to send a security access key (using even numbers as sub-function) is predefined once in the Renault Nissan Spec B/C ESD available from RFQ packages. It is marked final – ECU designers and suppliers shall not override or redefine this service in lower layers.

The service makes use of the predefined TABLE "SecurityAccess_sendKey" to specify security access related sub-function for the request. The TABLE in the Renault Nissan Spec B/C ESD encompasses the set of security access levels predefined by the RFQ package.

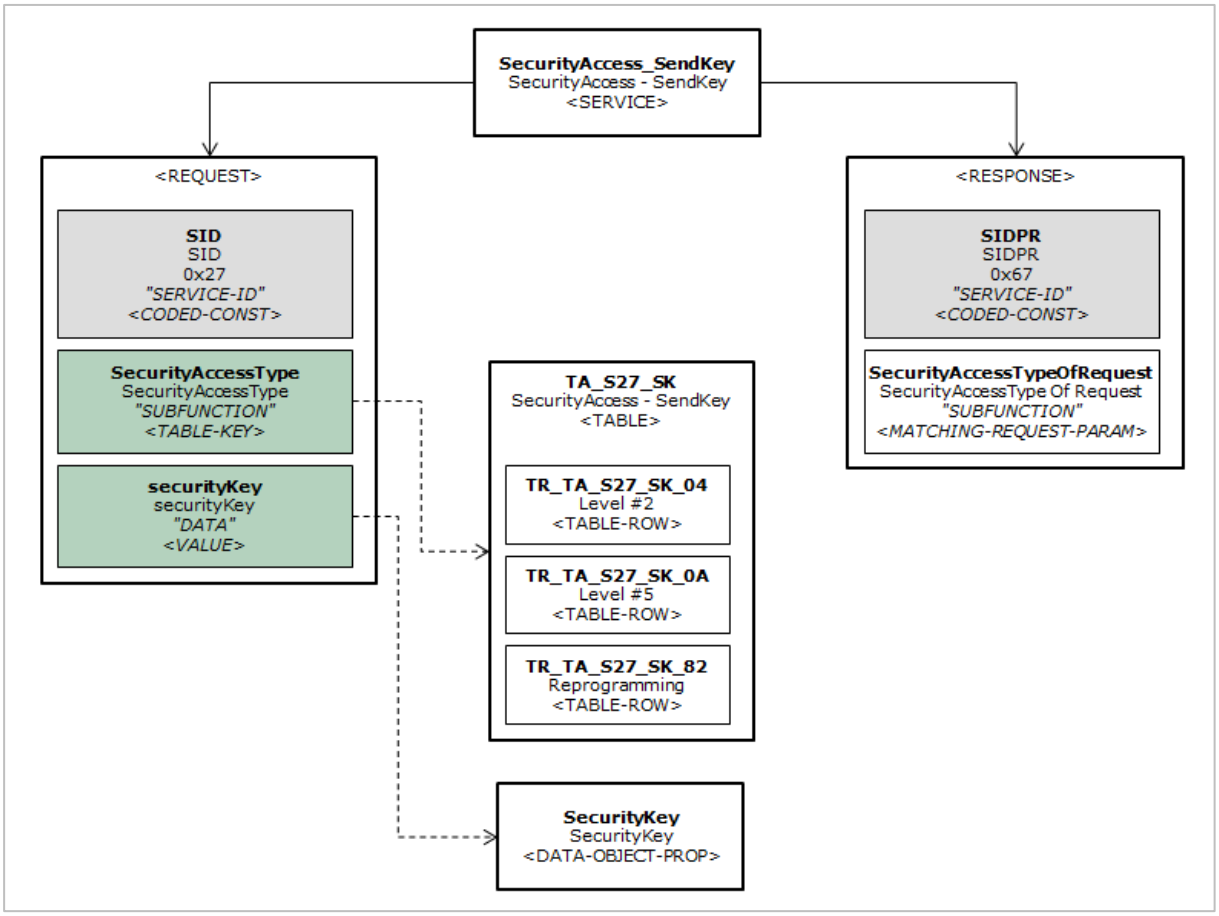


Figure 72 Structure of B/C DIAG-SERVICE SecurityAccess_sendKey ($27)

ECU designers may override the TABLE "TS_S27_SK" to tailor this predefined set of security levels for ECUs which support a different set of levels but only as long as the overridden set is **in accordance with one of the security models predefined by Renault / Nissan**. Please refer to the section 2.3.4.9 for detailed tailoring rules.

The request of the DIAG-SERVICE defines a VALUE "securityKey" which references the predefined DATA-OBJECT-PROP "SecurityKey" for all security access send key sub-functions. It is not possible to define different key lengths for different request seed sub-functions, but ECU designers may override the DATA-OBJECT-PROP "SecurityKey" to specify a different seed length for all sub functions. The used key length must be **in accordance with one of the security models predefined by Renault / Nissan**.

Implementation requirements:

[Rule] The SHORT-NAME of the DIAG-SERVICE shall be "SecurityAccess_sendKey".

[Rule] The LONG-NAME of the DIAG-SERVICE shall be "SecurityAccess - sendKey".

[Rule] The DIAG-SERVICE shall be marked as IS-FINAL="true".

[Rule] The SEMANTIC attribute of the DIAG-SERVICE shall be set to "SECURITY".

[Rule] The DIAGNOSTIC-CLASS attribute shall not be used.

[Rule] The DIAG-SERVICE shall reference the FUNCT-CLASS "SecurityAccess".

[Rule] The request of the DIAG-SERVICE shall reference the TABLE "TS_S27_SK" dynamically by SNREF.

2.3.6.15 EcuReset ($11)

Service $11 is predefined once in the Renault Nissan Spec B/C ESD available from RFQ packages. It is marked final – ECU designers and suppliers shall not override or redefine this service in lower layers.

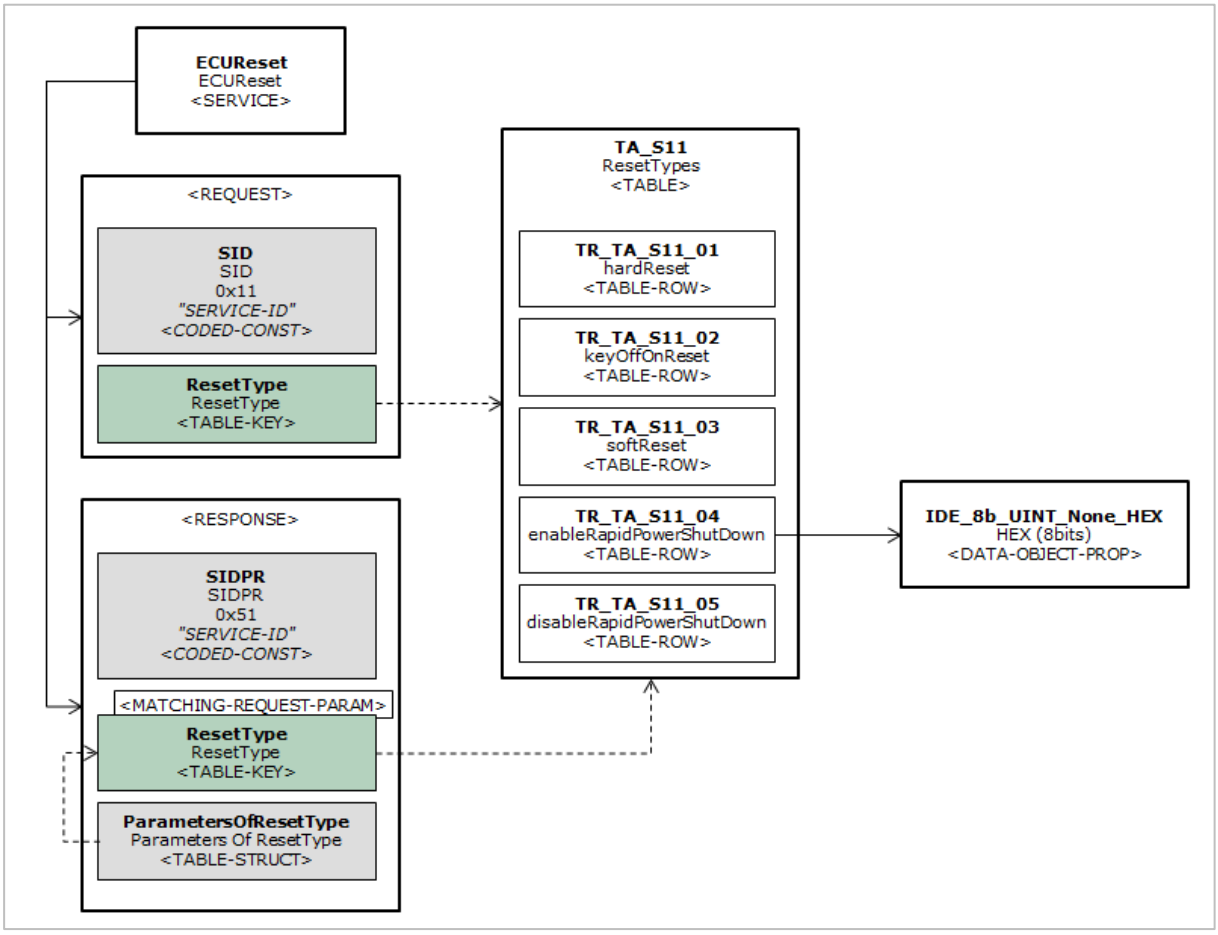


Figure 73 Structure of B/C DIAG-SERVICE ECUReset (0x11)

The service makes use of the predefined TABLE "TA_S11" to specify the supported reset types. The following reset types are thereby predefined:

- hardReset (0x01)
- keyOffOnReset (0x02)
- softReset (0x03)
- enableRapidPowerShutDown (0x04)
- disableRapidPowerShutDown (0x05)

ECU designers can override that TABLE to tailor these presets for ECUs which support a different set of reset types. Please refer to section 2.3.4.10 for the exact tailoring rules.

Implementation requirements:

[Rule] The SHORT- and LONG-NAME of the DIAG-SERVICE shall be "ECUReset".

[Rule] The SEMANTIC attribute of the DIAG-SERVICE shall be set to "FUNCTION"

[Rule] The DIAG-SERVICE shall not use the DIAGNOSTIC-CLASS attribute.

[Rule] It shall be marked as IS-FINAL="true".

[Rule] The DIAG-SERVICE shall reference the FUNCT-CLASS "ECUReset".

[Rule] The request and the response of the DIAG-SERVICE shall reference the TABLE "TS_S11" dynamically by SNREF.

2.3.6.16 TesterPresent ($3E)

Service $3E ("TesterPresent") is predefined once in the Renault Nissan Spec B/C ESD available from RFQ packages. It is marked final – ECU designers and suppliers shall not override or redefine this service in lower layers.

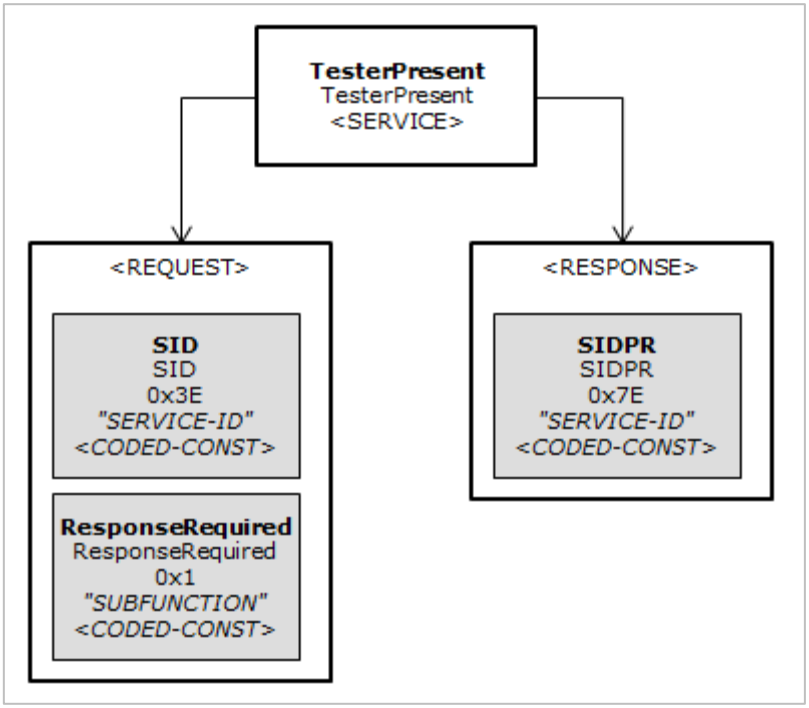


Figure 74 B/C DIAG-SERVICE TesterPresent ($3E)

Implementation requirements:

[Rule] The DIAG-SERVICE shall have the SHORT-/LONG-NAME "TesterPresent".

[Rule] The SEMANTIC attribute of the DIAG-SERVICE shall be set to "TESTERPRESENT".

[Rule] The DIAG-SERVICE shall not use the DIAGNOSTIC-CLASS attribute.

[Rule] The DIAG-SERVICE shall reference the FUNCT-CLASS "TesterPresent"

[Rule] It shall be marked as IS-FINAL="true".

[Rule] It shall be marked as IS-MANDATORY="true".

2.3.6.17 RequestDownload ($34)

Service $34 ("Request Download") is predefined once in the Renault Nissan Spec UDS ESD available from RFQ packages. It is marked final – ECU designers and suppliers shall not override or redefine this service in lower layers.

The ODX elements used by service $37 are shown in Figure 75.

Implementation requirements:

[Rule] The DIAG-SERVICE shall have the SHORT-NAME "RequestDownload".

[Rule] The DIAG-SERVICE shall have the LONG-NAME "Request Download".

[Rule] The DIAG-SERVICE shall be marked as IS-FINAL="true".

[Rule] The SEMANTIC attribute of the DIAG-SERVICE shall be set to "DOWNLOAD".

[Rule] The DIAGNOSTIC-CLASS attribute shall not be used.

[Rule] The DIAG-SERVICE shall reference the FUNCT-CLASS “Upload Download”.

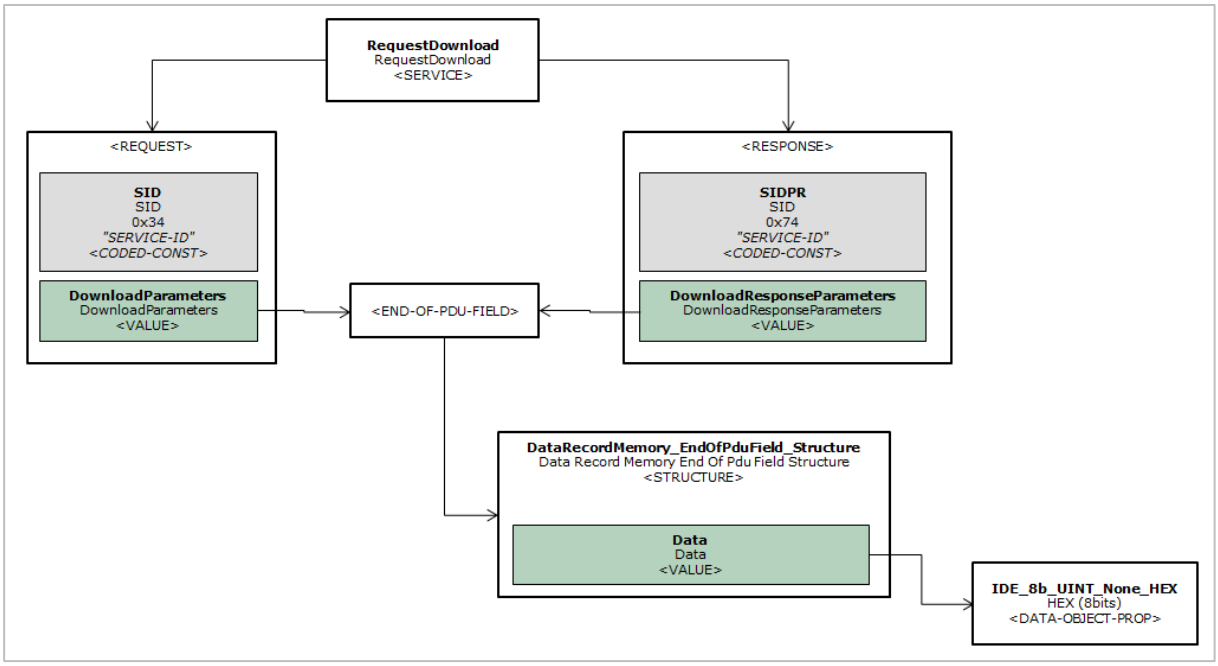


Figure 75 B/C DIAG-SERVICE RequestDownload ($34)

2.3.6.18 RequestUpload ($35)

Service $35 (“Request Upload”) is predefined once in the Renault Nissan Spec UDS ESD available from RFQ packages. It is marked final – ECU designers and suppliers shall not override or redefine this service in lower layers.

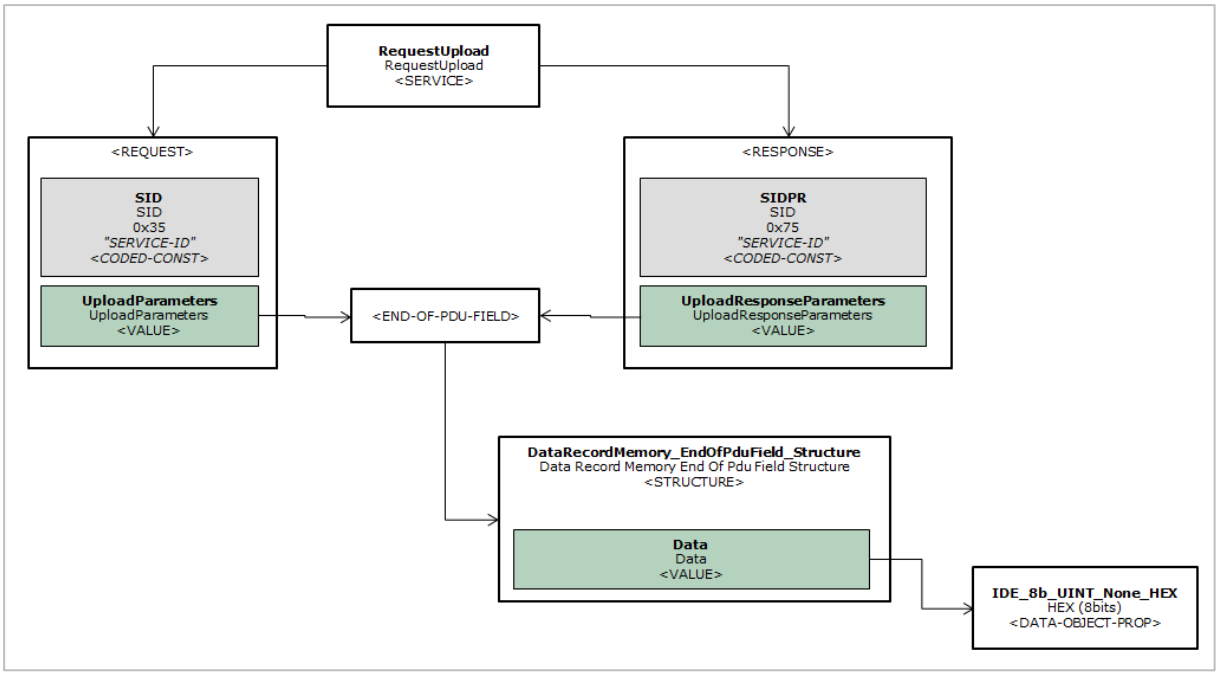


Figure 76 B/C DIAG-SERVICE RequestUpload ($35)

The ODX elements used by service $37 are shown in Figure 76.

Implementation requirements:

[Rule] The DIAG-SERVICE shall have the SHORT-NAME “RequestUpload”.

[Rule] The DIAG-SERVICE shall have the LONG-NAME “Request Upload”.

[Rule] The DIAG-SERVICE shall be marked as IS-FINAL=”true”.

[Rule] The SEMANTIC attribute of the DIAG-SERVICE shall be set to “UPLOAD”.

[Rule] The DIAGNOSTIC-CLASS attribute shall not be used.

[Rule] The DIAG-SERVICE shall reference the FUNCT-CLASS “Upload Download”.

2.3.6.19 TransferData ($36)

Service $36 (“Transfer Data”) is predefined once in the Renault Nissan Spec UDS ESD available from RFQ packages. It is marked final – ECU designers and suppliers shall not override or redefine this service in lower layers.

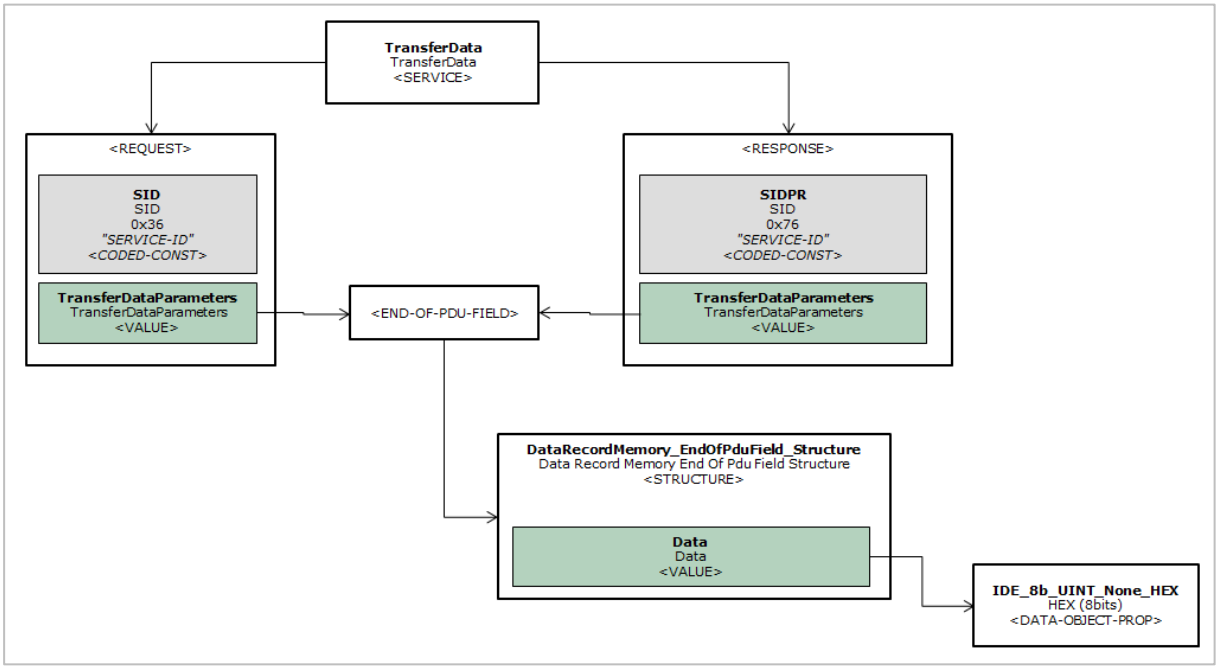


Figure 77 B/C DIAG-SERVICE TransferData ($36)

The ODX elements used by service $36 are shown in Figure 77.

Implementation requirements:

[Rule] The DIAG-SERVICE shall have the SHORT-NAME “TransferData”.

[Rule] The DIAG-SERVICE shall have the LONG-NAME “Transfer Data”.

[Rule] The DIAG-SERVICE shall be marked as IS-FINAL=”true”.

[Rule] The SEMANTIC attribute of the DIAG-SERVICE shall not be used.

[Rule] The DIAGNOSTIC-CLASS attribute shall not be used.

[Rule] The DIAG-SERVICE shall reference the FUNCT-CLASS “Upload Download”.

2.3.6.20 RequestTransferExit ($37)

Service $37 (“Request Transfer Exit”) is predefined once in the Renault Nissan Spec UDS ESD available from RFQ packages. It is marked final – ECU designers and suppliers shall not override or redefine this service in lower layers.

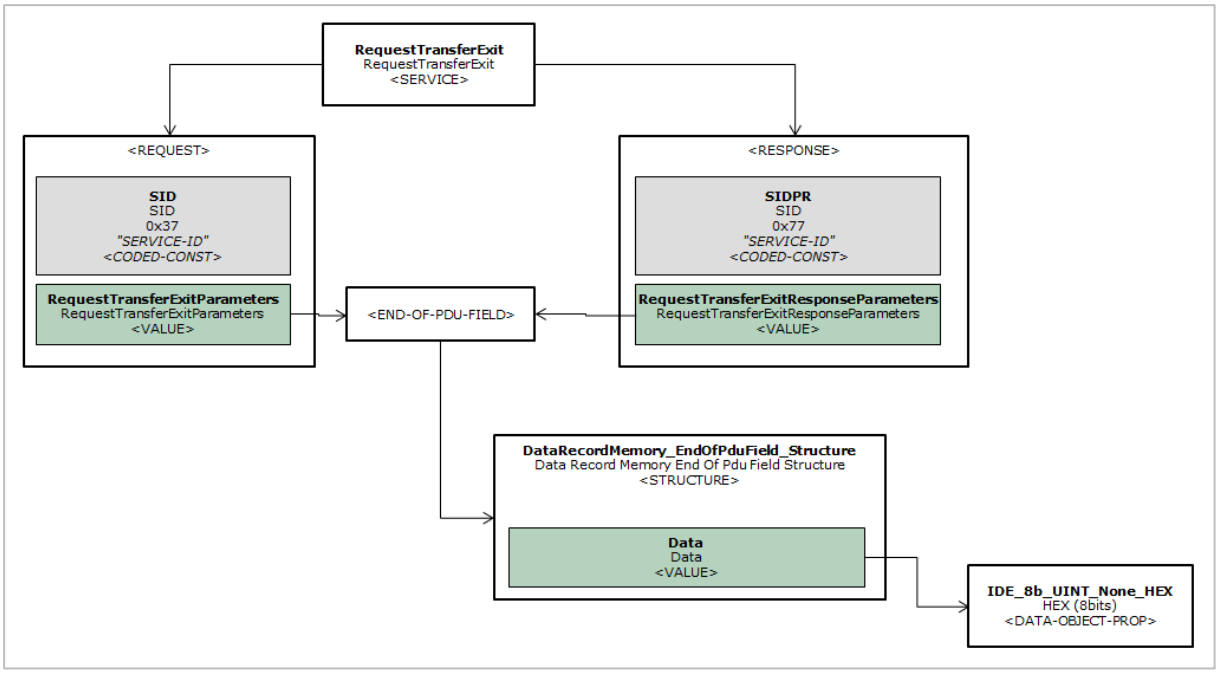


Figure 78 B/C DIAG-SERVICE RequestTransferExit ($37)

The ODX elements used by service $37 are shown in Figure 78.

Implementation requirements:

[Rule] The DIAG-SERVICE shall have the SHORT-NAME “RequestTransferExit”.

[Rule] The DIAG-SERVICE shall have the LONG-NAME “Request Transfer Exit”.

[Rule] The DIAG-SERVICE shall be marked as IS-FINAL=”true”.

[Rule] The SEMANTIC attribute of the DIAG-SERVICE shall not be used.

[Rule] The DIAGNOSTIC-CLASS attribute shall not be used.

[Rule] The DIAG-SERVICE shall reference the FUNCT-CLASS “Upload Download”.

2.3.6.21 CommunicationControl ($28)

Service $28 (“CommunicationControl”) is predefined once in the Renault Nissan Spec B/C ESD available from RFQ packages. It is marked final – ECU designers and suppliers shall not override or redefine this service in lower layers.

The service makes use of the predefined TABLE “TA_S28” to specify the supported sub-functions. The following sub-functions are thereby predefined:

- enableRxAndTx (0x00)
- enableRxAndDisableTx (0x01)
- disableRxAndEnableTx (0x02)
- disableRxAndTx(0x03)

ECU designers can override that TABLE to tailor these presets for ECUs which support a different set of sub-functions.

The ODX elements used by service $28 are shown in Figure 79.

Implementation requirements:

[Rule] The DIAG-SERVICE shall have the SHORT-/LONG-NAME “CommunicationControl”.

[Rule] The DIAG-SERVICE shall be marked as IS-FINAL=”true”.

[Rule] The SEMANTIC attribute of the DIAG-SERVICE shall be set to “COMMUNICATION”.

[Rule] The DIAGNOSTIC-CLASS attribute shall not be used.

[Rule] The DIAG-SERVICE shall reference the FUNCT-CLASS “CommunicationControl”.

[Rule] The request of the DIAG-SERVICE shall reference the TABLE “TA_S28” dynamically by SNREF.

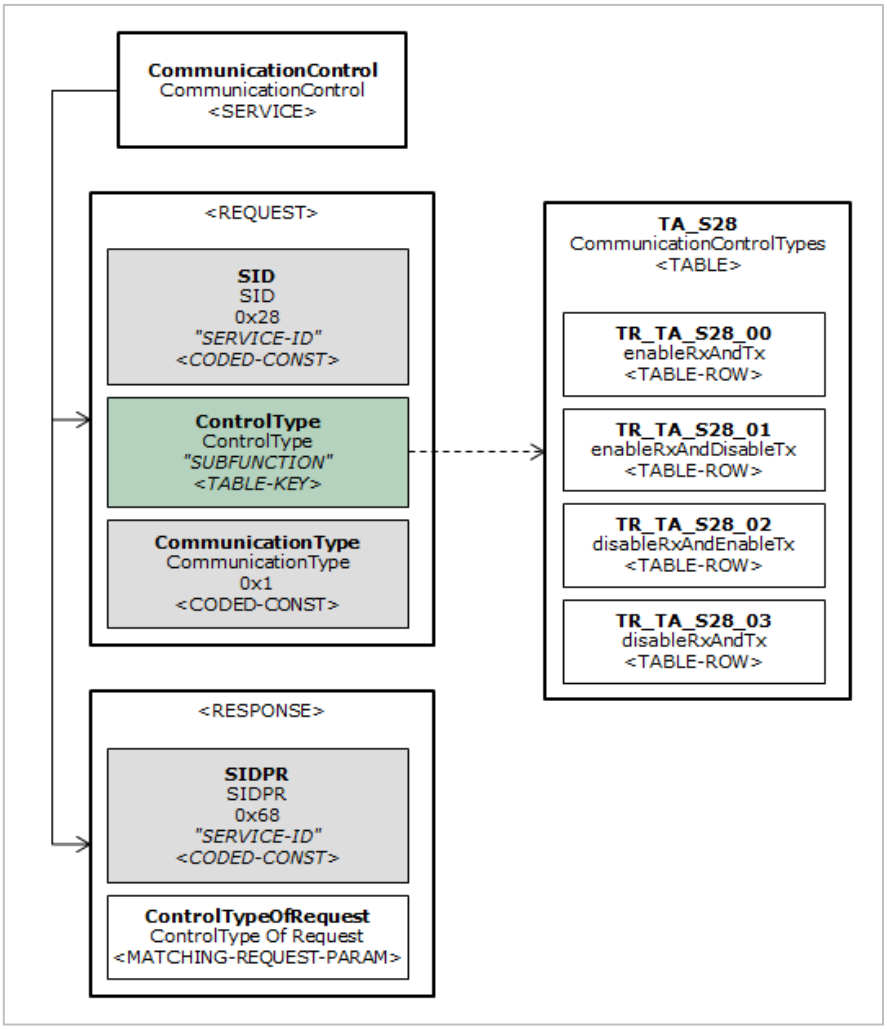


Figure 79 B/C DIAG-SERVICE CommunicationControl ($28)

2.3.6.22 ReadMemoryByAddress ($23)

Service $23 (“ReadMemoryByAddress”) is predefined once in the Renault Nissan Spec B/C ESD available from RFQ packages. It is marked final – ECU designers and suppliers shall not override or redefine this service in lower layers.

Implementation requirements:

[Rule] The DIAG-SERVICE shall have the SHORT-/LONG-NAME “ReadMemoryByAddress”.

[Rule] The DIAG-SERVICE shall be marked as IS-FINAL=”true”.

[Rule] The SEMANTIC attribute of the DIAG-SERVICE shall be set to “MEMORYREAD”.

[Rule] The DIAGNOSTIC-CLASS attribute shall not be used.

[Rule] The DIAG-SERVICE shall reference the FUNCT-CLASS “Memory”.

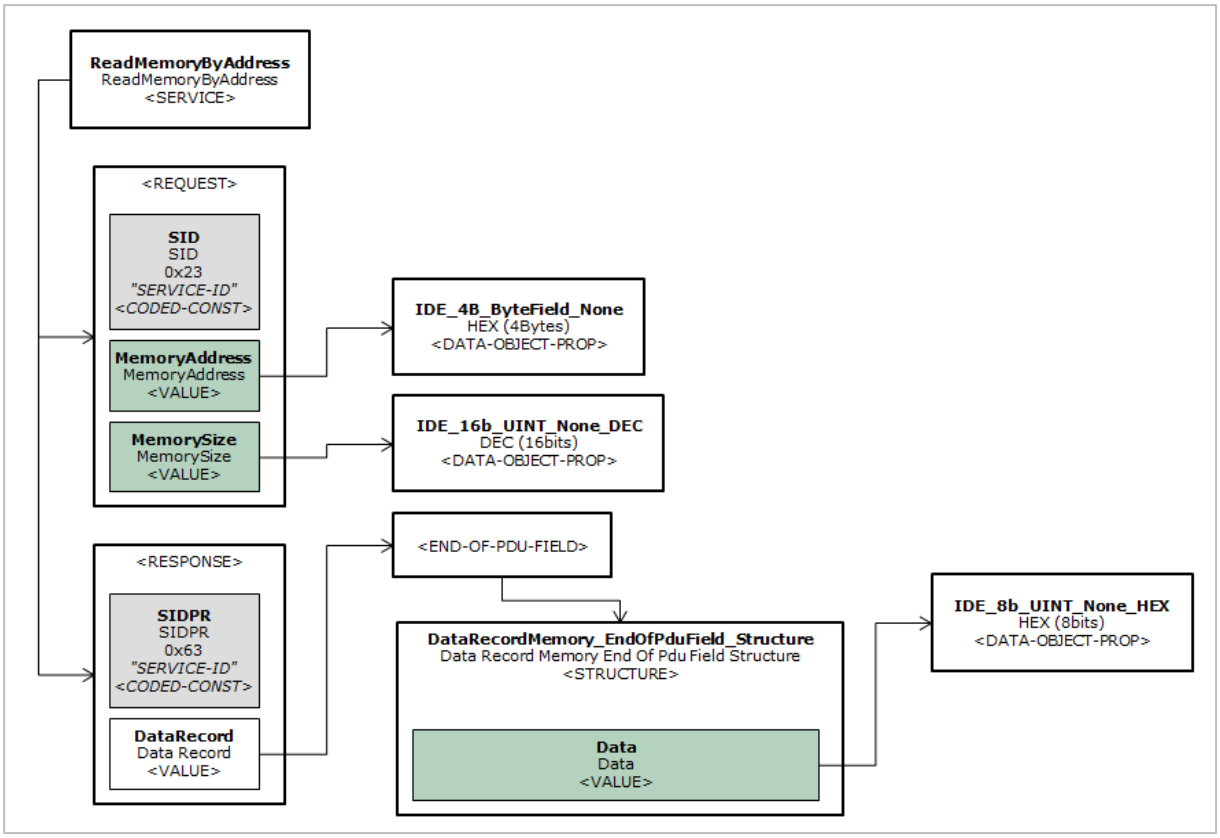


Figure 80 B/C DIAG-SERVICE ReadMemoryByAddress ($23)

2.3.6.23 WriteMemoryByAddress ($3D)

Service $3D ("WriteMemoryByAddress") is predefined once in the Renault Nissan Spec B/C ESD available from RFQ packages. It is marked final – ECU designers and suppliers shall not override or redefine this service in lower layers.

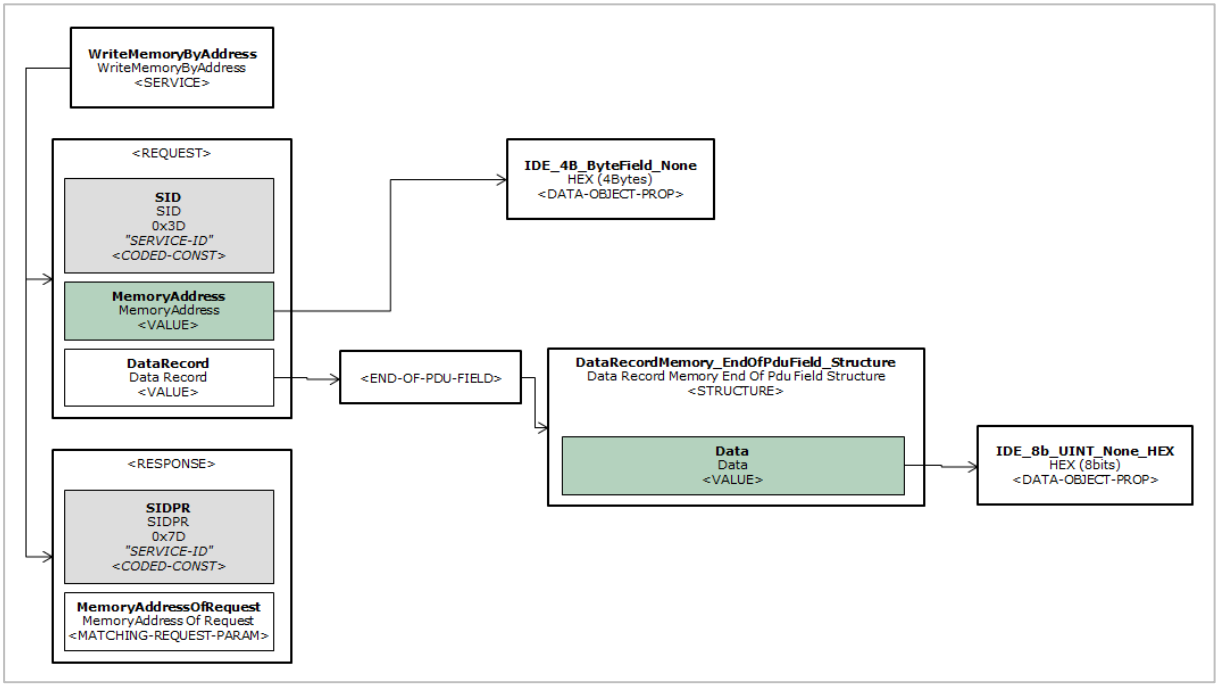


Figure 81 B/C DIAG-SERVICE WriteMemoryByAddress ($3D)

Implementation requirements:

[Rule] The DIAG-SERVICE shall have the SHORT-/LONG-NAME “WriteMemoryByAddress”.

[Rule] The DIAG-SERVICE shall be marked as IS-FINAL=”true”.

[Rule] The SEMANTIC attribute of the DIAG-SERVICE shall be set to “MEMORYWRITE”.

[Rule] The DIAGNOSTIC-CLASS attribute shall not be used.

[Rule] The DIAG-SERVICE shall reference the FUNCT-CLASS “Memory”.

2.3.7 GLOBAL-NEG-RESPONSE

Renault / Nissan does not deviate from ISO negative response handling. As a result, it is not allowed to define NEG-RESPONSE at DIAG-SERVICE level (cf. section 2.3.5.1). Instead, the diagnostic specification ESDs listed in Table 2 define GLOBAL-NEG-RESPONSEs as shown in Figure 82.

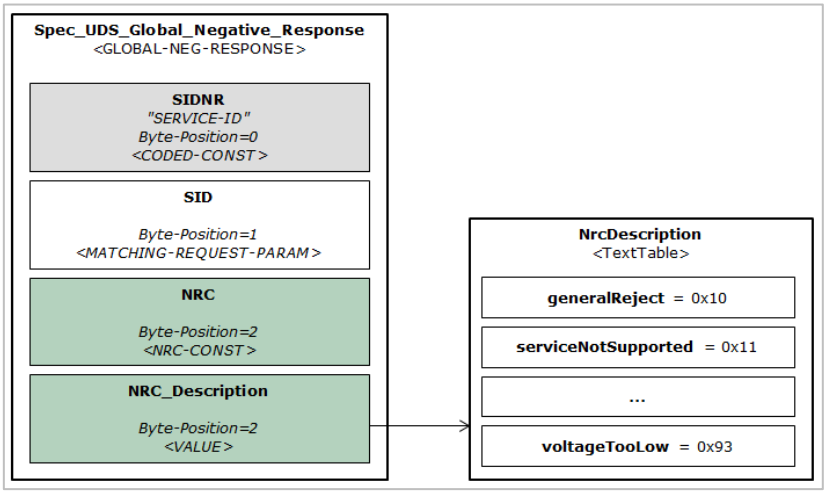


Figure 82 Structure of GLOBAL-NEG-RESPONSE

The MATCHING-REQUEST-PARAM at byte position 1 references the request service identifier. The NRC-CONST at byte position 2 defines the set of possible negative return codes while the text table referenced by the VALUE at the same byte position provides a textual representation of those values.

Implementation requirements:

[Rule] No GLOBAL-NEG-RESPONSE shall be defined outside of the diagnostic specification ESDs listed in Table 2.

[Rule] Each of these ESDs shall define exactly one GLOBAL-NEG-RESPONSE.

2.3.8 DIAG-DATA-DICTIONARY-SPEC / PARAM

2.3.8.1 General Rules for PARAMs

In Renault / Nissan ODX, the main role of PARAMs is to represent the actual data stored in the ECU or transmitted between the ECU and testers. This is particularly true for VALUE PARAMs used in requests and positive responses of services (including PARAMs indirectly used, e. g. when contained in STRUCTUREs which are used from requests or responses).

This fact is primarily reflected in the rules and recommendations for the PARAMs’ names which should be detached from peripheral information such as the conversions applied to those PARAMs by testers or the internal structure of the stored data. This strong, rather non-technical focus on the actual data represented by PARAMs results in the necessity to fulfill additional side conditions when modelling data with PARAMs – restricting the large design freedom given by ODX for that task in the first place.

On the one hand, these additional constraints shall make sure that the long proven philosophy concerning data modelling and management is also preserved in future ODX based development of Renault / Nissan ECUs; on the other side, it shall make sure that those descriptions can be converted without loss into the DDT format which already incorporates that philosophy to great extent.

Implementation requirements:

[Rule] The uppercase representation of an ODX PARAM's LONG-NAME shall either be unique amongst all uppercase PARAM LONG-NAME representations of the whole ODX model or all PARAMs having the same uppercase LONG-NAME representation shall either all reference the same DOP or DOPs which only differ in their byte orders.

[Reco] LONG-NAMEs of PARAMs should reflect the actual data stored in the ECU or transmitted between testers and the ECU which is represented by the PARAMs; LONG-NAMEs of PARAMs should not contain information specific to the structure of or the interpretation of or the conversion applied to the data they represent.

[Reco] Different PARAM names should not be introduced out of the mere wish to reflect different representations related to different physical measurement systems, e. g. representation of a speed in kilometres per hour vs miles per hour. Instead, appropriate DOPs should be referenced which make use of appropriate unit specs to reflect this.

2.3.8.2 General Recommendations for DOPs

In Renault / Nissan ODX, the main role of DOPs is actually to merely represent data interpretation rules, conversions or data formats. It is not their role to represent in any way actual data stored in the ECU or transmitted between the ECU and testers.

This is why many DOPs are predefined in Renault / Nissan ODX models intended for reuse in lower layers. The definition of ECU specific DOPs is possible, if needed, though reuse, preferably from the Datas Format ESD, should be preferred if at all possible.

Implementation requirements:

[Reco] DESCRIPTIONs should be avoided for DOPs used in models which are intended to be converted into DDT format, as that information will be lost during conversion.

[Reco] Definition of DOPs at BASE-VARIANT or ECU-VARIANT level should be avoided; if at all possible, DOPs should be reused from the predefined Datas Format ESD.

[Reco] LONG-NAMEs of DOPs should reflect the conversions / data formats modelled by that DOP and only that information; specifically, LONG-NAMEs of DOPs should not contain information specific to the actual data stored in the ECU or transmitted between testers and the ECU on which the conversion is applied. An exception to this is the naming of TEXTTABLEs which generally do contain data specific information as a matter of fact.

[Reco] Additional DOPs should not be introduced to only support different representations related to different physical measurement system, e. g. representation of a speed in kilometres per hour vs miles per hour. If at all possible, authors should revert to defining or reusing appropriate unit specs. Note in particular that authors may reuse predefined ODX RS unit specs and are encouraged to do so.

2.3.8.3 Summary of Special Rules for SHORT-NAMEs

[Rule] The SHORT-NAME of the TABLE "DIDs Readable" referenced by the DIAG-SERVICE with SHORT-NAME "ReadDataByIdentifier" shall be "TA_S22".

[Rule] The SHORT-NAME of the TABLE "LIDs Readable" referenced by the DIAG-SERVICE with SHORT-NAME "ReadDataByLocalIdentifier" shall be "TA_S21".

[Rule] The SHORT-NAME of the TABLE "DIDs Writable" referenced by the DIAG-SERVICE with SHORT-NAME "WriteDataByIdentifier" shall be "TA_S2E".

[Rule] The SHORT-NAME of the TABLE "LIDs Writeable" referenced by the DIAG-SERVICE with SHORT-NAME "WriteDataByLocalIdentifier" shall be "TA_S3B".

[Rule] The SHORT-NAME of the TABLE "Sessions" referenced by the DIAG-SERVICE with SHORT-NAME "DiagnosticSessionControl" shall be "TA_S10".

[Rule] The SHORT-NAME of the TABLE "ResetTypes" referenced by the DIAG-SERVICE with SHORT-NAME "ECUReset" shall be "TA_S11".

[Rule] The SHORT-NAME of the TABLE "SecurityAccess - requestSeed" referenced by the DIAG-SERVICE with SHORT-NAME "SecurityAccess_requestSeed" shall be "TA_S27_RS".

[Rule] The SHORT-NAME of the TABLE "SecurityAccess - sendKey" referenced by the DIAG-SERVICE with SHORT-NAME "SecurityAccess_sendKey" shall be "TA_S27_SK".

[Rule] The SHORT-NAME of the TABLE "CommunicationControlTypes" referenced by the DIAG-SERVICE with SHORT-NAME "CommunicationControl" shall be "TA_S28".

[Rule] The SHORT-NAME of the TABLE "DTCSettingTypes" referenced by the DIAG-SERVICE with SHORT-NAME "ControlDTCSetting" shall be "TA_S85".

[Rule] The SHORT-NAME of the TABLE "DIDs Supportable" referenced by the support check DIAG-SERVICEs for two byte identifiers shall be "TS_SC_DID".

[Rule] The SHORT-NAME of the TABLE "LIDs Supportable" referenced by the support check DIAG-SERVICEs for one byte identifiers shall be "TS_SC_LID".

[Rule] The SHORT-NAME of the TABLE "RIDs Startable RQ" referenced by the REQUEST of the DIAG-SERVICE with SHORT-NAME "Routine_Control_Start" shall be "TA_S31_ST_RQ".

[Rule] The SHORT-NAME of the TABLE "RIDs Startable PR" which is referenced by the POS-RESPONSE of the DIAG-SERVICE with SHORT-NAME "Routine_Control_Start" shall be "TA_S31_ST_PR".

[Rule] The SHORT-NAME of the TABLE "RIDs Stoppable RQ" referenced by the REQUEST of the DIAG-SERVICE with SHORT-NAME "Routine_Control_Stop" shall be "TA_S31_SP_RQ".

[Rule] The SHORT-NAME of the TABLE "RIDs Stoppable PR" which is referenced by the POS-RESPONSE of the DIAG-SERVICE with SHORT-NAME "Routine_Control_Stop" shall be "TA_S31_SP_PR".

[Rule] The SHORT-NAME of the TABLE "RIDs Resultable RQ" which is referenced by the REQUEST of the DIAG-SERVICE with SHORT-NAME "Routine_Control_Request_Results" shall be "TA_S31_RS_RQ".

[Rule] The SHORT-NAME of the TABLE "RIDs Resultable PR" which is referenced by the POS-RESPONSE of the DIAG-SERVICE with SHORT-NAME "Routine_Control_Request_Results" shall be "TA_S31_RS_PR".

[Rule] The SHORT-NAME of the TABLE "DIDs I/O Returnable" which is referenced by the REQUEST of the DIAG-SERVICE with SHORT-NAME "IOControl_returnControlToECU" shall be "TA_S2F_00".

[Rule] The SHORT-NAME of the TABLE "DIDs I/O Resetable" which is referenced by the REQUEST of the DIAG-SERVICE with SHORT-NAME "IOControl_resetToDefault" shall be "TA_S2F_01".

[Rule] The SHORT-NAME of the TABLE "DIDs I/O Freezable" which is referenced by the REQUEST of the DIAG-SERVICE with SHORT-NAME "IOControl_freezeCurrentState" shall be "TA_S2F_02".

[Rule] The SHORT-NAME of the TABLE "DIDs I/O Adjustable" which is referenced by the REQUEST of the DIAG-SERVICE with SHORT-NAME "IOControl_shotTermAdjustment" shall be "TA_S2F_03".

[Rule] The SHORT-NAME of TABLE "DIDs I/O Control Enable Mask Record" which is referenced via the four TABLEs with the SHORT-NAMEs "TA_S2F_00", "TA_S2F_01", "TA_S2F_02" and "TA_S2F_03" shall be "TA_S2F_CEMR".

[Rule] The SHORT-NAME of the TABLE "IO Control - Temporary" which is referenced by the REQUEST of the DIAG-SERVICE with SHORT-NAME "IOControlStartTemporary" shall be "TA_S30_00".

[Rule] The SHORT-NAME of the TABLE "IO Control - Permanent" which is referenced by the REQUEST of the DIAG-SERVICE with SHORT-NAME "IOControlStartPermanent" shall be "TA_S30_20".

[Rule] The SHORT-NAME of the TABLE "IO Control - Status" which is referenced by the REQUEST of the DIAG-SERVICE with SHORT-NAME "IOControlStatus" shall be "TA_S30_01".

[Rule] The SHORT-NAME of the TABLE "IO Control - Stop" which is referenced by the REQUEST of the DIAG-SERVICE with SHORT-NAME "IOControlStop" shall be "TA_S30_11".

[Rule] The SHORT-NAME of the TABLE "IO Control Other Control Status" which is referenced by the POS-RESPONSEs of the DIAG-SERVICEs with the SHORT-NAMEs "IOControlStart-Temporary", "IOControlStartPermanent", "IOControlStatus" and "IOControlStop" shall be "TA_S30_PR".

[Rule] The SHORT-NAME of TABLE "Routine – Start (Request)" referenced by the REQUEST of the DIAG-SERVICE with SHORT-NAME "StartRoutineByLocalIdentifier" shall be "TA_S31_RQ".

[Rule] The SHORT-NAME of TABLE "Routine – Start (Positive response)" referenced by the POS_RESPONSE of the DIAG-SERVICE with SHORT-NAME "StartRoutineByLocalIdentifier" shall be "TA_S31_PR".

[Rule] The SHORT-NAME of TABLE "Routine – Stop (Request)" referenced by the REQUEST of the DIAG-SERVICE with SHORT-NAME "StopRoutineByLocalIdentifier" shall be "TA_S32_RQ".

[Rule] The SHORT-NAME of TABLE "Routine – Stop (Positive response)" referenced by the REQUEST of the DIAG-SERVICE with SHORT-NAME "StopRoutineByLocalIdentifier" shall be "TA_S32_PR".

[Rule] The SHORT-NAME of the TABLE "DTCSettingTypes" referenced by the DIAG-SERVICE with SHORT-NAME "ControlDTCSetting" shall be "TA_S85".

[Rule] The SHORT-NAME of TABLE-ROWs contained in one of the TABLEs with SHORT-NAME "TA_S10", "TA_S11", "TA_S27_RS", "TA_S27_SK", "TA_S28" or "TA_S85" shall be "TR_{TA}_{NN}", where {TA} shall be replaced by the containing TABLE's SHORT-NAME and {NN} shall be replaced with the two digit hex code of the sub function (that is: session state, reset type…) represented by the TABLE-ROW.

[Rule] The SHORT-NAMEs of TABLE-ROWs contained in one of the DID related TABLEs (that is: one of the TABLEs with SHORT-NAME "TA_S22", "TA_S2E", "TA_S2F_FR", "TA_S2F_RS", "TA_S2F_RT", "TA_S2F_AD", "TA_S2F_03" or "TA_S2F_CEMR) shall be "TR_{TA}_{NNNN}", where {TA} shall be replaced by the containing TABLE's SHORT-NAME and {NNNN} shall be replaced with the four digit hex code of the DID represented by the TABLE-ROW.

[Rule] The SHORT-NAMEs of TABLE-ROW-REFs contained in one of the TABLEs with SHORT-NAME "TA_S2F_00", "TA_S2F_01" or "TA_S2F_02" shall be "TR_TA_S2F_CEMR_{NNNN}", where {NNNN} shall be replaced with the four-digit hex code of the DID represented by the TABLE-ROW-REF.

[Rule] The SHORT-NAMEs of TABLE-ROWs contained in one of the LID related TABLEs (that is: one of the TABLEs with SHORT-NAME "TA_S21", "TA_S3B", "TA_S30_00", "TA_S30_01", "TA_S30_11", "TA_S30_20" or "TA_S30_PR") shall be "TR_{TA}_{NN}", where {TA} shall be replaced by the containing TABLE's SHORT-NAME and {NN} shall be replaced with the two digit hex code of the LID represented by the TABLE-ROW.

[Rule] The SHORT-NAMEs of TABLE-ROWs contained in one of the routine related TABLEs (that is: one of the TABLEs with SHORT-NAME "TA_S31_SP_RQ", "TA_S31_SP_PR", "TA_S31_ST_RQ", "TA_S31_ST_PR", "TA_S31_RS_RQ" or "TA_S31_RS_PR") shall be "TR_{TA}_{NNNN}", where {TA} shall be replaced by the containing TABLE's SHORT-NAME and {NNNN} shall be replaced with the four digit hex code of the RID represented by the TABLE-ROW.

[Rule] If a STRUCTURE is referenced from one of the DID related TABLEs (that is: one of the TABLEs with SHORT-NAME "TA_S22", "TA_S2E", "TA_S2F_FR", "TA_S2F_RS", "TA_S2F_RT" or "TA_S2F_AD"), the SHORT-NAME of the STRUCTURE shall be "ST_{TA}_{NNNN}", where {TA} shall be replaced by the referencing TABLE's SHORT-NAME without its "TA_" prefix and {NNNN} shall be replaced with the four digit hex code of the DID whose data record is described by the STRUCTURE.

[Rule] If a STRUCTURE is referenced from the TABLE with SHORT-NAME "TA_S2F_CEMR", the SHORT-NAME of that STRUCTURE shall be "ST_S2F_{NNNN}_CEMR", where {NNNN} shall be replaced with the four-digit hex code of the IO controllable DID whose control enable mask record is described by the STRUCTURE.

[Rule] The SHORT-NAME of STRUCTUREs referenced by TABLE "TA_S2F_03" shall be "ST_S2F_{NNNN}_03", where {NNNN} shall be replaced by the four-digit hex code of the DID represented by the referencing TABLE-ROW.

[Rule] The SHORT-NAME of the STRUCTUREs representing control state records of IO controls shall be "ST_S2F_{NNNN}_03_CS", where {NNNN} shall be replaced by the four-digit hex code of the IO controllable DID whose control state record is represented by that STRUCTURE.

[Rule] If a STRUCTURE is referenced from one of the RID related TABLEs (that is: one of the TABLEs with SHORT-NAME "TA_S31_SP_RQ", "TA_S31_SP_PR", "TA_S31_ST_RQ", "TA_S31_ST_PR", "TA_S31_RS_RQ" or "TA_S31_RS_PR"), the SHORT-NAME of the STRUCTURE shall be "ST_{TA}_{NNNN}", where {TA} shall be replaced by the referencing TABLE's SHORT-NAME without its "TA_" prefix and {NNNN} shall be replaced with the four digit hex code of the RID whose routine control option record or routine info and status record is described by the STRUCTURE.

[Rule] The SHORT-NAME of a STRUCTURE referenced by a flat service's request shall be "ST_FS{SID}_{#NNNN}_RQ", where {SID} shall be replaced by the referencing service's SID and {#NNNN} can be optional used and replaced to represent a sub function and / or identifier if several flat services for the same SID need to be defined.

[Rule] The SHORT-NAME of a STRUCTURE referenced by a flat service's positive response shall be "ST_FS{SID}_{#NNNN}_PR", where {SID} shall be replaced by the referencing service's SID and {#NNNN} can be optional used and replaced to represent a sub function and / or identifier if several flat services for the same SID need to be defined.

[Rule] If a PARAM is contained in one of the STRUCTUREs touched by the special naming rules defined in this section, then its SHORT-NAME shall be "PA_{ST}_{NNN}", where {ST} shall be replaced with the SHORT-NAME of the containing STRUCTURE and {NNN} shall be replaced by the three-digit running number of the PARAM in the STRUCTURE (that is: the first PARAM shall be numbered 000, the second 001, the third 002 etc.).

2.3.9 STATE-CHART

In Renault / Nissan ODX models two state charts are mandatory, one for session states and one for security levels. These STATE-CHARTs are defined by the diagnostic specification ESDs listed in Table 2.

The STATE-CHARTs get inherited into each BASE-/ECU-VARIANT, allowing ECU designers to reuse the states and state transitions predefined therein, e. g. to specify the set of states and transitions supported by a concrete ECU and to specify state preconditions for reading or writing a DID etc. The detailed mechanics and rules for these use cases are described in section 2.3.4.

Implementation requirements:

[Rule] STATE-CHARTs must not be defined in DIAG-LAYERs other than the diagnostic specification ESDs listed in Table 2.

2.3.9.1 Session

The session related STATE-CHART shall predefine the following states:

LONG-NAME	SHORT-NAME
Default Session 01	Default_Session_01
Programming Session 02	Programming_Session_02
Extended Diagnostic Session 03	Extended_Diagnostic_Session_03
Safety System Diagnostic Session 04	Safety_System_Diagnostic_Session_04
Supplier Session 60	Supplier_Session_60
Supplier Session 61	Supplier_Session_61
Supplier Session 62	Supplier_Session_62
Supplier Session 63	Supplier_Session_63
Supplier Session 64	Supplier_Session_64
Supplier Session 65	Supplier_Session_65
Supplier Session 66	Supplier_Session_66
Supplier Session 67	Supplier_Session_67
Supplier Session 68	Supplier_Session_68
Supplier Session 69	Supplier_Session_69
Supplier Session 6A	Supplier_Session_6A
Supplier Session 6B	Supplier_Session_6B
Supplier Session 6C	Supplier_Session_6C
Supplier Session 6E	Supplier_Session_6E
Supplier Session 6D	Supplier_Session_6D
Supplier Session 6F	Supplier_Session_6F
Supplier Session 70	Supplier_Session_70
Supplier Session 71	Supplier_Session_71
Supplier Session 72	Supplier_Session_72
Supplier Session 73	Supplier_Session_73
Supplier Session 74	Supplier_Session_74
Supplier Session 75	Supplier_Session_75
Supplier Session 76	Supplier_Session_76
Supplier Session 77	Supplier_Session_77
Supplier Session 78	Supplier_Session_78
Supplier Session 79	Supplier_Session_79
Supplier Session 7A	Supplier_Session_7A
Supplier Session 7B	Supplier_Session_7B
Supplier Session 7C	Supplier_Session_7C
Supplier Session 7D	Supplier_Session_7D

Supplier Session 7E	Supplier_Session_7E

Table 11 Predefined Session States

Moreover, the session STATE-CHART shall predefine state transitions from each state to all other states listed in Table 11, ready to be reused by ECU designers as they see fit.

Implementation requirements:

[Rule] The SEMANTIC of the session state chart shall be set to "SESSION".

[Rule] The session STATE-CHART shall not define states different than the ones listed in Table 11.

2.3.9.2 Security

The security access related STATE-CHART shall predefine the following security levels:

LONG-NAME	SHORT-NAME
Locked	Locked
Unlocked #2	Unlocked_2
Unlocked #5	Unlocked_5
Unlocked #6	Unlocked_6

Table 12 Predefined Security Access Levels

Moreover, the security access STATE-CHART shall encompass the possible state transitions between the different security access levels, ready to be reused by ECU designers as they see fit.

Implementation requirements:

[Rule] The SEMANTIC of the security access state chart shall be set to "SECURITY".

[Rule] The security access STATE-CHART shall not define states different to the ones in Table 12.

© 2017 Vector Informatik GmbH

2.4 ODX-C

2.4.1 CAN

For CAN, Renault / Nissan ODX models support all communication parameters defined by the standard ISO COMPARAM-SPEC "ISO 15765-3 on ISO 15765-2 on ISO 11898-2 DWCAN", referencing the following set of comparam subsets:

- ISO_15765_3 for the application layer parameters
- ISO_15765_2 for the protocol layer parameters
- ISO_11898_2_DWCAN for the physical layer parameters

Implementation requirements:

[Rule] All default values defined in the above mentioned COMPARAM-SPEC are acceptable for Renault / Nissan ODX models.

[Rule] For Nissan ECUs, the following comparams may be overridden at BASE-VARIANT layer:

- CPS_Baudrate
- CPS_BlockSize
- CPS_BS_TX
- CPS_INIT_P2MAX
- CPS_InitMode
- CPS_ISO15765_BS
- CPS_ISO9141_K_LINE_ONLY
- CPS_P0
- CPS_P2max
- CPS_P3Min
- CPS_P4Min
- CPS_W1
- CPS_W2
- CPS_W3
- CPS_W4
- CPS_W4B

Comparams other than the ones defined above shall not be overridden at BASE-VARIANTs.

[Rule] Both CP_CanFuncReqId and CP_UniqueRespIdTable shall be overridden on FUNCTIONAL-GROUP level to define the CAN identifiers. Note that the FUNCTIONAL-GROUP layers are predefined by function DLCs (there is one such DLC per function, specification and RFQ version) and shall be reused as is.

[Rule] Apart from the comparams explicitly allowed to be overridden on FUNCTIONAL-GROUP or BASE-VARIANT layers no comparams may be overridden at any of the ODX-D layers in Renault / Nissan ODX models.

[Rule] Any comparam may be overridden in ODX-V at the ECU designer's solicitation.

2.4.2 Ethernet

For Ethernet, Renault / Nissan ODX models support all communication parameters defined by the ISO COMPARAM-SPEC "ISO 14229-5 on ISO 13400-2", referencing the following set of comparam subsets:

- ISO_14229_5 for the application layer parameters
- ISO_13400_2 for the protocol layer parameters
- IEEE_802_3 for the physical layer parameters

2.5ODX-V

The rules described in this Renault / Nissan AGL only proscribe the structure and contents of ODX-V files contained in solitary PDX containers describing a single ECU and intended for use in engineering.

Note that these rules do not apply for PDX containers describing multiple ECUs used for after sales.

2.5.1 Elements Used

Renault / Nissan ODX-V makes use of both INFO-COMPONENT and VEHICLE-INFORMATION elements, but only the following four sub element types are actually used:

- VEHICLE-MODEL, section see 2.5.3
- VEHICLE-CONNECTOR and PHYSICAL-VEHICLE-LINK, see section 2.5.4
- MEMBER-LOGICAL-LINK, see section 2.5.5

Implementation requirements:

[Rule] ODX-V element types other than VEHICLE-MODEL, VEHICLE-CONNTECTOR, PHYSICAL-VEHICLE-LINK and MEMBER-LOGICAL-LINK shall not be used.

2.5.2 Top-Level Structure

Renault / Nissan ODX-V files consist of three major building blocks:

- One or more VEHICLE-MODELs
- Exactly one VEHICLE-INFORMATION element named “Connectors”
- One or more VEHICLE-INFORMATION elements, each representing one ECU usage scenario[7]

The “Connectors” VEHILCE-INFORMATION shall thereby contain one or more VEHICLE-CONNECTOR and one or more PHYSICAL-VEHICLE-LINK elements while the VEHICLE-INFORMATION elements which represent ECU usage scenarios shall contain MEMBER-LOGICAL-LINK elements only.

Implementation requirements:

[Rule] ODX-V files shall contain at least one VEHICLE-MODEL element.

[Rule] ODX-V files shall contain exactly one VEHICLE-INFORMATION element named “Connectors”, encompassing at least one VEHICLE-CONNECTOR and one PHYSICAL-VEHICLE-LINK.

[Rule] The VEHICLE-INFORMATION element named “Connectors” shall not contain elements other than VEHICLE-CONNECTORs and PHYSICAL-VEHICLE-LINKs.

[Rule] ODX-V files shall contain at least one VEHICLE-INFORMATION element not named “Connectors” representing an ECU usage scenario.

[Rule] VEHICLE-INFORMATION elements representing an ECU usage scenario shall only contain MEMBER-LOGICAL-LINKs and no other vehicle information element types.

2.5.3 Vehicle Models

VEHICLE-MODELs in Renault / Nissan ODX-V are intended to be referenced by MEMBER-LOGICAL-LINKs which represent ECU usage scenarios. The role of the VEHICLE-MODELs is thereby to document which of the networks (e. g. CAN, Ethernet) supported by the described ECU are actually also used to diagnose the ECU in real vehicles and if so in what vehicles.

[7] An ECU usage scenario is as set of networks / protocols usable to diagnose the ECU when built into a concrete vehicle, e. g. one and the same UDS ECU could be diagnosable via CAN and DoIP in vehicles A, B and C (one usage scenario) but only diagnosable via CAN in vehicles D and E (a different usage scenario).

In this relationship between ECU usages and concrete vehicles, the VEHICLE-MODEL elements represent the vehicle side, providing the different bits of the vehicle-side information in their fields:

- The OEM of the vehicle is given by its LONG-NAME. e. g. Renault, Nissan, Dacia, Lada, Daimler…
- The name of the associated vehicle project is given by its SHORT-NAME (x06, x10ph2…).

Finally, the DESC element of the VEHICLE-MODEL provides these additional bits of information, where each bit is contained in a XHTML paragraph ( ) of its own, in the following sequence:

- First paragraph: The name of the model, e. g. Captur, Twingo, Twizy, Almera, Lodgy…
- Second paragraph: The model segment identifier (model category), e. g. I1, M1…
- Third to nth: The supported network types, e. g. DiagOnCAN, KWP2000 FastInit MonoPoint…

Implementation requirements:

[Rule] VEHICLE-MODEL elements shall give the OEM of the vehicle model represented in their LONG-NAME.

[Rule] The SHORT-NAME of VEHICLE-MODEL elements shall give the name of the associated vehicle project.

[Rule] The DESC field of VEHICLE-MODEL elements shall consist of a sequence of XHTML paragraphs.

[Rule] The first paragraph of VEHICLE-MODELs' DESC elements shall contain the actual name of the vehicle model.

[Rule] The second paragraph of VEHICLE-MODELs' DESC elements shall contain the model segment identifier / model category.

[Rule] Each paragraph of VEHICLE-MODELs' DESC elements beginning with the third shall contain the name of a network which can be used for diagnosing the represented vehicle.

2.5.4 Vehicle Information on Connectors

Each ODX-V file in solitary Renault / Nissan PDX containers contains one VEHICLE-INFORMATION element named "Connectors". It only contains sub elements of type VEHICLE-CONNECTOR and PHYSICAL-VEHICLE-LINK. It is used to specify all the network types supported in principle by the ECU described in the solitary PDX container.

The information given by that VEHICLE-INFORMATION element equally apply to all other VEHICLE-INFORMATION elements in the ODX-V file (of which there must exist at least one). To avoid redundant definitions this information, it is however only given once in the VEHICLE-INFORMATION element named "Connectors".

Implementation requirements:

[Rule] ODX-V files shall contain exactly one VEHICLE-INFORMATION element named "Connectors", encompassing at least one VEHICLE-CONNECTOR and one PHYSICAL-VEHICLE-LINK.

[Rule] The SHORT-NAME of the element with LONG-NAME "Connectors" shall be "HardWare".

[Rule] The VEHICLE-INFORMATION element named "Connectors" shall not contain elements other than VEHICLE-CONNECTORs and PHYSICAL-VEHICLE-LINKs.

2.5.5 Vehicle Information on ECU Usage Scenarios

Renault / Nissan ODX-V files in solitary PDX containers encompass one or more VEHICLE-INFORMATION elements containing MEMBER-LOGICAL-LINKs. Each of these VEHICLE-INFORMATION elements represents one possible usage scenario[7] of the ECU (e. g. "UDS on CAN 11 Bits" etc.). By referencing one or more VEHICLE-MODEL elements, the VEHICLE-INFORMATION element documents in what concrete vehicles the respective usage scenario is actually put into practice.

Implementation requirements:

[Rule] ODX-V files shall contain a VEHICLE-INFORMATION element for each possible usage scenario supported by the described ECU.

[Rule] VEHICLE-INFORMATION elements representing an ECU usage scenario shall only contain MEMBER-LOGICAL-LINKs and no other vehicle information element types.

[Rule] The SHORT- and LONG-NAME of a VEHICLE-INFORMATION element representing an ECU usage scenario shall correspond to the name of the protocol referenced by the contained MEMBER-LOGICAL-LINK (without the Renault Nissan prefix).

[Rule] The concrete vehicles in which a certain ECU usage scenario is put into practice shall be documented in the VEHICLE-INFORMATION element representing that usage scenario via corresponding INFO-COMPONENT-REFs to the respective VEHICLE-MODEL elements which represent those vehicles.

[Rule] MEMBER-LOGICAL-LINKs of VEHICLE-INFORMATION elements representing ECU usage scenarios shall reference PROTOCOLs plus BASE-VARIANTs, not FUNCTIONAL-GROUPs.

© 2017 Vector Informatik GmbH

2.6 Annexes

2.6.1 Prefixes for XML-IDs of ODX Object Instances

Element	Prefix
ADDITIONAL-AUDIENCE	AA
BASE-VARIANT	BV
CHECKSUM	CS
COMPANY-DATA	CD
COMPANY-DOC-INFO	CDI
all BASE-COMPARAM (COMPARAM, COMPLEX-COMPARAM)	CP
COMPARAM-SPEC	CPS
COMPARAM-SUBSET	CPSS
CONFIG-DATA	CFD
all CONFIG-ITEM (CONFIG-ID-ITEM, DATA-ID-ITEM, OPTION-ITEM, SYSTEM-ITEM)	CFI
CONFIG-RECORD	CFR
DATABLOCK	DB
DATA-RECORD	DR
all DIAG-COMM (DIAG-SERVICE, SINGLE-ECU-JOB)	DC
DIAG-LAYER-CONTAINER	DLC
DIAG-OBJECT-CONNECTOR	DOC
DIAG-VARIABLE	DV
all DOP-BASE (DATA-OBJECT-PROP, DTC-DOP, ENV-DATA-DESC, ENV-DATA, STRUCTURE, STATIC-FIELD, DYNAMIC-LENGTH-FIELD, DYNAMIC-ENDMARKER-FIELD, END-OF-PDU-FIELD, MUX)	DOP
DTC	DTC
ECU-CONFIG	EC
ECU-MEM	EM
ECU-MEM-CONNECTOR	EMC
ECU-SHARED-DATA	ESD
ECU-VARIANT	EV
EXPECTED-IDENT	EID
EXTERNAL-ACCESS-METHOD	EAM
FLASH	FL
FLASH-CLASS	FC
all FLASHDATA (EXTERN-FLASHDATA, INTERN-FLASHDATA)	FD
FUNCTIONAL-GROUP	FG
FUNCTION-DICTIONARY	FND
FUNCTION-NODE	FN
FUNCTION-NODE-GROUP	FNG
FUNCT-CLASS	FNC
GLOBAL-NEG-RESPONSE	GNR
all INFO-COMPONENT (ECU-PROXY, MODEL-YEAR, OEM, VEHICLE-MODEL, VEHICLE-TYPE)	IC
ITEM-VALUE	IV
LIBRARY	LIB
all LOGICAL-LINK (GATEWAY-LOGICAL-LINK, MEMBER-LOGICAL-LINK)	LL
MULTIPLE-ECU-JOB	MEJ

MULTIPLE-ECU-JOB-SPEC	MEJS
NEG-RESPONSE	NR
OUTPUT-PARAM	OP
OWN-IDENT	OID
all PARAM (CODED-CONST, DYNAMIC, LENGTH-KEY, MATCHING-REQUEST-PARAM, NRC-CONST, PHYS-CONST, RESERVED, SYSTEM, TABLE-ENTRY, TABLE-KEY, TABLE-STRUCT, VALUE)	PA
PHYSICAL-DIMENSION	PD
PHYSICAL-VEHICLE-LINK	PVL
PHYS-MEM	PM
all PHYS-SEGMENT (ADDRDEF-PHYS-SEGMENT, SIZEDEF-PHYS-SEGMENT)	PSG
POS-RESPONSE	PR
PROTOCOL	PT
PROT-STACK	PS
REQUEST	RQ
SDG-CAPTION	SDG
SEGMENT	SG
SESSION	SE
SESSION-DESC	SED
STATE	ST
STATE-CHART	STC
STATE-TRANSITION	STT
SUB-COMPONENT	SC
SUB-COMPONENT-PARAM-CONNECTOR	SCPC
TABLE	TA
TABLE-ROW	TR
TEAM-MEMBER	TM
UNIT	UN
VARIABLE-GROUP	VG
VEHICLE-CONNECTOR	VC
VEHICLE-CONNECTOR-PIN	VCP
VEHICLE-INFORMATION	VI
VEHICLE-INFO-SPEC	VIS

Table 13 Prefixes for XML-IDs of ODX Object Instances

© 2017 Vector Informatik GmbH

Index

$10 ..... 47, 81
$11 ..... 42, 71, 106
$14 ..... 65, 100
$19 ..... 60, 95
$1902 ..... 34, 60, 95
$1904 ..... 34, 36, 61, 65, 97, 100
$21 ..... 28, 33, 82, 84, 85
$2183 85
$2180 ..... 21, 84
$2183 21
$22 ..... 27, 29, 48, 86
$22F18A 21, 50
$22F194 ..... 21, 50
$22F195 ..... 21, 51
$22F1A0 ..... 21, 52
$23 ..... 78, 112
$27 ..... 41, 69, 70, 103, 104
$28 ..... 44, 77, 110
$2E 27, 53, 88
$2F 29, 31, 55, 56, 57, 58
$30 ..... 33, 91, 92, 93
$31 ..... 39, 40, 66
$32 40
$34 ..... 74, 108
$35 ..... 75, 108
$36 ..... 76, 109
$37 ..... 76, 109
$3B 28, 89
$3D 79, 112
$3E 73, 107
$80 84
$83 85
$85 ..... 43, 72
$F18A 21, 50
$F194 ..... 21, 50
$F195 ..... 21, 51
$F1A0 ..... 21, 52
AddressFormatIdentifier 74, 75
ADMIN-DATA 15
ALFID 74, 75
ALL-VALUE ENV-DATA 37
ASAM 17, 23
Auto Identification 21, 50, 51, 52, 84, 85
Base DTC Number 35
Base Variant 20
BASE-VARIANT 20, 123
CAN 120
Clear Diagnostic Information 65, 100
Code Generation 14
CommunicationControl 44, 77, 78, 110, 111
COMPARAM-SPEC 120
Connectors 121, 122
Control Enable Mask 29
Control Enable Mask Record 29
ControlDTCSetting 43, 72, 73
Conversion 15
Datas Format 15, 17, 22, 24, 114
DDT 15
Default Session 47
DESC 122
Description 15
Device 34, 62, 64, 97, 100
DIAG-DATA-DICTIONARY-SPEC 113
DIAG-LAYER 17
DIAG-LAYER-CONTAINER 23
Diagnostic Identification Code 22
Diagnostic Specification 17
Diagnostic Version 22, 52
DiagnosticSessionControl 47
DIAG-SERVICE 44, 80
DID 27, 36, 48, 53, 86, 88
DIDs Readable 27, 29, 37
DIDs Writable 27
DTC 34
DTC Status Availability Mask 60, 95
DTC Status Mask 60, 62, 63, 95, 97, 100
ECU Usage Scenario 122
ECU Variant 21
ECUReset 42, 71, 106
ECU-SHARED-DATA 17
ECU-VARIANT 21
ECU-VARIANT-PATTERN 21
Edition Number 22
ENV-DATA 36
Ethernet 120
Extended Data Record 38
Extended Diagnostic Session 47
Failure Type 34, 62, 64, 97, 100
Failure Type Byte 35
Fault Memory 34, 62, 66, 98, 101
File 23
Flat Service 21, 45, 91
Freeze Current State ($2F) 58
FTB 35
FUNCT-CLASS 25, 46, 80
FUNCT-GROUP 23
Functional Group 19
FUNCTIONAL-GROUP 23, 123
Global Snapshot 37
Global Negative Response 113

Global Snapshot Record 37
GLOBAL-NEGATIVE-RESPONSE 44, 80
GLOBAL-NEG-RESPONSE 113
I/O Controls 55, 56, 57, 58, 91, 92, 93
Identification 21, 50, 51, 52, 84, 85
Implementation State 21
INFO-COMPONENT.....121
INFO-COMPONENT-REF 123
Inheritance 17
Input Output Control Parameter.....30
InputOutputControlByIdentifier..55, 56, 57, 58
InputOutputControlByLocalIdentifier 91, 92
InputOutputControlByLocalIdentifier 93
Internationalization.....13
Layer 17
LengthFormatIdentifier 74, 75
LID.....28, 82, 89
LIDs Readable 28
LIDs Writable 28
Long Name.....14
MATCHING-PARAMETER 21
MEMBER-LOGICAL-LINK 121, 122
Memory 79, 80, 111, 113
Model Category (Vehicles).....122
Name 14, 25, 114
Negative Response 113
NEG-RESPONSE.....44, 80, 113
Network.....17, 18
Network Type 122
NOT-INHERITED 22
NRC 113
OBD.....34
ODX-RS 114
ODX-RS 17, 22, 23, 24
ODX-V 121
OEM 122
OID.....13
Other Status 90
Paragraph (XHTML).....122
PARAM.....113
PDX 121
PHYSICAL-VEHICLE-LINK 121, 122
POS-RESPONSE-SUPPRESSABLE 45, 73, 80
Prefix.....13, 124
Programming Session 47
PROTOCOL.....23, 123
Protocol Layer 18
Read Diagnostic Version (F1A0) 22
Read Identification (80).....22
Read Identification (80).....84
Read Identification (83).....22
Read Identification (83).....85
ReadDataByIdentifier..... 48, 86
ReadDataByLocalIdentifier 82, 84, 85
ReadDtcInformation 60, 95
ReadMemoryByAddress 78, 112
ReportDtcByStatusMask 60, 95
ReportDTCSnapshotRecordByDTCNumber . 61, 65, 97, 100
Request Routine Results ($31) 39, 68
Request Seed ($27)..... 41, 69, 103
RequestDownload 74, 108
RequestTransferExit..... 76, 109
RequestUpload 75, 108
Reset to Default ($2F)..... 57
ResetTypes..... 71, 106
Return Control to ECU ($2F) 56
RFQ Package 17
RID..... 39, 40, 66, 101, 103
Romanization..... 14
RoutineControl 39, 66
Safety System Diagnostic Session..... 47
SecurityAccess 41, 69, 70, 103, 104, 119
Segment Identifier..... 122
Semantic 12
Send Key ($27) 41, 70, 104
Session 26, 27, 28, 47, 81, 118
Short Name 14, 114
Short Term Adjustment ($2F) 32, 55
Snapshot 34, 36, 38, 62, 97
Software Number 22
Solitary PDX 121
SPRMIB..... 45, 73, 80
Snapshot 37
Start Permanent Control 90
Start Routine ($31) 39, 66
Start Temporary Control..... 90
StartDiagnosticSession 81
StartPermanentControl ($30)..... 92
StartRoutineByLocalIdentifier 40
StartRoutineByLocalIdentifier ($31) 40
StartRoutineByLocalIdentifier ($31) 40
StartRoutineByLocalIdentifier ($31) 101
StartTemporaryControl ($30) 91
State Chart 28, 118, 119
STATE-CHART..... 27, 118, 119
Status Request 90
StatusRequest ($30)..... 92
Stop Control 90
Stop Routine ($31)..... 39, 67
StopControl ($30) 93
StopRoutineByLocalIdentifier 40
StopRoutineByLocalIdentifier ($32)..... 40, 103
Supplier Number..... 22

Supplier Specific Affectation 90
Support Check 22, 44, 49, 54, 59, 83, 87, 89, 94, 102
System Supplier ECU Software Number..21, 50
System Supplier ECU Software Version Number.....21, 51
System Supplier Identifier 22, 50
TA_S21.....28
TA_S22.....27, 29, 37
TA_S2E.....27
TA_S30_00.....91
TA_S30_01.....92
TA_S30_11.....93
TA_S30_20.....92
TA_S30_PR.....33, 91, 92, 93
TA_S3B.....28
Tester.....12, 14
TesterPresent 73, 107
TID 13
TransferData 76, 109
Upload Download 74, 75, 76, 77, 108, 109, 110
Usage Scenario (of ECU) 122
Validation..... 12
Vehicle Model Category 122
Vehicle Project..... 122
VEHICLE-CONNECTOR..... 121, 122
VEHICLE-INFORMATION 121, 123
VEHICLE-MODEL..... 122
WriteDataByIdentifier 53, 88
WriteDataByLocalIdentifier 89
WriteMemoryByAddress 79, 112
XHTML..... 15, 122
XML ID..... 13
XML Schema 12

© 2017 Vector Informatik GmbH